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TECHNICAL SPECIFICATION



**Application integration at electric utilities – System interfaces for distribution management –
Part 14: MultiSpeak – CIM harmonization**



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IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland

Tel.: +41 22 919 02 11
Fax: +41 22 919 03 00
info@iec.ch
www.iec.ch

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TECHNICAL SPECIFICATION



**Application integration at electric utilities – System interfaces for distribution management –
Part 14: MultiSpeak – CIM harmonization**

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

ICS 33.200

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CONTENTS

FOREWORD.....	8
INTRODUCTION.....	10
1 Scope.....	12
2 Normative references.....	13
3 Terms, definitions, and abbreviations	14
3.1 Terms and definitions	14
3.2 Abbreviations	14
4 Reference model	15
4.1 General.....	15
4.2 CIM interface reference model.....	15
4.3 MultiSpeak reference model	15
4.4 MultiSpeak-CIM actor mapping	16
5 Document conventions.....	18
5.1 UML diagrams.....	18
6 Integration example	18
6.1 General.....	18
6.2 Message Header Mapping	19
7 Scope of the profile mappings.....	21
7.1 Profile mapping	21
8 1 st edition CIM meter reading to MultiSpeak 4.1.x profile mapping.....	22
8.1 Customer account config	22
8.2 Customer agreement config.....	24
8.3 Customer config	25
8.4 Customer meter data set	27
8.5 End device assets	27
8.6 ComFunction.....	28
8.7 ConnectDisconnectFunction	28
8.8 ElectricMeteringFunction	29
8.9 EndDeviceAsset.....	30
8.10 ReadingType.....	32
8.11 Register	33
8.12 End device controls.....	34
8.13 End device events	35
8.14 End device firmware	36
8.15 Meter asset config.....	38
8.16 Meter asset reading	40
8.17 Meter readings	42
8.18 Meter read schedule.....	44
8.19 Meter service requests	46
8.20 Meter readings	47
8.21 Activity records	47
8.22 ServiceDeliveryPoints	47
8.23 Meter system events	48
8.24 Pricing structure config.....	49
8.25 Receipt record	50
8.26 SDP Location config.....	53

8.27	Service category config	54
8.28	Service delivery point config	55
8.29	Service location config	57
8.30	Supplier config	59
8.31	Transaction record	63
9	IEC 61968-9:2013 to MultiSpeak v.5 (Release Candidate) profile mapping	65
9.1	General	65
9.2	Summary of changes between MultiSpeak Version 4.1.x and Version 5.0	65
9.3	Schema changes	67
9.4	Messaging changes	68
9.5	ExtensionsList	72
9.6	Mapping of IEC 61968-9:2013 to MultiSpeak v.5 (Release Candidate) Profile mapping	73
9.6.1	ActivityRecord	73
9.6.2	Status	74
9.6.3	ConfigurationEvents	74
9.6.4	CustomerAccountConfig	74
9.6.5	CustomerAgreementConfig	74
9.6.6	CustomerConfig	75
9.6.7	EndDeviceConfig	76
9.6.8	EndDeviceControls	80
9.6.9	EndDeviceEvents	81
9.6.10	PAN-related classes	83
Annex A	(normative) Embedded/Reused Class Mappings	87
Annex B	(informative) Example Mapping of MultiSpeak to CIM-based services	89
B.1	On Demand Read: MultiSpeak to CIM integration via ESB	89
B.2	Wiring the Integration	90
Annex C	(informative) UML of MultiSpeak Equivalent profiles	96
Annex D	(normative) MultiSpeak Equivalent CIM Profiles	105
Bibliography		259
Figure 1	– MultiSpeak interface reference model	16
Figure 2	– Systems integration via an ESB (Gray, Simmins, McNaughton, Haynes, Saint, McMorran, Lambert, 2011)	18
Figure 3	– CustomerAccountConfig CIM to MultiSpeak profile mapping	24
Figure 4	– CustomerAgreementConfig CIM to MultiSpeak mapping	25
Figure 5	– IEC 61968-9:2009 schema view of CustomerMeterDataSet profile	27
Figure 6	– CIM EndDeviceControls to MultiSpeak equivalent profile mapping	35
Figure 7	– Mapping of the CIM EndDeviceEvents to MultiSpeak mspEndDeviceEvents profile	36
Figure 8	– EndDeviceFirmware CIM to MultiSpeak profile mapping	38
Figure 9	– MeterAssetConfig CIM to MultiSpeak profile mapping	40
Figure 10	– MeterReadSchedule CIM to MultiSpeak profile mapping	46
Figure 11	– ServiceCategoryConfig profile mapping	55
Figure 12	– ServiceLocationConfig CIM to MultiSpeak mapping	59
Figure 13	– MultiSpeak Network class	59
Figure 14	– Supplier Config profile mapping	63

Figure 15 – Preferred form of request-response messages	69
Figure 16 – Preferred form for "Initiation of an Action Messages"	69
Figure 17 – Preferred form of publish-subscribe messages	69
Figure 18 – CIM and MultiSpeak both support the Names class and subclasses as an alternative means of identifying objects	72
Figure 19 – MultiSpeak extensionsList class, used for adding attributes to a class	73
Figure 20 – Attributes of CIM generic status class	74
Figure 21 – Change to the EndDeviceControl profile which added the EndDeviceControlType class.....	81
Figure 22 – MultiSpeak V5 endDeviceEvent "shorthand" relationship	83
Figure 23 – UML representing MultiSpeak v.5 priceTierStructure classes	86
Figure B.1 – Blank BPEL project canvas that is the starting point for mapping web services and message payloads.....	89
Figure B.2 – Src folder with subfolders for MultiSpeak and CIM resources to be used in the project.	90
Figure B.3 – Creation of a BPEL partner link for the InitiateMeterReadingsService.....	91
Figure B.4 – BPEL palette options.....	91
Figure B.5 – Completing the creation of the BPEL Receive widget	92
Figure B.6 – Creation of the second MultiSpeak partner link to utilize the ReadingChangedNotification service.	93
Figure B.7 – Creating the Reply widget to utilize the ReadingChangedNotification operation	93
Figure B.8 – The BPEL canvas with the Request-Reply pattern 'wired up" to utilize the InitiateMeterReadingsByMeterID and the ReadingChangedNotification operation	94
Figure B.9 – Method invoke property editor	94
Figure B.10 – "Wiring up the service in OpenESB	95
Figure C.1 – ConnectDisconnect portion of the UML of the MultiSpeak equivalent of the EndDeviceAssets profile.....	96
Figure C.2 – UML diagram of MultiSpeak equivalent of CIM EndDeviceAssets profile	97
Figure C.3 – UML diagram of MultiSpeak equivalent to CIM MeterAssetReading profile.....	97
Figure C.4 – UML for MultiSpeak equivalent of the CIM MeterReadings profile.....	98
Figure C.5 – UML of MultiSpeak equivalent of the CIM MeterServiceRequests profile.....	98
Figure C.6 – UML of the MultiSpeak equivalent to the CIM MeterSystemEvents profile	99
Figure C.7 – UML for the MultiSpeak equivalent of the CIM PricingStructureConfig	99
Figure C.8 – UML of the MultiSpeak equivalent of the CIM ReceiptRecord profile	100
Figure C.9 – UML of the MultiSpeak equivalent of the CIM SDPLocationConfig profile.....	101
Figure C.10 – UML for MultiSpeak equivalent of the CIM ServiceDeliveryPointConfig profile.....	102
Figure C.11 – UML of MultiSpeak equivalent ServiceSupplierConfig profile	103
Figure C.12 – UML of MultiSpeak equivalent of CIM TransactionRecord.....	104
Table 1 – Document overview for IEC 61968-14	13
Table 2 – MultiSpeak-CIM actor mapping	17
Table 3 – MultiSpeak and IEC 61968-100 message header mapping.....	19
Table 4 – CustomerAccountConfig correlations.....	22
Table 5 – CustomerAccountConfig Transformations.....	22

Table 6 – CustomerAccountConfig Gaps	23
Table 7 – CustomerAgreementConfig correlations	24
Table 8 – CustomerAgreementConfig transformations	24
Table 9 – CustomerAgreementConfig gaps.....	25
Table 10 – CustomerConfig correlations	26
Table 11 – CustomerConfig transformations	26
Table 12 – CustomerConfig gaps	26
Table 13 – EndDeviceAssets – ComFunction.....	28
Table 14 – EndDeviceEvents – ComFunction gaps	28
Table 15 – EndDeviceAssets – ConnectDisconnect correlations.....	29
Table 16 – EndDeviceAssets – ConnectDisconnect transformations.....	29
Table 17 – EndDeviceAssets – ConnectDisconnect gaps	29
Table 18 – EndDeviceAssets – ElectricMeteringFunction	30
Table 19 – EndDeviceAssets – ElectricMeteringFunction – Transformations.....	30
Table 20 – EndDeviceAssets – ElectricMeteringFunction gaps.....	30
Table 21 – EndDeviceAssets – EndDeviceAsset.....	31
Table 22 – EndDeviceAssets – EndDeviceAsset gaps.....	32
Table 23 – EndDeviceAssets – ReadingType correlations.....	33
Table 24 – EndDeviceAssets – ReadingType gaps	33
Table 25 – EndDeviceAssets – Register gaps.....	33
Table 26 – EndDeviceControls correlations	34
Table 27 – EndDeviceControls transformations	34
Table 28 – EndDeviceControls gaps.....	34
Table 29 – EndDeviceEvents correlations.....	35
Table 30 – EndDeviceEvents transformations.....	35
Table 31 – EndDeviceEvents gaps	36
Table 32 – EndDeviceFirmware correlations	36
Table 33 – EndDeviceFirmware transformations	37
Table 34 – EndDeviceFirmware gaps	37
Table 35 – MeterAssetConfig correlations	38
Table 36 – MeterAssetConfig transformations.....	38
Table 37 – MeterAssetConfig gaps.....	39
Table 38 – MeterAssetReading correlations.....	41
Table 39 – MeterAssetReading transformations.....	41
Table 40 – MeterAssetReading gaps	42
Table 41 – MeterReadings correlations	43
Table 42 – MeterReadings transformations.....	43
Table 43 – MeterReadings gaps.....	44
Table 44 – MeterReadSchedule correlations.....	45
Table 45 – MeterReadSchedule transformations.....	45
Table 46 – MeterReadSchedule gaps	45
Table 47 – MeterServiceRequests correlations	47
Table 48 – MeterServiceRequests transformations	47

Table 49 – MeterServiceRequests gaps	48
Table 50 – MeterSystemEvents correlations	48
Table 51 – PricingStructureConfig correlations	49
Table 52 – PricingStructureConfig transformations	49
Table 53 – PricingStructureConfig gaps.....	50
Table 54 – ReceiptRecord correlations	51
Table 55 – ReceiptRecord transformations	51
Table 56 – ReceiptRecord gaps	52
Table 57 – SDPLocationConfig correlations.....	53
Table 58 – SDPLocationConfig transformations	53
Table 59 – SDPLocationConfig gaps	54
Table 60 – ServiceCategoryConfig correlations	54
Table 61 – ServiceCategoryConfig transformations.....	54
Table 62 – ServiceCategoryConfig gaps.....	54
Table 63 – ServiceDeliveryPointConfig correlations	55
Table 64 – ServiceDeliveryPointConfig transformations	56
Table 65 – ServiceDeliveryPointConfig gaps	57
Table 66 – ServiceLocationConfig correlations	58
Table 67 – ServiceLocationConfig transformations	58
Table 68 – ServiceLocationConfig gaps.....	58
Table 69 – Service supplier correlations	61
Table 70 – ServiceSupplierConfig transformations.....	61
Table 71 – ServiceSupplierConfig gaps	62
Table 72 – TransactionRecord correlations.....	64
Table 73 – TransactionRecord transformations.....	64
Table 74 – TransactionRecord gaps	65
Table 75 – Relationships between Version 4.1 and Version 5.0 present-tense verb names	70
Table 76 – Relationships between Version 4.1 and Version 5.0 past-tense verb names	71
Table 77 – CIM to MultiSpeak Activity mapping	73
Table 78 – CustomerAccountConfig mapping changes for CIM 2nd edition, MSP v.5	75
Table 79 – CustomerConfig mapping changes for CIM 2 nd edition, MSP v.5	76
Table 80 – EndDeviceConfig mapping changes for CIM 2nd edition, MSP v.5.....	77
Table 81 – Mappings for classes added to EndDeviceConfig.....	78
Table 82 – SimpleEndDeviceFunction mapping for CIM 2nd edition to MSP v.5.....	80
Table 83 – CIM Seals class mapping to MSP v.5	80
Table 84 – CIM EndDeviceControlsType class mapping to MSP v.5	81
Table 85 – CIM EndDeviceControlsType class mapping to MSP v.5 with support for PAN classes	81
Table 86 – CIM 2nd edition – MSP v.5 EndDeviceEvent mapping and gaps.....	82
Table 87 – PANDemandResponse mapping from CIM to MSP v.5	84
Table 88 – PANDisplay class mapping	85
Table 89 – PANPricing class mapping	85
Table A.1 – Address Correlations	87

Table A.2 – Phone Number Correlations.....	88
Table A.3 – Service Location related transformations	88
Table A.4 – Electronic Address related transformations	88

INTERNATIONAL ELECTROTECHNICAL COMMISSION

**APPLICATION INTEGRATION AT ELECTRIC UTILITIES –
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Technical specifications are subject to review within three years of publication to decide whether they can be transformed into International Standards.

IEC 61968-14, which is a technical specification, has been prepared by IEC technical committee 57: Power systems management and associated information exchange.

This publication contains attached files in the form of xsd files. These are a copy of the text originally contained in Annex D of this publication. These files are intended to be used as a complement and do not form an integral part of the publication.

The text of this technical specification is based on the following documents:

Enquiry draft	Report on voting
57/1504/DTS	57/1555/RVC

Full information on the voting for the approval of this technical specification can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 61968 series, published under the general title *Application integration at electric utilities – System interfaces for distribution management*, can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

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INTRODUCTION

This part of IEC 61968 seeks to address the need for harmonization between the two leading standards for software interoperability that serve the electric utility industry; MultiSpeak™, which is maintained by the MultiSpeak Initiative, and the common information model (CIM), which is maintained by the International Electrotechnical Commission. While the CIM is an international standard, MultiSpeak has a significant footprint especially among municipal and cooperative utilities and vendors that serve these utilities, particularly in the United States. MultiSpeak is also used in fifteen countries worldwide.

The goal of this technical specification is to harmonize the two standards so that the capability currently inherent in MultiSpeak could eventually be expressed as a profile within the CIM. To that end this particular effort will map V4.1x of MultiSpeak with IEC 61968-9:2009, *Interfaces for meter reading and control* of the CIM. This mapping will occur at the CIM profile level, so that each CIM profile will have a mapping to a corresponding element in MultiSpeak. This will serve two ends; (a) for those that need to translate from interfaces based on one standard to another, this mapping will give a roadmap for how that is accomplished, and (b) this exercise will reveal where alignment currently exists between the two standards and where gaps exist. As each standard goes through successive releases, reviews of business requirements which necessitated the respective releases and updates can be made in both CIM and MultiSpeak to narrow existing gaps.

As part of this exercise this part of IEC 61968 will also give a practical example of the interoperability between the two standards using an archetype of one interface being mapped to another through an enterprise service bus (ESB). This proof of concept will utilize OpenESB as this is open source and freely available. This archetype will cover the on demand meter read use case, where a source application will initiate a request for an on demand read and this request will be served by a destination application. The applications themselves will not be developed; only the interface mapping through an ESB from a MultiSpeak based source to a CIM based destination will be developed.

Multiple standards that cover the same information domain present a problem for the vendor community when developing products, and for the customers that would use these products. The classic challenge becomes one of determining which standards to support or how best to support one or both standards. The problem for the customer is integrating products that follow different standards to work with each other.

As part of a harmonization effort both 1st edition and 2nd edition CIM profiles have been mapped to equivalent MultiSpeak profiles. The reason this approach was used is that the CIM and MultiSpeak models are too large to be compared in their entirety. It was determined that the profile, representing a snapshot of the respective models and something that has been determined to be useful in the exchange of information between systems, would be a better place to begin making comparisons.

The harmonization effort includes mapping of the profiles and an assessment detailing where the profiles correlate, where a transformation needs to occur, e.g. concatenating several fields to map to a single field in the corresponding profile, and where gaps exist.

In addition to the mapping of the profiles, Annex A lists the suggested changes to the respective standards for consideration in future releases.

IEC 61968, taken as a whole, defines interfaces for the major elements of an interface architecture for Distribution Management Systems (DMS). Part 1: Interface Architecture and General Recommendations, identifies and establishes requirements for standard interfaces based on an Interface Reference Model (IRM). Parts 3 to 9 of IEC 61968 define interfaces relevant to each of the major business functions described by the Interface Reference Model.

As used in IEC 61968, a DMS consists of various distributed application components for the utility to manage electrical distribution networks. These capabilities include monitoring and

control of equipment for power delivery, management processes to ensure system reliability, voltage management, demand-side management, outage management, work management, automated mapping and facilities management.

IEC 61968 is limited to the definition of interfaces and is implementation independent. They provide for interoperability among different computer systems, platforms, and languages. Methods and technologies used to implement functionality conforming to these interfaces are considered outside of the scope of IEC 61968; only the interface itself is specified in this standard.

APPLICATION INTEGRATION AT ELECTRIC UTILITIES – SYSTEM INTERFACES FOR DISTRIBUTION MANAGEMENT –

Part 14: MultiSpeak – CIM harmonization

1 Scope

This part of IEC 61968, which is a technical specification, explains how to do a mapping from CIM to MultiSpeak using an XML mapping tool, and the inclusion of an example project that gives practical guidance on the interoperability and mapping of source and destination applications through an ESB using an on demand meter read as the archetype.

This technical specification will also cover the mapping of IEC 61968-9:2009 profiles to equivalent MultiSpeak v.4.1x message profiles.

Each CIM profile will have a corresponding MultiSpeak profile with a prefix of “msp”. For example, CIM EndDeviceEvents matches the MultiSpeak mspEndDeviceEvents.

The following profiles from IEC 61968-9:2009 are harmonized:

AuxiliaryAgreementConfig
CustomerAccountConfig
CustomerAgreementConfig
CustomerConfig
CustomerMeterDataSet
EndDeviceAssets
EndDeviceControls
EndDeviceEvents
EndDeviceFirmware
MeterAssetConfig
MeterAssetReading
MeterReadings
MeterReadSchedule
MeterServiceRequests
MeterSystemEvents
PricingStructureConfig
ReceiptRecord
SDPLocationConfig
ServiceCategoryConfig
ServiceDeliveryPointConfig
ServiceLocationConfig
SupplierConfig
TransactionRecord

Although the profiles will have some changes as part of the IEC 61968-9:2013, the scope of the harmonization effort for the 2nd edition to MultiSpeak v.5 (release candidate), will be

limited to these same profiles. This will give the practitioner an ample set of mapping examples and allow them to make meaningful comparisons of the changes between IEC 61968-9:2009 and 2013 profiles and the respective mapping to MultiSpeak.

This part of IEC 61968 contains the clauses listed in Table 1.

Table 1 – Document overview for IEC 61968-14

Clause	Title	Purpose
1.	Scope	The scope and purpose of the document are described.
2.	References (Normative)	Documents that contain provisions which, through reference in this text, constitute provisions of this International Standard.
3.	Terms, definitions, and abbreviations	Establish the common terms used in this specification.
4.	Reference and Information Models	Description of general approach to metering system, reference model, use cases, interface reference model, meter reading and control functions and components, message type terms and static information model for both CIM and MultiSpeak.
5.	Document Conventions	Message types related to the exchange of information for documents related to maintenance and construction.
6.	Integration example	The main purpose of this standard is to provide guidance for integration. Since both MultiSpeak and CIM use web services for integration, this section addresses the needs to map the message headers as well as the profiles.
7.	Scope of the profile mappings	Enumerates the profiles that are mapped as part of this specification
8.	1st edition CIM Meter Reading to MultiSpeak 4.1.x Profile Mapping	Gives the reader specific guidance on CIM to MultiSpeak mapping.
9.	2 nd edition CIM Meter Reading to MultiSpeak 5.0.x Profile Mapping	Gives the reader specific guidance on CIM to MultiSpeak mapping.
Annex A	EMBEDDED/REUSED CLASS MAPPINGS	Rather than have mappings that appear in several profiles appear over and over, these repeated mappings are located in this annex for reference.
Annex B	Example Mapping of MultiSpeak to CIM-based services	This annex demonstrates one example of how to “wire” MultiSpeak to CIM-based web services through an enterprise service bus.
Annex C	UML of MultiSpeak Equivalent profiles	Provides examples of how to create MultiSpeak equivalents of various CIM profiles.
Annex D	MultiSpeak Equivalent CIM Profiles	This annex provides the XML for the MultiSpeak equivalent of the various CIM Profiles.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-300, *International Electrotechnical Vocabulary – Electrical and electronic measurements and measuring instruments*

IEC 61968-1, *Application integration at electric utilities – System interfaces for distribution management – Part 1: Interface architecture and general recommendations*

IEC TS 61968-2, *Application integration at electric utilities – System interfaces for distribution management – Part 2: Glossary*

IEC 61968-9:2013, *Application integration at electric utilities – System interfaces for distribution management – Part 9: Interfaces for meter reading and control*

IEC 61968-100, *Application integration at electric utilities – System interfaces for distribution management – Part 100: Implementation profiles*

IEC TR 62051, *Electricity metering – Glossary of terms*

IEC 62055-31, *Electricity metering – Payment systems – Part 31: Particular requirements – Static payment meters for active energy (classes 1 and 2)*

ISO 8601:2004, *Data Elements and Interchange Formats – Information Interchange – Representation of Dates and Times*

ISO 4217:2008, *Codes for the representation of currencies and funds.*

MultiSpeak Version 4.1.6

3 Terms, definitions, and abbreviations

3.1 Terms and definitions

For the purposes of this document, the terms and definitions given in given in IEC 60050-300, IEC TS 61968-2, IEC TR 62051, IEC 62055-31 apply.

Where there is a difference between the definitions in this standard and those contained in other referenced IEC standards, then those defined in IEC 61968-2 shall take precedence over the others listed, and those defined in IEC 61968-6 shall take precedence over those defined in IEC 61968-2.

3.2 Abbreviations

AM	Asset Management
AMR	Automated Meter Reading
AMI	Advanced Metering Infrastructure
BPEL	Business Process Execution Language
CIM	Common Information Model
CIS	Customer Information System
DMS	Distribution Management System
DR	Demand Response
DRMS	Demand Response Management System
HAN	Home area network
IEC	International Electrotechnical Commission
LC	Load Control
LMS	Load Management System
MAM	Meter Asset Management
MDM	Meter Data Management
MM	Meter Maintenance

MR	Meter Reading
MS	Metering System
NO	Network Operations
OMS	Outage Management System
POS	Point Of Sale
PAN	Premise Area Network (includes scope of HAN)
RF	Radio Frequency
UML	Unified Modeling Language
VEE	Validating, Editing, and Estimating
WM	Work Management
XSD	XML Schema Definition

4 Reference model

4.1 General

Both CIM and MultiSpeak utilize a reference model that describes the overarching view of their respective systems.

4.2 CIM interface reference model

The CIM interface reference model (IRM) is contained in IEC 61968-1.

4.3 MultiSpeak reference model

The MultiSpeak reference model contains a list of systems that have some logical groupings to represent various core functions within a utility as shown in Figure 1. While the CIM is a reference data model which requires further refinement and development of messaging profiles guidance on how to craft system and application interoperability, MultiSpeak system and application interoperability is fully expressed in a set of services and operations.

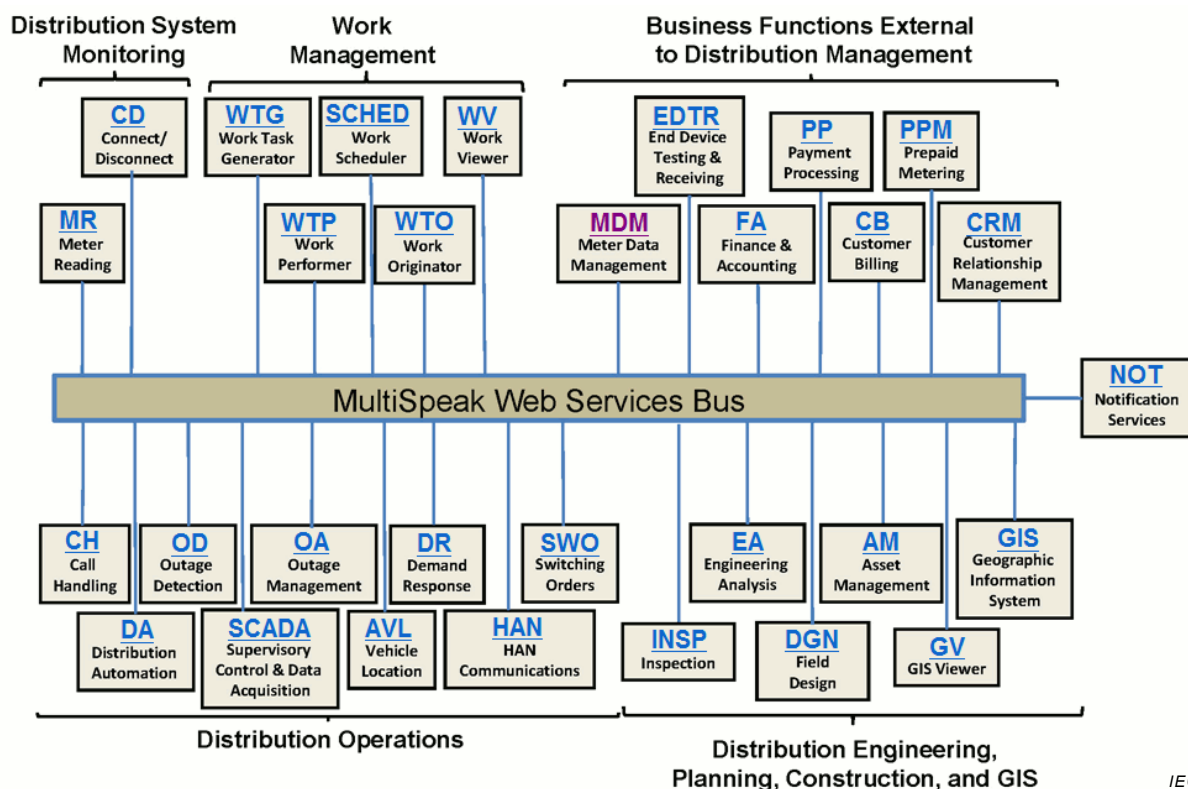


Figure 1 – MultiSpeak interface reference model

Each of the systems shown in the figure represent a service which may have hundreds of operations associated with it. Each service follows the naming convention of the system name with “_server” appended to it, e.g. MDM_server. See the MultiSpeak v.4.1 Technical Specification for more detailed information.

4.4 MultiSpeak-CIM actor mapping

Table 2 maps the MultiSpeak system actors to the respective system actors that are used by the CIM.

Table 2 – MultiSpeak-CIM actor mapping

MultiSpeak	Corresponding CIM	Comment
AM – Asset Management	AM-MAM	AM-MAM is a proposed actor for the future third edition of 61968-1
AVL – Vehicle Location	N/A	
CB – Customer Billing	CS	
CD – Connect/Disconnect	CS	
CH – Call Handling	CS	
CRM – Customer Relationship Management	CS	
DR – Demand Response	MR-LDC	
DA – Distribution Automation		
DGN – Field Design		
EA – Engineering Analysis	NE-NCLC	
EDTR – End Device Testing & Receiving		
FA – Finance & Accounting	CS	
GIS – Geographic Information Systems	AM-GINV	
GV – GIS Viewer		
HAN – HAN Communications	EXT-HAN	
INSP – Inspection		
MDM – Meter Data Management	MDMS	
MR – Meter Reading	MR	
NOT – Notification Services	N/A	The NOT server is a broadcast type of system used to distribute notification messages in a MultiSpeak based environment. Since this tends to be more implementation specific there is no CIM equivalent
OA – Outage Management	NO-OA	OA is a proposed CIM actor for the future third edition of 61968-1
OD – Outage Detection	NO-FLT	
PP – Payment Processing	MR	
PPM – Prepaid Metering	MR	
SCADA – Supervisory Control and Data Acquisition	NO-NMON	
SCHED – Work Scheduler	MC-SCHED	
SWO – Switching Orders	OP-SSC	

MultiSpeak	Corresponding CIM	Comment
WTG – Work Task Generator	MC-FRD	
WTO – Work Originator	CS-TCM	
WTP – Work Performer	MC-CON	
WV – Work Viewer		

5 Document conventions

5.1 UML diagrams

All UML-based sequence diagrams contained herein are to be considered as informative examples of how a message exchange could occur.

NOTE One of the strengths of both CIM and MultiSpeak is its flexibility. As technology advances, and new needs develop, new messages can be created and new logic models to capture the corresponding information. These new messages and models might involve additional systems (not pictured.) These new messages may leverage different options than the ones depicted in the example.

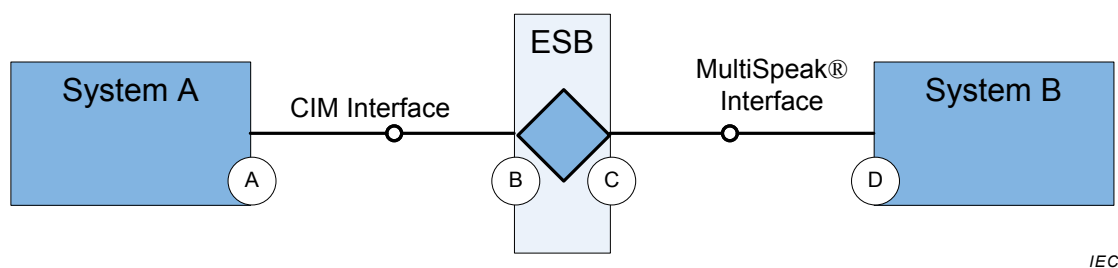
All UML-based communication diagrams and message flow diagrams contained herein are to be considered informative. See Annex D for the UML used to generate MultiSpeak equivalents of CIM-based profiles.

The reader is referred to IEC 61968-1 to locate the document that contains the normative definitions of the classes used in the CIM, or to the MultiSpeak interface reference model.

6 Integration example

6.1 General

The integration example shown in Figure 2 will illustrate how the on demand meter read use case may be realized in a system that must integrate using both MultiSpeak and CIM based interfaces. Figure 2 illustrates this pattern.



IEC

Figure 2 – Systems integration via an ESB (Gray, Simmins, McNaughton, Haynes, Saint, McMorran, Lambert, 2011)

A MultiSpeak interface may be provided on one side of the ESB, while a CIM interface is presented on the other. Within the ESB elements from the each payload must be mapped as well as the error handling from one system to another. For example, if an error occurs at interface B in the diagram from a call initiated by System A, there is a reasonable expectation that the interface will return the type of error that System A understands. However, a greater challenge is when an error is thrown by System B and needs to be propagated through the ESB to the calling system.

6.2 Message Header Mapping

In a message exchange there are attributes that are used to facilitate communication and which are common for every message interaction. Both CIM (based on IEC 61968-100) and MultiSpeak use message headers for their service integration. It is important to understand which attributes are used by each standard and how they correlate or where gaps exist.

In Table 3, N/A is used to indicate that this attribute is not used by the respective specification, e.g. in the table the CIM attribute “Verb” is used to indicate an action to be taken, but this attribute does not exist in the MultiSpeak message header.

The comments column is used to describe either the purpose of a given attribute, or additional information for those attributes that are not a one-to-one match, e.g. MultiSpeak has individual attributes to indicate the different aspects of a version, while the CIM only has a single attribute, thus an operation would need to be performed to map from one to the other.

Table 3 – MultiSpeak and IEC 61968-100 message header mapping

MultiSpeak attribute	CIM attribute	Comments
N/A – The verb is included as part of the operation naming.	Verb	Identifies the action to be taken
N/A – The noun is included as part of the operation naming.	Noun	Identifies the type of payload
MajorVersion MinorVersion Build Branch BuildString	Revision	For a mapping to occur the integer type version attributes need to be changed to string; the resulting attributes can then be concatenated and mapped to Revision
AppSource AppVersion Company	Source	identifying the source of the message, which should be the ID of the system or organization; for MultiSpeak the AppSource, AppVersion and Company may be concatenated and mapped to this field
MessageID	ReplayDetection (Nonce, Created)	A complexType made up of Nonce (a GUID or random number good for at least a day) and Created, a timestamp.
Timestamp	Timestamp	A timestamp generated by the source system to indicate when the message was created.
transactionID – (The transactionID is not included in the message header, rather it is included in the operation definition for those operations for which it is required.	CorrelationID	Can be supplied on a request, so that the client can correlate a corresponding reply message. The server will use the incoming correlation ID on the outgoing reply.
AuditID N/A	User (UserID, Organization)	User is a complex structure that identifies the user and associated organization. Should be supplied as it may be required for some interfaces, depending upon underlying implementations. MultiSpeak AuditID is mapped to UserID

MultiSpeak attribute	CIM attribute	Comments
responseURL – the responseURL is not included in the message header, rather it is included in the definition of the operation for those operations for which it is appropriate.	ReplyAddress	responseURL is not part of the MultiSpeakMsgHeader, but part of the payload, but can be mapped to ReplyAddress
N/A	Comment	A field to add optional information about the service
Context	Context	Indicates the message pattern, e.g. Request, Response
N/A (The necessity of acknowledgement is inherent in the definition of the operations and is not included in the message header).	AckRequired	A flag indicating the an acknowledgement should be sent to the calling system
N/A	Property (Name, Value)	A way to optionally extend the header with other attributes that are not included in IEC 61968-100
UserID	N/A	This is an authentication parameter for the system that has sent the message.
Pwd	N/A	This is an authentication parameter for the system that has sent the message.
DefaultCurrencyCode	N/A	ISO 4217:2008 code for the currency being used in the transaction
CSUnits	N/A	Any of the CS or coordinate system items are only appropriate if geographical data are being sent. In that case it is necessary to send the coordinate system name, CSUnits, and datum must be sent in order to be able to determine how to interpret the x,y,z coordinate data received.
CoordinateSystemName	N/A	
CoordinateSystemAuthority	N/A	
CoordinateSystemAuthority Code	N/A	
Datum	N/A	
SessionID	N/A	The SessionID and PreviousSessionID are for tightly coupled systems, e.g. instance engineering analysis (DMS) and GIS. In this case, it is desirable to batch sets of edits into sessions which can be dealt with as transactions. The PreviousSessionID is then used to track the progress of a transaction and throwing exceptions if a particular transaction arrives out of sequence. This is necessary in installations where delivery is not guaranteed. MultiSpeak does not assume messaging middleware is in place to guarantee delivery.
PreviousSessionID	N/A	
ObjectsRemaining	N/A	The LastSent, ObjectsRemaining along with a calling parameter on some methods, lastReceived, enables two systems to coordinate chunking of data payloads. The sending system sends a chunk of data along with the LastSent header attribute and optionally an ObjectsRemaining. The receiver returns the value that it received in the LastSent in its subsequent call by putting that value in the lastReceived calling parameter. This goes back and forth until the sender returns a set of data with ObjectsRemaining set to 0.
LastSent	N/A	

MultiSpeak attribute	CIM attribute	Comments
RegistrationID	N/A	The identifier for an instance of a subscription for service. A potential subscriber applies to a server for a registrationID, which the subscriber uses to specify the data being requested. Subsequent data published pursuant to the subscription for service carries the registrationID so that the subscriber knows under which of potentially many subscriptions it is receiving this data.
AuditPassword	N/A	Used in conjunction with AuditID in MultiSpeak web services to authenticate a human user of an automation service and to provide audit logging capabilities for actions taken by a specific user.
Any Attribute	Any Element	

7 Scope of the profile mappings

7.1 Profile mapping

The profile mapping begins with the message profiles available from IEC 61968-9:2013, Annex H which lists 23 standard profiles. A decision was made to limit the scope of this initial effort to some well-defined profiles that were used for application integration (which happens to be the primary intent of the MultiSpeak standard) as a way to determine key correlations. This decision was the result of attempting to map both models as ontologies in their entirety using web ontology language (OWL). An effort that was hampered at the time by the complexity of both ontologies, and the limitations of the reasoning available in the OWL tool. By limiting the scope to meter reading and control it was felt that this smaller subset would give a better indication, a “sample” of correlations between both standards, by breaking the harmonization effort into smaller pieces for analysis.

The following 61968-9:2013 profiles are evaluated as part of this effort:

AuxiliaryAgreementConfig
 CustomerAccountConfig
 CustomerAgreementConfig
 CustomerConfig
 CustomerMeterDataSet
 EndDeviceAssets
 EndDeviceControls
 EndDeviceEvents
 EndDeviceFirmware
 MeterAssetConfig
 MeterAssetReading
 MeterReadings
 MeterReadSchedule
 MeterServiceRequests
 MeterSystemEvents
 PricingStructureConfig
 ReceiptRecord
 SDPLocationConfig

ServiceCategoryConfig
ServiceDeliveryPointConfig
ServiceLocationConfig
SupplierConfig
TransactionRecord

As MultiSpeak v.4.1.x has only a single large XSD for reference which makes mapping to a single profile difficult, a corresponding MultiSpeak profile has been created to correspond to each CIM profile.

All translation mapping diagram contained herein are to be considered as normative examples of how mapping from CIM-based profiles, to MultiSpeak profiles could occur.

The reader should refer to the appropriate references for normative information regarding the definition of the respective IEC 61968-9 and MultiSpeak 4.1 standards.

8 1st edition CIM meter reading to MultiSpeak 4.1.x profile mapping

8.1 Customer account config

The correlations, transformations and gaps related to the CustomerAccountConfig profile are included in Tables 4, 5, and 6.

Table 4 – CustomerAccountConfig correlations

CIM	MultiSpeak	Comments
mRID	customerID	customerID is type ObjectID
budgetBill	budgetBill	Type:String in both cases
billingCycle	billingCycle	Type:String in both cases
description	Description	By using the mspCPSM:IdentifiedObject class in MultiSpeak the high level attributes from CIM can be mapped
name	Name	

Table 5 – CustomerAccountConfig Transformations

CIM	MultiSpeak	Comments
aliasName	objectName	
docStatus status	statusCode	statusCode is not normally associated with the account class. While there is no equivalent of docStatus in MultiSpeak, statusCode could have the CIM docStatus and Status classes concatenated and mapped to it.

Table 6 – CustomerAccountConfig Gaps

CIM	MultiSpeak	Comments
category		Does not appears in the CIM project. It may be ServiceCategory which is the type of service the customer uses.(14v11-10v25 and 15v13-11v-06)
lastModifiedDateTime		From the Document class
title		From the Document class
	accountsReceivable	A/R information for this account
	billDate	Date of last bill
	billingTerms	Describes the way that the account is billed. Suggested enumeration is Prepay, Budget, Cycle, or Other.
	calculatedCurrentBillAmount	Amount that the customer would owe if final billed today. This amount is based on the fixed charges, energyusage, fuel costs and taxes since the last bill.
	calculatedCurrentBillDateTime	Date and time that the calculated current bill amount was calculated.
	calculatedCurrentBillReadings	
	calculatedUsedYesterday	The value (expressed in currency) in energy, fuel costs, taxes, and prorated fixed charges the customer used yesterday. It may simply be the calculatedCurrentBillAmount from today minus the calculated current bill amount from yesterday.
	isPrePay	If the field is missing or false then the account is cycle or budget billed.
	lastBillAmount	The amount of the last bill to the customer.
	lastPaymentAmount	Amount of the last payment made.
	paymentDueDate	Date and time that the next payment is due.
	shutOffDate	Shut off date for non-payment.

Figure 3 shows a graphical mapping example, from the CIM CustomerAccountConfig profile, to the MultiSpeak equivalent.

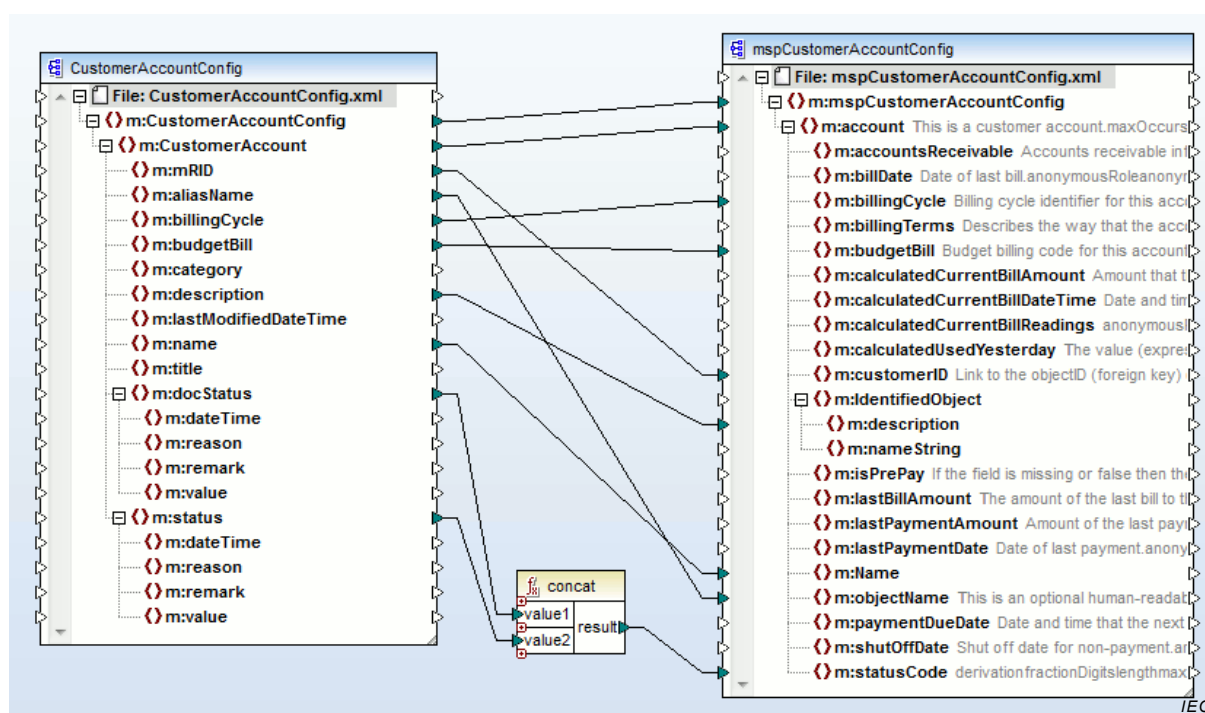


Figure 3 – CustomerAccountConfig CIM to MultiSpeak profile mapping

8.2 Customer agreement config

There are significant gaps in this particular profile as MultiSpeak does not support the generic document or agreement structures. However, customer, account, and service supplier (see ServiceSupplierConfig) can be linked. Tables 7 and 8 show the correlations and transformations, respectively, for the CustomerAccountConfig mapping to the MultiSpeak equivalent profile.

Table 7 – CustomerAgreementConfig correlations

CIM	MultiSpeak	Comments
Customer .mRID	Customer .objectID	
CustomerAccount .mRID	Account .objectID	

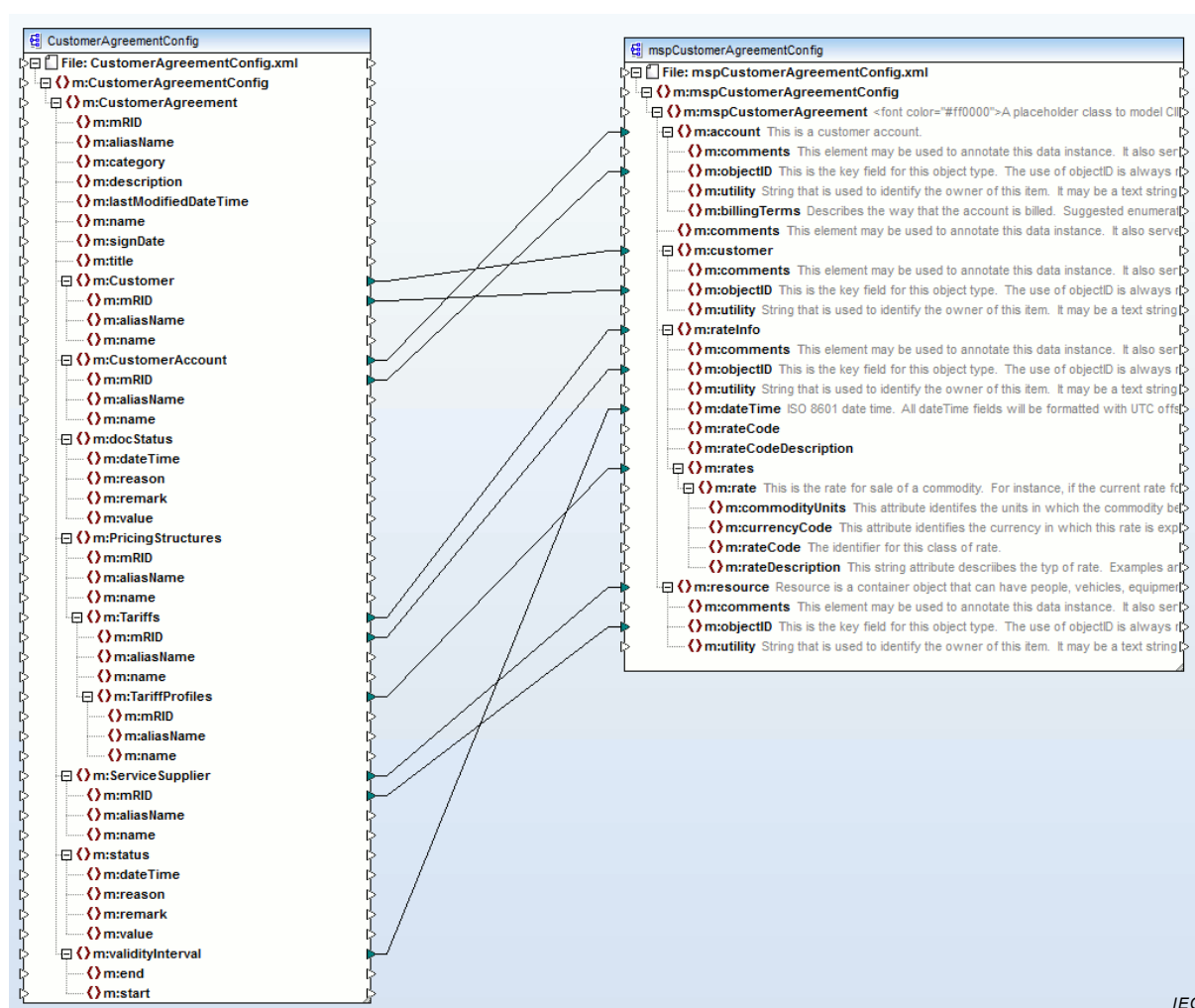
Table 8 – CustomerAgreementConfig transformations

CIM	MultiSpeak	Comments
Tariffs .mRID	rateInfo .objectID	
TariffProfiles .mRID	rates .objectID	
ServiceSupplier .mRID	resource .objectID	As noted in the ServiceSupplierConfig, this assumes an association is added from ContactInfo to resource to create an equivalent ServiceSupplierConfig
validityInterval	dateTime	Not a good match as validityInterval is for the length of the CustomerAgreement (start, end), while dateTime applies to a specific rateInfo attribute

Table 9 – CustomerAgreementConfig gaps

CIM	MultiSpeak	Comments
docStatus		This inherits from the generic status class and applies specifically to documents
PricingStructure		
Status		
	commodityUnits currencyCode rateCode rateCodeDescription	Attributes of rates that provide additional information that is not included in CIM
	billingTerms	Provides additional information about how an account is billed

Figure 4 shows a graphical mapping example, from the CIM CustomerAgreementConfig profile, to the MultiSpeak equivalent.

**Figure 4 – CustomerAgreementConfig CIM to MultiSpeak mapping**

8.3 Customer config

Tables 10, 11, and 12 show the correlations, transformations, and gaps, respectively, between the CIM CustomerConfig profile, and the MultiSpeak equivalent.

Table 10 – CustomerConfig correlations

CIM	MultiSpeak	Comments
Customer.mRID	Customer.objectID	
ElectronicAddresses email	eMailAddress.email	

For Address and Phone Number correlations, see Annex A.

Table 11 – CustomerConfig transformations

CIM	MultiSpeak	Comments
Customer .specialNeed .vip	Customer .specialNeeds	specialNeed and vip would need to be concatenated and mapped to specialNeeds
Customer.kind	revenueClass	Issue with this CIM profile in that normally you would need to drill down into a specific account to determine the kind
ElectronicAddresses .lan .name .password .radio .userID .web	otherContactInformation otherContactItem .details .infoType	All of the CIM attributes are optional, and can be mapped to otherContactItem.details, or concatenated if multiple attributes are populated

Table 12 – CustomerConfig gaps

CIM	MultiSpeak	Comments
pucNumber		
buildingName		
code		
name		
withinTownLimits		
Status class		
streetAddress		Nominally not a gap in that MultiSpeak allows for multiple addresses to be stored per the addressList; it can support many addresses instead of only
	detailedAddressFields .region	
townDetail.name		
townDetail.section		

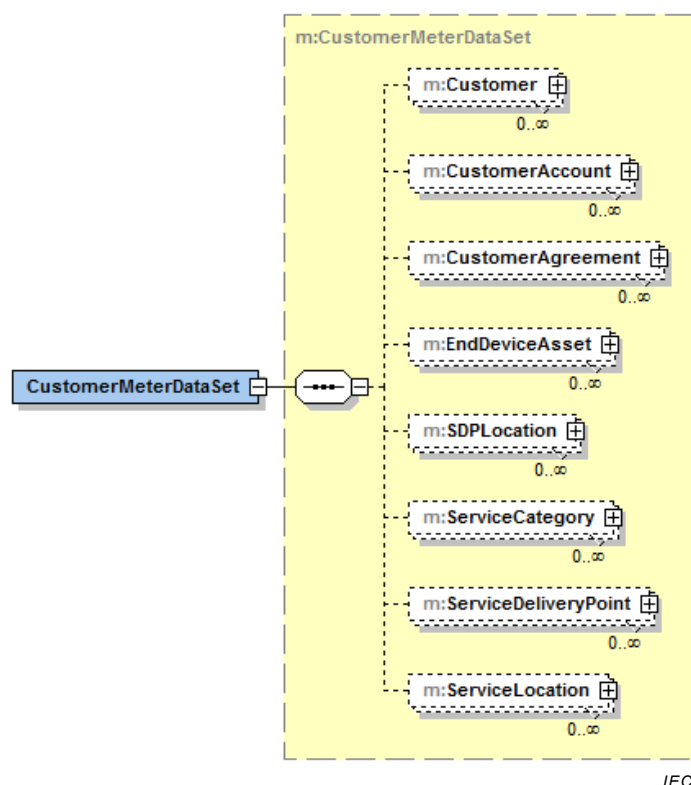
For phone number mappings see Table A.4, *Electronic Address related transformations*.

8.4 Customer meter data set

The CustomerMeterDataSet profile is an aggregation of several other profiles as can be seen in Figure 5.

Each of the high level structures shown in the schema view below are profiles themselves. Rather than replicate the mapping here the reader is referred to the respective paragraph for each profile.

Customer	– Customer Config
CustomerAccount	– Customer Account Config
CustomerAgreement	– Customer Agreement Config
EndDeviceAsset	– End device Assets
SDPLocation	– SDP Location Config
ServiceCategory	– Service Category Config
ServiceDeliveryPoint	– Service Delivery Point Config
ServiceLocation	– Service Location Config



IEC

Figure 5 – IEC 61968-9:2009 schema view of CustomerMeterDataSet profile

8.5 End device assets

There are six sections in the CIM EndDeviceAssets profile:

- ComFunction
- ConnectDisconnectFunction
- ElectricMeteringFunction
- EndDeviceAsset

- ReadingType
- Register

Due to the size of this profile the component parts will be examined individually.

8.6 ComFunction

In the CIM End Devices are not only meters, but other types of devices in the distribution network that have communication capability. This subclause details that capability. Tables 13 and 14 below show the correlations and gaps, respectively for the ComFunction portion of the EndDeviceAssets profile.

Table 13 – EndDeviceAssets – ComFunction

CIM	MultiSpeak	Comments
ComFunction	Module	
.firmwareID	.firmwareVersion	

The strength of the ComFunction in CIM is that it has several attributes that more fully define a communications function where MultiSpeak only has firmwareID. However, MultiSpeak uses a moduleList relationship to module which allows MultiSpeak to associate many modules (communications functions) with a given asset where in CIM it is limited to a single communications function per asset.

Table 14 – EndDeviceEvents – ComFunction gaps

CIM	MultiSpeak	Comments
mRID		Unique identifier for the ComFunction (or MultiSpeak module)
amrRouter		
configID		
hardwareID		
password		
programID		
twoWay		
	moduleList	In the EndDeviceAssets profile CIM has o to many ComFunction where in MultiSpeak there are o to many moduleLists which in turn have o to many modules

8.7 ConnectDisconnectFunction

This is the functionality that allows the meter (or other End Device) to be remotely connected or disconnected, or in the case of meters, to enable load limiting model.

For the purposes of mapping the ConnectDisconnect portion of this CIM profile, an abstract class mspConnectDisconnect was added to the UML to hold the many connect/disconnect or power limiting related classes. Tables 15, 16, and 17 shows the correlations, transformations, and gaps, respectively, for the EndDeviceAssets ConnectDisconnect section of the profile.

Table 15 – EndDeviceAssets – ConnectDisconnect correlations

CIM	MultiSpeak	Comments
ConnectDisconnectFunction .disabled	CDState .RCDState	

Table 16 – EndDeviceAssets – ConnectDisconnect transformations

CIM	MultiSpeak	Comments
ConnectDisconnectFunction .mRID	mSPConnectDisconnect .objectID	
rCDInfo .customerVoltageLimit .energyLimit .powerLimit	powerLimitationValue .units	Typically only one of these CIM attributes needs to be set at a time, therefore the inputs can use a logical OR to map to the MultiSpeak attribute

Table 17 – EndDeviceAssets – ConnectDisconnect gaps

CIM	MultiSpeak	Comments
eventCount		
isConnected		
isDelayedDiscon		
isLocalAutoDisconOp		
isLocalAutoReconOp		
isRemoteAutoDisconOp		
isRemoteAutoReconOp		
rCDInfo .armedTimeOut .energyUsageStartDateTime .isArmConnect .isArmDisconnect .isEnergyLimiting .needsPowerLimitCheck .needsVoltageLimitCheck .usePushButton		

8.8 ElectricMeteringFunction

This portion of the profile defines the set of electric metering functions of the end device asset. Tables 18, 19, and 20 show the correlations, transformations, and gaps for the ElectricMeteringFunction portion of the EndDeviceEvents profile.

Table 18 – EndDeviceAssets – ElectricMeteringFunction

CIM	MultiSpeak	Comments
ElectricMeteringFunction .mRID .firmwareID	electricMeter .MeterNo .meterologyFirmware_ Revision	
.currentRating .demandMultiplier .voltageRating	electricNameplate .maxAmperage .demandMult .elementsVoltage	

Table 19 – EndDeviceAssets – ElectricMeteringFunction – Transformations

CIM	MultiSpeak	Comments
ElectricMeteringFunction .disabled	electricMeter .meterConnectionStatus	
.kWhMultiplier .kWMultiplier	electricNameplate .multiplier	Logical OR the CIM attributes to the MultiSpeak multiplier attribute

Table 20 – EndDeviceAssets – ElectricMeteringFunction gaps

CIM	MultiSpeak	Comments
ElectricMetering_ Function .billingMultiplier .billingMultiplier_ Applied .configID .currentRating .demandMultiplier_ Applied .hardwareID .password .programID .transformerCTRatio .transformerRatios_ Applied .transformerVTRatio		
	electricMeter .meterType	

8.9 EndDeviceAsset

MultiSpeak's hierarchy of classes has a different focus, more meter centric where the CIM has generic classes. This is apparent in a CIM profile that uses ElectricMeterinFunction that is mapped to the MultiSpeak electricMeter. However, the generic EndDeviceAsset also maps to

the MultiSpeak electricMeter. MultiSpeak does not have a generic “end device” class so an end device must be mapped to the closest match, electricMeter. Tables 21 and 22 show the correlations and gaps, respectively, for the EndDeviceAsset portion of the EndDeviceAsset profile.

Table 21 – EndDeviceAssets – EndDeviceAsset

CIM	MultiSpeak	Comments
EndDeviceAsset	electricMeter	meterNo is of type objected
.mRID	.meterNo	
.amrSystem	.amrVendor	
.category	.amrDeviceType	
.description	.description	description class can be associated with the electricMeter class
.formNumber	.form	
.installationDate	.installedDate	
.serialNumber	.serialNumber	
	electricNamePlate	
.kH	.kh	
.kR	.kr	

For the ActivityRecords mapping, see 8.13.

Table 22 – EndDeviceAssets – EndDeviceAsset gaps

CIM	MultiSpeak	Comments
EndDeviceAsset .demandResponse .dstEnabled .loadControl .lotNumber .manufacturedDate .metrology .outageReport .purchasePrice .readRequest .relayCapable .reverseFlowHandling .timeZoneOffset		
	electricMeter .billingCycle .catalogNumber .manufacturer .meterType .meterStatusList .meterologyFirmware_ Revision .removedDate .meterBaseID	CIM has a generic status class, but this is one instance where it is not in a CIM profile
	electricNamePlate .accuracyClass .baseType .dials .frequency .numberOfElements .phases .regRatio .supplyVoltage .testAmperage .wires	

8.10 ReadingType

This subclause will cover the reading types associated with the end device assets profile. Tables 23, 24, and 25 show the correlations, transformations, and gaps, respectively, of the ReadType portion of the EndDeviceAssets profile.

Table 23 – EndDeviceAssets – ReadingType correlations

CIM	MultiSpeak	Comments
ReadingType .mRID .defaultQuality .defaultValueDataType .dynamicConfiguration .unit .channelNumber .reverseChronology	readingType .objectID .defaultQuality .defaultValueDataType .dynamicConfiguration .units .channel .forewardChronology	Logical NOT to map these attributes
.intervalLength .name	intervalProfile .intervalLength .name	
.kind .multiplier	readingTypeCode .measurementCategory .unitsMultiplier	

Table 24 – EndDeviceAssets – ReadingType gaps

CIM	MultiSpeak	Comments
	readingType .measTypeCategory .measurementType	
	intervalProfile .profileID	
	readingTypeCode .flowDirection .name	

8.11 Register

Table 25 – EndDeviceAssets – Register gaps

CIM	MultiSpeak	Comments
Register .mRID .leftDigitCount .rightDigitCount		
	registers .tier	

8.12 End device controls

There is a high degree of correlation between the CIM EndDeviceControls and the equivalent classes and attributes in MultiSpeak. Tables 26, 27, and 28 show the correlations, transformations, and gaps, respectively, of the EndDeviceControls profile.

Table 26 – EndDeviceControls correlations

CIM	MultiSpeak	Comments
mRID	objectID	customerID is type ObjectID
Description	description	Type:String in both cases
type	deviceType	Type:String in both cases
EndDeviceAsset:mRID	mSPDevice:objectID	
name	Name	MultiSpeak uses a CIM equivalent identifiedObject class to pass attributes such as Name

Table 27 – EndDeviceControls transformations

CIM	MultiSpeak	Comments
demandResponseProgram	demandResponseEvent	The challenge is the intent of each of these parent classes. One is pre-event, one is post-event.
scheduledInterval:start scheduledInterval:end	demandResponseEvent .duration	Nominally start/end can be used to calculate a duration but the question goes back to the intent of the information being captured
mRID	loadManagementID	loadManagementID is of type objectID
EndDeviceGroup	loadSection	The challenge here is a group of EndDevices versus what is an area of load within the distribution grid
EndDeviceGroup.mRID	loadGroupID	loadGroupID is of type objectID
groupAddress	loadGroup	CIM groupAddress is an integer, MultiSpeak loadGroup is a string

Table 28 – EndDeviceControls gaps

CIM	MultiSpeak	Comments
DemandResponseProgram .type		MultiSpeak has classes for demandResponse events, not programs
drProgramLevel		Level of a demand response program request, where 0=emergency.
drProgramMandatory		Boolean indicating whether the request is mandatory
priceSignal	ZigbeePublishPrice	There is not a direct correlation between the CIM attribute and the MultiSpeak class although nominally priceSignal (floatQuantity) might be transformed to rateLabel (string)

Figure 6 shows a graphical XSD mapping example of the EndDeviceControls profile.

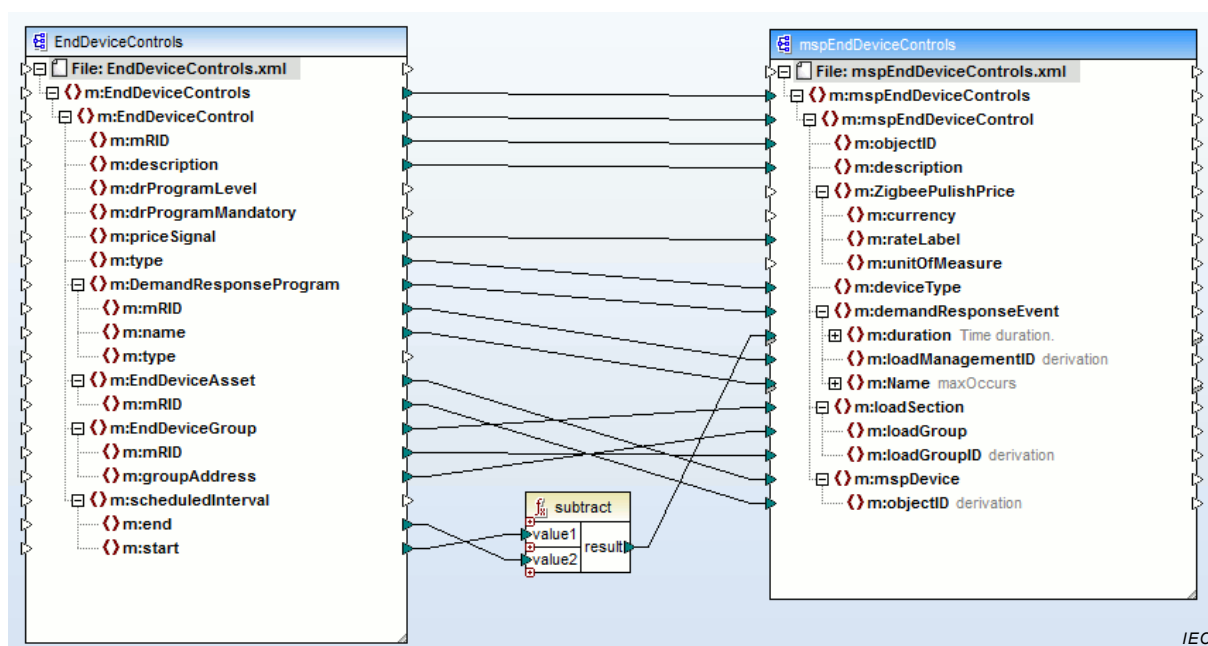


Figure 6 – CIM EndDeviceControls to MultiSpeak equivalent profile mapping

8.13 End device events

The biggest challenge with this profile is that CIM EndDeviceEvents is an abstraction meant to serve any “end device” whereas the event class in MultiSpeak is associated with metering. The other challenge is that MultiSpeak uses a wide variety of classes and enumerations to document different types of status depending on the context where CIM uses a generic, compound status class that can be in multiple contexts.

Another challenge is that there is not a good fit for CIM category attribute, although MultiSpeak's outageEventStatus appears that it could be leveraged to serve this purpose, however there is normally no relation in MultiSpeak between an "event (for metering) and an outageEvent which is normally associated with outage management.

Tables 29, 30, and 31 show the correlations, transformations, and gaps, respectively, of the EndDeviceEvents profile mapping.

Table 29 – EndDeviceEvents correlations

CIM	MultiSpeak	Comments
mRID	objectID	
argument	argument	
userID	userID	
timeStamp	dateTime	
reason	comments	MultiSpeak comments normally does not have an association to event

Table 30 – EndDeviceEvents transformations

CIM	MultiSpeak	Comments
category	outageEventStatus	No direct correlation but some attributes in the outageEventStatus might be able to be mapped to category

Table 31 – EndDeviceEvents gaps

CIM	MultiSpeak	Comments
severity		No MultiSpeak equivalent

Figure 7 shows a graphical XSD mapping example of the EndDeviceEvents profile.

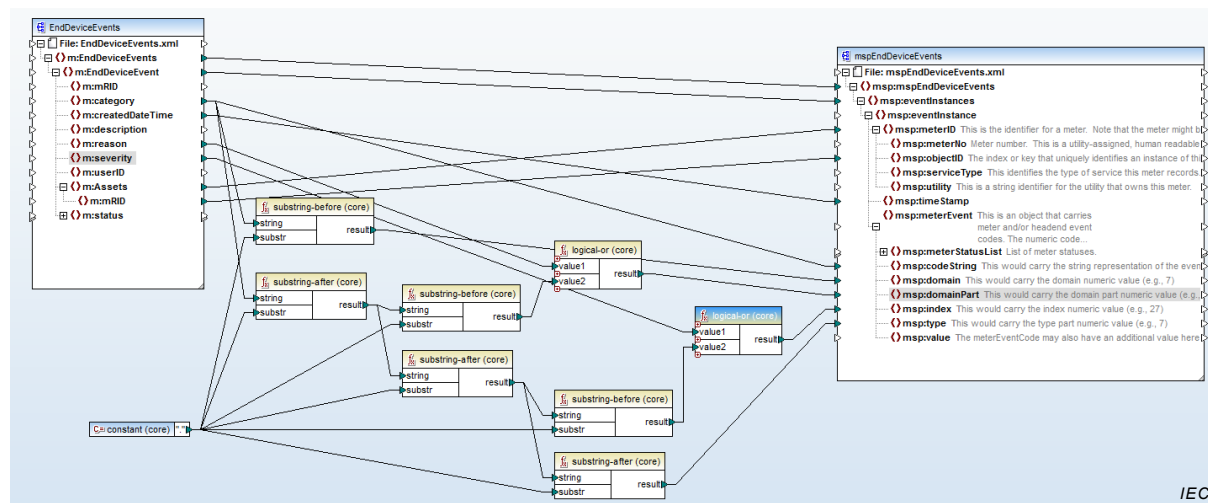


Figure 7 – Mapping of the CIM EndDeviceEvents to MultiSpeak mspEndDeviceEvents profile

8.14 End device firmware

The EndDeviceFirmware profile captures information about the particular functions that a device might have. This closely correlates to the MultiSpeak concept of Modules as shown in Tables 32, 33 and 34.

Table 32 – EndDeviceFirmware correlations

CIM	MultiSpeak	Comments
EndDeviceAsset	mspMeter	
mRID	objectID	
amrSystem	AMRVendor	
serialNumber	serialNumber	

Table 33 – EndDeviceFirmware transformations

CIM	MultiSpeak	Comments
DeviceFunctions	Module	MultiSpeak uses a parent class moduleList that allows for 1..* modules
mRID	objectID	
Category	moduleType	
ActivityRecords	mspAlarm	There is not a good match between these two classes although the intent appears to be similar
createdDateTime	eventTime	
reason severity category	eventCode	MultiSpeak has a specific code definition for events that are generated, however, it appears that the CIM attributes could be concatenated and mapped to this.
application configID disabled hardwareID password programID	description	While MultiSpeak allows for more than one instance of a module, the CIM DeviceFunction class has more granular information about a given DeviceFunction. However, these attributes can be concatenated and mapped to the MultiSpeak description attribute.
category	moduleType	Category of DeviceFunction is equivalent to moduleType of module

Table 34 – EndDeviceFirmware gaps

CIM	MultiSpeak	Comments
Status		CIM has a generic status class associated with the Activity records; no such status equivalent exists with mspAlarm
	sourceIdentifier	MultiSpeak has an additional identifier field for the device that has caused the alarm (CIM activity).

Figure 8 shows a graphical XSD mapping example of the EndDeviceFirmware profile.

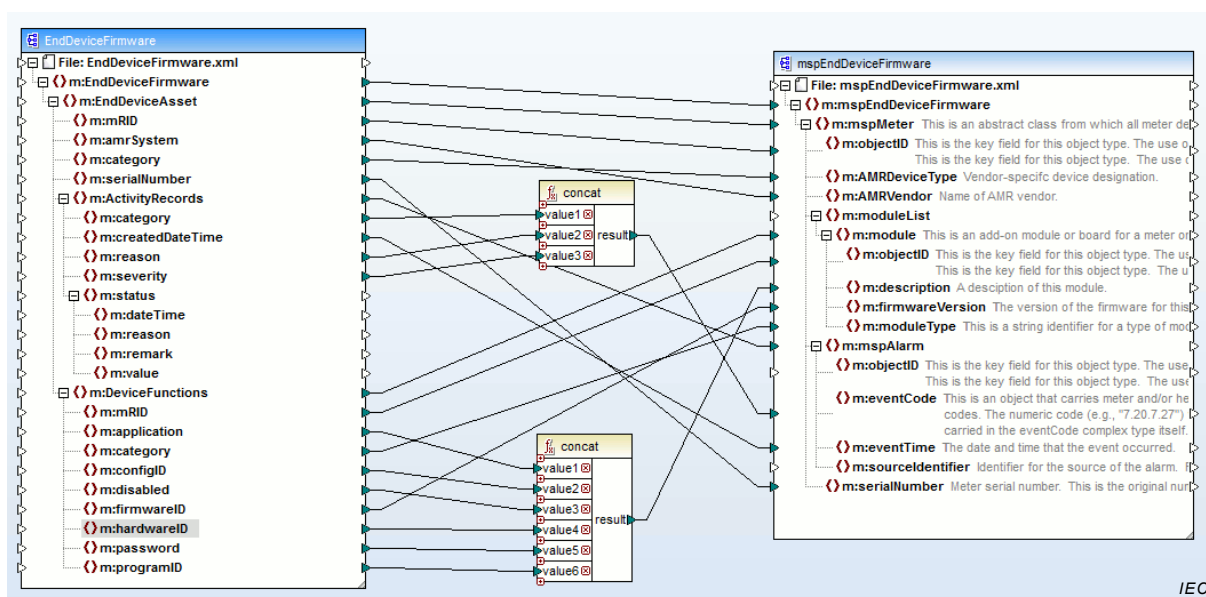


Figure 8 – EndDeviceFirmware CIM to MultiSpeak profile mapping

8.15 Meter asset config

The configuration information about a MeterAsset including its capabilities and in the CIM, Seals associated with the Meter. Tables 35, 36, and 37 show the correlations, transformation, and gaps, respectively, of the MeterAssetConfig profile mapping.

Table 35 – MeterAssetConfig correlations

CIM	MultiSpeak	Comments
MeterAsset mRID category installedDate serialNumber amrSystem	electricMeter objectID AMRDeviceType installedDate serialNumber amrVendor	
kH kR	electricNameplate kH kR	
ServiceLocation mRID	serviceLocation customerID	
sealNumber	sealNumber	

Table 36 – MeterAssetConfig transformations

CIM	MultiSpeak	Comments
None	None	

Table 37 – MeterAssetConfig gaps

CIM	MultiSpeak	Comments
Seals condition kind description appliedDateTime		In addition to sealNumber (which correlates to MultiSpeak sealNumber), CIM provides addition attributes that indicate when the seal was applied and enumerations for the seal condition and the type (kind) of seal that was applied
MeterAsset application demandResponse disconnect dstEnabled formNumber initialCondition initialLossOfLife loadControl lotNumber manufacturedDate metrology outageReport readRequest relayCapable reverseFlowHandling timeZoneOffset utcNumber	electricMeter billingCycle catalogNumber manufacturer meterNo meterStatusList meterType metrologyFirmwareRevision metrologyFirmwareVersion removedDate	
Status		There is no generic Status class in MultiSpeak

Figure 9 is a graphical XSD mapping example for the MeterAssetConfig profile.

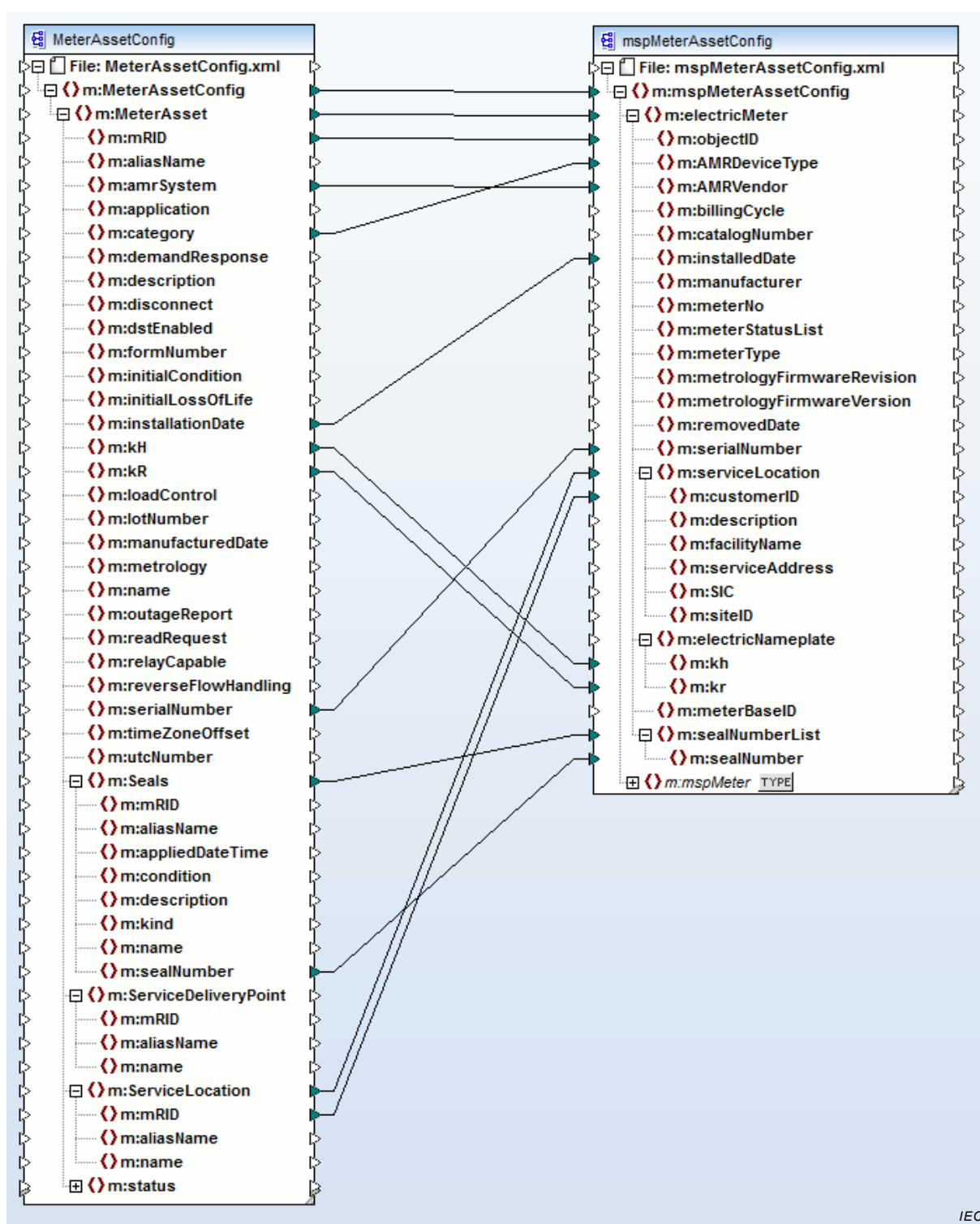


Figure 9 – MeterAssetConfig CIM to MultiSpeak profile mapping

8.16 Meter asset reading

Tables 38, 39, and 40 show the correlations, transformations, and gaps, respectively for the MeterAssetReading profile.

Table 38 – MeterAssetReading correlations

CIM	MultiSpeak	Comments
MeterAsset.mRID .serialNumber	electricMeter.objectID .serialNumber	
MeterReadings	MeterReading	
ReadingType.mRID	readingType.objectID	

MeterAssetReading uses address attributes that can be found in Annex B.

Table 39 – MeterAssetReading transformations

CIM	MultiSpeak	Comments
Readings .timestamp .value	readingValues readingValue.value	
ReadingType	readingType	
ReadingQualities .quality	readingType .defaultQuality	

MeterAssetReading uses Service Location transformations which can be found in Annex A.

Table 40 – MeterAssetReading gaps

CIM	MultiSpeak	Comments
MeterServiceWork .kind .priority .requestDateTime		
Status		
	readingType .channel .defaultValueDataType .dynamicConfiguration .forwardChronology .measTypeCategory .measurementType .units	
	readingTypeCode .accumulationBehavior .dataQualifier .displayableUOM .flowDirection .measurementCategory .name .phaseIndex .timeAttribute .unitsMultiplier .UOMCategory	

8.17 Meter readings

The CIM MeterReadings profile contains within it two main structures, MeterReading and ReadingType. Additionally, within the MeterReading there is also information about EndDeviceEvents, CustomerAgreement which identifies the customer, the MeterAsset which identifies the meter, IntervalBlocks, the Readings, and finally, the ServiceDeliveryPoint of the MeterReadings.

For the EndDeviceEvent mappings see 8.13.

In the CIM, MeterAsset has a status where in MultiSpeak the meter is identified first then the object carries the status.

Tables 41, 42, and 43 show the correlations, transformations, and gaps, respectively to map the MeterReadings profile.

Table 41 – MeterReadings correlations

CIM	MultiSpeak	Comments
ReadingType .mRID .channelNumber .defaultQuality .defaultValueDataType .dynamicConfiguration .multiplier .name .unit	readingType .objectID .channel .defaultQuality .defaultValueDataType .dynamicConfiguration .unitsMultiplier .name .units	
Readings .timestamp .value	readingValues .timestamp .value	

Table 42 – MeterReadings transformations

CIM	MultiSpeak	Comments
CustomerAgreement .mRID	customerID	A customerID in MultiSpeak is of type objectID, which can be used to identify a given customer
intervalLength	timeAttribute	
kind	measTypeCategory	
reverseChronology	forwardChronology	The intent is to indicate chronology, one is simply the inverse of the other
ReadingQualities .quality	readingStatusCode .codeCateogry .codeIndex .originatingSystemID	These correspond to IEC 61968-9:2013, Annex C
IntervalBlocks	blocks	

Table 43 – MeterReadings gaps

CIM	MultiSpeak	Comments
	readingTypeList	MultiSpeak has the capability through the list class, to associate many readingTypes.
Pending .multiplyBeforeAdd .offset .scalarDenominator .scalarFloat .scalarNumerator		
	intervalData .comment .intervalDelimiter .statusDelimiter	
	profiles	
	intervalProfile .intervalLength .name .profileID	
	channels	
	intervalChannel .fieldDescription .fieldName .index .units	

While both CIM and MultiSpeak handle interval blocks, they appear to be handled differently. CIM has the Pending class and then intervalReadings with values, timestamps and associated ReadingQualities, where MultiSpeak has many more attributes for intervalData, profiles, and channels.

8.18 Meter read schedule

This profile creates a schedule and in the CIM, has a customer account, customer agreement, end device asset, or end device group associated with it.

Tables 44, 45, and 46 show the correlations, transformations, and gaps, respectively, for the mapping of the MeterReadSchedule profile.

Table 44 – MeterReadSchedule correlations

CIM	MultiSpeak	Comments
CustomerAccount mRID	account objectID	
EndDeviceAsset mRID	mspMeter objectID	
EndDeviceGroup groupAddress	meterGroupList meterGroupID	In MultiSpeak the meterList can also be used to define which meters belong to a group
ReadingType mRID name	readingType readingTypeCode name	
TimeSchedule offset disabled	readingSchedule offset isEnabled	

Table 45 – MeterReadSchedule transformations

CIM	MultiSpeak	Comments
TimeSchedule recurrencePattern recurrencePeriod	readingSchedule periodicSchedule period timeUnits startTime	

Table 46 – MeterReadSchedule gaps

CIM	MultiSpeak	Comments
	serviceLocation	In MultiSpeak a serviceLocation can be associated with the readingSchedule which may provide important information about the area where the readings are occurring
window end start		The allowable window of time that the reading can occur
TimePoints absoluteTime relativeTimeInterval sequenceNumber		
CustomerAgreement mRID		
scheduleInterval end start		

Figure 10 is a graphical example of the XSD mapping of MeterReadSchedule.

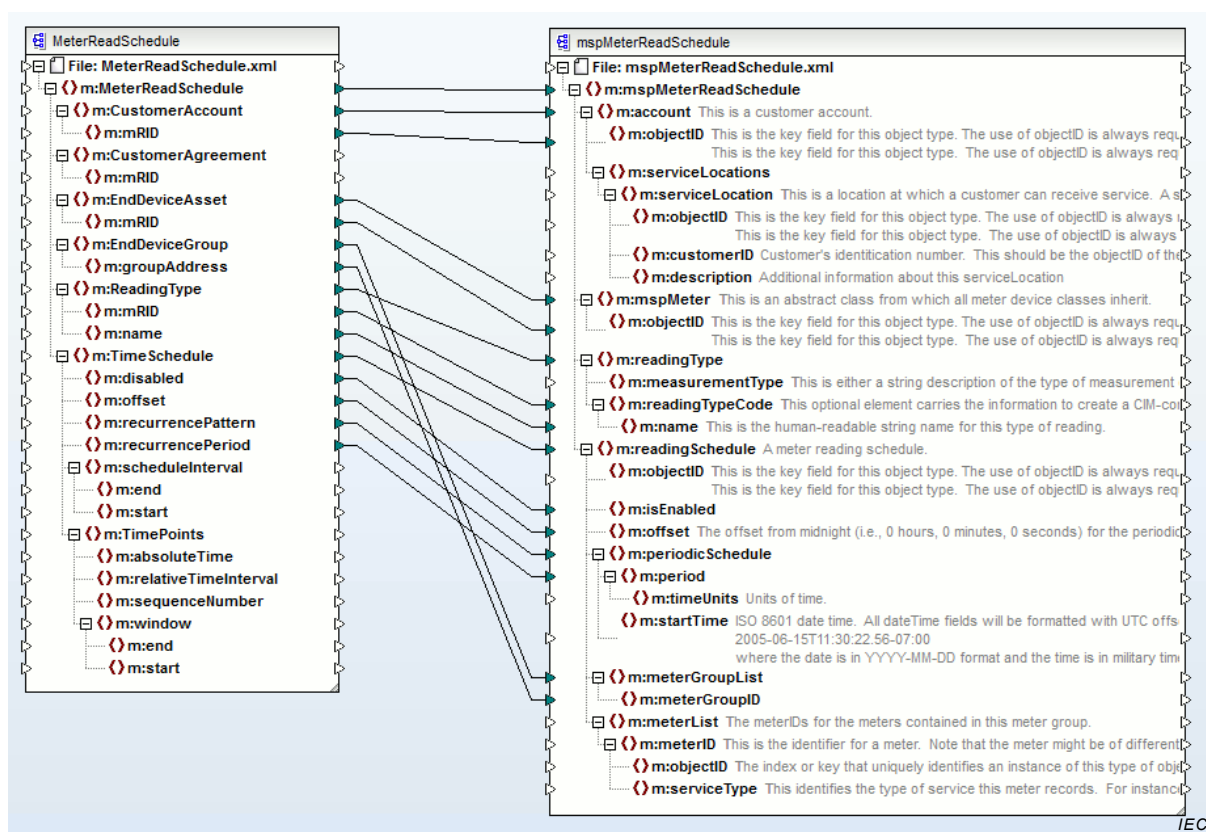


Figure 10 – MeterReadSchedule CIM to MultiSpeak profile mapping

8.19 Meter service requests

Meter Service Requests profile handles service requests for meters which may also include a meter swap-out, and meter readings for both the old (final) and new (initial) meters, if the meter needs to be replaced. This is a fairly common scenario as even if a meter can be repaired it is usually brought back to the meter shop for the repair and a replacement meter installed in the field as a result of the service work.

For the CIM, ReadingType is handled a bit differently in this profile than in the MeterReadings profile. In this profile instead of having the full structure that is seen in the MeterReadings profile, there is only an mRID, name, and optionally, a unit. For the MeterAsset (representing the new meter) and OldMeterAsset (for the old meter) the ReadingType in the associated MeterReadings is a ref.

Table 47 shows the correlations for the CIM to MultiSpeak equivalent MeterServiceRequests profile.

Table 47 – MeterServiceRequests correlations

CIM	MultiSpeak	Comments
ReadingType .mRID	readingType.objectID	
Customers .kind	customer .servType	
MeterAsset .amrSystem .serialNumber .kh .kr	electricMeter .amrVendor .serialNumber electricNameplate .kh .kr	

8.20 Meter readings

The CIM Meter Service Work profile also contains a set of meter reads for both the old and new meter (if appropriate for the profile). See 8.17 for the associated mappings.

Table 48 shows the transformations required to map from the MeterServiceRequests profile to the MultiSpeak equivalent.

Table 48 – MeterServiceRequests transformations

CIM	MultiSpeak	Comments
amrSystem	amrVendor	
MeterServiceWork .kind	serviceOrder .soType	
Customers .specialNeed .vip	customer .specialNeeds	Concatenate CIM specialNeed and vip into MultiSpeak specialNeeds
CustomerAgreement .mRID	serviceOrder .accountNumner	MultiSpeak does not support CustomerAgreement but the mRID can be mapped to an accountNumber to identify the customer

8.21 Activity records

The CIM Meter Service Work profile also contains a set of Activity records. See the paragraph on End Device Events to see how the category, severity, reason, could be mapped or substringed to the appropriate MultiSpeak classes and attributes.

8.22 ServiceDeliveryPoints

Addresses are included as part of the service delivery point information. For the address mappings see Table A.1, *Address correlations*.

For more information see subclause 8.28.

Table 49 shows the gaps in mapping from ServiceDeliveryPoints to the MultiSpeak equivalent.

Table 49 – MeterServiceRequests gaps

CIM	MultiSpeak	Comments
Customer .pucNumber		
Status		
MeterServiceWork .priority .category .lastModifiedDateTime		
MeterAsset .formNumber		

8.23 Meter system events

The intent of this profile is to capture events that may occur throughout the distribution network. There are three high level structures in this profile, ComMediaAsset (what the device uses to communicate), EndDeviceAsset, (the device itself), EndDeviceEvent (information relating to the event).

For EndDeviceEvent mappings see the paragraph End DeviceEvents.

The ComMediaAsset only contains an identifier for the communication device, and optionally, a serialnumber.

One challenge is that the “end device” in MultiSpeak is the meter, where in CIM it is meters or other devices that can communicate. In MultiSpeal mspMeter is an abstract class, which happens to contain everything needed to map to an EndDeviceAsset, but as an abstract class, would normally be inherited by electricMeter, which is not semantically the same.

Table 50 shows the correlations in mapping from the MeterSystemEvents profile to the MultiSpeak equivalent.

Table 50 – MeterSystemEvents correlations

CIM	MultiSpeak	Comments
ComMediaAsset .mRID .serialNumber	module .objectID .serialNumber	
EndDeviceAsset .mRID .serialNumber .amrSystem	mspMeter .objectID .serialNumber .amrVendor	

8.24 Pricing structure config

The PricingStructureConfig profile is what is used to communicate tariff related information. In addition to core tariff, CIM also supports consumption and time-based tariff intervals.

Tables 51, 52, and 53 show the correlations, transformations, and gaps, respectively in mapping from the PricingStructureConfig profile to the MultiSpeak equivalent.

Table 51 – PricingStructureConfig correlations

CIM	MultiSpeak	Comments
PricingStructure .description .code	mspPricingStructure .description .rateCode	
Tariffs.description	mspTariffs .rateComponentDescription	

Table 52 – PricingStructureConfig transformations

CIM	MultiSpeak	Comments
PricingStructure	mspPricingStructure	Each of these “msp” classes were abstract classes added to be placeholders to facilitate mapping to the CIM profile
Tariffs .mRID .startDate	mspTariffs .objectID .startTime	
	mspTariffIntervalList	
PricingStructure .revenueKind	mspPricingStructure .servType	Residential, Commercial, etc.
ServiceCategory .name	serviceType	serviceType in MultiSpeak is an enumeration of the many types of services that could be modeled
TariffProfiles	rateComponents	
fixedPortion .energyUnit .monetaryUnit	rateComponent .commodityUnits .currencyCode	
ConsumptionTariffIntervals	mspTariffIntervalList	Using this abstract class as a placeholder the rate, rateComponents, and rateComponent classes could be used to model consumption or timeTariff intervals
TimeTariffIntervals	mspTariffIntervalList	
ChildCharges.name	rateCode	
Charges.mRID	mspTariffIntervalList .objectID	

Table 53 – PricingStructureConfig gaps

CIM	MultiSpeak	Comments
Status		
docStatus		Is of type CIM Status
variablePortion		CIM allows for the distinction of a fixed and variable portion of a tariff
fixedPortion		
code revisionNumber title		Each of these attributes are in the CIM profile, but are not defined in the CIM specification, nor in the CIM model.
Tariffs .endDate .lastModifiedDateTime		Nominally there should not be an empty date in a tariff so the startDate of the next tariff is the endDate of the previous tariff
ChildCharges .kind .variablePortion		
fixedPortion .energyUnit .monetaryUnit .multiplier .value		

The key gap in MultiSpeak in relation to the variable and fixed portion of the CIM profile is the lack of a corresponding value and multiplier that the energy or monetary units apply to. MultiSpeak has a currencyCode and commodityUnits, but no attribute to say “how many”.

8.25 Receipt record

The ReceiptRecord profile is used to pass payment information from one system to another. The CIM profile defines both credit card and check transactions. MultiSpeak handles these two transaction types but also includes cash and ACH transactions. CIM also includes vendor or cashier information related to the transaction where MultiSpeak does not include this payment handling information.

The correlations, transformation, and gaps related to the ReceiptRecord profile are shown in Tables 54, 55, and 56, respectively.

Table 54 – ReceiptRecord correlations

CIM	MultiSpeak	Comments
Receipt .mRID	paymentTransaction .objectID	paymentTransaction has an objected if it inherits from mspObject (see UML above)
Card .expiryDate .cvNumber .pan	creditCardInfo .creditCardExpDate .cardVerificationValue .creditCardNumber	
Cheque .chequeNumber .micrNumber	checkInfo .checkingAccountNumber .bankRoutingNumber	
bankAccountDetail .holderName .accountNumber	checkInfo .nameOnCheck .checkNumber	
line .amount .dateTime .note	payableItem .currentAmount .dueDate .description	

Table 55 – ReceiptRecord transformations

CIM	MultiSpeak	Comments
Receipt.description	paymentTransaction .typeService	
Card .accountHolderName	creditCardInfo .firstName .lastName .mName .suffix	The CIM profile has only a single name field, which would need to be substringed into the appropriate sub-fields.

Table 56 – ReceiptRecord gaps

CIM	MultiSpeak	Comments
CashierShift		
VendorShift		
	paymentTransaction .accountNumber	Odd that the account for which a payment might be made is not included in the CIM profile.
Receipt.isBankable		
	creditCardInfo.type	Visa, Mastercard, etc
	creditCardInfo.address	
Cheque .date .kind		Nominally the kind attribute could be used to denote normal check or ACH
bankAccountDetail .bankName .branchCode .holderID		
	cashPayment	MultiSpeak has significant improvements over CIM for this profile with support for both ACH and cash transactions
	achPayment	
line.rounding		
	payableItem .chargeCode .disconnectDate .lastPaymentAmount .lastPaymentDate .pastDueAmount	
	payableItemList .total .utilityFee	
	tender .authorizationCode .financialTransactionID .paymentMode .responseCode .amount .convenienceFee	

This profile could be used to identify payers that may or may not be part of the utility for example, service stations or banks that take payments on behalf of the utility, or third party

aggregators. The CashierShift or VendorShift classes are used to capture who is taking the payment.

8.26 SDP Location config

The SDPLocation profile is somewhat different than Service Delivery Point configuration. SDPLocationConfig contains additional position point information in addition to mail address and contact information.

Tables 57, 58, and 59 show the correlation, transformations, and gaps, respectively, in mapping from the SDPLocationConfig profile, to the MultiSpeak equivalent.

Table 57 – SDPLocationConfig correlations

CIM	MultiSpeak	Comments
PositionPoints .xposition .yPosition .zPosition	gpsPoint .longitude .latitude .altitude	
SDPLocation .corporateCode	serviceLocation .SIC	

For Address and Phone related mappings, see Annex A.

Table 58 – SDPLocationConfig transformations

CIM	MultiSpeak	Comments
SDPLocation .mRID	mSPSDPLocation .objectID	A transformation as mSPSDPLocation is an added abstract class used only for this profile
SDPLocation .accessMethod .remark .siteAccessProblem .description	serviceLocation .description	The three CIM attributes can be concatenated and mapped to the MultiSpeak description attribute

For electronic address related transformation mappings, see Annex A.

Table 59 – SDPLocationConfig gaps

CIM	MultiSpeak	Comments
PositionPoints .sequenceNumber .isPolygon		The sequence number can be used to indicate a series of numbers and isPolygon, a Boolean, to indicate if the resulting set of coordinates closes.
SDPLocation .occupancyDate .category .direction .geoInfoReference		
Status		No generic Status class in MultiSpeak. Status is class (context) specific.

8.27 Service category config

The ServiceCategoryConfig profile is used to pass information about the different types of services that may be used by an organization, i.e., electric, gas, water. This is the smallest CIM profile and essentially reflects an enumeration of these different categories. The main difference with MultiSpeak is that the equivalent category, serviceType, is associated with objectRef. Within MultiSpeak objectRef can be used to refer to an account, a service, or a serviceLocation, all of which may have a serviceType associated with it.

Tables 60, 61, and 62 show the correlations, transformations, and gaps, respectively of mapping the ServiceCategoryConfig profile to the MultiSpeak equivalent.

Table 60 – ServiceCategoryConfig correlations

CIM	MultiSpeak	Comments
mRID	meterNo.objectID	
description	description	

Table 61 – ServiceCategoryConfig transformations

CIM	MultiSpeak	Comments
Kind	serviceType	Both attributes are enumerations of the types of services that are supported

Table 62 – ServiceCategoryConfig gaps

CIM	MultiSpeak	Comments
name aliasName		name and aliasName are both optional in this profile and nominally the Name class and identifiedObject class in MultiSpeak could be used to map these two attributes
	meterNo utility	Since serviceType is always associated with a meter, meterNo and utility are important attributes, especially as MultiSpeak supports the concept of meters being read by a utility on the behalf of another utility

Figure 11 shows a graphical XSD mapping example for the ServiceCategoryConfig profile.

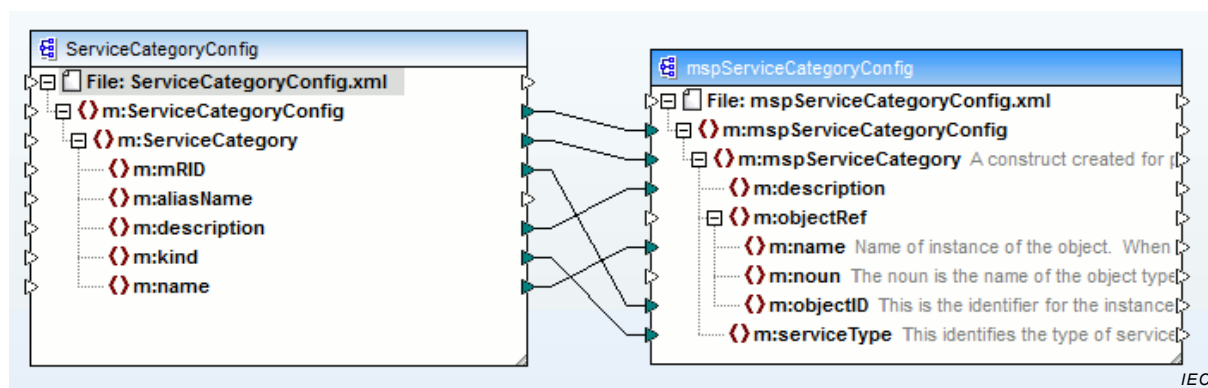


Figure 11 – ServiceCategoryConfig profile mapping

8.28 Service delivery point config

This profile contains information about a given service delivery point, e.g. phase, current, power, voltage, and related billing information. It also contains references to pricing structures (tariffs), SDPLocations, service category, service location, and service supplier.

Tables 63, 64, and 65 show the correlations, transformations, and gaps, respectively for mapping the ServiceDeliveryPointConfig profile to the MultiSpeak equivalent.

Table 63 – ServiceDeliveryPointConfig correlations

CIM	MultiSpeak	Comments
ServiceDeliveryPoint	serviceLocation	
.mRID	.objectID	
.billingCycle	electricService.billing_ Cycle	
.budgetBill	electricService.budget_ .description	Bill
.description	electricService.load_ Mgmt	
.loadMgmt	voltage.units	
.nominalService_ Voltage	current.units	
.ratedCurrent	realEnergy.units	
.ratedPower	.priority	
.servicePriority		

Table 64 – ServiceDeliveryPointConfig transformations

CIM	MultiSpeak	Comments
CustomerAgreement .mRID	account .customerID	MultiSpeak 4.1.5 does not support customer agreements, but the mRID of an agreement can be mapped to a customer account
ServiceDeliveryPoint .cptReference	instrumentTransformers .ct .pt	CIM uses a single attribute where MultiSpeak splits this between ct/pt.
.serviceDelivery_ Remark	serviceLocation .description	serviceDeliveryRemakr could be concatenated with description and mapped to the MultiSpeak description field
PricingStructures .mRID .name	rate .objectID .rateCodeDescription	
Tarriifs	rateComponents	CIM tariffs has an identifier (from identifiedObject) and name, where in MultiSpeak 4.1.5 this is a holder for one or more rateComponent
TariffProfiles .name	rateComponent .rateComponent_ Description	
ServiceSupplier._ mRID	resource.objectID	ServiceSupplier in CIM identifies personnel (utility or contractor) that provides a service with associated contact information. Resource in MultiSpeak 4.1.5 can be used to identify similar contact information (by adding an association) but nominally this is for physical assets, not people.
ServiceCategory._ name	serviceType	Normally in the CIM the mRID of a ServiceCategory would be used to uniquely identify the specific item.

See 8.30 to see how adding an association to MultiSpeak contact related classes very closely emulates the CIM ServiceSupplierConfig profile.

Table 65 – ServiceDeliveryPointConfig gaps

CIM	MultiSpeak	Comments
ServiceDeliveryPoint .checkBilling .consumptionRealEnergy .estimatedLoad .grounded .phaseConfig		checkBilling is a boolean that reflects that this is the result of of an inspection or that a previous billing was in error
Tariffs.mRID TariffProfiles.mRID		CIM uses mRID to identify tariffs and trariffsProfiles
	rate .rateCode .startTime rateComponent .commodityUnits .currencyCode .rateCode	MultiSpeak uses the attributes that contain the rate information rather than a identifier, e.g. objected like the CIM does
	serviceLocation .facilityID .GMLLocation .gpsLocation .gpsPoint .gridLocation .rotation	MultiSpeak has a variety of ways to identify a location beyond the address that the CIM uses.
SDPLocations.mRID		

8.29 Service location config

There is a high degree of correlation between the CIM ServiceLocationConfig and the equivalent MultiSPEak profile, mostly due to the commonality of address related information between the two standards.

Tables 66, 67, and 68 show the correlations, transformations, and gaps for mapping the ServiceLocationConfig profile to the MultiSpeak equivalent.

Table 66 – ServiceLocationConfig correlations

CIM	MultiSpeak	Comments
mRID	customerID	customerID is type objectID
City	city	
country	country	
postalCode	postalCode	
stateOrProvince	state	MultiSpeak “state” can be either a state or province
addressGeneral	addressGeneral	
suiteNumber	suiteNumber	

Table 67 – ServiceLocationConfig transformations

CIM	MultiSpeak	Comments
accessMethod siteAccessProblems	description	The two CIM attributes can be concatenated to map to the description attribute.
typeServiceLoc	SIC	typeServiceLoc is classification for the type of location according to the utility while SIC is a standard industry code. It may be more appropriate to use CIM Organisation.code to made to MultiSpeak SIC

Table 68 – ServiceLocationConfig gaps

CIM	MultiSpeak	Comments
inspectReq		There is an inspection class in MultiSpeak but this is more appropriate for capturing data related to an inspection, not flagging that one needs to occur.
	facilityName	
	townCode	townCode is part of
	buildingNumber postOfficeBox region streetNumber streetPrefix streetSuffix streetType	The detailedAddressFields provide more address information and while used in other CIM profiles, e.g. ServiceSupplierConfig, this additional information is not used in this profile.

Figure 12 shows a graphical XSD mapping example for the ServiceLocationConfig profile.

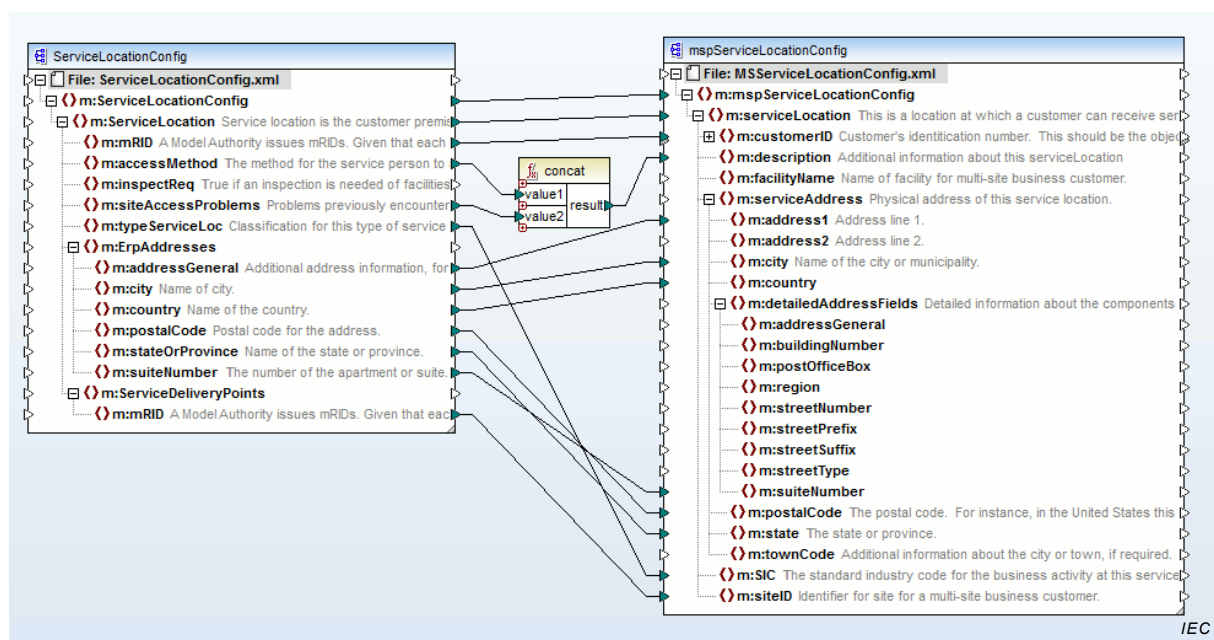


Figure 12 – ServiceLocationConfig CIM to MultiSpeak mapping

Of interest to the ServiceLocationConfig harmonization discussion is the Network class from MultiSpeak (Figure 13) offers additional information that may be of use in reference to a service location although there are no CIM equivalents.

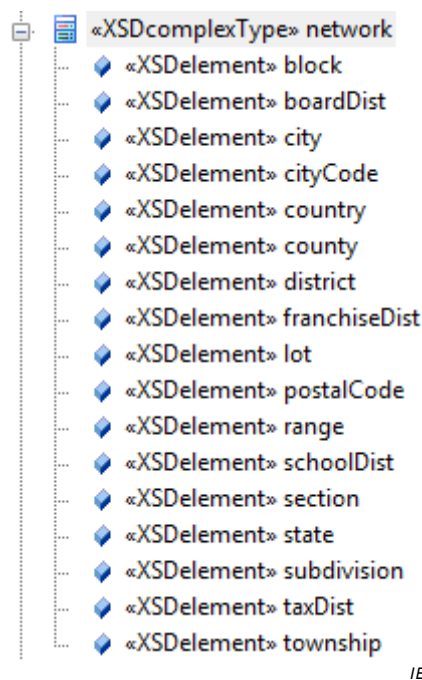


Figure 13 – MultiSpeak Network class

8.30 Supplier config

The SupplierConfig profile in CIM is used to transfer information about the service supplier, whether they work for the utility, a retailer, or other entity, and all of the contact information associated with the supplier. Since this information does not have an equivalent in MultiSpeak it would need to be concatenated and mapped to the utility field that resource inherits from mspObject.

MultiSpeak does not have this concept of a service supplier organization, but has numerous classes to capture both information related to a work item and classes to capture contact information. However, the contact information is associated with the high level mspPerson. Useful for mapping purposes is a container class, resource, available that is used for work related items that also inherits from mspObject. Creating a MultiSpeak equivalent of Supplier Config profile is simply a matter of adding an association from the resource class to the contactinfo class shown here. A notable gap, since MultiSpeak does not have this Supplier entity concept, is the issuerIdentificationNumber used by CIM to identify this organization (per ISO/IEC 7812-1 and ISO/IEC 7812-2) although nominally this could be mapped into the MultiSpeak objectId (concatenated with the mRID mapping from CIM).

In this profile both MultiSpeak and CIM use complete addressing information (where in the ServiceLocationConfig the CIM used limited address information), which maps very well. MultiSpeak using a construct where there can be many types of addresses which are tracked by an address type so it makes for a fairly clean hierarchy. CIM on the other hand calls out two types of addresses, street address and postal address in its profile. To map this second class MultiSpeak can simply instantiate a new instance of address and assign it a new type.

The other challenge with this profile is that the CIM captures other types of electronic address besides email addresses. LAN, radio, and web, which are not included in MultiSpeak. However, Multispeak has a generic otherContactInformation class that can be used to capture this information, again, it can be instantiated with a type to indicate the type of information being captured and the information, LAN address, web address, or radio address, mapped to the details attribute.

The final challenge is that the CIM profile uses the Status class construct and mRID, name, description fields (from identifiedObject) when it is not clear what value these classes and fields add to this profile. By tying the contact information to the high level ServiceSupplier this provides all the identification that the contact information should need to associate a given instance of contact information with its organization.

Tables 69, 70, and 71 are used to show the correlations, transformations, and gaps, respectively, for mapping the ServiceSupplierConfig profile to the MultiSpeak equivalent.

Table 69 – Service supplier correlations

CIM	MultiSpeak	Comments
mRID	resource.objectID	
email	eMail	
poBox	postOfficeBox	
postalCode	postalCode	
addressGeneral	addressGeneral	
buildName	buildingNumber	Both fields are strings
number	streetNumber	
prefix	streetPrefix	
suffix	streetSuffix	
suiteNumber	suiteNumber	
type	streetType	
code	TownCode	
country	country	
name	city	CIM uses name although in the annotation it is clear that city name is the intent
stateOrProvince	state	State in MultiSpeak can be either a state or province.
areaCode	areaCode	
cityCode	cityCode	
countryCode	countryCode	
extension	extension	
localNumber	localNumber	

Table 70 – ServiceSupplierConfig transformations

CIM	MultiSpeak	Comments
kind name	resource.utility	Kind identifies whether the service supplier is a utility, retailer, or other. This information would need to be concatenated and placed in the utility attribute that resource inherits
postalAddress / streetAddress	address	Not a transformation per se, but CIM calls out a specific class while MultiSpeak uses a concept where additional addresses can be added by instantiating a new class with the type being noted addressType
LAN web radio	otherContactItem	MultiSpeak does not have these equivalent fields but allows for additional contact information to be added by instantiating new otherContactItems
Telephone.description	phoneType	Description is inherited in the CIM class, but the phoneType in MultiSpeak is not a direct correlation but is a construct that allows MultiSpeak to add new phone records associated with an entity.

Table 71 – ServiceSupplierConfig gaps

CIM	MultiSpeak	Comments
Status		It is not clear what value having a Status associated with ServiceSupplier adds for this profile
issuerIdentification_ Number		This is used by CIM per ISO/IEC 7812-1 and ISO/IEC 7812-2, prefix number issued to an entity by the International Standards Organisation for the purpose of tagging onto electronic financial transactions
description, name, mRID, aliasName		In some cases this inheritance is passed down to subordinate classes as required where it appears this information could be optional.
	addressType phoneType infoType emailType	MultiSpeak uses these attributes to add additional instantiations of a given class of contact information
	priorityCode	MultiSpeak provides for the ability to have multiple types of contact information, but give information regarding the order in which they should be used

Figure 14 shows a graphical XSD mapping example for the ServiceSupplierConfig profile.

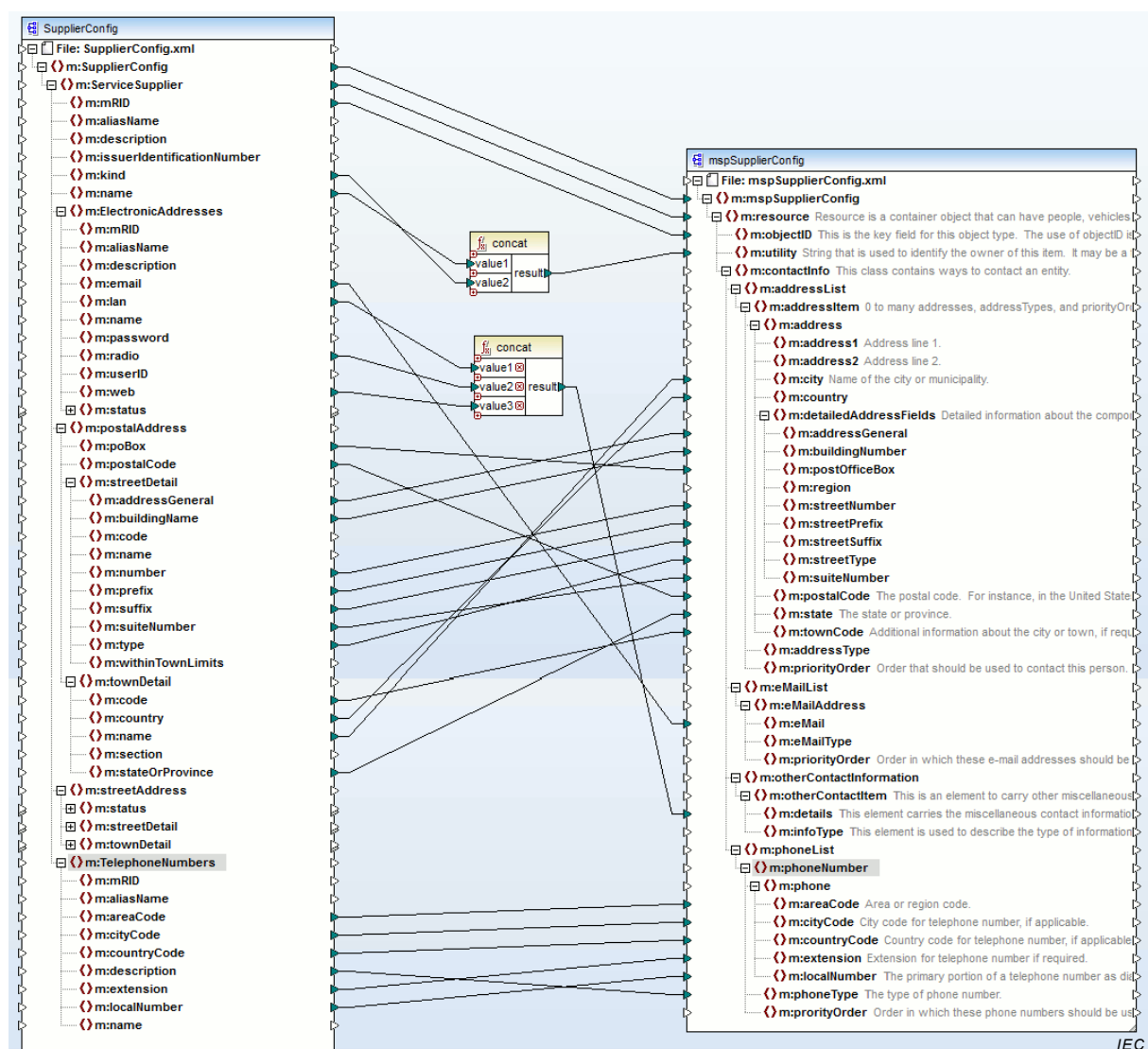


Figure 14 – Supplier Config profile mapping

8.31 Transaction record

The transaction record as a concept does not exist in MultiSpeak 4.1.6 so it can be expected that there are some significant gaps in this profile. Additionally, as there is no equivalent root class for this profile in MultiSpeak, one was added (shown in red in the figure) to be the placeholder for the associations that MultiSpeak does support.

This profile could be used to identify payers that are not part of the utility, for example service stations or banks that take payments on behalf of the utility, or third party aggregators.

In CIM an account is associated with one rate tariff. In MultiSpeak an account can have multiple service locations that can have multiple services. Each service is associated with one rate tariff.

Tables 72, 73, and 74 show the correlations, transformations, and gaps for mapping the TransactionRecord profile to the MultiSpeak equivalent.

Table 72 – TransactionRecord correlations

CIM	MultiSpeak	Comments
description	description	
CustomerAccount .mRID	account .accountNumber	
MeterAsset .mRID	meterID .meterNo	

Table 73 – TransactionRecord transformations

CIM	MultiSpeak	Comments
Transaction .mRID	mSPTransaction .objectID	While this would normally be a correlation, mSPTransaction is an addition to the model to facilitate the mapping of this profile
Transaction .kind	allowableTransactionTypes .transactionType	
Transaction .serviceUnits_ Energy .serviceUnitsError	mSPTransaction .realEnergy	The transaction may be for energy, or a correction to the energy transaction so this is mapped using a logical OR
PricingStructures .mRID .name	rate .objectID .rateCodeDescription	
Tariffs	rateComponents	CIM tariffs has an identifier (from identifiedObject) and name, where in MultiSpeak 4.1.5 this is a holder for one or more rateComponent
TariffProfiles .name	rateComponent .rateComponentDescription	
Charges	acctsReceivable	
.description	.description	
Receipt .mRID .description	paymentTransaction .accountNumber .typeService	
line.amount	acctsReceivable .receivableAmount	

Table 74 – TransactionRecord gaps

CIM	MultiSpeak	Comments
VendorShift		
CashierShift		
UserAttributes		
Transaction .diverseReference .donorReference .receiverReference .reversedId		
AuxiliaryAccount		
	meterID .serviceType .utility	MultiSpeak provides more information to identify the meter beyond just the meterNo/objected
ChildCharges		
	acctsReceivable .receivableType .serviceLocationID .serviceType	Nominally CIM line class maps to MultiSpeak acctsReceivable, however MultiSpeak provides some additional attributes for receivables
line .note .rounding		

9 IEC 61968-9:2013 to MultiSpeak v.5 (Release Candidate) profile mapping

9.1 General

There were significant changes from Version 4.1.x of MultiSpeak to Version 5. When attempting to harmonize with the CIM and understand the changes that occurred from IEC 61968-9:2009 to IEC 61968-9:2013, it will be important to also understand the corresponding MultiSpeak changes.

9.2 Summary of changes between MultiSpeak Version 4.1.x and Version 5.0

- 1) Moved to a contract-first approach to build the WSDLs. This results in cleaner WSDLs that support importing schemas rather than embedding schemas. As a result, numerous problems with element cardinality in the schemas have been corrected.
- 2) A mechanism to build the WSDLs has been adopted that will result in greater consistency among the various endpoint schemas and simpler WSDL maintenance.
- 3) Issues with circular schema references have been corrected.
- 4) Most enumerations have been hard-enumerated with the addition of an extensibility mechanism. This approach clearly identifies to implementers what values are expected for enumerations, which encourages simpler and much tighter integration among products. However, the extensibility mechanism permits the use of custom enumerations where necessary.

- 5) The UML class model has been extensively improved. Annotations have been significantly expanded, updated and improved. Numerous class diagrams have been added and significant improvements have been made to existing class diagrams.
- 6) Work management interfaces have been completely refactored to permit more flexible interface development. Significant discoverability has been added to the work interfaces to permit a broader range of work management applications to interoperate with less custom development. The work management endpoints for Version 5.0 will include:
 - **Work Requester (WR)** – WR identifies and raises work to be managed. Was called Work Task Originator (WTO) in Version 4.1.
 - **Work Owner (WO)** – Authority for workRequests and location to query the status of that work. New in Version 5.0.
 - **Work Generator (WG)** – Accepts workRequests from Work Owner, divides them into atomic work parts and will return 1-to-many workRequests to WO. Was called Work Task Generator (WTG) in Version 4.1.
 - **Scheduling and Assignment (SA)** – Responsible for scheduling future activities and assigning current activities to resources. Was called Work Scheduler (SCHED) in Version 4.1.
 - **Resource Management (RM)** – Manages what vehicles, equipment, workers, work groups, and crews may be assigned work. New in Version 5.0.
 - **Work Performer (WP)** – The head-end system that manages the work of the resources in the field. Was called Work Task Performer (WTP) in Version 4.1.
 - **Work Viewer (WV)** – WV can display the status of current work and history of completed work. Unchanged from Version 4.1.
- 7) All publish messages (Notification-type methods) are hosted on a Notification (NOT) endpoint. Any application, by hosting a Notification endpoint, can now subscribe to any published data, regardless of which endpoint is the publisher. This adds flexibility to implementations, centralizes the notification methods, and simplifies all other endpoints.
- 8) Subscription management is now centralized on a new Publisher endpoint, which can be hosted by any application that intends to publish data. This action centralizes the subscription management methods, thus simplifying all other endpoints. Dynamic subscription management is **required** in Version 5.0, whereas it was optional in Version 4.1.
- 9) In addition to the Notification and Publisher endpoints, ten new endpoints have been added in Version 5.0, including:
 - **Assembly (ASM)** – This endpoint is responsible for the definition of what material and labor is required to build construction assemblies. This functionality was included in the Finance and Accounting (FA) endpoint in Version 4.1.
 - **Commissioning and Provisioning (CP)** – This endpoint is responsible for commissioning and provisioning customer premises area networks (PANs) and PAN devices. This functionality was included in the Customer Billing (CB) endpoint in Version 4.1.
 - **Demand Management (DM)** – This endpoint is responsible for initiating demand response and pricing events. The DM is the owner of demand response and price programs. This functionality was included in the Customer Billing (CB) endpoint in Version 4.1.
 - **Material Inventory (INV)** – This endpoint manages the material items in physical inventory. This functionality was included in the Finance and Accounting (FA) endpoint in Version 4.1.
 - **Message Management (MM)** – This endpoint is responsible for sending messages to PAN devices. This functionality was included in the Customer Billing (CB) endpoint in Version 4.1.
 - **One-Call/Underground Locations (LOC)** – This endpoint manages the exchange of information about installed utility facilities with a centralized “one-call” system. Initially,

this endpoint will be empty; functionality will be added in future releases of Version 5.0 as it is developed.

- **Power System Model Exchange (MOD)** – The exchange of power system models (called “connectivity” in MultiSpeak) was previously supported in both the Engineering Analysis (EA) and the Geographic Information System (GIS) endpoint. In Version 5.0, it has been abstracted out as its own endpoint. Engineering Analysis, DMS, GIS, or other applications that intend to serve up power system model information will do so by implementing a MOD endpoint in Version 5.0.
- **Resource Management (RM)** – Manages what vehicles, equipment, workers, work groups, and crews may be assigned work. This functionality is sometimes called crew composition.
- **Weather (WEA)** – This is the source for weather data. Initially, this endpoint will be empty; functionality will be added in future releases of Version 5.0 as it is developed.
- **Work Owner (WO)** – The owner of work and work requests. This is the central location for discovering the status of work owned by this entity.

- 10) Capabilities have been added to manage tags and clearances in the Switching Order server. Switching orders and switching steps are now completely integrated with work management and with tagging, lockout and clearance management.
- 11) The Inspection (INSP) endpoint has been eliminated in Version 5.0, since inspections and assessments are types of work that can be managed using the same work management framework as is used for other types of work.
- 12) The GIS Viewer (GV) and Customer Relationship Management (CRM) endpoints were eliminated since they were subscribers of data only and hence can be replaced in function by the Notification (NOT) endpoint.
- 13) New abstract classes have been added to improve messaging, including: mspReturnable and mspExtensible.
- 14) New abstract classes have been added to improve asset management, including: mspAsset and mspAssetHistoryEvent.
- 15) Modified the object inheritance hierarchy to ensure that assets can carry their own asset information and asset event history.
- 16) Added a generic asset object that can be used by an asset management system to manage any type of asset so that it is possible to link the asset management system's view of that asset with the concrete data instance of that asset.
- 17) A new abstract class mspOutage has been added to permit management of outages of service types other than electricity.
- 18) New abstract classes have been added to better model multiple service types as well as to better harmonize MultiSpeak serviceLocations and services with their CIM equivalents. Abstract classes added for this purpose include: mspUsagePoint, mspElectricUsagePoint, mspGasUsagePoint, mspWaterUsagePoint, mspPropaneUsagePoint, and mspOtherUsagePoint.
- 19) Other new abstract classes include: mspEvent and mspVehicleDeviceCodes.
- 20) Message naming has been standardized to clarify the intent of messages.
- 21) An unsolicited notification method, called SystemStateNotification has been added so that applications can publish information to subscribers about errors or other critical state information.
- 22) Extensive use has been made throughout the schema of generic reusable structures such as supplementalNotes, locationPriorities, locationHazards, locationAssociations, and organizationalAssociations to add detail about objects in a standardized manner.

9.3 Schema changes

- 1) In Version 5.0, all objects that are to be persisted inherit from the mspObject; objects that do not inherit from mspObject, but are to be sent as part of a message payload inherit from a new abstract class, mspReturnable. Objects that do not inherit from either mspObject or mspReturnable inherit from a new abstract class, mspExtensible.

- 2) The verb and replaceID attributes on the mspObject have been removed. The verb is no longer necessary since in web services the verb is included in the method name. In batch file implementations, four new object containers have been added to the MultiSpeak object to clarify the intended “verb” action: (i) createdContainer, (ii) changedContainer, (iii) deletedContainer, and (iv) responseContainer. For instance, objects sent in the createdContainer are intended to be created. This is similar to a Version 3.0 or V4.1 object carrying the mspObject verb “New”. Similarly, the changedContainer replaces the use of the “Change” verb and the deletedContainer replaces the use of the “Delete” verb. The “replace” action is replaced by sending a xxxExchange object in the createdContainer, where “xxx” may be replaced with any of a number of asset-type objects, such as electricMeter. The responseContainer is intended for use in response to Get-type web services where the type of the returned objects is not known in advance.
- 3) New human readable identifiers for data instances: primaryIdentifier and secondaryIdentifier have been added to the mspObject and replace the objectName element on the mspObject.
- 4) A new mechanism was added to ensure unique naming of data instances. In Version 3.0 and 4.1, it was necessary to specify a noun type and an objectID. This remains the minimum required naming in Version 5.0. However in some circumstances this approach does not ensure uniqueness. It was decided to support a fully qualified object name of the form: GlobalDomain.RegisteredName.SystemName.Noun.objectID. The GlobalDomain for MultiSpeak data exchanges will be “MultiSpeak”. The RegisteredName will be a domain of uniqueness similar to an IP domain. It is suggested that either the utility manages this domain or a vendor manages this domain for its products (these approaches are not mutually exclusive, but will result in two unique names for each data instance). It is further suggested that the RegisteredName be the IP domain for the registering entity since this is ensured to be unique. The SystemName is an additional domain level. If the registeredName is managed by the utility, it is suggested that the system name refer to the vendor’s product; if the RegisteredName refers to the vendor, it is suggested that the system name refer to a product or utility name. Uniqueness of this name is managed by the entity referred to in the RegisteredName. Noun is the MultiSpeak noun type and objectID is the MultiSpeak objectID for the data instance.
- 5) Added mspUsagePoint, mspElectricUsagePoint, mspGasUsagePoint, mspWaterUsagePoint, mspPropaneUsagePoint, and mspOtherUsagePoint. Added the concept of XXXMeteringPoint which, along with XXXServicePoint, inherit from the appropriate XXXUsagePoint (where XXX = electric, gas, water, propane or other). Also as part of this change, meterBase was taken out of the connectivity model for electric services (as was the case for Version 4.1); electricServicePoint was inserted into connectivity model in the place of meterBase.
- 6) In Version 5.0, all optional lists are of cardinality 1-many instead of 0-many as was the practice in Version 4.1.
- 7) The materialManagementAssembly, assembly, and CPR objects have been significantly refactored.
- 8) The outageEvent and all of its children have been significantly refactored. Added the concept of outages for service types other than electricity (e.g. gas, water, etc.).
- 9) Numerous other minor changes.

9.4 Messaging changes

- 1) In order to clarify which information is to be used for different types of messages, separate message headers have been defined for request and response message types. The batch message header has been updated to reflect the changes made to the web service message headers.
- 2) A new reply object, which is intended to exchange information about the success or failure of a message, is included in all response messages. The reply object replaces the array of errorObject that was used in Versions 3.0 and 4.1.
- 3) DefaultSystemName, DefaultRegisteredName, and a codedName structure were added to facilitate abbreviated fully-qualified names of the type described in schema changes section above.

- 4) Most messages have been recast to carry an array of objects as a payload and most Get-type methods have been changed to support an array of calling parameters rather than a single parameter instance.
- 5) Message naming has been modified for greater standardization. Figure 15 shows the preferred form of request/response (Get-type) messages. Figure 16 shows the preferred form of initiation of an action (Initiate-type) messages. Lastly, Figure 17 shows the preferred form of publish-subscribe method. In each of these figures, “NOUN” refers to a MultiSpeak object that is the payload of the message or is referred to using a pointer in the message, “VERB” refers to one of the suggested verbs, and “PARAMETERS” refers to parameters that limit the scope of the message.

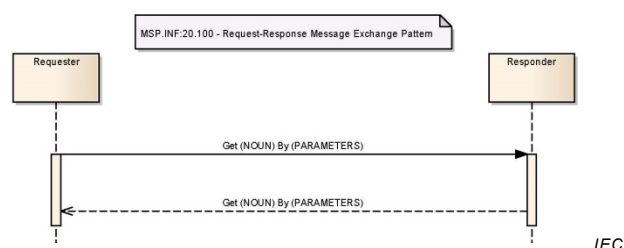


Figure 15 – Preferred form of request-response messages

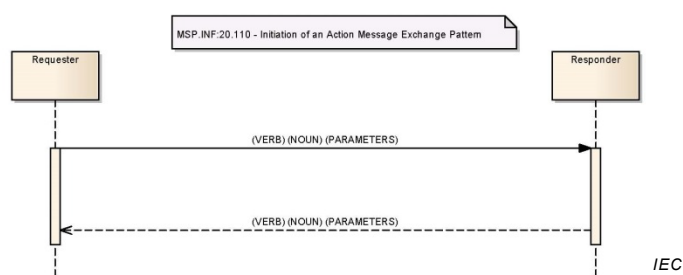


Figure 16 – Preferred form for "Initiation of an Action Messages"

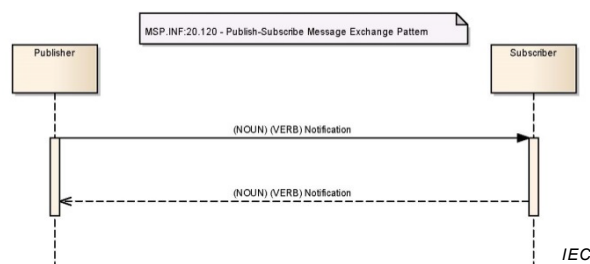


Figure 17 – Preferred form of publish-subscribe messages

Table 75 shows the preferred usage for present tense verbs (typically used in request/response (Get-type) messages or action requests (Initiate-type messages)). Table 76 shows the preferred past tense verbs. Past tense verbs are usually used as qualifiers in published messages (Notification-type) messages.

Table 75 – Relationships between Version 4.1 and Version 5.0 present-tense verb names

Version 4.1 verb	Version 5.0 name(s)	Description
Get	Get	Request information from the system of record.
Initiate	Initiate Set	- The Requester asks the Responder to take an action. - The Requester asks the Responder to change the status of a record or to set a flag for a record. This form is used if it is possible to send a pointer to a data instance and a flag; if a full object is required, then the Initiate style message is preferred.
Cancel	Cancel Clear	- The Requester asks the Responder to cancel a previously requested action. - The Requester asks the Responder to clear a previously set status for a record or to clear a previously set flag for a record. This form is used if it is possible to send a pointer to a data instance and a flag; if a full object is required, then the Cancel style message is preferred.
Add	Create Link	- Requester asks the Responder to create a data instance of an object. - Requester asks the Responder to associate two previously unassociated persistent objects.
Delete	Delete Unlink	- Requester asks the Responder to delete a data instance of an object. - Requester asks the Responder to unassociate two previously associated persistent objects.
Discard	Delete	Requester asks the Responder to delete a data instance of an object.
Insert	Insert	Requester asks the Responder to insert an object into a previously created class or group.
Remove	Remove	Requester asks the Responder to remove an object from a class or group.
Establish	Create	Requester asks the Responder to create a defined set of instructions that will subsequently be enabled.
Delete	Delete	Requester asks the Responder to delete a previously-defined set of instructions.
Enable	Enable	Requester asks the Responder to start acting on a previously created set of instructions.
Disable	Disable	Requester asks the Responder to stop acting on a previously-enabled set of instructions.
Enroll	Enroll	Adds a customer or account to a program.
Unenroll	Unenroll	Removes a customer or account from a program.
Modify	Change	Requests the system of record to make a change in a record for which it is responsible.
Update	Change	Modifies the database record for an object.
Ping	Ping	Determines the state of a web service endpoint.
Register	Register	Subscribes for notifications of changes in data.
Unregister	Unregister	Unsubscribes for notifications of changes in data.
Request	Get	Requests that the system of record provide data.

Version 4.1 verb	Version 5.0 name(s)	Description
Adjust	Change	Makes changes in customer account balance.
Extend	Change	Creates or extends a customer account payment agreement.
Commit	Change	Activates a transaction.
Assign	None	Assigns work to a crew or resource. Will be performed with an AssignmentNotification in Version 5.0.
Unassign	None	Unassigns work from a crew or resource. Will be performed with a WorkUnassignmentNotification in Version 5.0.
Write	Link	Adds a comment to a customer account record.
Close	Set	Changes the status of a customer call record to be “Closed”.

**Table 76 – Relationships between Version 4.1
and Version 5.0 past-tense verb names**

Version 4.1 verb	Version 5.0 name(s)	Description
Add	Created	Publisher notifies Subscriber that a new record of type NOUN has been added to the database of the system of record.
N/A	Deleted	Publisher notifies Subscriber that a record of type NOUN has been removed from the database of the system of record.
Changed	Changed	Publisher notifies Subscriber that changes have been made in a record of type NOUN in the database of the system of record.
Remove	Removed	Publisher notifies Subscriber that an asset of type NOUN has been removed from service and placed into inventory subject to evaluation of suitability for reuse.
Retire	Retired	Publisher notifies Subscriber that an asset of type NOUN has been removed from service and will be retired from service.
Exchanged	Exchanged	Publisher notifies Subscriber that one data instance of an asset has been removed from service and replaced by another data instance of the same asset.
Installed	Installed	Publisher notifies Subscriber that an asset of type NOUN has been physically placed into service.
<event>	<event>	In the case that a Publisher notifies Subscribers of events or conditions, no verb is used. For instance, if a new meterEvent is being issued, the message would be MeterEventNotification.

The strategy for this section will not be to provide entire XSD to XSD mappings as the files in some cases are so large as to be impractical for viewing as a Microsoft Word document or Adobe PDF file. Instead the component parts of each profile will be identified and those will be mapped. In this way, duplicates of sections do not have to be mapped repeatedly mapped wherever they appear in a profile. For example, the Names.Name class and subclasses are used to identify objects in numerous places in any given profile. Once the user has understands how to map this part of a profile it does not need to be repeated. This will lend itself to both a more succinct harmonization document, and since these component parts can be shown more readily in the documentation due to their smaller size, should contribute to the reader’s understanding. As a further example, the profile EndDeviceEvents is made up of:

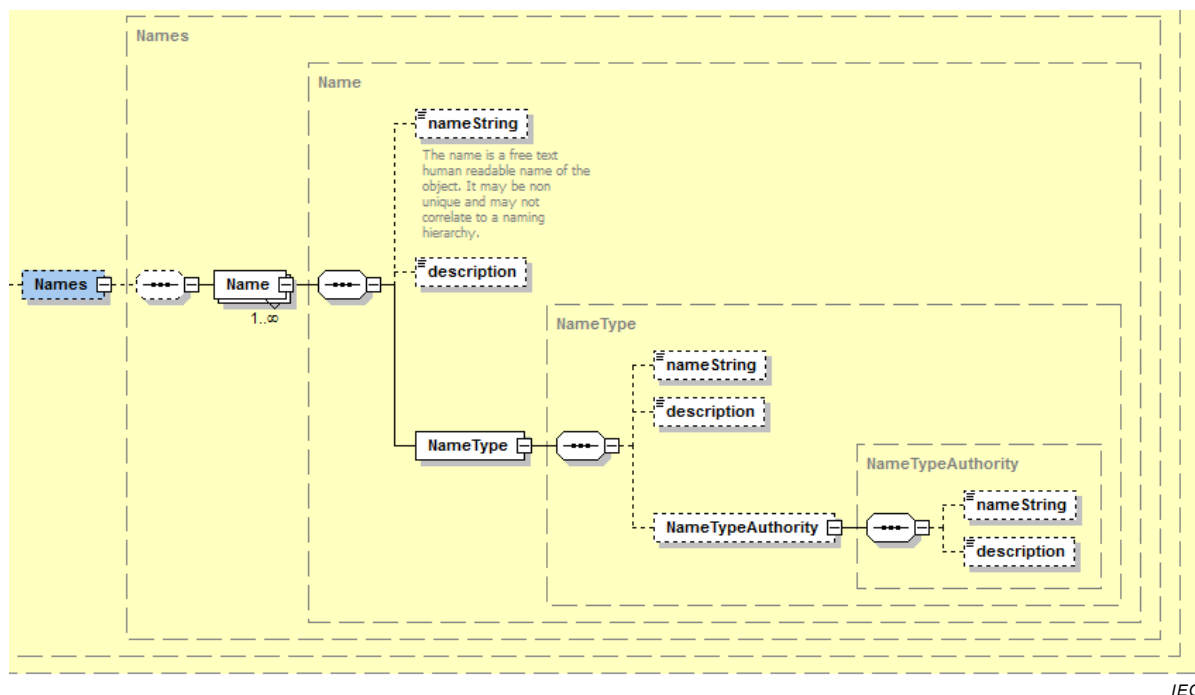
- EndDeviceEvent
- Assets reference
- EndDeviceEventDetails
- EndDeviceEventType
- Names
- Status
- UsagePoint

These component parts will be decomposed and handled individually for mapping purposes.

Parent class relationships

As with the CIM 61968-9 to MultiSpeak 4.1.x mapping, it is appropriate to begin the harmonization with an examination of the highest level classes (referred to as such because most classes in the respective models inherit from these classes); identifiedObject for the CIM and mspObject for MultiSpeak. For v.5 of MultiSpeak the mspObject class has seen some significant changes relative to v4.1.x. MultiSpeak's objectID is still the equivalent of the CIM's identifiedObject, but mspObject also optionally supports the CIM equivalent of the Names class construct.

Figure 18 shows the XML schema view of the Names class in the CIM which is also directly supported in MultiSpeak.



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Figure 18 – CIM and MultiSpeak both support the Names class and subclasses as an alternative means of identifying objects

9.5 ExtensionsList

MultiSpeak builds extensibility into its objects by using the ExtensionsList object shown in Figure 19. By using this class, if there is some data that needs to be passed that is not specified in a version of the MultiSpeak standard, the extensionsList class is the mechanism that is provided to allow vendors to extend the standard in a flexible, predictable way. Profiles in the CIM do not have the same mechanism. Therefore if a CIM profile has been extended it

can be mapped to the extensionsList of a given MultiSpeak class, however, because the CIM does not provide an equivalent mechanism, whoever has extended the CIM profile will have to document the specifics if a CIM profile has to be mapped to an equivalent MultiSpeak profile.

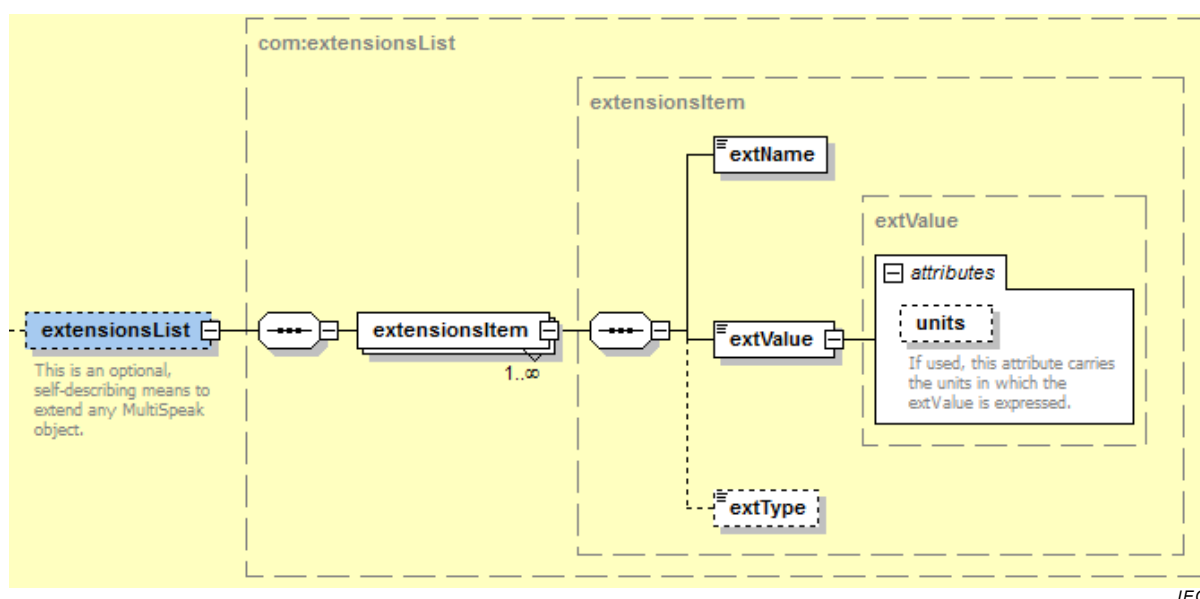


Figure 19 – MultiSpeak extensionsList class, used for adding attributes to a class

9.6 Mapping of IEC 61968-9:2013 to MultiSpeak v.5 (Release Candidate) Profile mapping

9.6.1 ActivityRecord

This class is used in a variety of profiles. In the CIM 1st edition mapping there was no good fit within MultiSpeak although the mspAlarm (abstract class and related classes) seemed to be the best fit.

Table 77 shows the correlation between the CIM and MultiSepak mapping for Activity records.

Table 77 – CIM to MultiSpeak Activity mapping

CIM	MultiSpeak	Comments
ActivityRecord .createdDateTime .reason	endDeviceEvent	mspAlarm in MSP v.5 has changed and should not be mapped to the CIM ActivityRecord The IEC 61968-9 specification lacks clarity on the different between ActivityRecord and EndDeviceEvents
.severity .type	endDeviceEvent .eventSeverity endDeviceEventTypeOption .endDeviceEventType	
Status class		See mapping paragraph on mapping the CIM Status class

It may also be informative for the reader to review the CIM EndDeviceEvent mapping.

9.6.2 Status

The CIM uses a generic class for all status related information needed by any object (Figure 20). In MultiSpeak there is typically a status attribute associated with the context in which the MultiSpeak object is being used. For CIM to MultiSpeak harmonization any given CIM status will then need to be mapped to the appropriate MultiSpeak attribute, and conversely, any given context-based MultiSpeak status information will need to be mapped to the CIM status class.

Name	Type	Initial Value
dateTime	Date Time	
reason	String	
remark	String	
value	String	

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Figure 20 – Attributes of CIM generic status class

9.6.3 ConfigurationEvents

Configuration events were added to the “config” type profiles in the IEC 61968-9:2013, and are used to indicate:

- The date and time at which the configuration or configuration change became effective or is to become effective,
- The entity performing the configuration operation, and
- Free-form remarks providing additional business information related to the configuration event.

9.6.4 CustomerAccountConfig

In the CIM for IEC 61968-9:2013, the only additions were the Names and ConfigurationEvents classes. Other than these two additional classes, the strategy for mapping the CIM CustomerAccount to the MultiSpeak customer remains unchanged from 1st edition CIM; the same correlations and gaps remain.

9.6.5 CustomerAgreementConfig

Changes from CIM 1st edition

Classes added:

- ConfigurationEvents
- DemandResponsePrograms
- ServiceCategory

Classes removed:

- Customer
- CustomerAccount
- PricingStructures
- ServiceSupplier

Table 78 shows the changes and correlations for the mapping of the CustomerAccountConfig to MultiSpeak.

Table 78 – CustomerAccountConfig mapping changes for CIM 2nd edition, MSP v.5

CIM	MultiSpeak	Comments
DemandResponseProgram .mRID	DRProgramEnrollment .customerID	customerID is of type objectID
ServiceCategory .mRID .kind	serviceKind	serviceKind is an enumeration in MSP
CustomerAccountConfig .createdDateTime		No equivalent CustomerAccountConfig in MSP

9.6.6 CustomerConfig

CIM changes from 1st to 2nd edition

Attribute(s) removed

- description
- name

Classes Moved

- ElectronicAddresses
- postalAddress
- TelephoneNumbers

Classes Added

- ConfigurationEvents
- Organization (Note: In IEC 61968-9:2013 profile the Organization class now holds electronic and street address information)
- Names

Table 79 shows the changes and correlations for the mapping of the CustomerConfig to MultiSpeak.

Table 79 – CustomerConfig mapping changes for CIM 2nd edition, MSP v.5

CIM	MultiSpeak	Comments
pucNumber		No MSP equivalent
specialNeed vip	servicePriority .description .servicePriorityType .servicePrioritySubType	MSP v.5 added a robust class to address special need identification
Organisation	organization	MSP v.5 has added an organization class with an association to contactinfo See the 1 st edition mapping for electronic address, postal addresses, and telephone numbers
ConfigurationEvents .createdDateTime .effectiveDateTime .modifiedBy .reason .remark .severity .type		Not supported in MSP v.5 CIM has several attributes in this class that pertain to information about the customer

9.6.7 EndDeviceConfig

NOTE The EndDeviceConfig profile was EndDeviceAssets in IEC 61968-9:2009.

CIM changes from 1st to 2nd edition

Classes Added

- SimpleEndDeviceFunction

Classes Removed

- Register
- ElectricMeteringFunction

Classes Changed

- ComFunction added attributes
- EndDeviceAsset became EndDevice
- ReadingType added attributes

Table 80 shows the correlation and mapping changes of the second edition of Meter Reading and Control standard, EndDeviceConfig mappings, the the MultiSpeak v.5 equivalent profile.

Table 80 – EndDeviceConfig mapping changes for CIM 2nd edition, MSP v.5

CIM	MultiSpeak	Comments
ComFunction .direction .enabled		No equivalent direction or enabled attributes in MSP v.5 Module or related classes CIM direction is the kind of communication direction (replaces “twoWay” attribute)
ConnectDisconnectFunction .enabled	CDEnabledItem .isCDEnabled	CIM .disabled has been changed to .enabled name and password attributes and Registers class has been removed Names class has been added
rcdInfo	remoteReconnectSettingKind remoteReconnectString	See 1 st edition mapping There is not a one to one mapping of rcdInfo to MSP v.5, however, the SettingKind enumeration and reconnectString allow for various settings to be mapped. Also see meterConnectionStatusKind
EndDevice Added attributes: .initialCondition .initialLossOfLife .installCode .isPan .isVirtual .type .utcNumber	mSPMeter Initial condition can be mapped to meterConditions class No match for lossOfLife MultiSpeak has	Several attribute changes have been made to this class within this profile for 2 nd edition
ElectronicAddress class		see 1 st edition mapping

There were several classes added within the EndDevice profile. These mappings are shown in Table 81.

Table 81 – Mappings for classes added to EndDeviceConfig

CIM	MultiSpeak	Comments
ActivityRecords class .createdDateTime .reason .severity .type		No equivalent in MSP v.5 No clear in the CIM specification how this differs from EndDeviceEvent
ConfigurationEvents class .createdDateTime .effectiveDateTime .modifiedBy .reason .severity .remark .severity .type	configurationStatusItem .timestamp .configurationStatus	MSP v.5 uses the configuration status as a string. There would need to be concatenation to put the CIM attributes into this string
EndDeviceInfo .isSolidState .phaseCount .ratedCurrent .ratedVoltage AssetModel class (see below) Capability class (see below)	Not supported Not supported CurrentLimit VoltageLimit	
AssetModel .corporateStandardKind .modelName .modelVersion .usageKind Manufacturer (type mRID)	Not supported Not supported Not supported Not directly supported Organization (type objectID)	MultiSpeak supports and makes a distinction among meter types (electric, gas, water, propane)

CIM	MultiSpeak	Comments
capability .autonomousDst .communication .connectDisconnect .demandResponse .electricMetering .gasMetering .metrology .onRequestRead .outageHistory .pressureCompensation .pricingInfo .pulseOutput .relaysProgramming .reverseFlow .superCompressability_ Compensation .temperature_ Compensation .textMessage .waterMetering	timeZone.DSTEnabled Not supported (assumed) Not supported Not supported electricMeter class gasMeter class Not supported (assumed) requestByType Not supported (assumed) pressureCompensationType Not supported Not supported Not supported Not supported Not supported temperatureCompensation_ type Not supported waterMeter class	While Multispeak supports these capabilities, there is no class that represents these capabilities specifically MSP Enumeration (includes immediate) Event history is managed with meterHistoryEvent class MSP enumeration MSP enumeration
lifecycle .installationDate .manufacturedDate .purchaseDate .receivedDate .removalDate .retiredDate	meterInstallationHistory .meterHistoryEvent	In MSP v.5 the meterInstallationHistory can hold all of the related dates as History Events

One of the changes to the EndDeviceConfig profile in IEC 61968-9:2013 was the addition of a SimpleEndDeviceFunction class. Some attributes existed in the 1st edition profile but are now included in this new and expanded class. The mappings for this class are shown in Tables 82 and 83.

Table 82 – SimpleEndDeviceFunction mapping for CIM 2nd edition to MSP v.5

CIM	MultiSpeak	Comments
SimpleEndDeviceFunction .mRID .configID .enabled .firmwareID .hardwareID .kind .password .programID	Module .objectID Map to .description Map to .description .firmwareVersion Map to .description .moduleType Map to .description Map to .description	The various unsupported attributes will have to be concatenated and mapped to MSP v.5 description attribute.
Registers .mRID .isVirtual .leftDigitCount .rightDigitCount .tenthTierName	Registers .objectID Not supported Not supported Not supported .tier	In MSP v.5 Total registers are identified by the absence of the 'tier' attribute. The first tier is identified by the 'tier' attribute set to 1, the second tier is identified by the 'tier' attribute set to 2, etc.
Channels .mRID .isVirtual .ReadingType	channel .objectID Not supported readingType.objectID	MSP has several classes for managing channels, channelList, channel, blocklist, channelBlock
tenthTier .end .start	priceTierStructure .startTime	

Table 83 – CIM Seals class mapping to MSP v.5

CIM	MultiSpeak	Comments
Seals .appliedDateTime .condition .kind .sealNumber	meterSeal .datePlaced Not supported .sealColor .sealNumber	Seals are now supported in MSP v.5

9.6.8 EndDeviceControls

In IEC 61968-9:2013 the EndDeviceControls profile saw a significant change as this profile has been split into two main parts: EndDeviceControl and EndDeviceControlType. Figure 21 shows the XML schema view of this change.

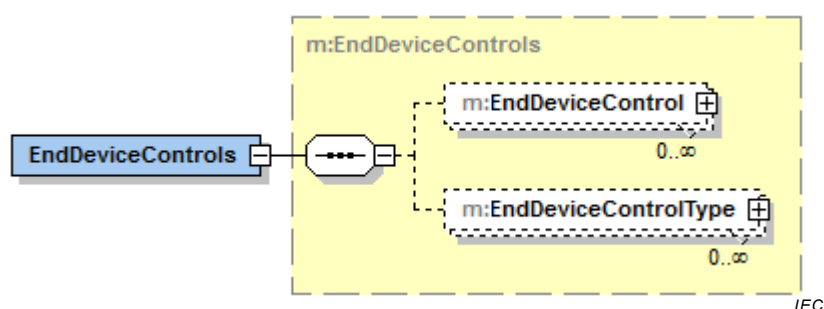


Figure 21 – Change to the EndDeviceControl profile which added the EndDeviceControlType class

Table 84 – CIM EndDeviceControlsType class mapping to MSP v.5

CIM	MultiSpeak	Comments
EndDeviceControlType .domain .eventOrAction .subDomain .type		

The EndDeviceControl profile saw numerous changes for IEC 61968-9:2013, including the support for PAN devices, UsagePoints, and UsagePointGroups.

Table 85 – CIM EndDeviceControlsType class mapping to MSP v.5 with support for PAN classes

CIM	MultiSpeak	Comments
EndDeviceControl .drProgramLevel .drProgramMandatory .issuerID .issuerTrackingID .priceSignal .reason PanDemandResponse PanDisplay PanPricing		 See the table of mapping PAN classes

9.6.9 EndDeviceEvents

Table 86 shows the correlation and changes that occurred in the Meter Reading and Control, second edition to MultiSpeak v.5 profile equivalent.

Table 86 – CIM 2nd edition – MSP v.5 EndDeviceEvent mapping and gaps

CIM	MultiSpeak	Comments
UsagePoint	mspUsagePoint	Optionally identifies a UsagePoint for this event using identifiedObject or the Names class
	mspEvent	This is an abstract class is the parent for mspReturnable
	mspReturnable	Abstract class that is the parent for endDeviceEventList
	endDeviceEventList	The “list” is a holder class in MultiSpeak that allows one-to-many children, in this case, endDeviceEvent
endDeviceEvent .mRID .userID .severity Assets.mRID	endDeviceEvent .objectID .userID .eventSeverity .deviceReference	
endDeviceEventType domain eventOrAction subDomain type	endDeviceEventType endDeviceDomain eventOrAction endDeviceSubdomain endDeviceType codeString	codeString is used to hold the string representation of the event code.
endDeviceEvent .createdDateTime .issuerID .issuerTrackingID .reason	endDeviceEvent .argument .associatedMeasurement	
endDeviceEventDetails .name .value		The CIM provides a sub class mechanism to extend the event with additional details using a name-value pair combination
	endDeviceEventTypeList	Holder class for a set of endDeviceEventTypes
	endDeviceEventTypeltem .endDeviceEventType .endDeviceEventTypeRef	

In MultiSpeak v.5 a “shorthand” capability for events was added. The intent was that this would be used as an alternative to link a set of endDeviceEvent to an endDeviceEventType without having to specify the entire endDeviceEventType string. In this usage, there could be an abbreviation form (the endDeviceEventTypeRef) that would be defined in an endDeviceEvents container object that could be used rather than using the whole event type string. For example, if the endDeviceEventType were 1.2.3.4.5 then rather than associating this with each instance of an endDeviceEvent, an alias could be defined, e.g. “A” to be the shorthand reference for “1.2.3.4.5” and include a cross-reference table in the endDeviceEvents.endDeviceEventTypeList.

Figure 22 illustrates how the parent class endDeviceEvent might have an endDeviceEventList associated with it (cardinality of 0 to 1). This list of events is then contained in the endDeviceEventTypeltem class.

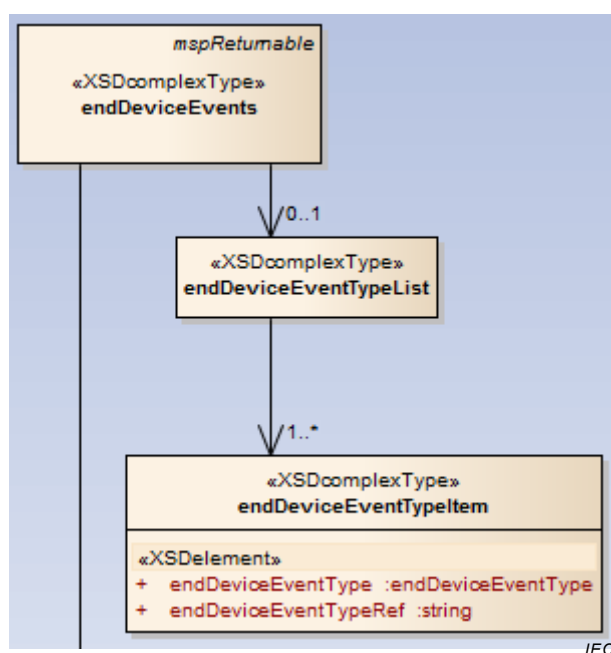


Figure 22 – MultiSpeak V5 endDeviceEvent "shorthand" relationship

9.6.10 PAN-related classes

Since IEC 61968-9:2013 was released both CIM and MultiSpeak have seen support for PAN devices. The mappings are documented in Tables 87, 88, and 89.

Table 87 – PANDemandResponse mapping from CIM to MSP v.5

CIM	MultiSpeak	Comments
PanDemandResponse .avgLoadAdjustment .cancelControlMode .cancelDateTime .cancelNow .command .coolingOffset .coolingSetPoint .criticalityLevel .duration .durationIndefinite .dutyCycle .enrollmentGroup .heatingOffset .heatingSetPoint .startDateTime	demandResponseEvent .averageLoadAdjustment Not Supported Not Supported Not Supported tunnelCommentContent .commandContent temperatureTier .coolingOffset thermostatScheduleTime .coolingSetPoint .criticality .eventDuration Not Supported .dutyCycle PANAddressing_ Group,name .heatingOffSet .heatingSetPoint .randomizedEventStart	
appliance .isElectricVehicle .isExteriorLighting .isGenerationSystem .isHvacCompressorOr_ Furnace .isInteriorLighting .isIrrigationPump .isManagedCommercial_ Industrial .isPoolPumpSpaJacuzzi .isSimpleMiscLoad .isSmartAppliance .isStripAndBaseboard_ Heater .isWaterHeater	Not Supported Not Supported Not Supported Not Supported Not Supported Not Supported Not Supported Not Supported Not Supported Not Supported Not Supported Not Supported	MSP v.5 has a very robust set of classes and attributes to support PAN devices, but does not attempt to create a class of Boolean-based attributes in an attempt to address every potential type of device that might be addressed. In MSP v.5 there is the ability to indicate PANDeviceType and the description suggests that the Table 5.11 of the SEP1.0 Specification be used, however, the specific types are not defined.

Table 88 – PANDisplay class mapping

CIM	MultiSpeak	Comments
PANDisplay		
.command	tunnelCommentContent .commandContent	
.confirmationRequired	Not Supported	
.duration	demandResponseEvent .duration	
.durationIndefinite	Not Supported	
.priority	Not Supported	
.startDateTime		
.textMessage	premisesDisplayMessage_ StatusKind	
.transmissionMode		

Table 89 – PANPricing class mapping

CIM	MultiSpeak	Comments
PANDisplay		
.command	pricingEvent Not Supported	
.duration	.eventDuration	
.durationIndefinite	Not Supported	
.providerID	.eventID	
.startDateTime	pricingPeriod .timePeriod	
PANPricingDetails		
.alternateCostDelivered	Not Supported	MSP v.5 approaches the priceTier in a slightly different manner so that priceTierCount is not directly supported, but the priceTier is contained within a priceTiers class. See Figure below.
.alternateCostUnit	Not Supported	
.currentTimeDate	Not Supported	
.generationPrice	Not Supported	
.price	priceTier.price	
.priceRatio	Not Supported	
.priceTier	priceTier	
.priceTierCount	Not Supported	
.priceTierLabel	priceTier.rateIdentifier	
.rateLabel	priceTier.rateLabel	
.registerTier	priceTier.register	
.unitOfMeasure	priceTier.units	

Figure 23 shows the UML class diagram that shows the classes, relationships, and attributes for the priceTierStructure and related classes.

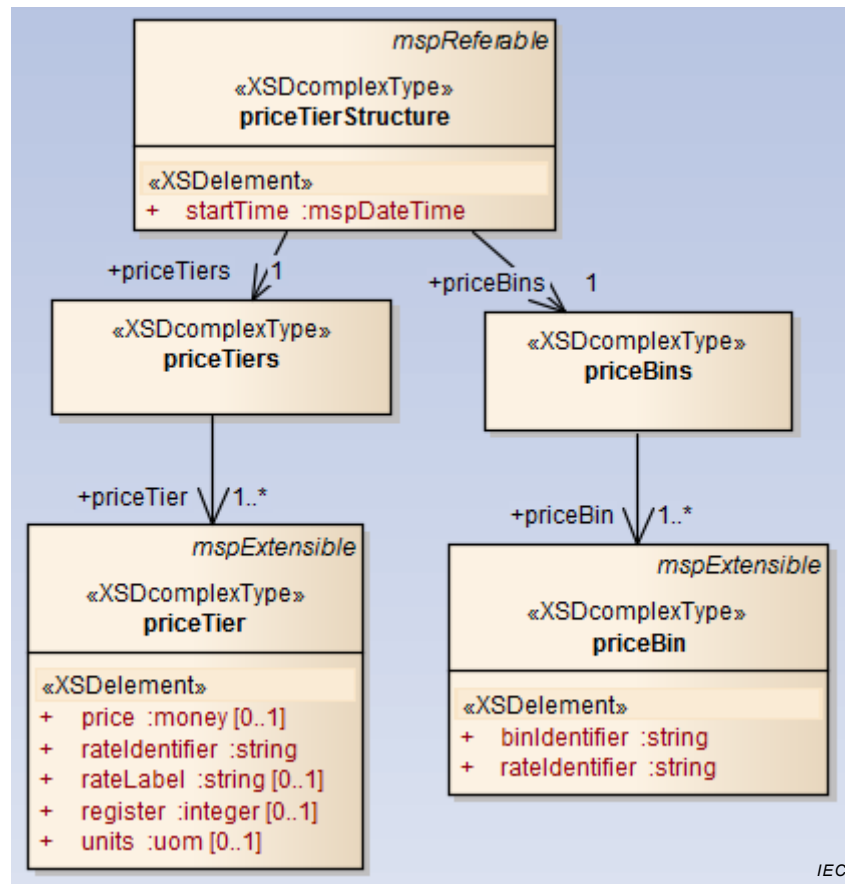


Figure 23 – UML representing MultiSpeak v.5 priceTierStructure classes

Annex A (normative)

Embedded/Reused Class Mappings

The purpose of this annex is to document structures that are used repeatedly, e.g. Address related classes. Often these classes and attributes are used repeatedly throughout many profiles. Rather than repeatedly document their use throughout the report, those structures will be documented once in this annex and referred to here.

These embedded structures that appear repeatedly throughout the respective profiles are classes that are used for addresses, phone numbers, service location, and electronic addresses. These are mapped in the Tables A.1 to A.4.

Table A.1 – Address Correlations

CIM	MultiSpeak	Comments
poBox	postOfficeBox	
postalCode	postalCode	
addressGeneral	addressGeneral	
buildName	buildingNumber	Both fields are strings
number	streetNumber	
prefix	streetPrefix	
suffix	streetSuffix	
suiteNumber	suiteNumber	
type	streetType	
code	TownCode	
country	country	
name	city	CIM uses name although in the annotation it is clear that city name is the intent
stateOrProvince	state	State in MultiSpeak can be either a state or province.
areaCode	areaCode	
cityCode	cityCode	
countryCode	countryCode	
extension	extension	
localNumber	localNumber	

Table A.2 – Phone Number Correlations

CIM	MultiSpeak	Comments
TelephoneNumbers.	phoneNumber.	
areaCode	areaCode	
countryCode	countryCode	
extension	extension	
localNumber	localNumber	

Table A.3 – Service Location related transformations

CIM	MultiSpeak	Comments
ServiceLocation .accessMethod .needsInspection .siteAccessProblem	ServiceLocation .description	The three CIM attributes can be concatenated and mapped to the MultiSpeak description attribute.

Table A.4 – Electronic Address related transformations

CIM	MultiSpeak	Comments
email	emailAddress .eMail .eMailType .priorityOrder	CIM only specifies a single email address. MultiSpeak uses an eMailList as a holder for 0 to many email addresses which can be ordered by type (personal, business, alternate, other) and priority (to specify the order in which they are used)
lan radio	otherContactItem .details .infoType	MultiSpeak does not support LAN or radio directly. It uses the other ContactInformation class as a holder for many of types of contact information. In this case the infoType would be set to "LAN", with the details the LAN attribute from CIM
TelephoneNumbers .areaCode .cityCode .countryCode .extension .localNumber	phoneNumber .areaCode .cityCode .countryCode .extension .localNumber	CIM allow multiple phone numbers. MultiSpeak allows multiple phone numbers through the phoneList class with a 0 to many phoneNumbers. MultiSpeak also includes phoneType and priorityOrder so that if more than one phone number is used the priority in which they are to be used is captured.

Annex B (informative)

Example Mapping of MultiSpeak to CIM-based services

B.1 On Demand Read: MultiSpeak to CIM integration via ESB

In the following example used to support the on demand read use case, the initiating system will use a MultiSpeak interface with the responding system using a CIM based interface. While the specific operations to support this use case are available in MultiSpeak, a corresponding service to support the CIM based interface will need to be crafted. To do this the implementation guidance found in IEC 61968-100 is utilized.

This proof of concept will use an ESB project as an example. This will facilitate the demonstration of using both MultiSpeak-based, and CIM-based services and to develop BPEL that will show the mapping of both message header and payload elements from one system to another.

To begin with create a new BPEL project which will display the BPEL design canvas, as shown in Figure B.1.

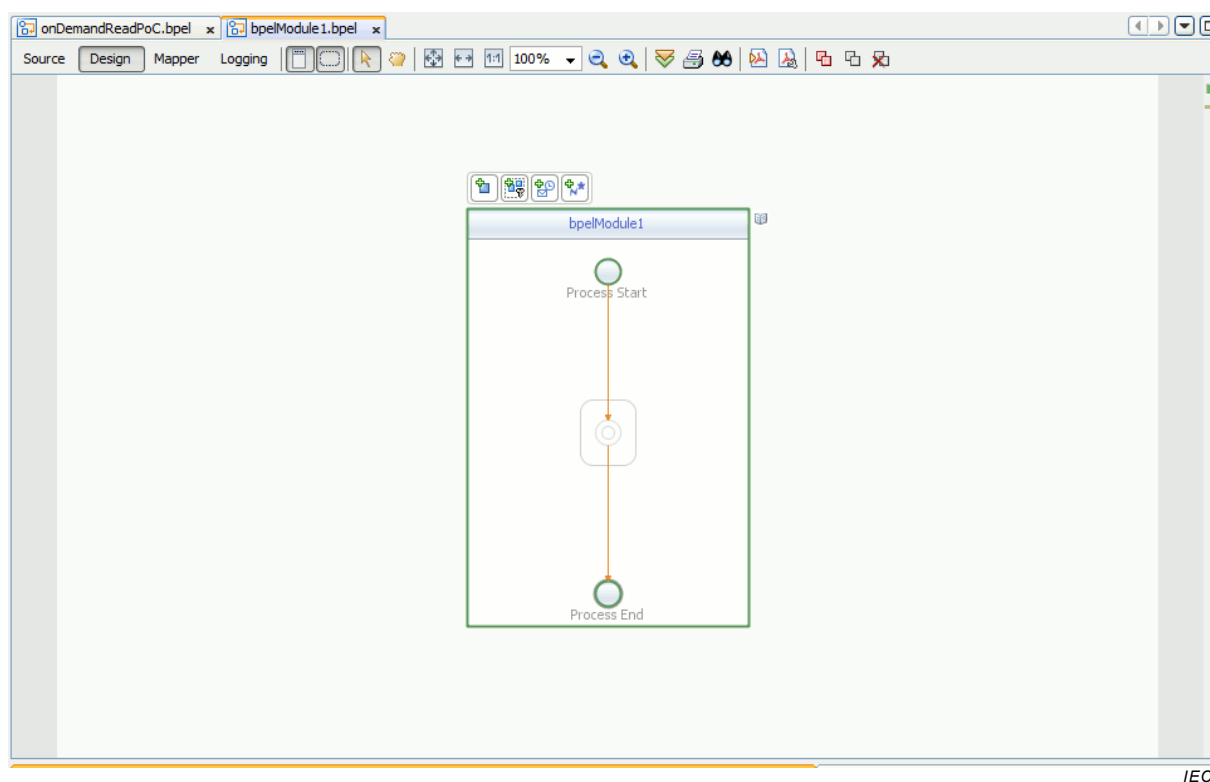


Figure B.1 – Blank BPEL project canvas that is the starting point for mapping web services and message payloads.

The systems integration will utilize the following services and operations:

MultiSpeak

MR_Server

InitiateMeterReadingsByMeterID

CB_Server

ReadingChangedNotification

CIM

GetMeterReadings

NOTE CIM web service and operations are developed according to the IEC 61968-100:2014 Implementation guide.

To begin using these resources the WSDLs and XSDs for each service are placed in the src folder of the project. To assist with the organization separate folders will be used for the MultiSpeak and CIM resources, as shown in Figure B.2.

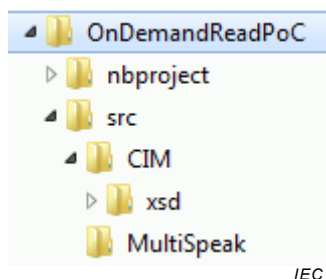


Figure B.2 – Src folder with subfolders for MultiSpeak and CIM resources to be used in the project.

Note that the CIM folder has a subfolder for XSD files. This is per the IEC 61968-100:2014 guidance on developing WSDLs and XSDs.

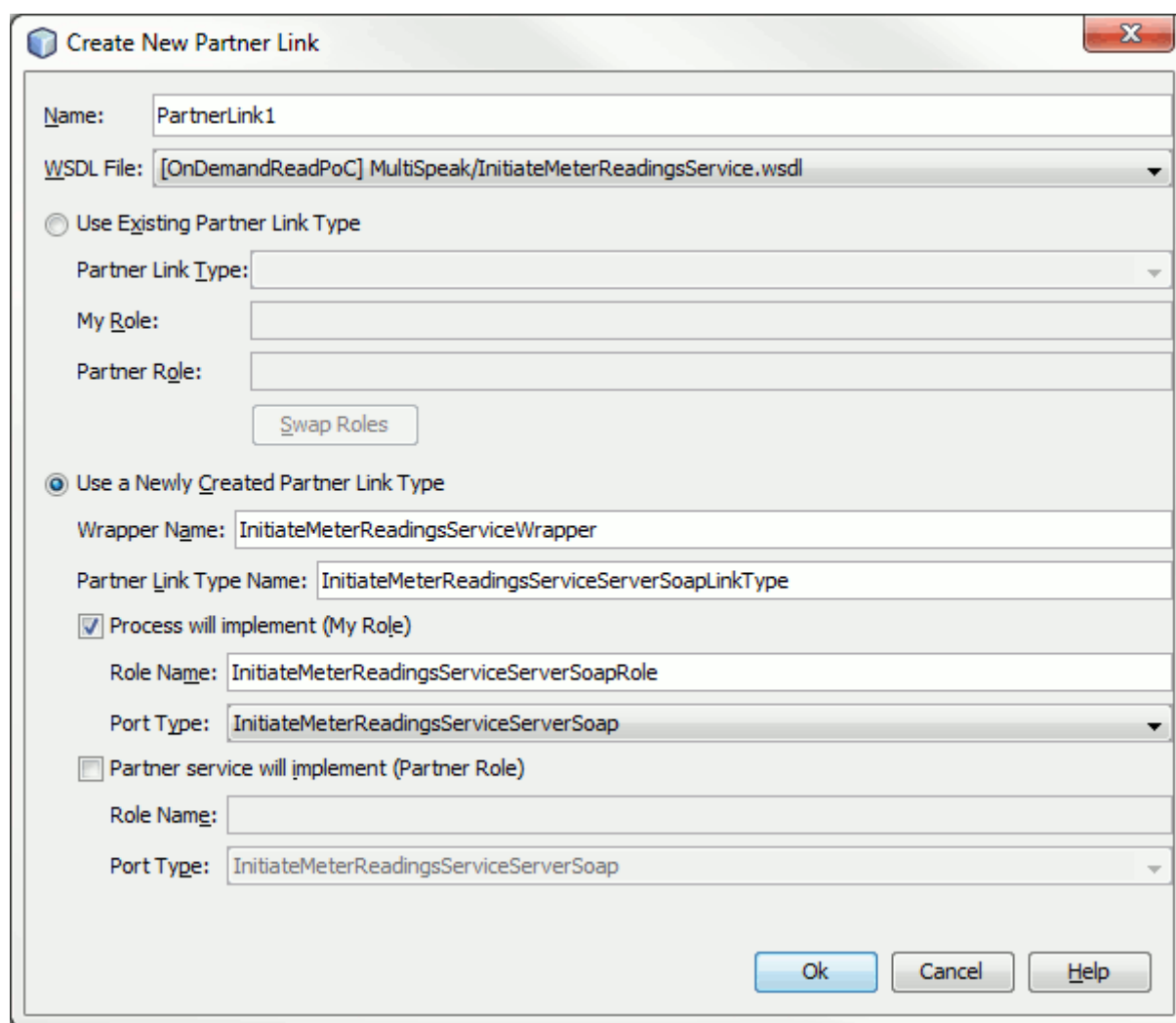
B.2 Wiring the Integration

Once the resources are available to the project in the src folder they can be utilized in the design canvas. The MultiSpeak service will be mapped on the left side of the canvas, CIM on the right. Dragging and dropping the MR_Server onto the left-side of the canvas prompts for the creation of a PartnerLink.

NOTE Since the MR_server contains hundreds of operations, a smaller WSDL has been created that contains only the pertinent operations for this exercise and renamed “InitiateMeterReadingsService” shown in the figure below. This is used to keep the BPEL design canvas to a useful size for this illustration.

The PartnerLink dialog will create partner link name, wrapper, partner link type, role name, and port name automatically on the user's behalf.

Clicking OK will complete the creation of the partner link, as shown in Figure B.3.



Create New Partner Link

Name: PartnerLink1

WSDL File: [OnDemandReadPoC] MultiSpeak/InitiateMeterReadingsService.wsdl

☐ Use Existing Partner Link Type

Partner Link Type:

My Role:

Partner Role:

Swap Roles

☒ Use a Newly Created Partner Link Type

Wrapper Name: InitiateMeterReadingsServiceWrapper

Partner Link Type Name: InitiateMeterReadingsServiceServerSoapLinkType

☒ Process will implement (My Role)

Role Name: InitiateMeterReadingsServiceServerSoapRole

Port Type: InitiateMeterReadingsServiceServerSoap

☐ Partner service will implement (Partner Role)

Role Name:

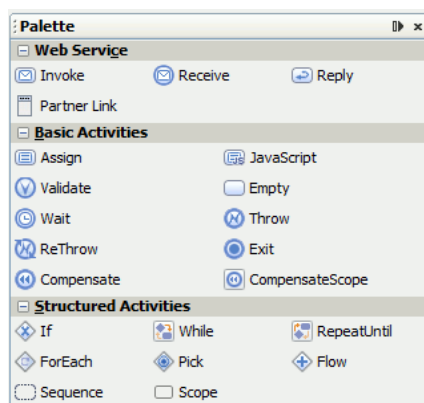
Port Type: InitiateMeterReadingsServiceServerSoap

Ok Cancel Help

IEC

Figure B.3 – Creation of a BPEL partner link for the InitiateMeterReadingsService.

Once the partner link has been created the Request-Reply pattern can be developed. This is accomplished by dragging and dropping a Receive icon from the BPEL palette shown in Figure B.4.



IEC

Figure B.4 – BPEL palette options

In the Receive widget the name and partner link are created automatically. The desired operation and input variable can be selected by using the operation dropdown and Create... button respectively, as shown in Figure B.5.

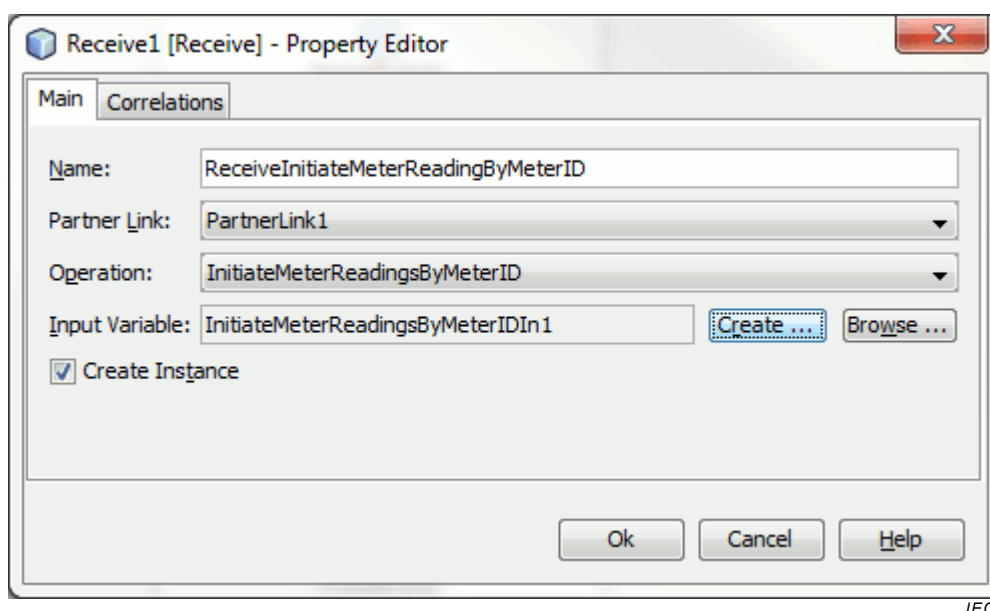
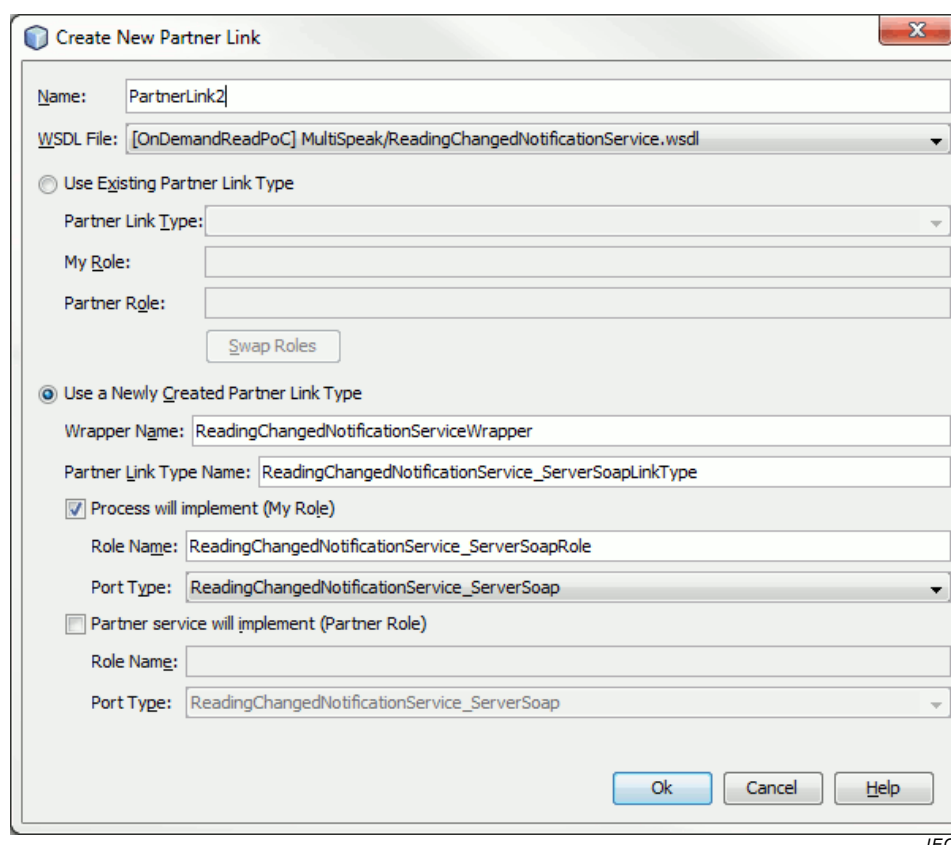


Figure B.5 – Completing the creation of the BPEL Receive widget

To create the Reply pattern the CB_Server WSDL is dragged onto the BPEL canvas to create the second partner link.

NOTE For this illustration, since the CB_server contains hundreds of operations, the WSDL has been edited to contain only the operation that is used for this example; ReadingChangedNotification.

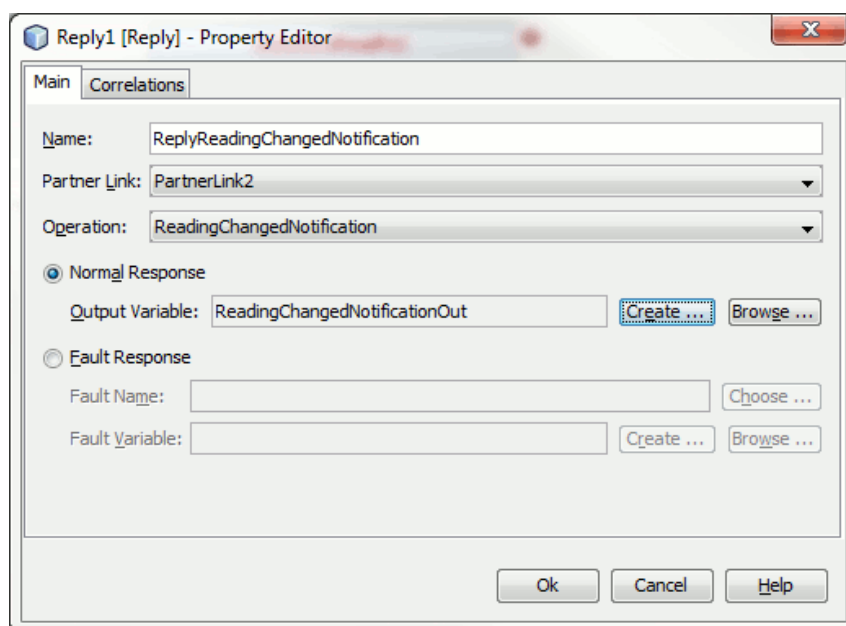
As before, the Partner Link name, wrapper, partner link type name, role name and port have all been automatically filled in. Clicking OK makes this second PartnerLink available on the BPEL canvas, as shown in Figure B.6.



IEC

Figure B.6 – Creation of the second MultiSpeak partner link to utilize the ReadingChangedNotification service.

Selecting the Reply widget and dragging it onto the BPEL canvas allows the ReadingChangedNotification to be wired up to complete the Request-Reply operation. The set of steps to accomplish this action is shown in figures B.7, B.8, B.9 and B.10 successively.



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Figure B.7 – Creating the Reply widget to utilize the ReadingChangedNotification operation

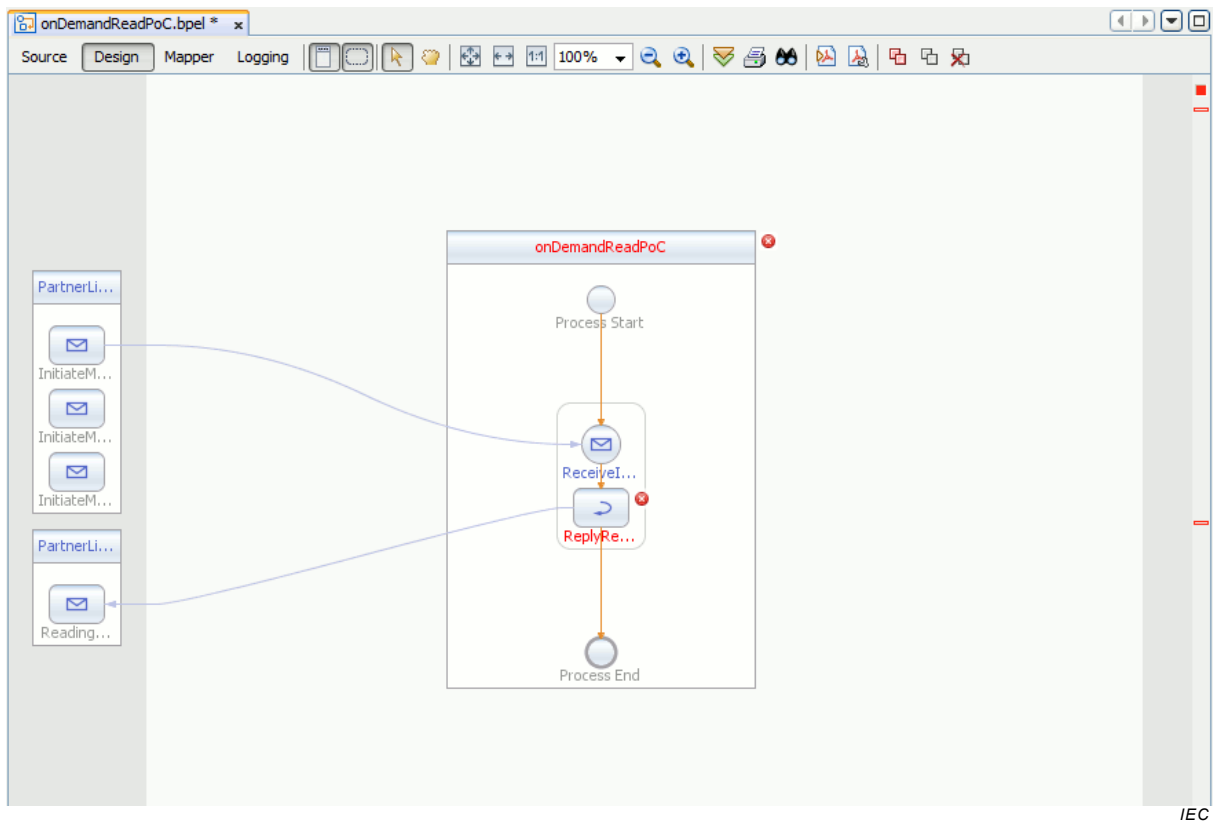
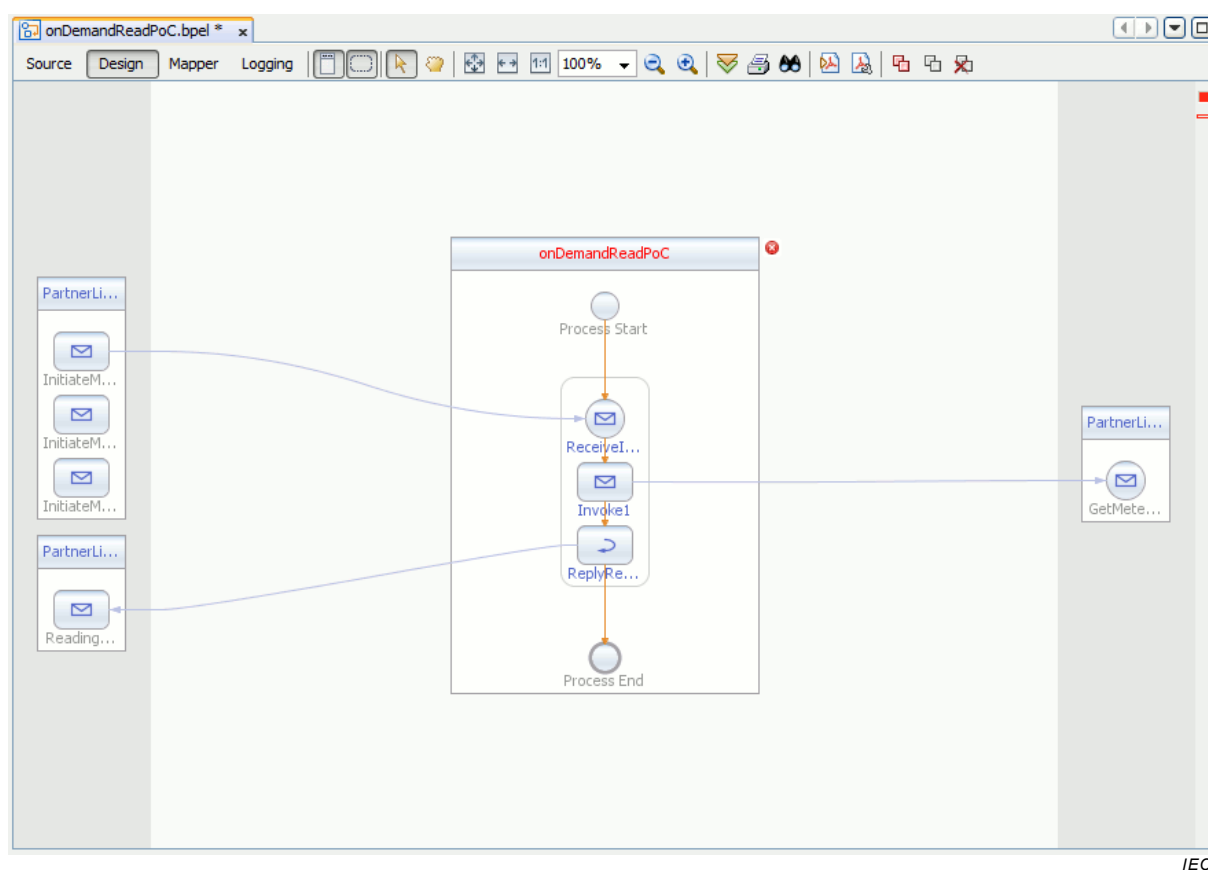


Figure B.8 – The BPEL canvas with the Request-Reply pattern 'wired up" to utilize the InitiateMeterReadingsByMeterID and the ReadingChangedNotification operation

Figure B.9 – Method invoke property editor



IEC

Figure B.10 – "Wiring up the service in OpenESB"

Annex C (informative)

UML of MultiSpeak Equivalent profiles

These diagrams illustrate how MultiSpeak equivalents of the CIM profiles were created. MultiSpeak does not use the same “profile” concept as the CIM, so profiles have to be created before an appropriate mapping can occur. The benefit of this approach is that instead of attempting to compare the entire models of both standards, they can be compared in the context of the profile; the subset of the model that is used for work or messaging. The goal behind the generation of these profiles is that when using the XSD generation add-in to Enterprise Architect, that the equivalent profiles could be generated in one pass, without any required editing. However, this sometimes required that some associations or classes had to be added to the MultiSpeak profile to accomplish this. These additions have been highlighted in red to distinguish them from the core MultiSpeak class and associations.

These examples are shown in Figures C.1 through C.12 for the following profiles:

- ConnectDisconnect, Figure C.1
- EndDeviceAssets, Figure C.2
- MeterAssetReading, Figure C.3
- MeterReadings, Figure C.4
- MeterServiceRequests, Figure C.5
- MeterSystemEvents, Figure C.6
- PricingStructureConfig, Figure C.7
- ReceiptRecord, Figure C.8
- SDPLocationConfig, Figure C.9
- ServiceDeliveryPointConfig, Figure C.10
- ServiceSupplierConfig, Figure C.11
- TransactionRecord, Figure C.12

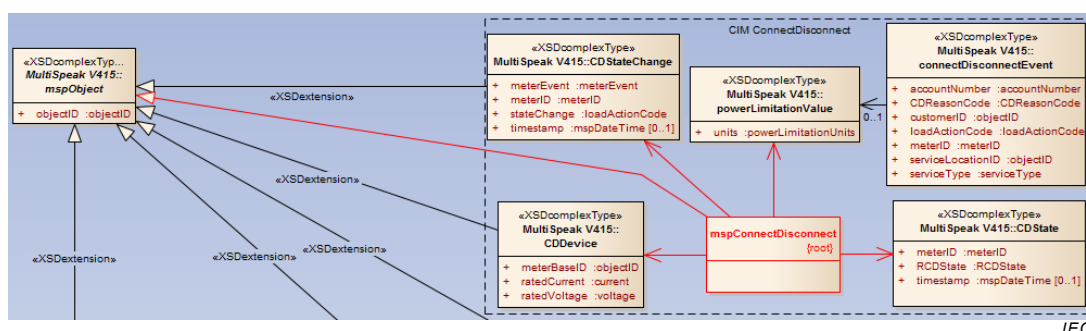


Figure C.1 – ConnectDisconnect portion of the UML of the MultiSpeak equivalent of the EndDeviceAssets profile

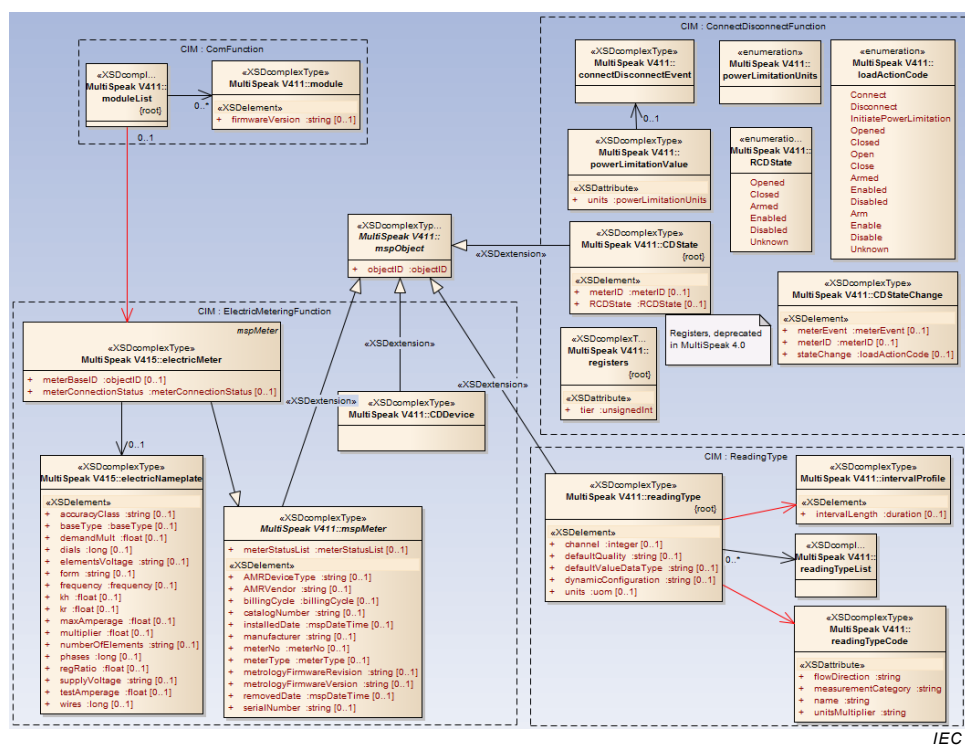


Figure C.2 – UML diagram of MultiSpeak equivalent of CIM EndDeviceAssets profile

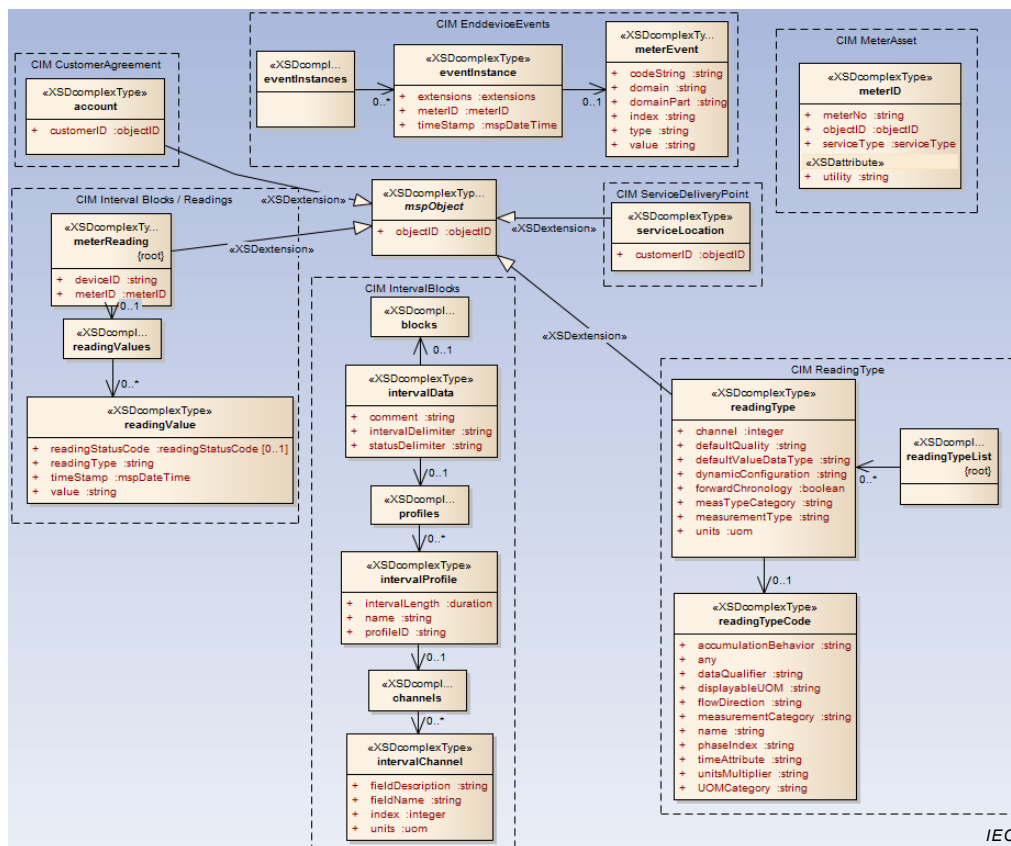


Figure C.3 – UML diagram of MultiSpeak equivalent to CIM MeterAssetReading profile

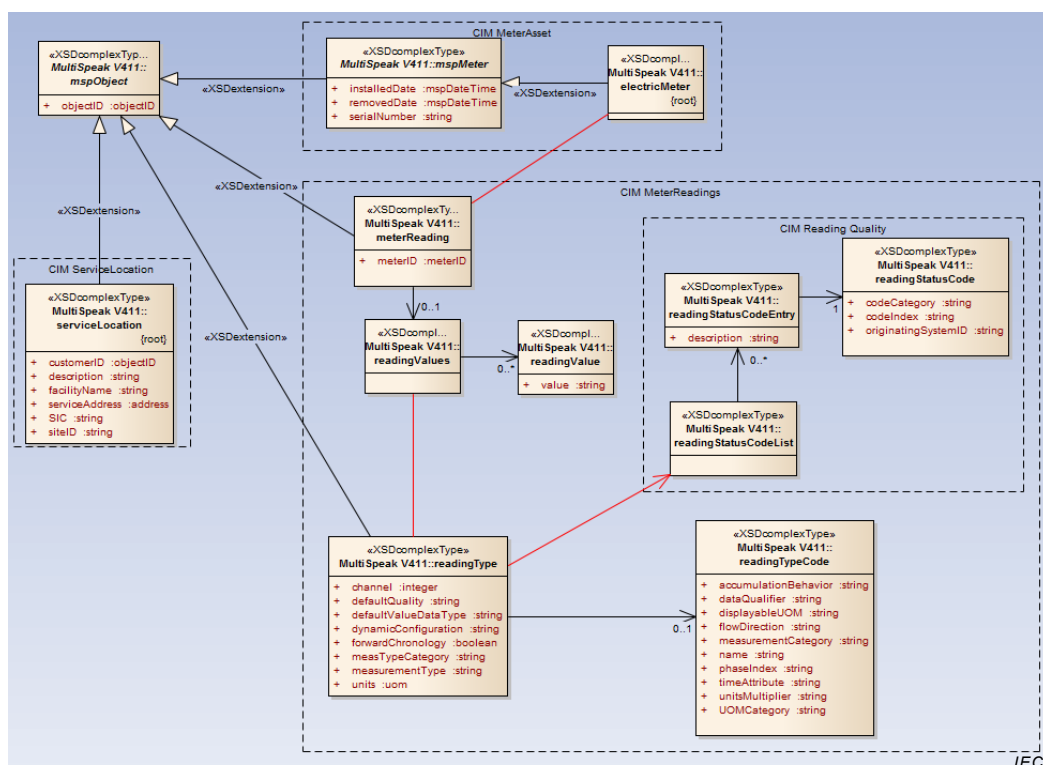


Figure C.4 – UML for MultiSpeak equivalent of the CIM MeterReadings profile

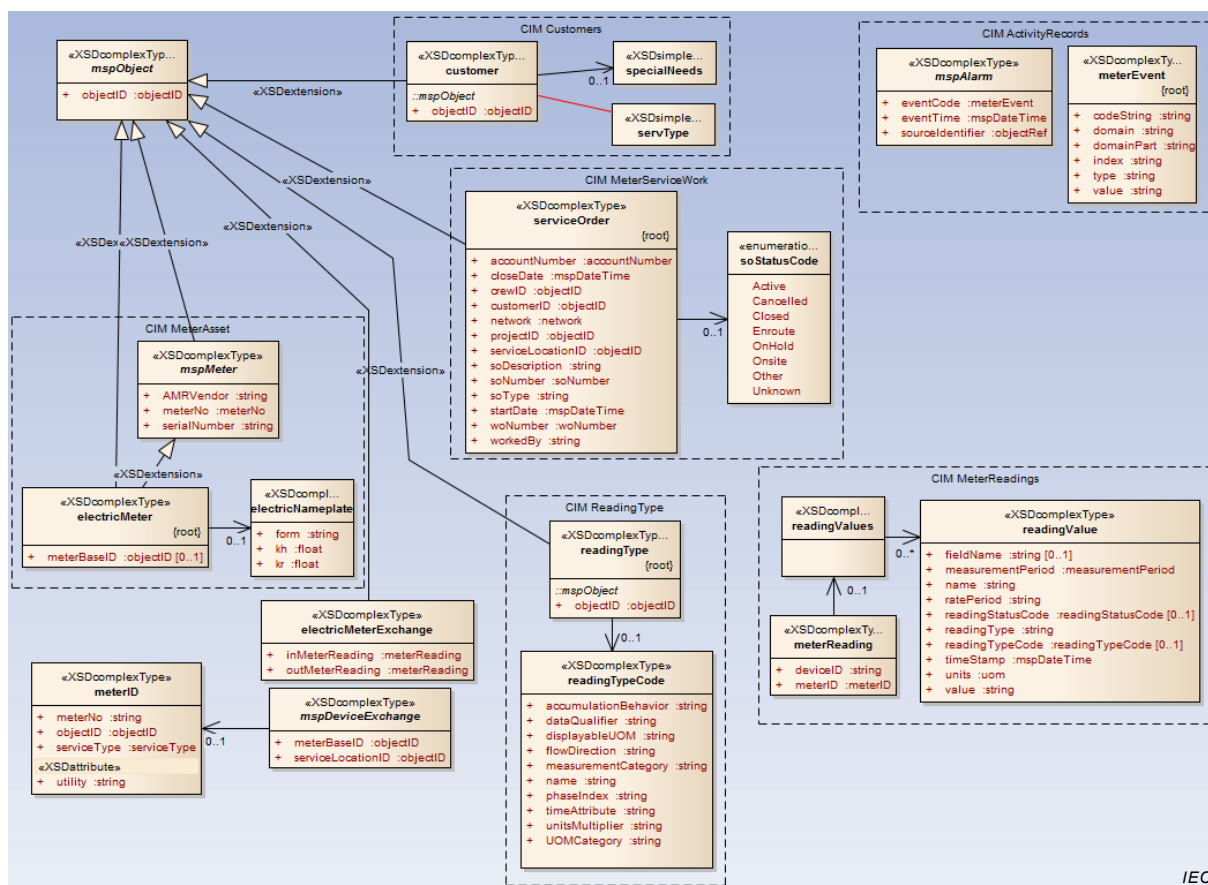


Figure C.5 – UML of MultiSpeak equivalent of the CIM MeterServiceRequests profile

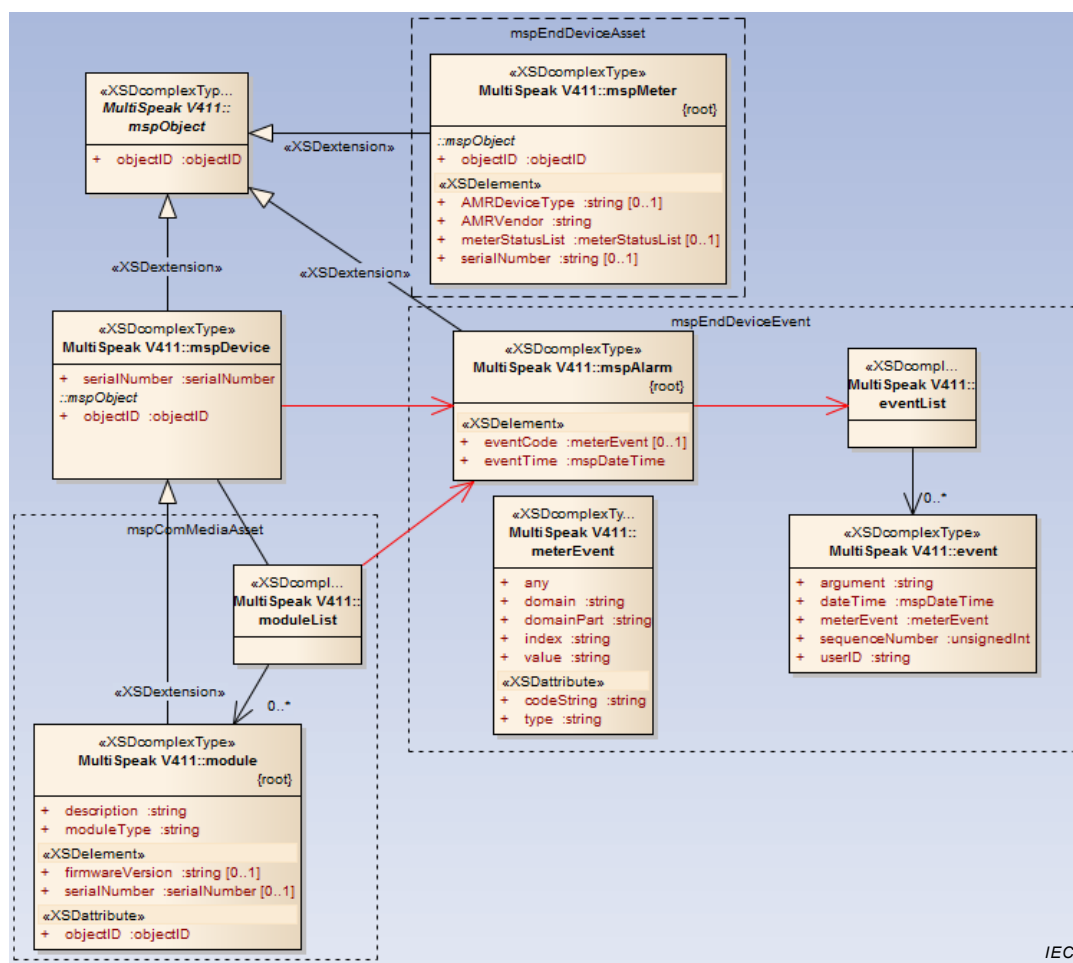


Figure C.6 – UML of the MultiSpeak equivalent to the CIM MeterSystemEvents profile

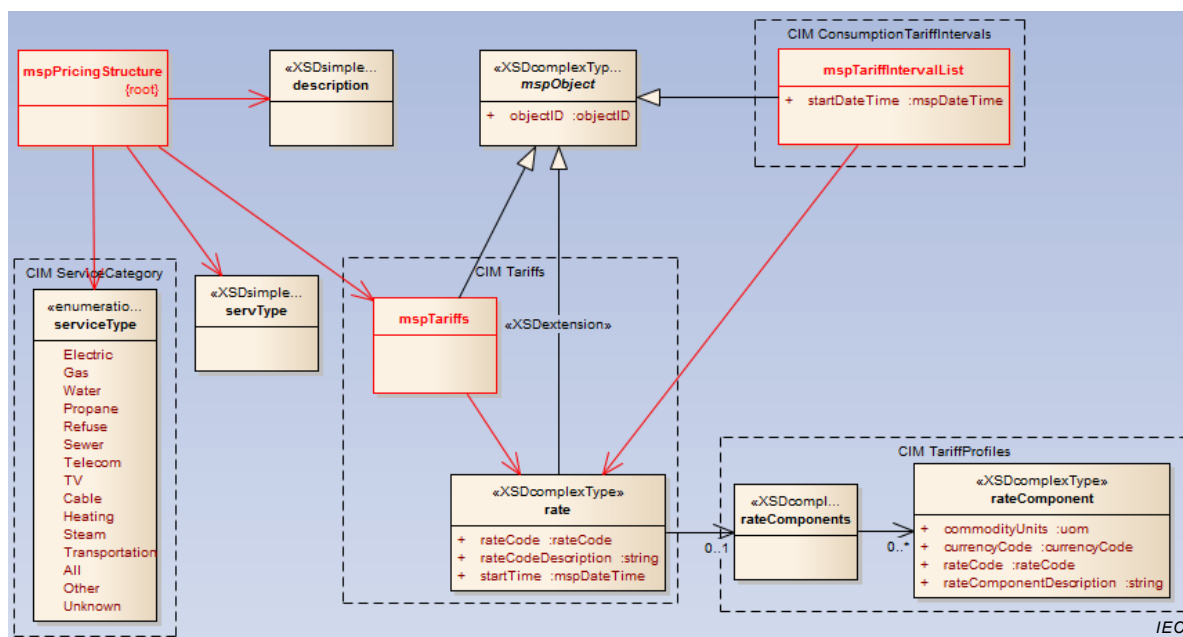


Figure C.7 – UML for the MultiSpeak equivalent of the CIM PricingStructureConfig

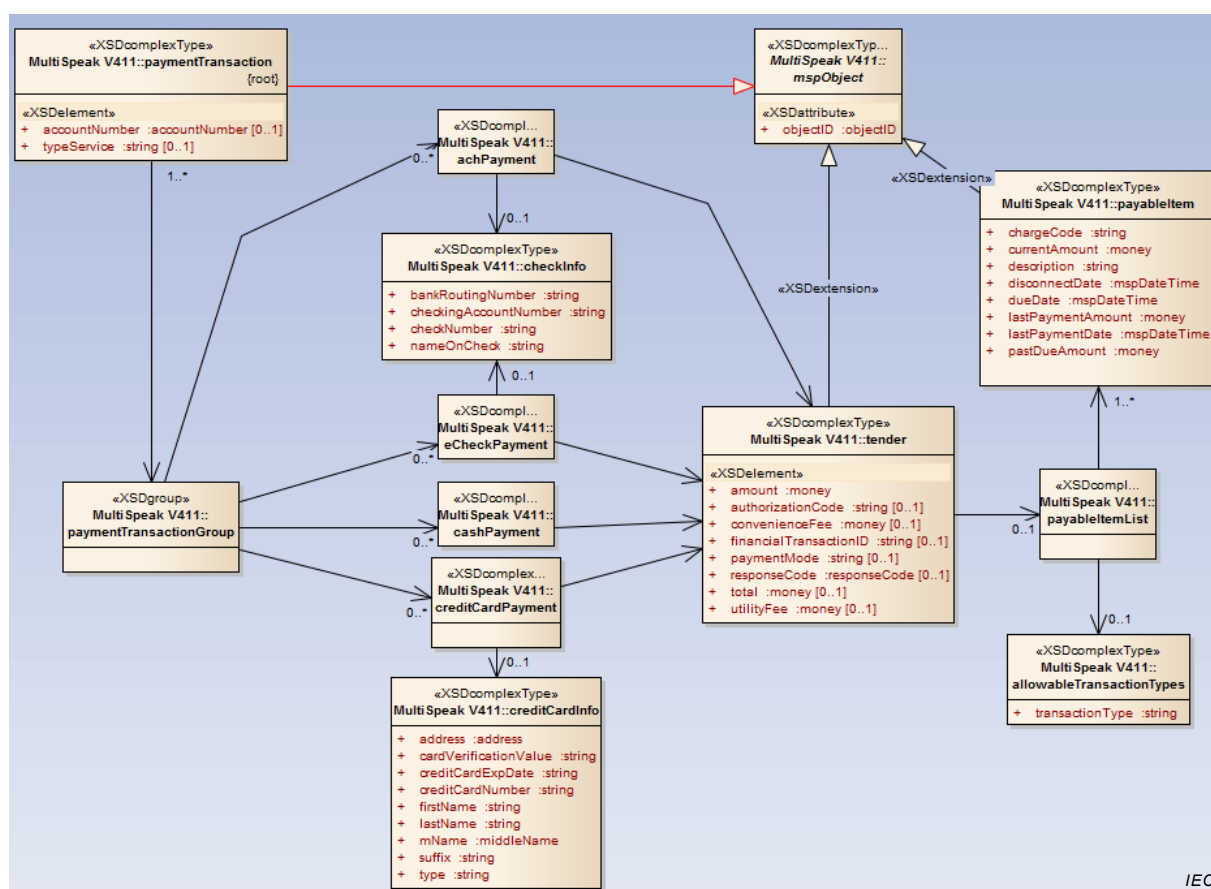


Figure C.8 – UML of the MultiSpeak equivalent of the CIM ReceiptRecord profile

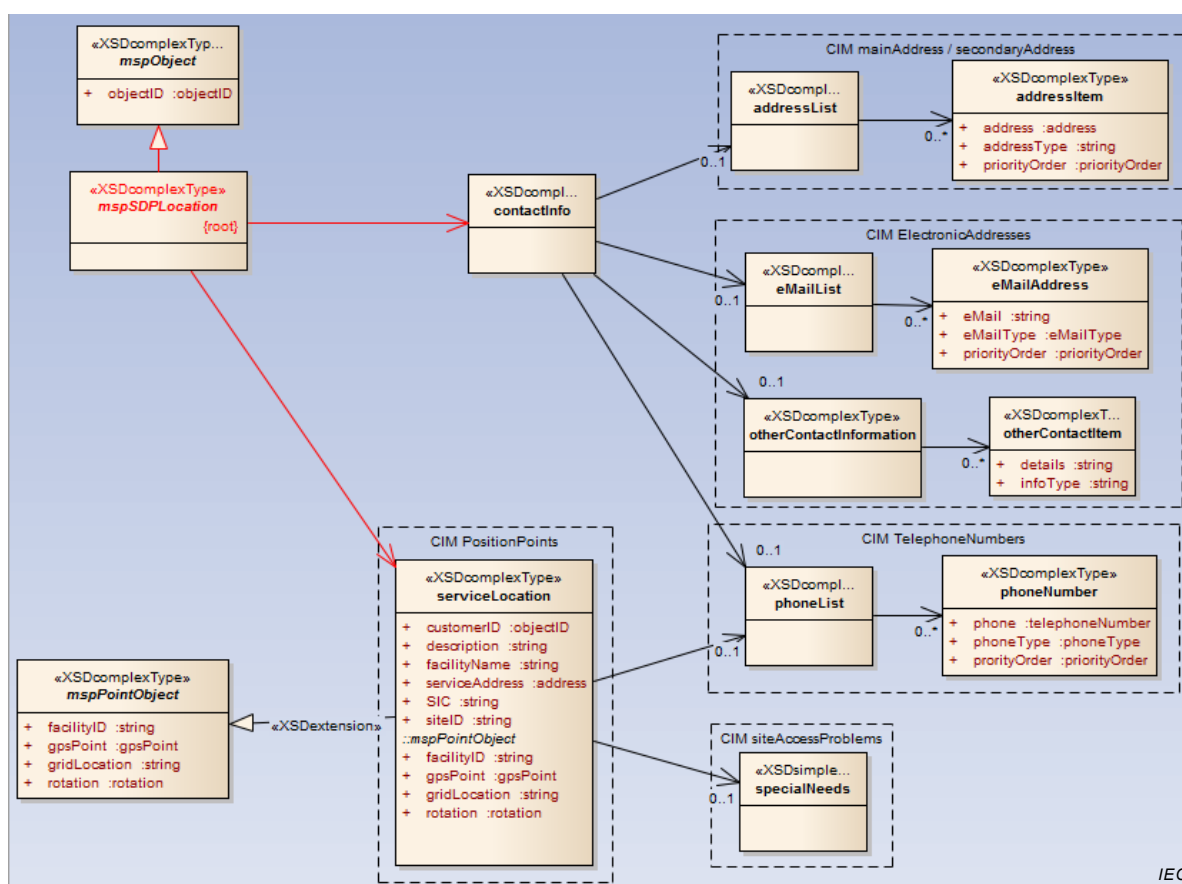


Figure C.9 – UML of the MultiSpeak equivalent of the CIM SDPLocationConfig profile

This profile uses an abstract class `mspSDILocation` as a placeholder to tie the other MultiSpeak related classes used to build this profile.

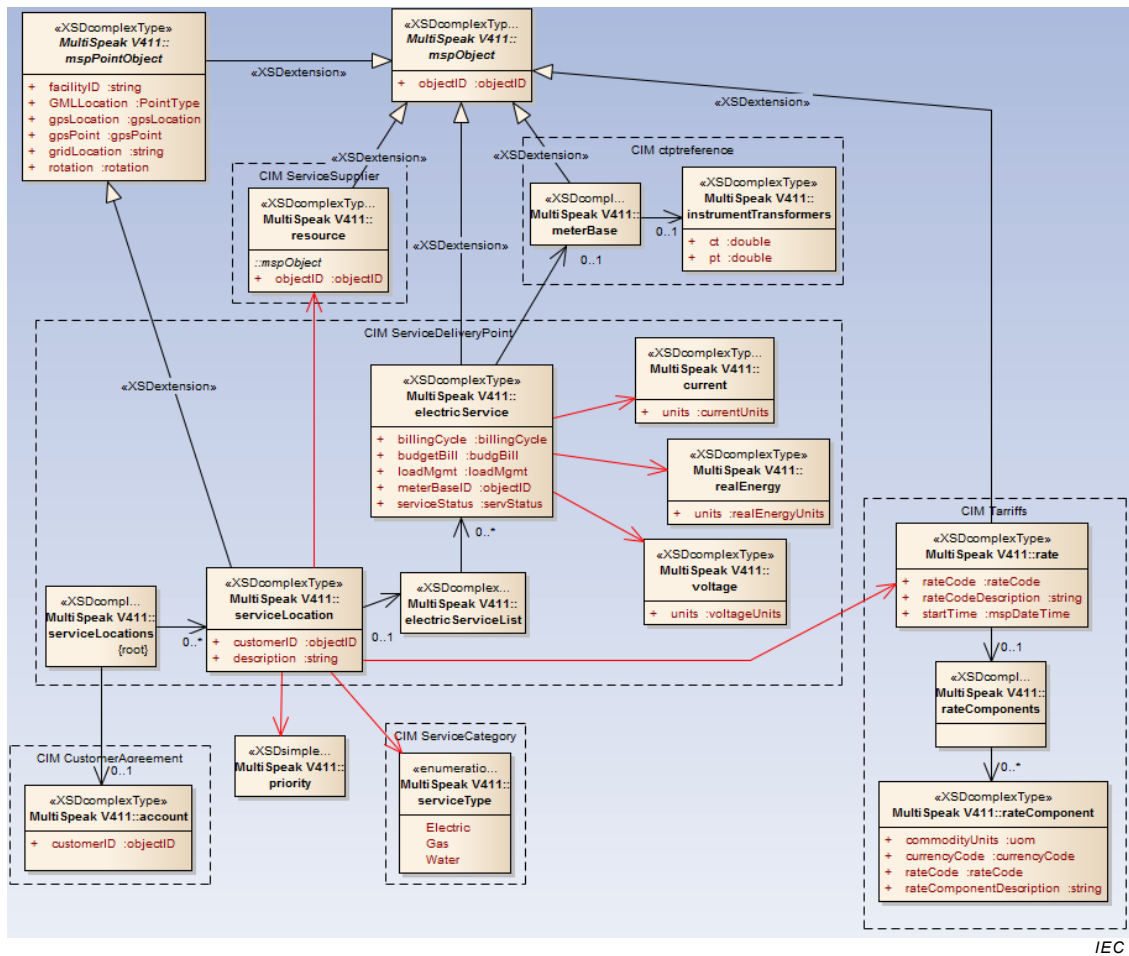
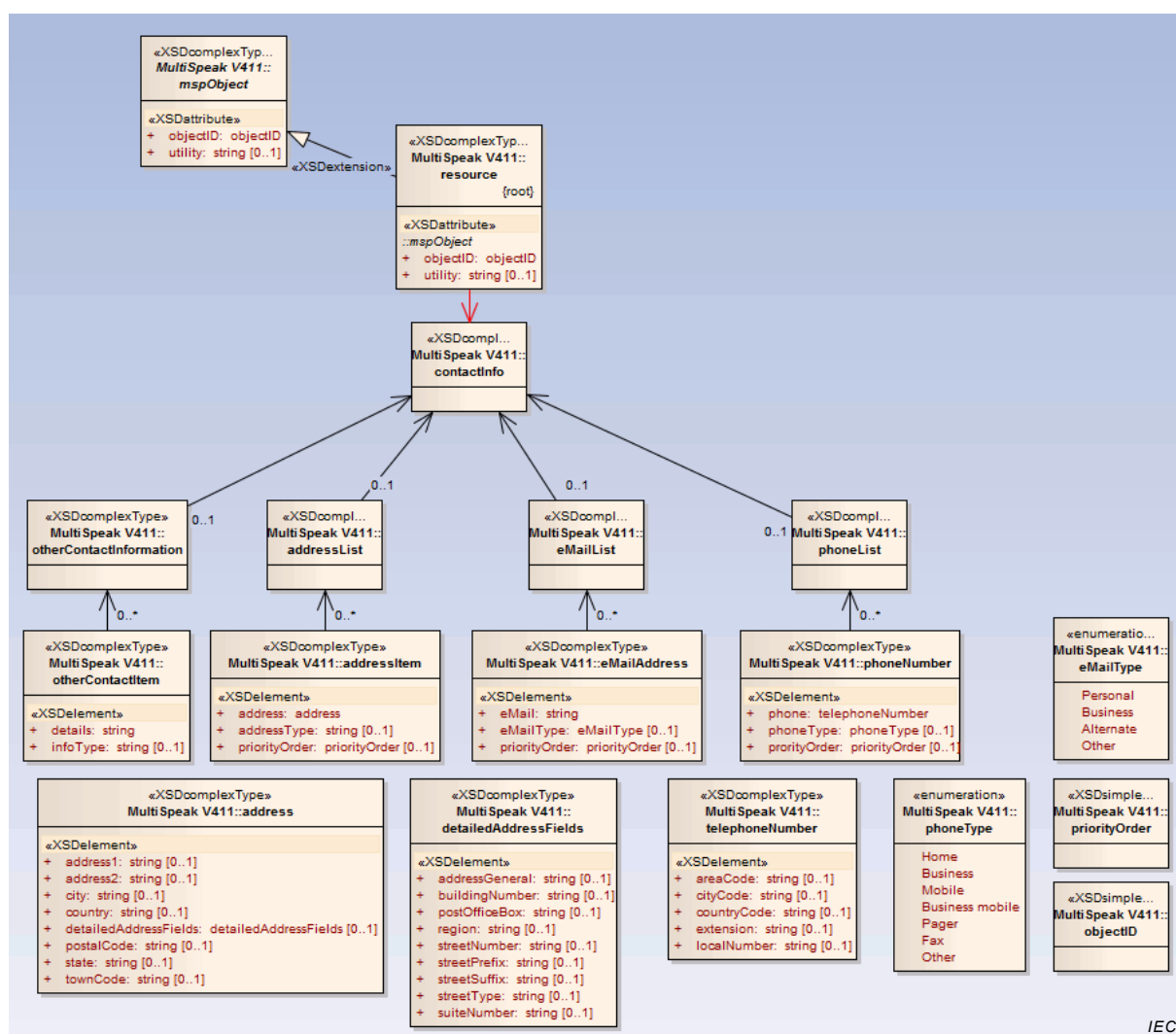


Figure C.10 – UML for MultiSpeak equivalent of the CIM ServiceDeliveryPointConfig profile



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Figure C.11 – UML of MultiSpeak equivalent ServiceSupplierConfig profile

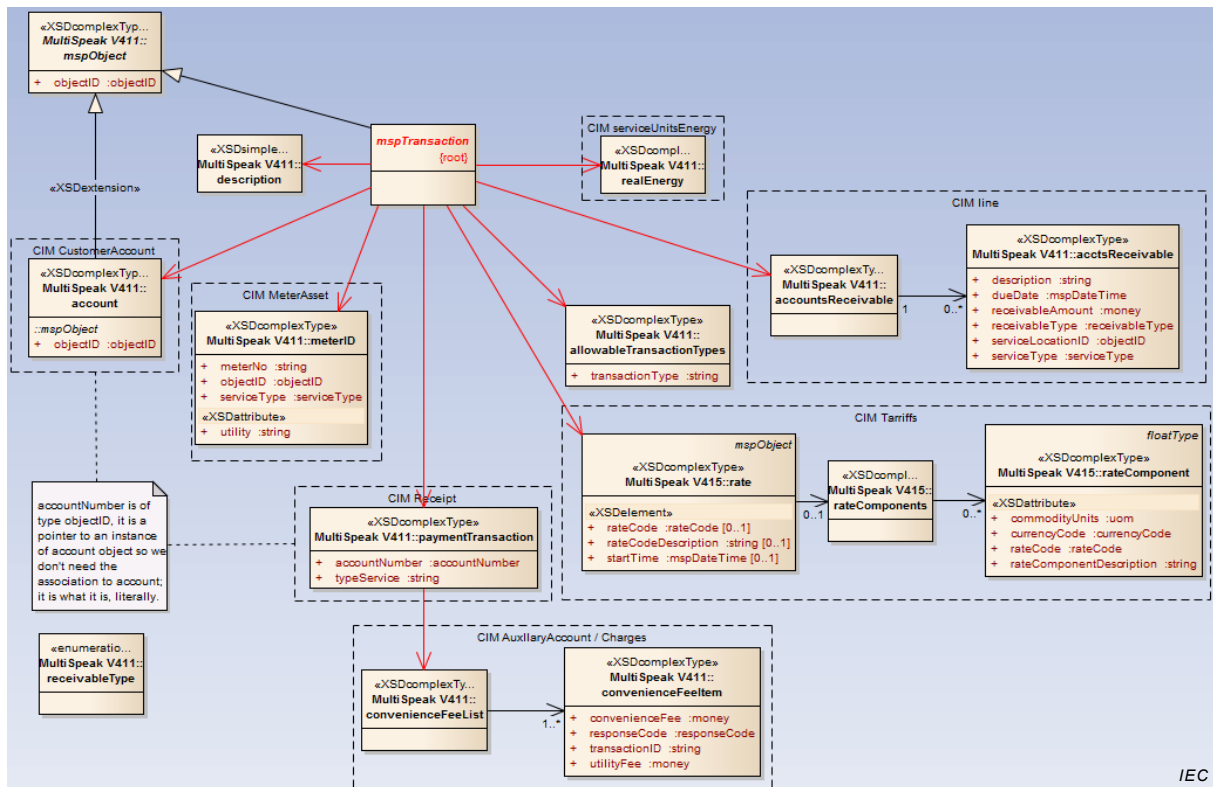


Figure C.12 – UML of MultiSpeak equivalent of CIM TransactionRecord

Annex D (normative)

MultiSpeak Equivalent CIM Profiles

These equivalent profiles are based on MultiSpeak 4.1.6, and the CIM as it was represented in IEC 61968-9:2009, Meter Reading and Control.

mSPAuxiliaryAgreementConfig

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL" xmlns:xs="http://www.w3.org/2001/XMLSchema"
  xmlns:cimp="http://www.iec.org/CimProfile" xmlns="http://www.multispeak.org/2012/mSPAuxiliaryAgreementConfig"
  xmlns:mSP="http://www.multispeak.org/2012/mSPAuxiliaryAgreementConfig"
  targetNamespace="http://www.multispeak.org/2012/mSPAuxiliaryAgreementConfig" elementFormDefault="qualified"
  attributeFormDefault="unqualified" xml:base="http://www.multispeak.org/2012/mSPAuxiliaryAgreementConfig">
  <xs:element name="mSPAuxiliaryAgreementConfig" type="mSPAuxiliaryAgreementConfig"/>
  <xs:complexType name="mSPAuxiliaryAgreementConfig">
    <xs:sequence>
      <xs:element name="AuxiliaryAgreement" type="AuxiliaryAgreement" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="account" sawSDL:modelReference="MultiSpeak v411 /mSPAuxiliaryAgreementConfig/account">
    <xs:annotation>
      <xs:documentation>This is a customer account.</xs:documentation>
    </xs:annotation>
    <xs:complexContent>
      <xs:extension base="mSPObject">
        <xs:sequence>
          <xs:element name="accountsReceivable" type="accountsReceivable" minOccurs="1"
            sawSDL:modelReference="MultiSpeak v411 /mSPAuxiliaryAgreementConfig/account.accountsReceivable">
            <xs:annotation>
              <xs:documentation>Accounts receivable information for this account.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="billDate" type="mSPDateTime" minOccurs="1" sawSDL:modelReference="MultiSpeak v411
            /mSPAuxiliaryAgreementConfig/account.billDate">
            <xs:annotation>
              <xs:documentation>Date of last bill.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="billingTerms" type="xs:string" minOccurs="1" sawSDL:modelReference="MultiSpeak v411
            /mSPAuxiliaryAgreementConfig/account.billingTerms">
            <xs:annotation>
              <xs:documentation>Describes the way that the account is billed. Suggested enumeration is Prepay, Budget, Cycle,
            or Other.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="budgetBill" type="budgBill" minOccurs="1" sawSDL:modelReference="MultiSpeak v411
            /mSPAuxiliaryAgreementConfig/account.budgetBill">
            <xs:annotation>
              <xs:documentation>Budget billing code for this account</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="customerID" type="objectID" minOccurs="1" sawSDL:modelReference="MultiSpeak v411
            /mSPAuxiliaryAgreementConfig/account.customerID">
            <xs:annotation>
              <xs:documentation>Link to the objectID (foreign key) of the customer that has this account.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="lastBillAmount" type="money" minOccurs="1" sawSDL:modelReference="MultiSpeak v411
            /mSPAuxiliaryAgreementConfig/account.lastBillAmount">
            <xs:annotation>
              <xs:documentation>The amount of the last bill to the customer.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="lastPaymentAmount" type="money" minOccurs="1" sawSDL:modelReference="MultiSpeak v411
            /mSPAuxiliaryAgreementConfig/account.lastPaymentAmount">
            <xs:annotation>
              <xs:documentation>Amount of the last payment made.</xs:documentation>
            </xs:annotation>
          </xs:element>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
```

```

</xs:element>
<xs:element name="lastPaymentDate" type="mspDateTime" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxiliaryAgreementConfig/account.lastPaymentDate">
  <xs:annotation>
    <xs:documentation>Date of last payment.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="paymentDueDate" type="mspDateTime" minOccurs="1" sawsdl:modelReference="MultiSpeak
v411 /mspAuxiliaryAgreementConfig/account.paymentDueDate">
  <xs:annotation>
    <xs:documentation>Date and time that the next payment is due.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="shutOffDate" type="mspDateTime" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxiliaryAgreementConfig/account.shutOffDate">
  <xs:annotation>
    <xs:documentation>Shut off date for non-payment.</xs:documentation>
  </xs:annotation>
</xs:element>
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="accountNumber" sawsdl:modelReference="MultiSpeak v411
/mspAuxiliaryAgreementConfig/accountNumber">
  <xs:annotation>
    <xs:documentation>Account number for this service location. Depending on the system of record, this may be either the
account.objectID or the account.objectName.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="accounts" sawsdl:modelReference="MultiSpeak v411 /mspAuxiliaryAgreementConfig/accounts">
  <xs:sequence>
    <xs:element name="account" type="account" minOccurs="0" maxOccurs="unbounded" sawsdl:modelReference="MultiSpeak
v411 /mspAuxiliaryAgreementConfig/accounts.account"/>
    <xs:element name="AuxiliaryAgreement" type="AuxiliaryAgreement" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="MultiSpeak v411 /mspAuxiliaryAgreementConfig/accounts.AuxiliaryAgreement"/>
    <xs:element name="remainingBalance" type="remainingBalance" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="MultiSpeak v411 /mspAuxiliaryAgreementConfig/accounts.remainingBalance"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="accountsReceivable" sawsdl:modelReference="MultiSpeak v411
/mspAuxiliaryAgreementConfig/accountsReceivable">
  <xs:annotation>
    <xs:documentation>Money due on a customer account.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="acctsReceivable" type="acctsReceivable" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="MultiSpeak v411 /mspAuxiliaryAgreementConfig/accountsReceivable.acctsReceivable"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="acctsReceivable" sawsdl:modelReference="MultiSpeak v411
/mspAuxiliaryAgreementConfig/acctsReceivable">
  <xs:sequence>
    <xs:element name="description" type="xs:string" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxiliaryAgreementConfig/acctsReceivable.description"/>
    <xs:element name="dueDate" type="mspDateTime" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxiliaryAgreementConfig/acctsReceivable.dueDate">
      <xs:annotation>
        <xs:documentation>The date that the receivable is (or was) due.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="receivableAmount" type="money" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxiliaryAgreementConfig/acctsReceivable.receivableAmount">
      <xs:annotation>
        <xs:documentation>Amount of money that is due for this receivable, expressed in default currency unless the
currencyCode attribute is set to a different currency type.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="AuxiliaryAgreement" abstract="true" sawsdl:modelReference="MultiSpeak v411
/mspAuxiliaryAgreementConfig/AuxiliaryAgreement">
  <xs:sequence>
    <xs:complexType name="billingCycle" sawsdl:modelReference="MultiSpeak v411 /mspAuxiliaryAgreementConfig/billingCycle">
      <xs:restriction base="xs:string"/>
    </xs:complexType>
  </xs:sequence>

```

```

</xs:simpleType>
<xs:complexType name="billingData" sawsdl:modelReference="MultiSpeak v411 /mspAuxilliaryAgreementConfig/billingData">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="accountNumber" type="accountNumber" minOccurs="1" sawsdl:modelReference="MultiSpeak
v411 /mspAuxilliaryAgreementConfig/billingData.accountNumber">
          <xs:annotation>
            <xs:documentation>Account number for this service location. Depending on the system of record, this may be
either the account.objectID or the account.objectName.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="billingDate" type="mspDateTime" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/billingData.billingDate"/>
        <xs:element name="AuxilliaryAgreement" type="AuxilliaryAgreement" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="MultiSpeak v411 /mspAuxilliaryAgreementConfig/billingData.AuxilliaryAgreement"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="billingDetail" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/billingDetail">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="accountNumber" type="accountNumber" minOccurs="1" sawsdl:modelReference="MultiSpeak
v411 /mspAuxilliaryAgreementConfig/billingDetail.accountNumber">
          <xs:annotation>
            <xs:documentation>Account number for this service location. Depending on the system of record, this may be
either the account.objectID or the account.objectName.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="billingCycle" type="billingCycle" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/billingDetail.billingCycle"/>
        <xs:element name="creditCharge" type="money" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/billingDetail.creditCharge"/>
        <xs:element name="debitCharge" type="money" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/billingDetail.debitCharge"/>
        <xs:element name="payment" type="money" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/billingDetail.payment"/>
        <xs:element name="priorBalance" type="money" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/billingDetail.priorBalance"/>
        <xs:element name="AuxilliaryAgreement" type="AuxilliaryAgreement" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="MultiSpeak v411 /mspAuxilliaryAgreementConfig/billingDetail.AuxilliaryAgreement"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:simpleType name="budgBill" sawsdl:modelReference="MultiSpeak v411 /mspAuxilliaryAgreementConfig/budgBill">
  <xs:annotation>
    <xs:documentation>Budget billing code.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="currencyCode" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/currencyCode">
  <xs:annotation>
    <xs:documentation>Three letter currency code as standardized in ISO4217-2001. Typically is the two letter Internet domain
code for the country followed by a one letter designator for the currency; for instance the USA dollar is
USD.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string">
    <xs:enumeration value="USD" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/currencyCode.USD"/>
    <xs:enumeration value="EUR" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/currencyCode.EUR"/>
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="customer" sawsdl:modelReference="MultiSpeak v411 /mspAuxilliaryAgreementConfig/customer">
  <xs:sequence>
    <xs:element name="accounts" type="accounts" minOccurs="0" maxOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/customer.accounts"/>
    <xs:element name="AuxilliaryAgreement" type="AuxilliaryAgreement" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="MultiSpeak v411 /mspAuxilliaryAgreementConfig/customer.AuxilliaryAgreement"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="description" sawsdl:modelReference="MultiSpeak v411 /mspAuxilliaryAgreementConfig/description">

```



```

<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="money" sawsdl:modelReference="MultiSpeak v411 /mspAuxilliaryAgreementConfig/money">
  <xs:sequence>
    <xs:element name="currencyCode" type="currencyCode" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/money.currencyCode">
      <xs:annotation>
        <xs:documentation>Three letter currency code as standardized in ISO4217-2001. Typically is the two letter Internet
domain code for the country followed by a one letter designator for the currency; for instance the USA dollar is
USD.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="mspDateTime" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable
to denote UTC or "Zulu" time.all dates will be formatted as: 2005-06-15T11:30:22.56-07:00 where the date is in YYYY-MM-DD
format and the time is in military time, beginning after the date-time separator &ldquo;T&rdquo;, in hh:mm:ss.ss
format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds
is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian)
or negative offset (west of the prime meridian). ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The
use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:2005-06-15T11:30:22.56-07:00 where the
date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss
format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds
is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian)
or negative offset (west of the prime meridian).</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspObject" abstract="true" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either directly or
indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectID" type="objectID" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/mspObject.objectID">
      <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectID is always required.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="objectID" sawsdl:modelReference="MultiSpeak v411 /mspAuxilliaryAgreementConfig/objectID">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="priorityType" sawsdl:modelReference="MultiSpeak v411 /mspAuxilliaryAgreementConfig/priorityType">
  <xs:annotation>
    <xs:documentation>Designation of importance level of this event.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string">
    <xs:enumeration value="Normal" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/priorityType.Normal"/>
    <xs:enumeration value="NeedsAttention" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/priorityType.NeedsAttention"/>
    <xs:enumeration value="Urgent" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/priorityType.Urgent"/>
    <xs:enumeration value="Other" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/priorityType.Other"/>
    <xs:enumeration value="Unknown" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/priorityType.Unknown"/>
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="remainingBalance" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/remainingBalance">
  <xs:sequence>
    <xs:element name="balance" type="money" minOccurs="1" sawsdl:modelReference="MultiSpeak v411
/mspAuxilliaryAgreementConfig/remainingBalance.balance"/>
    <xs:element name="accounts" type="accounts" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="MultiSpeak v411 /mspAuxilliaryAgreementConfig/remainingBalance.accounts"/>
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

mspCustomerAccountConfig

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL" xmlns:xs="http://www.w3.org/2001/XMLSchema"
  xmlns:cimp="http://www.iec.org/CimProfile" xmlns="http://www.epri.net/2012/mspCustomerAccountConfig"
  xmlns:msp="http://www.epri.net/2012/mspCustomerAccountConfig"
  targetNamespace="http://www.epri.net/2012/mspCustomerAccountConfig" elementFormDefault="qualified"
  attributeFormDefault="unqualified" xml:base="http://www.epri.net/2012/mspCustomerAccountConfig">
  <xs:element name="mspCustomerAccountConfig" type="mspCustomerAccountConfig"/>
  <xs:complexType name="mspCustomerAccountConfig">
    <xs:sequence>
      <xs:element name="account" type="account" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="account" sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account">
    <xs:annotation>
      <xs:documentation>This is a customer account.</xs:documentation>
    </xs:annotation>
    <xs:sequence>
      <xs:element name="accountsReceivable" type="accountsReceivable" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.accountsReceivable">
        <xs:annotation>
          <xs:documentation>Accounts receivable information for this account.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="billDate" type="mspDateTime" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak V411/mspCustomerAccountConfig/account.billDate">
        <xs:annotation>
          <xs:documentation>Date of last bill.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="billingCycle" type="billingCycle" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak V411/mspCustomerAccountConfig/account.billingCycle">
        <xs:annotation>
          <xs:documentation>Billing cycle identifier for this account.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="billingTerms" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.billingTerms">
        <xs:annotation>
          <xs:documentation>Describes the way that the account is billed. Suggested enumeration is Prepay, Budget, Cycle, or
Other.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="budgetBill" type="budgBill" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak V411/mspCustomerAccountConfig/account.budgetBill">
        <xs:annotation>
          <xs:documentation>Budget billing code for this account</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="calculatedCurrentBillAmount" type="money" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.calculatedCurrentBillAmount">
        <xs:annotation>
          <xs:documentation>Amount that the customer would owe if final billed today. This amount is based on the fixed
charges, energyuseage, fuel costs and taxes since the last bill.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="calculatedCurrentBillDateTime" type="mspDateTime" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.calculatedCurrentBillDateTime">
        <xs:annotation>
          <xs:documentation>Date and time that the calculated current bill amount was calculated.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="calculatedCurrentBillReadings" type="calculatedCurrentBillReadings" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.calculatedCurrentBillReadings"/>
      <xs:element name="calculatedUsedYesterday" type="money" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.calculatedUsedYesterday">
        <xs:annotation>

```



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        <xs:documentation>The value (expressed in currency) in energy, fuel costs, taxes, and prorated fixed charges the
        customer used yesterday. It may simply be the calculatedCurrentBillAmount from today minus the calculated current bill amount
        from yesterday.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="customerID" type="objectID" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak V411/mspCustomerAccountConfig/account.customerID">
    <xs:annotation>
        <xs:documentation>Link to the objectID (foreign key) of the customer that has this account.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="isPrePay" type="xs:boolean" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak V411/mspCustomerAccountConfig/account.isPrePay">
    <xs:annotation>
        <xs:documentation>If the field is missing or false then the account is cycle or budget billed.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="lastBillAmount" type="money" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.lastBillAmount">
    <xs:annotation>
        <xs:documentation>The amount of the last bill to the customer.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="lastPaymentAmount" type="money" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.lastPaymentAmount">
    <xs:annotation>
        <xs:documentation>Amount of the last payment made.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="lastPaymentDate" type="mspDateTime" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.lastPaymentDate">
    <xs:annotation>
        <xs:documentation>Date of last payment.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="paymentDueDate" type="mspDateTime" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.paymentDueDate">
    <xs:annotation>
        <xs:documentation>Date and time that the next payment is due.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="shutOffDate" type="mspDateTime" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.shutOffDate">
    <xs:annotation>
        <xs:documentation>Shut off date for non-payment.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="objectName" type="objectName" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.objectName">
    <xs:element name="statusCode" type="statusCode" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/account.statusCode"/>
</xs:sequence>
</xs:complexType>
<xs:complexType name="accountsReceivable" sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/accountsReceivable">
    <xs:annotation>
        <xs:documentation>Money due on a customer account.</xs:documentation>
    </xs:annotation>
    <xs:sequence/>
</xs:complexType>
    <xs:simpleType name="billingCycle" sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/billingCycle">
    <xs:restriction base="xs:string"/>
</xs:simpleType>
    <xs:simpleType name="budgBill" sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/budgBill">
    <xs:annotation>
        <xs:documentation>Budget billing code.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>

```

```

</xs:simpleType>
<xs:complexType name="calculatedCurrentBillReadings"
sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/calculatedCurrentBillReadings">
  <xs:sequence/>
</xs:complexType>
<xs:complexType name="money" sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/money">
  <xs:sequence/>
</xs:complexType>
<xs:simpleType name="mspDateTime" sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable
to denote UTC or "Zulu" time.all dates will be formatted as:

```

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "-" in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).</xs:documentation>

```

  <xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:annotation>
</xs:simpleType>
<xs:complexType name="mspObject" abstract="true" sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either directly or
indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence/>
</xs:complexType>
<xs:simpleType name="objectID" sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/objectID">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="objectName" sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/objectName">
  <xs:annotation>
    <xs:documentation>This is an optional human-readable name for this instance of this object.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="statusCode" sawSDL:modelReference="http://www.epri.net/2012/cim/ea/MultiSpeak
V411/mspCustomerAccountConfig/statusCode">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
</xs:schema>

```

mspCustomerAgreementConfig

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xml:base="http://www.multispeak.org/2012/mspCustomerAgreementConfig" xmlns:xs="http://www.w3.org/2001/XMLSchema"
xmlns:cimp="http://www.iec.org/CimProfile" xmlns="http://www.multispeak.org/2012/mspCustomerAgreementConfig"
targetNamespace="http://www.multispeak.org/2012/mspCustomerAgreementConfig"
xmlns:msp="http://www.multispeak.org/2012/mspCustomerAgreementConfig" elementFormDefault="qualified"
attributeFormDefault="unqualified">
  <xs:element name="mspCustomerAgreementConfig" type="mspCustomerAgreementConfig" />
  <xs:complexType name="mspCustomerAgreementConfig">
    <xs:sequence />
  </xs:complexType>

```

```

<xs:complexType name="account" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/account">
  <xs:annotation>
    <xs:documentation>This is a customer account.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="billingTerms" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/account.billingTerms">
          <xs:annotation>
            <xs:documentation>Describes the way that the account is billed. Suggested enumeration is Prepay, Budget, Cycle, or
Other.</xs:documentation>
          </xs:annotation>
          </xs:element>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="comments" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/comments">
    <xs:annotation>
      <xs:documentation>This element may be used to annotate this data instance. It also serves to separate extensions made
with the extensions class from those made using the extensionsList class.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string" />
  </xs:simpleType>
  <xs:simpleType name="currencyCode" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode">
    <xs:annotation>
      <xs:documentation>Three letter currency code as standardized in ISO4217-2001. Typically is the two letter Internet domain
code for the country followed by a one letter designator for the currency; for instance the USA dollar is
USD.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string">
      <xs:enumeration value="USD" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.USD" />
      <xs:enumeration value="EUR" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.EUR" />
      <xs:enumeration value="ADF" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.ADF" />
      <xs:enumeration value="ADP" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.ADP" />
      <xs:enumeration value="AED" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.AED" />
      <xs:enumeration value="AFN" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.AFN" />
      <xs:enumeration value="ALL" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.ALL" />
      <xs:enumeration value="AMD" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.AMD" />
      <xs:enumeration value="ANG" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.ANG" />
      <xs:enumeration value="AOA" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.AOA" />
      <xs:enumeration value="ARS" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.ARS" />
      <xs:enumeration value="ATS" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.ATS" />
      <xs:enumeration value="AUD" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.AUD" />
      <xs:enumeration value="AWG" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.AWG" />
      <xs:enumeration value="AZN" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.AZN" />
      <xs:enumeration value="BAM" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.BAM" />
      <xs:enumeration value="BBD" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.BBD" />
      <xs:enumeration value="BDT" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.BDT" />
      <xs:enumeration value="BEF" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.BEF" />
      <xs:enumeration value="BGN" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.BGN" />
    </xs:restriction>
  </xs:simpleType>

```

[illegible]

[illegible]

[illegible]

[illegible]


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    <xs:enumeration value="UZS" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.UZS" />
    <xs:enumeration value="VAL" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.VAL" />
    <xs:enumeration value="VEF" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.VEF" />
    <xs:enumeration value="VND" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.VND" />
    <xs:enumeration value="VUV" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.VUV" />
    <xs:enumeration value="WST" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.WST" />
    <xs:enumeration value="XAF" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XAF" />
    <xs:enumeration value="XAG" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XAG" />
    <xs:enumeration value="XAU" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XAU" />
    <xs:enumeration value="XBA" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XBA" />
    <xs:enumeration value="XBB" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XBB" />
    <xs:enumeration value="XBC" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XBC" />
    <xs:enumeration value="XBD" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XBD" />
    <xs:enumeration value="XCD" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XCD" />
    <xs:enumeration value="XDR" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XDR" />
    <xs:enumeration value="XEU" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XEU" />
    <xs:enumeration value="XFU" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XFU" />
    <xs:enumeration value="XOF" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XOF" />
    <xs:enumeration value="XPD" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XPD" />
    <xs:enumeration value="XPF" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XPF" />
    <xs:enumeration value="XPT" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XPT" />
    <xs:enumeration value="XTS" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XTS" />
    <xs:enumeration value="XXX" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.XXX" />
    <xs:enumeration value="YER" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.YER" />
    <xs:enumeration value="ZAR" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.ZAR" />
    <xs:enumeration value="ZMK" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.ZMK" />
    <xs:enumeration value="ZWD" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.ZWD" />
    <xs:enumeration value="Other" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/currencyCode.Other" />
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="customer" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/customer">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence />
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="extensions" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/extensions">
  <xs:annotation>
    <xs:documentation>The extensions container is one way that MultiSpeak objects can be extended using XSD
elements.</xs:documentation>
  </xs:annotation>
  <xs:sequence />
</xs:complexType>
<xs:complexType name="extensionsList" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/extensionsList">
  <xs:annotation>

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<xs:documentation>This is an optional, self-describing means to extend any MultiSpeak object.</xs:documentation>
</xs:annotation>
<xs:sequence />
</xs:complexType>
<xs:simpleType name="mspDateTime" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable
to denote UTC or "Zulu" time.all dates will be formatted as:

```

```
2005-06-15T11:30:22.56-07:00
```

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

```
2005-06-15T11:30:22.56-07:00
```

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).</xs:documentation>

```

</xs:annotation>
<xs:restriction base="xs:string" />
</xs:simpleType>
<xs:complexType name="mspObject" abstract="true" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either directly or
indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectId" type="objectId" minOccurs="1"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/mspObject.objectID">
      <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectId is always required.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="utility" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/mspObject.utility">
      <xs:annotation>
        <xs:documentation>String that is used to identify the owner of this item. It may be a text string or a pointer to an
organization.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="comments" type="comments" minOccurs="0" maxOccurs="1"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/mspObject.comments">
      <xs:annotation>
        <xs:documentation>This element may be used to annotate this data instance. It also serves to separate extensions made
with the extensions class from those made using the extensionsList class.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="objectId" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/objectID">
  <xs:restriction base="xs:string" />
</xs:simpleType>
<xs:complexType name="rate" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/rate">
  <xs:annotation>
    <xs:documentation>This is the rate for sale of a commodity. For instance, if the current rate for electrical energy is
$0.10/kWh, then the float value 0.10 is carried in the rate base element and the optional attributes currency code would carry
"USD" for dollars, the commodityUnits would carry "kWh", and the rateDescription would carry "Current".

```

This is the rate for sale of a commodity. For instance, if the current rate for electrical energy is \$0.10/kWh, then the float value 0.10 is carried in the rate base element and the optional attributes currency code would carry "USD" for dollars, the commodityUnits would carry "kWh", and the rateDescription would carry "Current".</xs:documentation>

```
</xs:annotation>
```

```

<xs:sequence>
  <xs:element name="commodityUnits" type="uom" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rate.commodityUnits">
    <xs:annotation>
      <xs:documentation>This attribute identifies the units in which the commodity being sold is expressed. For instance, for
electrical energy sold at retail this might be "kWh".</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="currencyCode" type="currencyCode" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rate.currencyCode">
    <xs:annotation>
      <xs:documentation>This attribute identifies the currency in which this rate is expressed, as defined in ISO 4217. For
instance, "USD" is United States Dollars.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="rateCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rate.rateCode">
    <xs:annotation>
      <xs:documentation>The string identifier for this rate.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="rateDescription" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rate.rateDescription" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="rateInfo" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/rateInfo">
  <xs:sequence>
    <xs:element name="dateTime" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rateInfo.dateTime" />
    <xs:element name="extensions" type="extensions" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rateInfo.extensions" />
    <xs:element name="extensionsList" type="extensionsList" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rateInfo.extensionsList" />
    <xs:element name="rateCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rateInfo.rateCode">
      <xs:annotation>
        <xs:documentation>This is the string identifier for this rate.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="rateCodeDescription" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rateInfo.rateCodeDescription">
      <xs:annotation>
        <xs:documentation>This string attribute describes the typ of rate. Examples are "Current" for the currently effective rate.
"Peak" for the critical peak pricing rate. "AverageThisPeriod" for the average rate during the current billing period.
"AverageLastPeriod" for the average during the previous billing period. "MinimumThisPeriod"
"MaximumThisPeriod", etc.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="CustomerAgreement" type="CustomerAgreement" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rateInfo.CustomerAgreement" />
    <xs:element name="rates" type="rates" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rateInfo.rates" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="rates" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/rates">
  <xs:sequence>
    <xs:element name="rate" type="rate" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/rates.rate" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="resource" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/resource">
  <xs:annotation>

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    <xs:documentation>Resource is a container object that can have people, vehicles, equipment, and crews to perform work
tasks.</xs:documentation>
</xs:annotation>
<xs:complexContent>
  <xs:extension base="mspObject">
    <xs:sequence />
  </xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="uom" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom">
  <xs:annotation>
    <xs:documentation>This is the list of units of measure from ANSI C12.19</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string">
    <xs:enumeration value="Acre_ft" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Acre ft" />
    <xs:enumeration value="Amps" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Amps" />
    <xs:enumeration value="Amps_RMS" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Amps RMS" />
    <xs:enumeration value="Amps_squared_RMS" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Amps squared RMS" />
    <xs:enumeration value="atmospheres" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.atmospheres" />
    <xs:enumeration value="Bar" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Bar" />
    <xs:enumeration value="ccf" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.ccf" />
    <xs:enumeration value="ccf_per_Hr" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.ccf per Hr" />
    <xs:enumeration value="cm" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.cm" />
    <xs:enumeration value="Corrosion" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Corrosion" />
    <xs:enumeration value="Counter" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Counter" />
    <xs:enumeration value="cubic_feet" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.cubic feet" />
    <xs:enumeration value="cubic_feet_corrected" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.cubic feet corrected" />
    <xs:enumeration value="cubic_feet_per_Hr" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.cubic feet per Hr" />
    <xs:enumeration value="cubic_feet_per_Hr_corrected"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/uom.cubic feet per Hr corrected" />
    <xs:enumeration value="cubic_feet_per_Hr_uncorrected"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/uom.cubic feet per Hr uncorrected" />
    <xs:enumeration value="cubic_feet_uncorrected" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.cubic feet uncorrected" />
    <xs:enumeration value="cubic_meter" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.cubic meter" />
    <xs:enumeration value="cubic_meter_per_Hr" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.cubic meter per Hr" />
    <xs:enumeration value="cubic_meters_corrected" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.cubic meters corrected" />
    <xs:enumeration value="cubic_meters_per_Hr_corrected"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/uom.cubic meters per Hr corrected" />
    <xs:enumeration value="cubic_meters_per_Hr_uncorrected"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/uom.cubic meters per Hr uncorrected" />
    <xs:enumeration value="cubic_meters_uncorrected" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.cubic meters uncorrected" />
    <xs:enumeration value="Current_phase_angle" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Current phase angle" />
    <xs:enumeration value="Date" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Date" />
    <xs:enumeration value="Date_time" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Date time" />
    <xs:enumeration value="deg_C" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.deg C" />
    <xs:enumeration value="deg_F" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.deg F" />
    <xs:enumeration value="deg_K" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.deg K" />

```

```
<xs:enumeration value="Differential_pascal" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Differential pascal" />
<xs:enumeration value="Differential_pound_per_square_inch"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/uom.Differential pound per square inch" />
<xs:enumeration value="dm" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.dm" />
<xs:enumeration value="Dollar" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Dollar" />
<xs:enumeration value="feet_per_second" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.feet per second" />
<xs:enumeration value="Frequency" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Frequency" />
<xs:enumeration value="ft" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.ft" />
<xs:enumeration value="ft_of_water" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.ft of water" />
<xs:enumeration value="GQ" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.GQ" />
<xs:enumeration value="GQh" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.GQh" />
<xs:enumeration value="Gram_square_cm" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Gram square cm" />
<xs:enumeration value="GVA" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.GVA" />
<xs:enumeration value="GVAh" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.GVAh" />
<xs:enumeration value="GVAR" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.GVAR" />
<xs:enumeration value="GVARh" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.GVARh" />
<xs:enumeration value="GW" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.GW" />
<xs:enumeration value="GWh" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.GWh" />
<xs:enumeration value="IMP_gl" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.IMP gl" />
<xs:enumeration value="IMP_gl_per_Hr" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.IMP gl per Hr" />
<xs:enumeration value="inches" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.inches" />
<xs:enumeration value="Inches_H2O_column" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Inches H2O column" />
<xs:enumeration value="Inches_HG_column" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Inches HG column" />
<xs:enumeration value="inches_of_water" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.inches of water" />
<xs:enumeration value="Interval_timer" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Interval timer" />
<xs:enumeration value="Ionization" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Ionization" />
<xs:enumeration value="Joules" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Joules" />
<xs:enumeration value="Joules_per_Hr" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Joules per Hr" />
<xs:enumeration value="kAmps" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.kAmps" />
<xs:enumeration value="kAmps_RMS" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.kAmps RMS" />
<xs:enumeration value="kAmps_squared_RMS" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.kAmps squared RMS" />
<xs:enumeration value="kJoules" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.kJoules" />
<xs:enumeration value="kJoules_per_Hr" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.kJoules per Hr" />
<xs:enumeration value="km" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.km" />
<xs:enumeration value="km_per_hour" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.km per hour" />
<xs:enumeration value="km_per_second" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.km per second" />
<xs:enumeration value="kQ" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.kQ" />
<xs:enumeration value="kQh" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.kQh" />
<xs:enumeration value="kV" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.kV" />
```


[illegible]

[illegible]

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<xs:enumeration value="Nbr_of_reverse_rotation" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Nbr of reverse rotation" />
<xs:enumeration value="Nbr_of_sag" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Nbr of sag" />
<xs:enumeration value="Nbr_of_swells" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Nbr of swells" />
<xs:enumeration value="Nbr_of_times_programmed" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Nbr of times programmed" />
<xs:enumeration value="Nbr_of_V_T.H.D._excess" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Nbr of V T.H.D. excess" />
<xs:enumeration value="Nbr_of_voltage_unbalance" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Nbr of voltage unbalance" />
<xs:enumeration value="Nbr_of_watchdog" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Nbr of watchdog" />
<xs:enumeration value="Normal_voltage_period" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Normal voltage period" />
<xs:enumeration value="percent" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.percent" />
<xs:enumeration value="percent_relative_humidity" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.percent relative humidity" />
<xs:enumeration value="PH_factor" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.PH factor" />
<xs:enumeration value="pounds_per_square_foot" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.pounds per square foot" />
<xs:enumeration value="Power_factor" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Power factor" />
<xs:enumeration value="PPM_chlorine" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.PPM chlorine" />
<xs:enumeration value="PPM_lead" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.PPM lead" />
<xs:enumeration value="PPM_odorant" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.PPM odorant" />
<xs:enumeration value="PPM_SO2" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.PPM SO2" />
<xs:enumeration value="Q" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Q" />
<xs:enumeration value="Qh" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Qh" />
<xs:enumeration value="Sense_input" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Sense input" />
<xs:enumeration value="Static_pascal" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Static pascal" />
<xs:enumeration value="Static_pound_per_square_inch"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspCustomerAgreementConfig/MultiSpeak
V411/uom.Static pound per square inch" />
<xs:enumeration value="T.H.D._Current" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.T.H.D. Current" />
<xs:enumeration value="T.H.D._Voltage" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.T.H.D. Voltage" />
<xs:enumeration value="Therm" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Therm" />
<xs:enumeration value="Therm_per_Hr" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Therm per Hr" />
<xs:enumeration value="Time" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Time" />
<xs:enumeration value="turbidity" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.turbidity" />
<xs:enumeration value="US_gl" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.US gl" />
<xs:enumeration value="US_gl_per_Hr" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.US gl per Hr" />
<xs:enumeration value="V" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.V" />
<xs:enumeration value="V_RMS" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.V RMS" />
<xs:enumeration value="VA" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.VA" />
<xs:enumeration value="VAh" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.VAh" />
<xs:enumeration value="VAR" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.VAR" />
<xs:enumeration value="VARh" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.VARh" />
<xs:enumeration value="Voltage_phase_angle" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Voltage phase angle" />
<xs:enumeration value="Vsquared_RMS" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Vsquared RMS" />

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    <xs:enumeration value="W" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.W" />
    <xs:enumeration value="Wh" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/MultiSpeak V411/uom.Wh" />
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="CustomerAgreement" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/CustomerAgreement">
  <xs:annotation>
    <xs:documentation>&lt;font color="#ff0000"&gt;&A placeholder class to model CIM CustomerAgreementConfig
equivalency.&lt;/font&gt;</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="account" type="account" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/CustomerAgreement.account" />
    <xs:element name="comments" type="comments" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/CustomerAgreement.comments" />
    <xs:element name="customer" type="customer" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/CustomerAgreement.customer" />
    <xs:element name="resource" type="resource" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspCustomerAgreementConfig/CustomerAgreement.resource" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

mspCustomerMeterDataSet

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawsdl="http://www.w3.org/ns/sawsdl"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mspCustomerMeterDataSet"
xmlns:msp="http://www.multispeak.org/2012/mspCustomerMeterDataSet"
targetNamespace="http://www.multispeak.org/2012/mspCustomerMeterDataSet"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mspCustomerMeterDataSet">
  <xs:element name="mspCustomerMeterDataSet" type="mspCustomerMeterDataSet"/>
  <xs:complexType name="mspCustomerMeterDataSet">
    <xs:sequence/>
  </xs:complexType>
  <xs:complexType name="account" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/account">
    <xs:annotation>
      <xs:documentation>This is a customer account.</xs:documentation>
    </xs:annotation>
    <xs:complexContent>
      <xs:extension base="mspObject">
        <xs:sequence>
          <xs:element name="billingCycle" type="billingCycle" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/account.billingCycle">
            <xs:annotation>
              <xs:documentation>Billing cycle identifier for this account.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="budgetBill" type="budgBill" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/account.budgetBill">
            <xs:annotation>
              <xs:documentation>Budget billing code for this account</xs:documentation>
            </xs:annotation>
          </xs:element>

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    <xs:element name="customerID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/account.customerID">
    <xs:annotation>
    <xs:documentation>Link to the objectID (foreign key) of the customer that has this
account.</xs:documentation>
    </xs:annotation>
    </xs:element>
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="accountNumber"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/accountNumber">
    <xs:annotation>
    <xs:documentation>Account number for this service location. Depending on the system of record,
this may be either the account.objectID or the account.objectName.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="address" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/address">
    <xs:sequence>
    <xs:element name="address1" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/address.address1">
    <xs:annotation>
    <xs:documentation>Address line 1.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="address2" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/address.address2">
    <xs:annotation>
    <xs:documentation>Address line 2.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="city" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/address.city">
    <xs:annotation>
    <xs:documentation>Name of the city or municipality.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="country" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/address.country"/>
    <xs:element name="detailedAddressFields" type="detailedAddressFields" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/address.detailedAddressFields">
    <xs:annotation>
    <xs:documentation>Detailed information about the components of the composite address
lines.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="postalCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/address.postalCode">
    <xs:annotation>
    <xs:documentation>The postal code. For instance, in the United States this is the zip
code.</xs:documentation>

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        </xs:annotation>
      </xs:element>
      <xs:element name="state" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/address.state">
        <xs:annotation>
          <xs:documentation>The state or province.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="townCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/address.townCode">
        <xs:annotation>
          <xs:documentation>Additional information about the city or town, if
required.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="billingCycle" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/billingCycle">
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="budgBill" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/budgBill">
    <xs:annotation>
      <xs:documentation>Budget billing code.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="customer" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/customer">
    <xs:complexContent>
      <xs:extension base="mspObject">
        <xs:sequence>
          <xs:element name="servType" type="servType" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/customer.servType"/>
          <xs:element name="specialNeeds" type="specialNeeds" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/customer.specialNeeds"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="detailedAddressFields"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/detailedAddressFields">
    <xs:sequence>
      <xs:element name="addressGeneral" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/detailedAddressFields.addressGeneral"/>
      <xs:element name="buildingNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/detailedAddressFields.buildingNumber"/>
      <xs:element name="postOfficeBox" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/detailedAddressFields.postOfficeBox"/>
      <xs:element name="region" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/detailedAddressFields.region"/>
    </xs:sequence>
  </xs:complexType>

```

```

        <xs:element name="streetNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/detailedAddressFields.streetNumber"/>
        <xs:element name="streetPrefix" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/detailedAddressFields.streetPrefix"/>
        <xs:element name="streetSuffix" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/detailedAddressFields.streetSuffix"/>
        <xs:element name="streetType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/detailedAddressFields.streetType"/>
        <xs:element name="suiteNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/detailedAddressFields.suiteNumber"/>
    </xs:sequence>
</xs:complexType>
<xs:complexType name="electricMeter"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricMeter">
    <xs:sequence>
        <xs:element name="meterBaseID" type="objectID" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricMeter.meterBaseID">
            <xs:annotation>
                <xs:documentation>This is the objectID of the meterBase with which this electric meter is
associated, if any.</xs:documentation>
            </xs:annotation>
        </xs:element>
        <xs:element name="electricNameplate" type="electricNameplate" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricMeter.electricNameplate"/>
    </xs:sequence>
</xs:complexType>
<xs:complexType name="electricNameplate"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricNameplate">
    <xs:sequence>
        <xs:element name="maxAmperage" type="xs:float" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricNameplate.maxAmperage">
            <xs:annotation>
                <xs:documentation>The maximum of the meter load range in amperes.</xs:documentation>
            </xs:annotation>
        </xs:element>
        <xs:element name="phases" type="xs:long" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricNameplate.phases">
            <xs:annotation>
                <xs:documentation>The number of phases metered by this device.</xs:documentation>
            </xs:annotation>
        </xs:element>
    </xs:sequence>
</xs:complexType>
<xs:complexType name="electricService"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricService">
    <xs:annotation>
        <xs:documentation>A point of service for receiving electrical utility service.</xs:documentation>
    </xs:annotation>
    <xs:complexContent>
        <xs:extension base="mspObject">

```

```

<xs:sequence>
  <xs:element name="accountNumber" type="accountNumber" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricService.accountNumber">
    <xs:annotation>
      <xs:documentation>Account number for this service location. Depending on the system of
record, this may be either the account.objectID or the account.objectName.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="billingCycle" type="billingCycle" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricService.billingCycle"/>
  <xs:element name="budgetBill" type="budgetBill" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricService.budgetBill">
    <xs:annotation>
      <xs:documentation>Budget bill code. </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="customerID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricService.customerID">
    <xs:annotation>
      <xs:documentation>A foreign key for the customer associated with this electric
service.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="electricMeterID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricService.electricMeterID">
    <xs:annotation>
      <xs:documentation>This is the ID of the electricMeter that is installed at this
electricService</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="loadMgmt" type="loadMgmt" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricService.loadMgmt">
    <xs:annotation>
      <xs:documentation>Load management code.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="serviceLocationID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricService.serviceLocationID">
    <xs:annotation>
      <xs:documentation>This is a pointer to the serviceLocation that includes this
electricService.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="priorityType" type="priorityType" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricService.priorityType"/>
  <xs:element name="realEnergy" type="realEnergy" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/electricService.realEnergy"/>
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>

```

```

<xs:complexType name="instrumentTransformers"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/instrumentTransformers">
  <xs:annotation>
    <xs:documentation>A set of CT and PT ratios that may be associated with an electric meter
base.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="ct" type="xs:double" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/instrumentTransformers.ct">
      <xs:annotation>
        <xs:documentation>Ratio of the current transformers installed at this meter
base.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="pt" type="xs:double" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/instrumentTransformers.pt">
      <xs:annotation>
        <xs:documentation>Ratio of the potential transformers installed at this meter
base.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="loadMgmt" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/loadMgmt">
  <xs:annotation>
    <xs:documentation>Load management code.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="meterBase" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/meterBase">
  <xs:annotation>
    <xs:documentation>This is an electric meter base. Meter base encapsulates all devices and
characteristics of an electric service that remain unchanged when an electric meter is replaced, such
as intrument transformers, connect/disconnect devices, etc.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="electricMeter" type="electricMeter" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/meterBase.electricMeter"/>
    <xs:element name="instrumentTransformers" type="instrumentTransformers" minOccurs="0"
maxOccurs="1" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/meterBase.instrumentTransformers"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="meterNo" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/meterNo">
  <xs:annotation>
    <xs:documentation>Meter number (utility-defined identifier usually called a Company
Number).</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspMeter" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/mspMeter">
  <xs:annotation>

```



```

    <xs:documentation>This is an abstract class from which all meter device classes
inherit.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="meterNo" type="meterNo" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/mspMeter.meterNo">
          <xs:annotation>
            <xs:documentation>Meter number (utility-defined identifier usually called a Company
Number).</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexType>
  <xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/mspObject">
    <xs:annotation>
      <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
    </xs:annotation>
    <xs:sequence>
      <xs:element name="objectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/mspObject.objectID">
        <xs:annotation>
          <xs:documentation>This is the key field for this object type. The use of objectID is always
required.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="mspServiceCategory"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/mspServiceCategory">
    <xs:complexContent>
      <xs:extension base="mspObject">
        <xs:sequence>
          <xs:element name="serviceType" type="serviceType" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/mspServiceCategory.serviceType"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="objectID" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/objectID">
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="priorityType" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/priorityType">
    <xs:annotation>
      <xs:documentation>Designation of importance level of this event.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string">
      <xs:enumeration value="Normal" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/priorityType.Normal"/>

```

```

        <xs:enumeration value="NeedsAttention"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/priorityType.NeedsAttention"/>
        <xs:enumeration value="Urgent" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/priorityType.Urgent"/>
        <xs:enumeration value="Other" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/priorityType.Other"/>
        <xs:enumeration value="Unknown"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/priorityType.Unknown"/>
    </xs:restriction>
</xs:simpleType>
<xs:complexType name="realEnergy" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergy">
    <xs:sequence>
        <xs:element name="electricService" type="electricService" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergy.electricService"/>
    </xs:sequence>
</xs:complexType>
<xs:simpleType name="realEnergyUnits"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits">
    <xs:restriction base="xs:string">
        <xs:enumeration value="Wh" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.Wh"/>
        <xs:enumeration value="kWh" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.kWh"/>
        <xs:enumeration value="MWh" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.MWh"/>
        <xs:enumeration value="GWh" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.GWh"/>
        <xs:enumeration value="mWh" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.mWh"/>
        <xs:enumeration value="microWh"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.microWh"/>
        <xs:enumeration value="TWh" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.TWh"/>
        <xs:enumeration value="PWh" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.PWh"/>
        <xs:enumeration value="BTU" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.BTU"/>
        <xs:enumeration value="Therm" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.Therm"/>
        <xs:enumeration value="Erg" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.Erg"/>
        <xs:enumeration value="Calorie" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.Calorie"/>
        <xs:enumeration value="eV" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.eV"/>
        <xs:enumeration value="Joule" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/realEnergyUnits.Joule"/>
    </xs:restriction>
</xs:simpleType>
<xs:complexType name="serviceLocation"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceLocation">
    <xs:annotation>
        <xs:documentation>This is a location at which a customer can receive service. A serviceLocation
can have utility services of several different serviceTypes.</xs:documentation>
    </xs:annotation>

```

```

<xs:complexContent>
  <xs:extension base="mspObject">
    <xs:sequence>
      <xs:element name="customerID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceLocation.customerID">
        <xs:annotation>
          <xs:documentation>Customer's identification number. This should be the objectID of the
customer.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="description" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceLocation.description">
        <xs:annotation>
          <xs:documentation>Additional information about this serviceLocation</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="serviceAddress" type="address" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceLocation.serviceAddress">
        <xs:annotation>
          <xs:documentation>Physical address of this service location.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="serviceLocations"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceLocations">
  <xs:sequence>
    <xs:element name="serviceLocation" type="serviceLocation" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceLocations.serviceLocation"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="serviceType" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceType">
  <xs:annotation>
    <xs:documentation>The type of utility service.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string">
    <xs:enumeration value="Electric" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceType.Electric"/>
    <xs:enumeration value="Gas" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceType.Gas"/>
    <xs:enumeration value="Water" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceType.Water"/>
    <xs:enumeration value="Propane" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceType.Propane"/>
    <xs:enumeration value="Steam" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/serviceType.Steam"/>
  </xs:restriction>
</xs:simpleType>
<xs:simpleType name="servType" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/servType">
  <xs:annotation>
    <xs:documentation>Type of customer service. Example enumerations are: "Residential",
"Commerical", "Industrial", etc.</xs:documentation>
  </xs:annotation>

```



```

    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="specialNeeds"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerMeterDataSet/specialNeeds">
    <xs:annotation>
      <xs:documentation>A descriptor for any special needs that this customer has, such as life
support equipment.

```

A descriptor for any special needs that this customer has, such as life support equipment. THIS SIMPLE TYPE HAS BEEN DEPRECATED IN V4.1 AND WILL BE REPLACED WITH AN ARRAY OF NAME-VALUE PAIRS DESCRIBING TYPES OF SPECIAL NEEDS IN V4.2. </xs:documentation>

```

    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
</xs:schema>

```

mspCustomerConfig

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mspCustomerConfig"
xmlns:msp="http://www.multispeak.org/2012/mspCustomerConfig"
targetNamespace="http://www.multispeak.org/2012/mspCustomerConfig"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mspCustomerConfig">
  <xs:element name="mspCustomerConfig" type="mspCustomerConfig"/>
  <xs:complexType name="mspCustomerConfig">
    <xs:sequence>
      <xs:element name="customer" type="customer" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="account" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/account">
    <xs:annotation>
      <xs:documentation>This is a customer account.</xs:documentation>
    </xs:annotation>
    <xs:sequence>
      <xs:element name="billingCycle" type="billingCycle" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/account.billingCycle">
        <xs:annotation>
          <xs:documentation>Billing cycle identifier for this account.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="budgetBill" type="budgBill" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/account.budgetBill">
        <xs:annotation>
          <xs:documentation>Budget billing code for this account</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="customerID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/account.customerID">
        <xs:annotation>
          <xs:documentation>Link to the objectID (foreign key) of the customer that has this
account.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>

```

```

</xs:complexType>
<xs:complexType name="accounts" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/accounts">
  <xs:sequence>
    <xs:element name="account" type="account" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/accounts.account"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="address" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/address">
  <xs:sequence>
    <xs:element name="address1" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/address.address1">
      <xs:annotation>
        <xs:documentation>Address line 1.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="address2" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/address.address2">
      <xs:annotation>
        <xs:documentation>Address line 2.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="city" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/address.city">
      <xs:annotation>
        <xs:documentation>Name of the city or municipality.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="country" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/address.country">
      <xs:element name="detailedAddressFields" type="detailedAddressFields" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/address.detailedAddressFields">
        <xs:annotation>
          <xs:documentation>Detailed information about the components of the composite address
lines.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="postalCode" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/address.postalCode">
        <xs:annotation>
          <xs:documentation>The postal code. For instance, in the United States this is the zip
code.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="state" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/address.state">
        <xs:annotation>
          <xs:documentation>The state or province.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="townCode" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/address.townCode">

```

```

    <xs:annotation>
      <xs:documentation>Additional information about the city or town, if
required.</xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="addressItem"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/addressItem">
  <xs:sequence>
    <xs:element name="address" type="address" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/addressItem.address"/>
    <xs:element name="addressType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/addressItem.addressType">
      <xs:annotation>
        <xs:documentation>Type of address. For instance, billing, mailing, service location,
etc.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="priorityOrder" type="priorityOrder" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/addressItem.priorityOrder">
      <xs:annotation>
        <xs:documentation>Order that should be used to contact this person. First = 1, second = 2,
etc.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="addressList"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/addressList">
  <xs:sequence>
    <xs:element name="addressItem" type="addressItem" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/addressList.addressItem"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="billingCycle" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/billingCycle">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="budgBill" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/budgBill">
  <xs:annotation>
    <xs:documentation>Budget billing code.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="contactInfo"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/contactInfo">
  <xs:annotation>
    <xs:documentation>This class contains ways to contact an entity.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="addressList" type="addressList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/contactInfo.addressList"/>

```

```

    <xs:element name="eMailList" type="eMailList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/contactInfo.eMailList"/>
    <xs:element name="otherContactInformation" type="otherContactInformation" minOccurs="0"
maxOccurs="1" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/contactInfo.otherContactInformation"/>
    <xs:element name="phoneList" type="phoneList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/contactInfo.phoneList"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="customer" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/customer">
  <xs:complexContent>
    <xs:extension base="mspPerson">
      <xs:sequence>
        <xs:element name="accounts" type="accounts" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/customer.accounts">
          <xs:annotation>
            <xs:documentation>List of accounts associated with this customer.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="revenueClass" type="revenueClass" minOccurs="1"
maxOccurs="unbounded" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/customer.revenueClass"/>
        <xs:element name="specialNeeds" type="specialNeeds" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/customer.specialNeeds">
          <xs:annotation>
            <xs:documentation>A descriptor for any special needs that this customer has, such as life
support equipment.</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="detailedAddressFields"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/detailedAddressFields">
  <xs:sequence/>
</xs:complexType>
<xs:complexType name="eMailAddress"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/eMailAddress">
  <xs:sequence>
    <xs:element name="eMail" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/eMailAddress.eMail"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="eMailList" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/eMailList">
  <xs:sequence>
    <xs:element name="eMailAddress" type="eMailAddress" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/eMailList.eMailAddress"/>
  </xs:sequence>
</xs:complexType>

```

```

<xs:complexType name="mspDevice" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspDevice">
  <xs:sequence>
    <xs:element name="deviceClass" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspDevice.deviceClass">
      <xs:annotation>
        <xs:documentation>A high-level description of this type of object (e.g., "meter",
"communication", "connect/disconnect", "demand response"etc.).</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="facilityID" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspDevice.facilityID">
      <xs:annotation>
        <xs:documentation>A utility-defined string designation for this device.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="serialNumber" type="serialNumber" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspDevice.serialNumber">
      <xs:annotation>
        <xs:documentation>Serial number for this device.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectId" type="objectId" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspObject.objectID">
      <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectId is always
required.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="mspPerson" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspPerson">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="firstName" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspPerson.firstName">
          <xs:annotation>
            <xs:documentation>First name of person.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="governmentID" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspPerson.governmentID">

```



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    <xs:annotation>
      <xs:documentation>Unique identifier for person relative to it's governing authority, for
example a federal tax identifier (such as a Social Security number in the United
States).</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="lastName" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspPerson.lastName">
    <xs:annotation>
      <xs:documentation>Last name of person.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="mName" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspPerson.mName">
    <xs:annotation>
      <xs:documentation>Middle name of person.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="prefix" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspPerson.prefix"/>
  <xs:element name="suffix" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspPerson.suffix"/>
  <xs:element name="contactInfo" type="contactInfo" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/mspPerson.contactInfo"/>
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="objectID" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/objectID">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="otherContactInformation"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/otherContactInformation">
  <xs:sequence>
    <xs:element name="otherContactItem" type="otherContactItem" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/otherContactInformation.otherContactItem">
      <xs:annotation>
        <xs:documentation>This is an element to carry other miscellaneous contact information, such
as web site address, truck radio number, etc.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="otherContactItem"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/otherContactItem">
  <xs:annotation>
    <xs:documentation>This is an element to carry other miscellaneous contact information, such as
web site address, truck radio number, etc.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="details" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/otherContactItem.details">

```

```

    <xs:annotation>
      <xs:documentation>This element carries the miscellaneous contact
information.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="infoType" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/otherContactItem.infoType">
    <xs:annotation>
      <xs:documentation>This element is used to describe the type of information stored in the
"details" element. Examples might be: IP address, truck number, radio number.</xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="phoneList" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/phoneList">
  <xs:sequence>
    <xs:element name="phoneNumber" type="phoneNumber" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/phoneList.phoneNumber"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="phoneNumber"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/phoneNumber">
    <xs:sequence>
      <xs:element name="phone" type="telephoneNumber" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/phoneNumber.phone"/>
      <xs:element name="phoneType" type="phoneType" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/phoneNumber.phoneType">
        <xs:annotation>
          <xs:documentation>The type of phone number.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="priorityOrder" type="priorityOrder" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/phoneNumber.priorityOrder"/>
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="phoneType" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/phoneType">
    <xs:annotation>
      <xs:documentation>Type of phone number.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string">
      <xs:enumeration value="Home" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/phoneType.Home"/>
      <xs:enumeration value="Business"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/phoneType.Business"/>
      <xs:enumeration value="Mobile" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/phoneType.Mobile"/>
      <xs:enumeration value="Business_mobile"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/phoneType.Business mobile"/>
      <xs:enumeration value="Pager" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/phoneType.Pager"/>
      <xs:enumeration value="Fax" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/phoneType.Fax"/>
    </xs:restriction>
  </xs:simpleType>

```

```

    <xs:enumeration value="Other" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/phoneType.Other"/>
  </xs:restriction>
</xs:simpleType>
  <xs:simpleType name="priorityOrder" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspCustomerConfig/MultiSpeak V411/priorityOrder">
    <xs:annotation>
      <xs:documentation>Order in which this contact information type should be used to contact this
person. First = 1, second =2, etc.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="revenueClass"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/revenueClass">
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="serialNumber"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/serialNumber">
    <xs:annotation>
      <xs:documentation>Manufacturer-defined serial number for this device.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="specialNeeds"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/specialNeeds">
    <xs:annotation>
      <xs:documentation>A descriptor for any special needs that this customer has, such as life
support equipment.

```

A descriptor for any special needs that this customer has, such as life support equipment. THIS SIMPLE TYPE HAS BEEN DEPRECATED IN V4.1 AND WILL BE REPLACED WITH AN ARRAY OF NAME-VALUE PAIRS DESCRIBING TYPES OF SPECIAL NEEDS IN V4.2.</xs:documentation>

```

    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="telephoneNumber"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/telephoneNumber">
    <xs:sequence>
      <xs:element name="areaCode" type="xs:string" minOccurs="0"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/telephoneNumber.areaCode">
        <xs:annotation>
          <xs:documentation>Area or region code.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="countryCode" type="xs:string" minOccurs="0"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/telephoneNumber.countryCode">
        <xs:annotation>
          <xs:documentation>Country code for telephone number, if applicable.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="extension" type="xs:string" minOccurs="0"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/telephoneNumber.extension">
        <xs:annotation>
          <xs:documentation>Extension for telephone number if required.</xs:documentation>
        </xs:annotation>

```



```

</xs:element>
<xs:element name="localNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspCustomerConfig/MultiSpeak
V411/telephoneNumber.localNumber">
  <xs:annotation>
    <xs:documentation>The primary portion of a telephone number as dialed in a local calling
area.</xs:documentation>
  </xs:annotation>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:schema>

```

mspEndDeviceAssets

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/cim" xmlns:msp="http://www.multispeak.org/2012/cim"
targetNamespace="http://www.multispeak.org/2012/cim" elementFormDefault="qualified"
attributeFormDefault="unqualified" xml:base="http://www.multispeak.org/2012/cim">
  <xs:element name="mspEndDeviceAssets" type="mspEndDeviceAssets"/>
  <xs:complexType name="mspEndDeviceAssets">
    <xs:sequence/>
  </xs:complexType>
  <xs:simpleType name="billingCycle"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/billingCycle">
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="CDDevice"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/CDDevice">
    <xs:annotation>
      <xs:documentation>This is a connect/disconnect device. When installed, it is usually associated
with a meter base at a customer service location.</xs:documentation>
    </xs:annotation>
    <xs:complexContent>
      <xs:extension base="mspObject">
        <xs:sequence>
          <xs:element name="meterBaseID" type="objectID" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/CDDevice.meterBaseID">
            <xs:annotation>
              <xs:documentation>This is the objectID of the meterBase at which this CDDevice is
installed, if any.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="ratedCurrent" type="current" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/CDDevice.ratedCurrent">
            <xs:annotation>
              <xs:documentation>Rated current of the connect/disconnect device.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="ratedVoltage" type="voltage" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/CDDevice.ratedVoltage">
            <xs:annotation>
              <xs:documentation>Rated voltage of the load disconnect device.</xs:documentation>
            </xs:annotation>
          </xs:element>

```

```

    </xs:sequence>
  </xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="CDState"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/CDState">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="meterID" type="meterID" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/CDState.meterID"/>
        <xs:element name="RCDState" type="RCDState" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/CDState.RCDState">
          <xs:annotation>
            <xs:documentation>The current state of this remote connect/disconnect device
(CDDevice).</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="CDStateChange"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/CDStateChange">
  <xs:sequence>
    <xs:element name="meterEvent" type="meterEvent" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/CDStateChange.meterEvent">
      <xs:annotation>
        <xs:documentation>This is an object that carries
meter and/or headend event
codes. The numeric code
(e.g., "7.20.7.27") would be
carried in the meterEvent
complex type itself. This is used as the mRID in IEC 61968-9 standard EndDeviceEvent instances.
The domain, domainPart, type, index, value, and codeString representation
could optionally be sent in
attributes on the complex type.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="meterID" type="meterID" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/CDStateChange.meterID">
      <xs:annotation>
        <xs:documentation>The meterIdentifier of the meter to which this CDStateChange is
associated.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="stateChange" type="loadActionCode" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/CDStateChange.stateChange">
      <xs:annotation>
        <xs:documentation>Action to be taken with this event or state of this customer service
(Actions = Connect, Disconnect, Arm, Enable, Disable, InitiatePowerLimitation, Open, Close States =
Opened, Closed, Armed, Enabled, Disabled).</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>

```

```

</xs:complexType>
<xs:complexType name="connectDisconnectEvent"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/connectDisconnectEvent">
  <xs:sequence/>
</xs:complexType>
<xs:complexType name="current"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/current">
  <xs:sequence/>
</xs:complexType>
<xs:simpleType name="deviceStatus"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/deviceStatus">
  <xs:restriction base="xs:string">
    <xs:enumeration value="On"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/deviceStatus.On"/>
    <xs:enumeration value="Off"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/deviceStatus.Off"/>
    <xs:enumeration value="OutOfService"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/deviceStatus.OutOfService"/>
    <xs:enumeration value="PowerLimitation"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/deviceStatus.PowerLimitation"/>
    <xs:enumeration value="Other"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/deviceStatus.Other"/>
    <xs:enumeration value="Unknown"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/deviceStatus.Unknown"/>
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="duration"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/duration">
  <xs:annotation>
    <xs:documentation>Time duration.</xs:documentation>
  </xs:annotation>
  <xs:sequence/>
</xs:complexType>
<xs:complexType name="intervalProfile"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/intervalProfile">
  <xs:sequence>
    <xs:element name="intervalLength" type="duration" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/intervalProfile.intervalLength">
      <xs:annotation>
        <xs:documentation>Interval length, typically measured in minutes.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="loadActionCode"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode">
  <xs:annotation>
    <xs:documentation>This enumeration has codes for (i) actions to be taken at this customer
connect/disconnect device and (ii) to return the state of this CDDevice.</xs:documentation>
  </xs:annotation>

```

```

    </xs:annotation>
    <xs:restriction base="xs:string">
      <xs:enumeration value="Connect"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Connect"/>
      <xs:enumeration value="Disconnect"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Disconnect"/>
      <xs:enumeration value="InitiatePowerLimitation"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.InitiatePowerLimitation"/>
      <xs:enumeration value="Opened"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Opened"/>
      <xs:enumeration value="Closed"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Closed"/>
      <xs:enumeration value="Open"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Open"/>
      <xs:enumeration value="Close"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Close"/>
      <xs:enumeration value="Armed"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Armed"/>
      <xs:enumeration value="Enabled"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Enabled"/>
      <xs:enumeration value="Disabled"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Disabled"/>
      <xs:enumeration value="Arm"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Arm"/>
      <xs:enumeration value="Enable"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Enable"/>
      <xs:enumeration value="Disable"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Disable"/>
      <xs:enumeration value="Unknown"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/loadActionCode.Unknown"/>
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="meterEvent"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/meterEvent">
    <xs:annotation>
      <xs:documentation>This is an object that carries
meter and/or headend event
codes. The numeric code
(e.g., "7.20.7.27") would be
carried in the meterEvent
complex type itself. This is used as the mRID in IEC 61968-9 standard EndDeviceEvent instances.
The domain, domainPart, type, index, value, and codeString representation
could optionally be sent in
attributes on the complex type.</xs:documentation>
    </xs:annotation>
    <xs:sequence/>
  </xs:complexType>

```

meter and/or headend event codes. The numeric code (e.g., "7.20.7.27") would be carried in the meterEvent complex type itself. This is used as the mRID in IEC 61968-9 standard EndDeviceEvent instances. The domain, domainPart, type, index, value, and codeString representation could optionally be sent in attributes on the complex type.

```

<xs:complexType name="meterID"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/meterID">
  <xs:annotation>
    <xs:documentation>This is the identifier for a meter. Note that the meter might be of different
    service types (for example: electric, gas, propane, water, etc.). Since objectIDs are guaranteed to be
    unique only over the domain of one type of noun (class or object), it is necessary to specify both a
    serviceType and an objectID in order to ensure a unique instance of a meter. The value of the meterID
    should contain a string concatenated as utility.serviceType.objectID</xs:documentation>
  </xs:annotation>
</xs:sequence>
</xs:complexType>
<xs:simpleType name="meterNo"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/meterNo">
  <xs:annotation>
    <xs:documentation>Meter number (utility-defined identifier usually called a Company
    Number).</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="meterStatusList"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/meterStatusList">
  <xs:annotation>
    <xs:documentation>List of meter statuses.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
</xs:complexType>
<xs:simpleType name="meterType"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/meterType">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="module"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/module">
  <xs:annotation>
    <xs:documentation>This is an add-on module or board for a meter or other
    device.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="firmwareVersion" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/module.firmwareVersion">
      <xs:annotation>
        <xs:documentation>The version of the firmware for this module.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="moduleList" type="moduleList" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/module.moduleList"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="moduleList"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/moduleList">
  <xs:sequence>
    <xs:element name="mspMeter" type="mspMeter" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/moduleList.mspMeter"/>
  </xs:sequence>

```



```

</xs:complexType>
<xs:simpleType name="mspDateTime"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspDateTime">

```

```

<xs:annotation>

```

```

<xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets.
The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

```

```

2005-06-15T11:30:22.56-07:00

```

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

```

2005-06-15T11:30:22.56-07:00

```

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

```

</xs:documentation>
</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspMeter" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter">
<xs:annotation>
<xs:documentation>This is an abstract class from which all meter device classes
inherit.</xs:documentation>
</xs:annotation>
<xs:complexContent>
<xs:extension base="mspObject">
<xs:sequence>
<xs:element name="AMRDeviceType" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.AMRDeviceType">
<xs:annotation>
<xs:documentation>Vendor-specific device designation.</xs:documentation>
</xs:annotation>
</xs:element>
<xs:element name="AMRVendor" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.AMRVendor">
<xs:annotation>
<xs:documentation>Name of AMR vendor.</xs:documentation>
</xs:annotation>
</xs:element>
<xs:element name="billingCycle" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.billingCycle"/>
<xs:element name="catalogNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.catalogNumber"/>

```

```

    <xs:element name="installedDate" type="mspDateTime" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.installedDate"/>
    <xs:element name="manufacturer" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.manufacturer">
        <xs:annotation>
            <xs:documentation>Meter manufacturer.</xs:documentation>
        </xs:annotation>
    </xs:element>
    <xs:element name="meterNo" type="meterNo" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.meterNo">
        <xs:annotation>
            <xs:documentation>Meter number (utility-defined identifier usually called a Company
Number).</xs:documentation>
        </xs:annotation>
    </xs:element>
    <xs:element name="meterStatusList" type="meterStatusList" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.meterStatusList"/>
    <xs:element name="meterType" type="meterType" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.meterType"/>
    <xs:element name="metrologyFirmwareRevision" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.metrologyFirmwareRevision"/>
    <xs:element name="metrologyFirmwareVersion" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.metrologyFirmwareVersion"/>
    <xs:element name="removedDate" type="mspDateTime" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.removedDate"/>
    <xs:element name="serialNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspMeter.serialNumber">
        <xs:annotation>
            <xs:documentation>Meter serial number. This is the original number assigned to the meter
by the manufacturer.</xs:documentation>
        </xs:annotation>
    </xs:element>
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/mspObject">
    <xs:annotation>
        <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
    </xs:annotation>
    <xs:sequence/>
</xs:complexType>
<xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/objectID">
    <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="powerLimitationUnits"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/powerLimitationUnits">

```

```

<xs:annotation>
  <xs:documentation>Units for the powerLimitationValue.</xs:documentation>
</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="powerLimitationValue"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/powerLimitationValue">
  <xs:sequence>
    <xs:element name="units" type="powerLimitationUnits" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/powerLimitationValue.units">
      <xs:annotation>
        <xs:documentation>Units in which powerLimitationValue is expressed.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="connectDisconnectEvent" type="connectDisconnectEvent" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/powerLimitationValue.connectDisconnectEvent"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="RCDState"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/RCDState">
  <xs:annotation>
    <xs:documentation>The current state of this remote connect/disconnect device
(CDDevice).</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string">
    <xs:enumeration value="Opened"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/RCDState.Opened"/>
    <xs:enumeration value="Closed"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/RCDState.Closed"/>
    <xs:enumeration value="Armed"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/RCDState.Armed"/>
    <xs:enumeration value="Enabled"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/RCDState.Enabled"/>
    <xs:enumeration value="Disabled"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/RCDState.Disabled"/>
    <xs:enumeration value="Unknown"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/RCDState.Unknown"/>
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="readingType"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingType">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="channel" type="xs:integer" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingType.channel">
          <xs:annotation>
            <xs:documentation>Logical positioning of this measurement data.</xs:documentation>
          </xs:annotation>
        </xs:element>

```



```

<xs:element name="defaultQuality" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingType.defaultQuality">
  <xs:annotation>
    <xs:documentation>This quality code is to be supplied to each block.value unless a
block.validityStatus is supplied instead. (e.g. status = "Good").</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="defaultValueDataType" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingType.defaultValueDataType">
  <xs:annotation>
    <xs:documentation>The numeric type to be expected for the associated Block.value (e.g.
dataType= unsignedInteger).</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="dynamicConfiguration" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingType.dynamicConfiguration">
  <xs:annotation>
    <xs:documentation>Demand configuration such as block, rolling, logarithmic and sizes such
as 15 minutes, 30 minutes, 5 minutes subinterval.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="units" type="uom" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingType.units">
  <xs:annotation>
    <xs:documentation>The units of measure that this readingType are described
in.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="intervalProfile" type="intervalProfile" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingType.intervalProfile"/>
<xs:element name="readingTypeCode" type="readingTypeCode" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingType.readingTypeCode"/>
<xs:element name="readingTypeList" type="readingTypeList" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingType.readingTypeList"/>
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="readingTypeCode"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingTypeCode">
  <xs:annotation>
    <xs:documentation>This is the readingType identifier as recommended in Annex C of IEC 61968-
9. An example of the readingTypeCode would be "2.6.7.1.0.12.0.0.0.3.72". The readingTypeCode is
built up by concatenating the numerical values of each of the attributes (except "name") in this
element. Note that several of the attributes as defined in IEC61968-9 are compound, so that the total
number of attributes does not match the total number of concatenated values. The readingTypeCode
given above corresponds to "15-minute IntervalData Forward Energy, (kWh)".</xs:documentation>
  </xs:annotation>
</xs:sequence>
<xs:element name="flowDirection" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingTypeCode.flowDirection">
  <xs:annotation>

```

<xs:documentation>This attribute describes the direction of flow of the quantify measured. Examples are: 1=Forward, 2=Lagging, 3=Leading, etc. It is suggested that these values are numerical and match the recommended enumerations shown in Annex C of IEC61968-9.**</xs:documentation>**

</xs:annotation>
</xs:element>
<xs:element name="measurementCategory" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingTypeCode.measurementCategory">

<xs:annotation>
<xs:documentation>It is suggested that these values are numerical and match the recommended enumerations shown in Annex C of IEC61968-9.**</xs:documentation>**

</xs:annotation>
</xs:element>
<xs:element name="name" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingTypeCode.name">

<xs:annotation>
<xs:documentation>This is the human-readable string name for this type of reading.**</xs:documentation>**

</xs:annotation>
</xs:element>
<xs:element name="unitsMultiplier" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingTypeCode.unitsMultiplier">

<xs:annotation>
<xs:documentation>It is suggested that these values are numerical and match the recommended enumerations shown in Annex C of IEC61968-9.**</xs:documentation>**

</xs:annotation>
</xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="readingTypeList"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/readingTypeList">

<xs:annotation>
<xs:documentation>The definition of readingType comes from Annex C of IEC 61968-9.**</xs:documentation>**

</xs:annotation>
</xs:sequence>
</xs:complexType>
<xs:complexType name="registers"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/registers">

<xs:annotation>
<xs:documentation>One 'registers' element is created for each billing tier of a reading and contains all registers for that tier. Total registers are identified by the absence of the 'tier' attribute. The first tier is identified by the 'tier' attribute set to 1, the second tier is identify by the 'tier' attribute set to 2.**</xs:documentation>**

</xs:annotation>
<xs:sequence>
<xs:element name="tier" type="xs:unsignedInt" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/registers.tier"/>

</xs:sequence>
</xs:complexType>
<xs:simpleType name="uom" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/uom">

<xs:annotation>
<xs:documentation>This is the list of units of measure from ANSI C12.19**</xs:documentation>**
</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>

```

<xs:complexType name="voltage"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceAssets/MultiSpeak V411/voltage">
  <xs:sequence/>
</xs:complexType>
</xs:schema>

```

mspEndDeviceControls

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2011/cim" xmlns:msp="http://www.multispeak.org/2011/cim"
targetNamespace="http://www.multispeak.org/2011/cim" elementFormDefault="qualified"
attributeFormDefault="unqualified" xml:base="http://www.multispeak.org/2011/cim">
  <xs:element name="mspEndDeviceControls" type="mspEndDeviceControls"/>
  <xs:complexType name="mspEndDeviceControls">
    <xs:sequence>
      <xs:element name="mspObject" type="mspObject" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="demandResponseEvent"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/demandResponseEvent">
    <xs:annotation>
      <xs:documentation>NEEDS WORK ESP ON ZIGBEE ALIGHMENT.

```

This element has been deprecated as of V4.1.4 and will be removed in V4.2. It should not be implemented, rather the demandResponseEvent shall be used. </xs:documentation>

```

    </xs:annotation>
    <xs:complexContent>
      <xs:extension base="mspObject">
        <xs:sequence>
          <xs:element name="duration" type="duration" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/demandResponseEvent.duration">
            <xs:annotation>
              <xs:documentation>Time in minutes over which to control (0=continue until alternate
instruction).</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="loadManagementID" type="objectID" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/demandResponseEvent.loadManagementID"/>
          <xs:element name="Name" type="Name" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/demandResponseEvent.Name"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="description"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/description">
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="duration"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/duration">
    <xs:annotation>
      <xs:documentation>Time duration.</xs:documentation>

```

```

</xs:annotation>
<xs:sequence/>
</xs:complexType>
<xs:complexType name="loadManagementDevice"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/loadManagementDevice">
  <xs:annotation>
    <xs:documentation>This is a device that permits utility control of customer load. The device ID is
contained in the object ID for each device.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="deviceType" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/loadManagementDevice.deviceType">
          <xs:annotation>
            <xs:documentation>Load management device type.</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="loadSection"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/loadSection">
  <xs:sequence>
    <xs:element name="loadGroup" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/loadSection.loadGroup">
      <xs:annotation>
        <xs:documentation>Load group name. This is only meaningful if there is more than one load
section.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="loadGroupID" type="objectID" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/loadSection.loadGroupID">
      <xs:annotation>
        <xs:documentation>Pointer to a grouping that includes this load (a ConformLoadGroup or
NonconformLoadGroup in the CPSM profile). The use of this element is not required for MultiSpeak
transfers.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="mspDevice" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/mspDevice">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="objectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/mspDevice.objectID">
          <xs:annotation>
            <xs:documentation>This is the key field for this object type. The use of objectID is always
required.</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>

```

```

</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectId" type="objectId" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/mspObject.objectID">
      <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectId is always
required.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="Class1" type="Class1" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/mspObject.Class1"/>
      <xs:element name="description" type="description" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/mspObject.description"/>
      <xs:element name="loadSection" type="loadSection" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/mspObject.loadSection"/>
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="objectId"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/objectID">
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="ZigBeePublishPrice"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/ZigBeePublishPrice">
    <xs:annotation>
      <xs:documentation>This is a draft class for discussion.

```

This element has been deprecated as of the V4.1.4 release and will be removed at V4.2. It should not be implemented.</xs:documentation>

```

</xs:annotation>
<xs:complexContent>
  <xs:extension base="mspObject">
    <xs:sequence>
      <xs:element name="currency" type="xs:integer" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/ZigBeePublishPrice.currency"/>
      <xs:element name="rateLabel" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/ZigBeePublishPrice.rateLabel"/>
      <xs:element name="unitOfMeasure" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/MultiSpeak V411/ZigBeePublishPrice.unitOfMeasure">
        <xs:annotation>
          <xs:documentation>See Table D.17 for enumeration of this element.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:extension>

```



```

    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="Name"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/mspCPSM/mspCPSM/Name">
    <xs:sequence>
      <xs:element name="nameString" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/mspCPSM/mspCPSM/Name.nameString"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="Class1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/Class1">
    <xs:sequence>
      <xs:element name="mspObject" type="mspObject" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspEndDeviceControls/Class1.mspObject"/>
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

mspEndDeviceEvents

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xml:base="http://www.multispeak.org/2011/mspEndDeviceEvents"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2011/mspEndDeviceEvents"
targetNamespace="http://www.multispeak.org/2011/mspEndDeviceEvents"
xmlns:msp="http://www.multispeak.org/2011/mspEndDeviceEvents" elementFormDefault="qualified"
attributeFormDefault="unqualified">
  <xs:element name="mspEndDeviceEvents" type="mspEndDeviceEvents" />
  <xs:complexType name="mspEndDeviceEvents">
    <xs:sequence>
      <xs:element name="eventInstances" type="eventInstances" minOccurs="0"
maxOccurs="unbounded" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="eventInstance"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/eventInstance">
    <xs:sequence>
      <xs:element name="meterID" type="meterID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/eventInstance.meterID" />
      <xs:element name="timeStamp" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/eventInstance.timeStamp" />
      <xs:element name="meterEvent" type="meterEvent" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/eventInstance.meterEvent" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="eventInstances"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/eventInstances">
    <xs:sequence>
      <xs:element name="eventInstance" type="eventInstance" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
v415.EAP#CIM Profiles/mspEndDeviceEvents/eventInstances.eventInstance" />
    </xs:sequence>
  </xs:complexType>

```

```

</xs:sequence>
</xs:complexType>
<xs:complexType name="meterEvent"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterEvent">
  <xs:annotation>
    <xs:documentation>This is an object that carries meter and/or headend event codes. The numeric
code (e.g., "7.20.7.27") would be carried in the meterEvent complex type itself. This is used as the
mRID in IEC 61968-9 standard EndDeviceEvent instances. The domain, domainPart, type, index,
value, and codeString representation could optionally be sent in attributes on the complex
type.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="codeString" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterEvent.codeString">
      <xs:annotation>
        <xs:documentation>This would carry the string representation of the event code. For instance,
the code string for event code 7.20.7.27 is "Meter change-out" as defined in IEC 61968-9. This is
included primarily for extensibility.)</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="domain" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterEvent.domain">
      <xs:annotation>
        <xs:documentation>This would carry the domain numeric value (e.g., 7)</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="domainPart" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterEvent.domainPart">
      <xs:annotation>
        <xs:documentation>This would carry the domain part numeric value (e.g.,
20)</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="index" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterEvent.index">
      <xs:annotation>
        <xs:documentation>This would carry the index numeric value (e.g., 27)</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="type" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterEvent.type">
      <xs:annotation>
        <xs:documentation>This would carry the type part numeric value (e.g., 7)</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="value" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterEvent.value">
      <xs:annotation>
        <xs:documentation>The meterEventCode may also have an additional value here to
differentiate event conditions of a type.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>

```

```

<xs:complexType name="meterID"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterID">
  <xs:annotation>
    <xs:documentation>This is the identifier for a meter. Note that the meter might be of different
    service types (for example: electric, gas, propane, water, etc.). Since objectIDs are guaranteed to be
    unique only over the domain of one type of noun (class or object), it is necessary to specify both a
    serviceType and an objectID in order to ensure a unique instance of a meter. The value of the meterID
    should contain a string concatenated as utility.serviceType.objectID</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="meterNo" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterID.meterNo">
      <xs:annotation>
        <xs:documentation>Meter number. This is a utility-assigned, human readable number. The
        objectID of the meter is called meterID.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="objectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterID.objectID">
      <xs:annotation>
        <xs:documentation>The index or key that uniquely identifies an instance of this type of
        object.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="serviceType" type="serviceType" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterID.serviceType">
      <xs:annotation>
        <xs:documentation>This identifies the type of service this meter records. For instance for an
        electricMeter, the serviceType would be "Electric".</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="utility" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/meterID.utility">
      <xs:annotation>
        <xs:documentation>This is a string identifier for the utility that owns this
        meter.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="mspDateTime"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets.
    The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as: 2005-06-
    15T11:30:22.56-07:00 where the date is in YYYY-MM-DD format and the time is in military time,
    beginning after the date-time separator &ldquo;T&rdquo;, in hh:mm:ss.ss format, followed by
    the time zone offset. The use of fractional seconds is optional and any level of decimal precision in
    seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive
    offset (east of the prime meridian) or negative offset (west of the prime meridian). ISO 8601 date time.
    All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or
    "Zulu" time.all dates will be formatted as: 2005-06-15T11:30:22.56-07:00 where the date is in YYYY-
    MM-DD format and the time is in military time, beginning after the date-time separator "T", in
    hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and
    any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset
  </xs:documentation>
  </xs:annotation>

```


from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).</xs:documentation>

```

</xs:annotation>
<xs:restriction base="xs:string" />
</xs:simpleType>
<xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/objectID">
  <xs:restriction base="xs:string" />
</xs:simpleType>
<xs:simpleType name="serviceType"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak v415.EAP#CIM
Profiles/mspEndDeviceEvents/serviceType">
  <xs:annotation>
    <xs:documentation>The type of utility service.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string" />
</xs:simpleType>
</xs:schema>

```

mspEndDeviceFirmware

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mspEndDeviceFirmware"
xmlns:msp="http://www.multispeak.org/2012/mspEndDeviceFirmware"
targetNamespace="http://www.multispeak.org/2012/mspEndDeviceFirmware"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mspEndDeviceFirmware">
  <xs:element name="mspEndDeviceFirmware" type="mspEndDeviceFirmware"/>
  <xs:complexType name="mspEndDeviceFirmware">
    <xs:sequence>
      <xs:element name="mspMeter" type="mspMeter" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="manufacturer"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/manufacturer">
    <xs:annotation>
      <xs:documentation>Manufacturer of this asset.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="meterEvent"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/meterEvent">
    <xs:annotation>
      <xs:documentation>This is an object that carries
meter and/or headend event
codes. The numeric code
(e.g., "7.20.7.27") would be
carried in the meterEvent
complex type itself. This is used as the mRID in IEC 61968-9 standard EndDeviceEvent instances.
The domain, domainPart, type, index, value, and codeString representation
could optionally be sent in
attributes on the complex type.</xs:documentation>
    </xs:annotation>
    <xs:sequence/>
  </xs:complexType>

```

```

<xs:complexType name="module"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/module">
  <xs:annotation>
    <xs:documentation>This is an add-on module or board for a meter or other
device.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="description" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/module.description">
          <xs:annotation>
            <xs:documentation>A description of this module.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="firmwareVersion" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/module.firmwareVersion">
          <xs:annotation>
            <xs:documentation>The version of the firmware for this module.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="moduleType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/module.moduleType">
          <xs:annotation>
            <xs:documentation>This is a string identifier for a type of module.</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="moduleList"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/moduleList">
  <xs:sequence>
    <xs:element name="module" type="module" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/moduleList.module"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="mspAlarm"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspAlarm">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="eventCode" type="meterEvent" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspAlarm.eventCode">
          <xs:annotation>
            <xs:documentation>This is an object that carries meter and/or headend event
codes. The numeric code (e.g., "7.20.7.27") would be
carried in the eventCode complex type itself. This is used as the mRID in IEC 61968-9 standard
EndDeviceEvent instances. The domain, domainPart, type, index, value, and codeString
representation could optionally be sent in
attributes on the complex type.</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>

```

```

<xs:element name="eventTime" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspAlarm.eventTime">
  <xs:annotation>
    <xs:documentation>The date and time that the event occurred.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="sourceIdentifier" type="objectRef" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspAlarm.sourceIdentifier">
  <xs:annotation>
    <xs:documentation>Identifier for the source of the alarm. For instance, this might be a
meter, in which case the objectID should be the concatenated string used in the meterID (that includes
the noun and the objectID), or it could be a SCADA point, in which case the noun should be
SCADAPoint and objectID would be the SCADAPointID.</xs:documentation>
  </xs:annotation>
</xs:element>
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="mspDateTime"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets.
The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

```

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).</xs:documentation>

```

</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspDevice" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspDevice">
  <xs:sequence>
    <xs:element name="deviceClass" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspDevice.deviceClass">
      <xs:annotation>
        <xs:documentation>A high-level description of this type of object (e.g., "meter",
"communication", "connect/disconnect", "demand response"etc.).</xs:documentation>
      </xs:annotation>
    </xs:element>

```

```

    <xs:element name="facilityID" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspDevice.facilityID">
    <xs:annotation>
    <xs:documentation>A utility-defined string designation for this device.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="inServiceDate" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspDevice.inServiceDate">
    <xs:annotation>
    <xs:documentation>The date and time that a device was placed into active
service.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="manufacturer" type="manufacturer" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspDevice.manufacturer">
    <xs:annotation>
    <xs:documentation>The manufacturer of this device.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="outServiceDate" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspDevice.outServiceDate">
    <xs:annotation>
    <xs:documentation>The date and time that a device was removed from active
service.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="serialNumber" type="serialNumber" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspDevice.serialNumber">
    <xs:annotation>
    <xs:documentation>Serial number for this device.</xs:documentation>
    </xs:annotation>
    </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="mspMeter"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspMeter">
    <xs:annotation>
    <xs:documentation>This is an abstract class from which all meter device classes
inherit.</xs:documentation>
    </xs:annotation>
    <xs:complexContent>
    <xs:extension base="mspObject">
    <xs:sequence>
    <xs:element name="AMRDeviceType" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspMeter.AMRDeviceType">
    <xs:annotation>
    <xs:documentation>Vendor-specific device designation.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="AMRVendor" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspMeter.AMRVendor">
    <xs:annotation>
    <xs:documentation>Name of AMR vendor.</xs:documentation>
    </xs:annotation>

```

```

        </xs:element>
        <xs:element name="serialNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspMeter.serialNumber">
        <xs:annotation>
        <xs:documentation>Meter serial number. This is the original number assigned to the meter
by the manufacturer.</xs:documentation>
        </xs:annotation>
        </xs:element>
        <xs:element name="moduleList" type="moduleList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspMeter.moduleList"/>
        <xs:element name="mspAlarm" type="mspAlarm" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspMeter.mspAlarm"/>
        </xs:sequence>
        </xs:extension>
        </xs:complexContent>
        </xs:complexType>
        <xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspObject">
        <xs:annotation>
        <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
        </xs:annotation>
        <xs:sequence>
        <xs:element name="objectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/mspObject.objectID">
        <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectID is always
required.

```

This is the key field for this object type. The use of objectID is always required.</xs:documentation>

```

        </xs:annotation>
        </xs:element>
        </xs:sequence>
        </xs:complexType>
        <xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/objectID">
        <xs:restriction base="xs:string"/>
        </xs:simpleType>
        <xs:complexType name="objectRef"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/objectRef">
        <xs:sequence/>
        </xs:complexType>
        <xs:simpleType name="serialNumber"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/mspEndDeviceFirmware/MultiSpeak V411/serialNumber">
        <xs:annotation>
        <xs:documentation>Manufacturer-defined serial number for this device.</xs:documentation>
        </xs:annotation>
        <xs:restriction base="xs:string"/>
        </xs:simpleType>
</xs:schema>

```

mspMeterAssetConfig

```
<?xml version="1.0" encoding="utf-8"?>
```



```

<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
  xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
  xmlns="http://www.multispeak.org/2012/mspMeterAssetConfig"
  xmlns:msp="http://www.multispeak.org/2012/mspMeterAssetConfig"
  targetNamespace="http://www.multispeak.org/2012/mspMeterAssetConfig"
  elementFormDefault="qualified" attributeFormDefault="unqualified"
  xml:base="http://www.multispeak.org/2012/mspMeterAssetConfig">
  <xs:element name="mspMeterAssetConfig" type="mspMeterAssetConfig"/>
  <xs:complexType name="mspMeterAssetConfig">
    <xs:sequence>
      <xs:element name="electricMeter" type="electricMeter" minOccurs="0"
maxOccurs="unbounded"/>
      <xs:element name="mspMeter" type="mspMeter" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="address" sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/address">
    <xs:sequence/>
  </xs:complexType>
  <xs:simpleType name="billingCycle" sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/billingCycle">
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="current" sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/current">
    <xs:sequence/>
  </xs:complexType>
  <xs:simpleType name="deviceStatus"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/deviceStatus">
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="electricMeter"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/electricMeter">
    <xs:complexContent>
      <xs:extension base="mspMeter">
        <xs:sequence>
          <xs:element name="meterBaseID" type="objectID" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/electricMeter.meterBaseID">
            <xs:annotation>
              <xs:documentation>This is the objectID of the meterBase with which this electric meter is
associated, if any.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="electricNameplate" type="electricNameplate" minOccurs="0"
maxOccurs="1" sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/electricMeter.electricNameplate"/>
          <xs:element name="sealNumberList" type="sealNumberList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/electricMeter.sealNumberList"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="electricNameplate"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/electricNameplate">
    <xs:sequence>

```

```

    <xs:element name="kh" type="xs:float" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/electricNameplate.kh">
    <xs:annotation>
    <xs:documentation>Meter kh (watthour) constant. This constant is the number of watthours
that must be applied to the meter to cause one disk revolution for an electromechanical meter or the
number of watthours represented by one increment pulse for an electronic meter.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="kr" type="xs:float" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/electricNameplate.kr">
    <xs:annotation>
    <xs:documentation>Watthour meter register constant. The multiplier used to convert the
register reading to kilowatthours (or other suitable units).</xs:documentation>
    </xs:annotation>
    </xs:element>
</xs:sequence>
</xs:complexType>
<xs:simpleType name="manufacturer"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/manufacturer">
    <xs:annotation>
    <xs:documentation>Manufacturer of this asset.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="meterID" sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/meterID">
    <xs:annotation>
    <xs:documentation>This is the identifier for a meter. Note that the meter might be of different
service types (for example: electric, gas, propane, water, etc.). Since objectIDs are guaranteed to be
unique only over the domain of one type of noun (class or object), it is necessary to specify both a
serviceType and an objectID in order to ensure a unique instance of a meter. The value of the meterID
should contain a string concatenated as utility.serviceType.objectID</xs:documentation>
    </xs:annotation>
    <xs:sequence/>
</xs:complexType>
<xs:simpleType name="meterNo" sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/meterNo">
    <xs:annotation>
    <xs:documentation>Meter number (utility-defined identifier usually called a Company
Number).</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="meterStatusList"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/meterStatusList">
    <xs:annotation>
    <xs:documentation>List of meter statuses.</xs:documentation>
    </xs:annotation>
    <xs:sequence/>
</xs:complexType>
<xs:simpleType name="meterType" sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/meterType">
    <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="mspDateTime"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspDateTime">
    <xs:annotation>

```

<xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

```

</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspMeter" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter">
  <xs:annotation>
    <xs:documentation>This is an abstract class from which all meter device classes
inherit.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="AMRDeviceType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.AMRDeviceType">
          <xs:annotation>
            <xs:documentation>Vendor-specific device designation.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="AMRVendor" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.AMRVendor">
          <xs:annotation>
            <xs:documentation>Name of AMR vendor.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="billingCycle" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.billingCycle"/>
        <xs:element name="catalogNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.catalogNumber"/>
        <xs:element name="installedDate" type="xs:dateTime" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.installedDate"/>
        <xs:element name="manufacturer" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.manufacturer">
          <xs:annotation>

```



```

        <xs:documentation>Meter manufacturer.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="meterNo" type="meterNo" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.meterNo">
      <xs:annotation>
        <xs:documentation>Meter number (utility-defined identifier usually called a Company
Number).</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="meterStatusList" type="meterStatusList" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.meterStatusList"/>
    <xs:element name="meterType" type="meterType" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.meterType"/>
    <xs:element name="metrologyFirmwareRevision" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.metrologyFirmwareRevision"/>
    <xs:element name="metrologyFirmwareVersion" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.metrologyFirmwareVersion"/>
    <xs:element name="removedDate" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.removedDate"/>
    <xs:element name="serialNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.serialNumber">
      <xs:annotation>
        <xs:documentation>Meter serial number. This is the original number assigned to the meter
by the manufacturer.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="serviceLocation" type="serviceLocation" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspMeter.serviceLocation"/>
  </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectId" type="objectId" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/mspObject.objectID">
      <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectId is always
required.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="objectId" sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/objectID">
  <xs:restriction base="xs:string"/>

```

```

</xs:simpleType>
<xs:simpleType name="sealNumber" sawsdl:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/sealNumber">
  <xs:annotation>
    <xs:documentation>Identifier for a seal applied to a revenue meter.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="sealNumberList"
sawsdl:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/sealNumberList">
  <xs:sequence>
    <xs:element name="sealNumber" type="sealNumber" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/sealNumberList.sealNumber"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="serialNumber"
sawsdl:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/serialNumber">
  <xs:annotation>
    <xs:documentation>Manufacturer-defined serial number for this device.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="serviceLocation"
sawsdl:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/serviceLocation">
  <xs:annotation>
    <xs:documentation>This is a location at which a customer can receive service. A serviceLocation
can have utility services of several different serviceTypes.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="customerID" type="objectID" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/serviceLocation.customerID">
      <xs:annotation>
        <xs:documentation>Customer's identification number. This should be the objectID of the
customer.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="description" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/serviceLocation.description">
      <xs:annotation>
        <xs:documentation>Additional information about this serviceLocation</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="facilityName" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/serviceLocation.facilityName">
      <xs:annotation>
        <xs:documentation>Name of facility for multi-site business customer.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="serviceAddress" type="address" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/serviceLocation.serviceAddress">
      <xs:annotation>
        <xs:documentation>Physical address of this service location.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>

```

```

    <xs:element name="SIC" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/serviceLocation.SIC">
    <xs:annotation>
    <xs:documentation>The standard industry code for the business activity at this service
location, if this service location is commercial.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="siteID" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012//MultiSpeak
V411/mspMeterAssetConfig/MultiSpeak V411/serviceLocation.siteID">
    <xs:annotation>
    <xs:documentation>Identifier for site for a multi-site business customer.</xs:documentation>
    </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

mspMeterAssetReading

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mspMeterAssetReading"
xmlns:msp="http://www.multispeak.org/2012/mspMeterAssetReading"
targetNamespace="http://www.multispeak.org/2012/mspMeterAssetReading"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mspMeterAssetReading">
  <xs:element name="mspMeterAssetReading" type="mspMeterAssetReading"/>
  <xs:complexType name="mspMeterAssetReading">
    <xs:sequence>
    <xs:element name="mspMeterAssetReading" type="mspMeterAssetReading" minOccurs="0"
maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="address" sawSDL:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/address">
    <xs:sequence>
    <xs:element name="address1" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/address.address1">
    <xs:annotation>
    <xs:documentation>Address line 1.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="address2" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/address.address2">
    <xs:annotation>
    <xs:documentation>Address line 2.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="city" type="xs:string" minOccurs="1" sawSDL:modelReference="MultiSpeak
v415 Harmonization Project.EAP/MultiSpeak V411/address.city">
    <xs:annotation>
    <xs:documentation>Name of the city or municipality.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="country" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/address.country"/>

```

```

    <xs:element name="detailedAddressFields" type="detailedAddressFields" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/address.detailedAddressFields">
    <xs:annotation>
    <xs:documentation>Detailed information about the components of the composite address
lines.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="postalCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/address.postalCode">
    <xs:annotation>
    <xs:documentation>The postal code. For instance, in the United States this is the zip
code.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="state" type="xs:string" minOccurs="1" sawSDL:modelReference="MultiSpeak
v415 Harmonization Project.EAP/MultiSpeak V411/address.state">
    <xs:annotation>
    <xs:documentation>The state or province.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="townCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/address.townCode">
    <xs:annotation>
    <xs:documentation>Additional information about the city or town, if
required.</xs:documentation>
    </xs:annotation>
    </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="detailedAddressFields" sawSDL:modelReference="MultiSpeak v415
Harmonization Project.EAP/MultiSpeak V411/detailedAddressFields">
    <xs:sequence>
    <xs:element name="addressGeneral" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/detailedAddressFields.addressGeneral"/>
    <xs:element name="buildingNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/detailedAddressFields.buildingNumber"/>
    <xs:element name="postOfficeBox" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/detailedAddressFields.postOfficeBox"/>
    <xs:element name="region" type="xs:string" minOccurs="1" sawSDL:modelReference="MultiSpeak
v415 Harmonization Project.EAP/MultiSpeak V411/detailedAddressFields.region"/>
    <xs:element name="streetNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/detailedAddressFields.streetNumber"/>
    <xs:element name="streetPrefix" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/detailedAddressFields.streetPrefix"/>
    <xs:element name="streetSuffix" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/detailedAddressFields.streetSuffix"/>
    <xs:element name="streetType" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/detailedAddressFields.streetType"/>
    <xs:element name="suiteNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/detailedAddressFields.suiteNumber"/>
    </xs:sequence>

```

```

</xs:complexType>
<xs:complexType name="electricMeter" sawsdl:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/electricMeter">
  <xs:complexContent>
    <xs:extension base="mspMeter">
      <xs:sequence>
        <xs:element name="meterReading" type="meterReading" minOccurs="0"
maxOccurs="unbounded" sawsdl:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/electricMeter.meterReading"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="meterID" sawsdl:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/meterID">
  <xs:annotation>
    <xs:documentation>This is the identifier for a meter. Note that the meter might be of different
service types (for example: electric, gas, propane, water, etc.). Since objectIDs are guaranteed to be
unique only over the domain of one type of noun (class or object), it is necessary to specify both a
serviceType and an objectID in order to ensure a unique instance of a meter. The value of the meterID
should contain a string concatenated as utility.serviceType.objectID</xs:documentation>
  </xs:annotation>
</xs:sequence>
</xs:complexType>
<xs:complexType name="meterReading" sawsdl:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/meterReading">
  <xs:annotation>
    <xs:documentation>This is one or more readings made on a single meter. The meter could be a
revenue meter or some other type of meter. The readings can be any type of value that can be
measured by the meter.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="meterID" type="meterID" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/meterReading.meterID">
          <xs:annotation>
            <xs:documentation>This is the identifier for a meter. Note that the meter might be of
different service types (for example: electric, gas, propane, water, etc.). Since objectIDs are
guaranteed to be unique only over the domain of one type of noun (class or object), it is necessary to
specify both a serviceType and an objectID in order to ensure a unique instance of a
meter.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="electricMeter" type="electricMeter" minOccurs="0"
maxOccurs="unbounded" sawsdl:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/meterReading.electricMeter"/>
        <xs:element name="readingValues" type="readingValues" minOccurs="0" maxOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/meterReading.readingValues"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:simpleType name="mspDateTime" sawsdl:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets.
The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

```

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

```

</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspMeter" abstract="true" sawsdl:modelReference="MultiSpeak v415
Harmonization Project.EAP/MultiSpeak V411/mspMeter">
  <xs:annotation>
    <xs:documentation>This is an abstract class from which all meter device classes
inherit.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="installedDate" type="mspDateTime" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/mspMeter.installedDate"/>
        <xs:element name="removedDate" type="mspDateTime" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/mspMeter.removedDate"/>
        <xs:element name="serialNumber" type="xs:string" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/mspMeter.serialNumber">
          <xs:annotation>
            <xs:documentation>Meter serial number. This is the original number assigned to the meter
by the manufacturer.</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="mspObject" abstract="true" sawsdl:modelReference="MultiSpeak v415
Harmonization Project.EAP/MultiSpeak V411/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectId" type="objectId" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/mspObject.objectID">
      <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectId is always
required.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>

```

```

    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="objectID" sawsdl:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/objectID">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="readingStatusCode" sawsdl:modelReference="MultiSpeak v415
Harmonization Project.EAP/MultiSpeak V411/readingStatusCode">
  <xs:annotation>
    <xs:documentation>This is a three-part code that describes the status or quality of a meter
reading. For additional information about these reading statusCode definitions, see the definitions for
QualityCodes in IEC 61968-9, Annex D. meanings. The readingStatusCode that is included in the data
field shall consist of a dotted notation where the readingStatusCode value =
originatingSystemID.codeCategory.codeIndex.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="any" minOccurs="1" sawsdl:modelReference="MultiSpeak v415
Harmonization Project.EAP/MultiSpeak V411/readingStatusCode.any"/>
    <xs:element name="codeCategory" type="xs:string" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingStatusCode.codeCategory">
      <xs:annotation>
        <xs:documentation>This string categorizes quality or status codes. The values of this string
should match those given in IEC61968-9, Annex D, Table D.2.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="codeIndex" type="xs:string" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingStatusCode.codeIndex">
      <xs:annotation>
        <xs:documentation>These values are index numbers that further identify quality or status
codes. These values should match those included in IEC61968-9, Annex D, Table
D.3.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="originatingSystemID" type="xs:string" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingStatusCode.originatingSystemID">
      <xs:annotation>
        <xs:documentation>This is a three digit code that identifies the system that is commenting on
the quality or status of the meter reading. Values should be those given in IEC61968-9, Annex D,
Table D.1. For instance, 0 – Not applicable or unavailable, 1 – End device, 2 – Metering system, and
3- Meter data management system.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="readingStatusCodeEntry" sawsdl:modelReference="MultiSpeak v415
Harmonization Project.EAP/MultiSpeak V411/readingStatusCodeEntry">
  <xs:annotation>
    <xs:documentation>This is an entry that documents the status codes for meter readings and
intervalData blocks along with their meanings. For additional information about these reading
statusCode definitions, see the definitions for QualityCodes in IEC 61968-9, Annex D.
meanings.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="description" type="xs:string" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingStatusCodeEntry.description"/>

```

```

    <xs:element name="readingStatusCode" type="readingStatusCode" minOccurs="1"
maxOccurs="1" sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingStatusCodeEntry.readingStatusCode"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="readingStatusCodeList" sawsdl:modelReference="MultiSpeak v415
Harmonization Project.EAP/MultiSpeak V411/readingStatusCodeList">
  <xs:annotation>
    <xs:documentation>List of the codes used to describe the status of meter readings and their
meanings. For additional details on these codes, see the definitions for QualityCode in IEC 61968-9,
Annex D.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="readingStatusCodeEntry" type="readingStatusCodeEntry" minOccurs="0"
maxOccurs="unbounded" sawsdl:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/readingStatusCodeList.readingStatusCodeEntry"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="readingType" sawsdl:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/readingType">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="channel" type="xs:integer" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingType.channel">
          <xs:annotation>
            <xs:documentation>Logical positioning of this measurement data.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="defaultQuality" type="xs:string" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingType.defaultQuality">
          <xs:annotation>
            <xs:documentation>This quality code is to be supplied to each block.value unless a
block.validityStatus is supplied instead. (e.g. status = "Good").</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="defaultValueDataType" type="xs:string" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingType.defaultValueDataType">
          <xs:annotation>
            <xs:documentation>The numeric type to be expected for the associated Block.value (e.g.
dataType= unsignedInteger).</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="dynamicConfiguration" type="xs:string" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingType.dynamicConfiguration">
          <xs:annotation>
            <xs:documentation>Demand configuration such as block, rolling, logarithmic and sizes such
as 15 minutes, 30 minutes, 5 minutes subinterval.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="forwardChronology" type="xs:boolean" minOccurs="1"
sawsdl:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingType.forwardChronology">
          <xs:annotation>
            <xs:documentation>By default (true), intervals are presented in "forward" chronological
order. This field is provided for systems that must operate in "reverse" chronological
order.</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>

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```

</xs:element>
<xs:element name="measTypeCategory" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingType.measTypeCategory">
  <xs:annotation>
    <xs:documentation>This attribute can be used as a further descriptor for a TOU rate or
some other condition which characterizes the metering. Typical categories include: Energy, Power,
Demand, Voltage, Current, Voltage Angle, Current Angle, Phase Angle, Power Factor, Pressure,
Volume, Date, Time.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="measurementType" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingType.measurementType">
  <xs:annotation>
    <xs:documentation>This is either a string description of the type of measurement (i.e.,
Demand, energy, indoor temperature, etc) or a pointer to an instance of a CIM
MeasurementType.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="units" type="uom" minOccurs="1" sawSDL:modelReference="MultiSpeak
v415 Harmonization Project.EAP/MultiSpeak V411/readingType.units">
  <xs:annotation>
    <xs:documentation>The units of measure that this readingType are described
in.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="readingStatusCodeList" type="readingStatusCodeList" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/readingType.readingStatusCodeList"/>
<xs:element name="readingTypeCode" type="readingTypeCode" minOccurs="0"
maxOccurs="1" sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingType.readingTypeCode"/>
<xs:element name="readingValues" type="readingValues" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/readingType.readingValues"/>
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="readingTypeCode" sawSDL:modelReference="MultiSpeak v415
Harmonization Project.EAP/MultiSpeak V411/readingTypeCode">
  <xs:annotation>
    <xs:documentation>This is the readingType identifier as recommended in Annex C of IEC 61968-
9. An example of the readingTypeCode would be "2.6.7.1.0.12.0.0.0.3.72". The readingTypeCode is
built up by concatenating the numerical values of each of the attributes (except "name") in this
element. Note that several of the attributes as defined in IEC61968-9 are compound, so that the total
number of attributes does not match the total number of concatenated values. The readingTypeCode
given above corresponds to "15-minute IntervalData Forward Energy, (kWh)".</xs:documentation>
  </xs:annotation>
</xs:sequence>
  <xs:element name="accumulationBehavior" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingTypeCode.accumulationBehavior">
    <xs:annotation>
      <xs:documentation>This attribute describes how the reading was accumulated. Examples are:
1=Bulk Quality, 2=Continuous Cumulative, etc. It is suggested that these values are numerical and
match the recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="any" minOccurs="1" sawSDL:modelReference="MultiSpeak v415
Harmonization Project.EAP/MultiSpeak V411/readingTypeCode.any"/>

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    <xs:element name="dataQualifier" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingTypeCode.dataQualifier">
    <xs:annotation>
    <xs:documentation>This attribute describes the type of data (examples are 1=Absolute,
2=Average, ..8=Maximum, etc.).It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="displayableUOM" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingTypeCode.displayableUOM">
    <xs:annotation>
    <xs:documentation>It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="flowDirection" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingTypeCode.flowDirection">
    <xs:annotation>
    <xs:documentation>This attribute describes the direction of flow of the quantify measured.
Examples are: 1=Forward, 2=Lagging, 3=Leading, etc. It is suggested that these values are numerical
and match the recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="measurementCategory" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingTypeCode.measurementCategory">
    <xs:annotation>
    <xs:documentation>It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="name" type="xs:string" minOccurs="1" sawSDL:modelReference="MultiSpeak
v415 Harmonization Project.EAP/MultiSpeak V411/readingTypeCode.name">
    <xs:annotation>
    <xs:documentation>This is the human-readable string name for this type of
reading.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="phaseIndex" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingTypeCode.phaseIndex">
    <xs:annotation>
    <xs:documentation>It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="timeAttribute" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingTypeCode.timeAttribute">
    <xs:annotation>
    <xs:documentation>This attribute describes the time period of interest. Examples are: 1="10-
minute", 2="15-minute", etc. It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="unitsMultiplier" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingTypeCode.unitsMultiplier">
    <xs:annotation>

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    <xs:documentation>It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
  </xs:annotation>
</xs:element>
  <xs:element name="UOMCategory" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingTypeCode.UOMCategory">
    <xs:annotation>
      <xs:documentation>It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="readingValue" sawSDL:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/readingValue">
  <xs:annotation>
    <xs:documentation>A reading taken by a meter.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="value" type="xs:string" minOccurs="1" sawSDL:modelReference="MultiSpeak
v415 Harmonization Project.EAP/MultiSpeak V411/readingValue.value">
      <xs:annotation>
        <xs:documentation>Value of reading. Units for this reading are contained in the "units"
element.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="readingValues" sawSDL:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/readingValues">
  <xs:sequence>
    <xs:element name="readingType" type="readingType" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/readingValues.readingType"/>
    <xs:element name="readingValue" type="readingValue" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/readingValues.readingValue"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="serviceLocation" sawSDL:modelReference="MultiSpeak v415
Harmonization Project.EAP/MultiSpeak V411/serviceLocation">
  <xs:annotation>
    <xs:documentation>This is a location at which a customer can receive service. A serviceLocation
can have utility services of several different serviceTypes.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="customerID" type="objectID" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/serviceLocation.customerID">
          <xs:annotation>
            <xs:documentation>Customer's identification number. This should be the objectID of the
customer.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="description" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/serviceLocation.description">
          <xs:annotation>
            <xs:documentation>Additional information about this serviceLocation</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>

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```

        </xs:annotation>
      </xs:element>
      <xs:element name="facilityName" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/serviceLocation.facilityName">
        <xs:annotation>
          <xs:documentation>Name of facility for multi-site business customer.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="serviceAddress" type="address" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/serviceLocation.serviceAddress">
        <xs:annotation>
          <xs:documentation>Physical address of this service location.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="SIC" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/serviceLocation.SIC">
        <xs:annotation>
          <xs:documentation>The standard industry code for the business activity at this service
location, if this service location is commercial,</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="siteID" type="xs:string" minOccurs="1"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/MultiSpeak
V411/serviceLocation.siteID">
        <xs:annotation>
          <xs:documentation>Identifier for site for a multi-site business
customer.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="uom" sawSDL:modelReference="MultiSpeak v415 Harmonization
Project.EAP/MultiSpeak V411/uom">
  <xs:annotation>
    <xs:documentation>This is the list of units of measure from ANSI C12.19</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="__mspMeterAssetReading" abstract="true"
sawSDL:modelReference="MultiSpeak v415 Harmonization Project.EAP/mspMeterAssetReading">
  <xs:annotation>
    <xs:documentation>A placeholder abstract class to match the equivalent
CIM.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="mspMeter" type="mspMeter" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="MultiSpeak v415 Harmonization
Project.EAP/mspMeterAssetReading.mspMeter"/>
    <xs:element name="serviceLocation" type="serviceLocation" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="MultiSpeak v415 Harmonization
Project.EAP/mspMeterAssetReading.serviceLocation"/>
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

mSPMeterReadings

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mSPMeterAssetReading"
xmlns:mSP="http://www.multispeak.org/2012/mSPMeterAssetReading"
targetNamespace="http://www.multispeak.org/2012/mSPMeterAssetReading"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mSPMeterAssetReading">
  <xs:element name="mSPMeterReadings" type="mSPMeterReadings"/>
  <xs:complexType name="mSPMeterReadings">
    <xs:sequence/>
  </xs:complexType>
  <xs:complexType name="account" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mSPMeterReadings/account">
    <xs:annotation>
      <xs:documentation>This is a customer account.</xs:documentation>
    </xs:annotation>
    <xs:complexContent>
      <xs:extension base="mSPObject">
        <xs:sequence>
          <xs:element name="customerID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mSPMeterReadings/account.customerID">
            <xs:annotation>
              <xs:documentation>Link to the objectID (foreign key) of the customer that has this
account.</xs:documentation>
            </xs:annotation>
          </xs:element>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="blocks" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mSPMeterReadings/blocks">
    <xs:sequence/>
  </xs:complexType>
  <xs:complexType name="channels" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mSPMeterReadings/channels">
    <xs:annotation>
      <xs:documentation>There is one channel for each value being measured (e.g., kWh in, kWh out,
kWh net would be three channels).</xs:documentation>
    </xs:annotation>
    <xs:sequence>
      <xs:element name="readingType" type="readingType" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mSPMeterReadings/channels.readingType"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="duration" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mSPMeterReadings/duration">
    <xs:annotation>
      <xs:documentation>Time duration.</xs:documentation>
    </xs:annotation>
    <xs:sequence/>
  </xs:complexType>
  <xs:complexType name="intervalBlock"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mSPMeterReadings/intervalBlock">
    <xs:sequence>

```



```

    <xs:element name="intervalStart" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/intervalBlock.intervalStart">
    <xs:annotation>
    <xs:documentation>The start time for the first interval.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="logTime" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/intervalBlock.logTime">
    <xs:annotation>
    <xs:documentation>When the readings were received at the head end (logged to the
database).</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="meterID" type="meterID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/intervalBlock.meterID">
    <xs:annotation>
    <xs:documentation>Identifier for the meter that made this measurement.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="profileID" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/intervalBlock.profileID">
    <xs:annotation>
    <xs:documentation>Identifier for the profile that describes this block of
data.</xs:documentation>
    </xs:annotation>
    </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="intervalData"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/intervalData">
    <xs:sequence>
    <xs:element name="blocks" type="blocks" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/intervalData.blocks"/>
    </xs:sequence>
</xs:complexType>
<xs:complexType name="intervalProfile"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/intervalProfile">
    <xs:sequence>
    <xs:element name="intervalLength" type="duration" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/intervalProfile.intervalLength">
    <xs:annotation>
    <xs:documentation>Interval length, typically measured in minutes.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="channels" type="channels" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/intervalProfile.channels">
    <xs:annotation>
    <xs:documentation>There is one channel for each value being measured (e.g., kWh in, kWh
out, kWh net would be three channels).</xs:documentation>
    </xs:annotation>
    </xs:element>
    </xs:sequence>
</xs:complexType>

```

```

<xs:complexType name="meterEvent"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/meterEvent">
  <xs:annotation>
    <xs:documentation>This is an object that carries
meter and/or headend event
codes. The numeric code
(e.g., "7.20.7.27") would be
carried in the meterEvent
complex type itself. This is used as the mRID in IEC 61968-9 standard EndDeviceEvent instances.
The domain, domainPart, type, index, value, and codeString representation
could optionally be sent in
attributes on the complex type.</xs:documentation>
  </xs:annotation>
  <xs:sequence/>
</xs:complexType>
<xs:complexType name="meterID" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/meterID">
  <xs:annotation>
    <xs:documentation>This is the identifier for a meter. Note that the meter might be of different
service types (for example: electric, gas, propane, water, etc.). Since objectIDs are guaranteed to be
unique only over the domain of one type of noun (class or object), it is necessary to specify both a
serviceType and an objectID in order to ensure a unique instance of a meter. The value of the meterID
should contain a string concatenated as utility.serviceType.objectID</xs:documentation>
  </xs:annotation>
  <xs:sequence/>
</xs:complexType>
<xs:simpleType name="meterNo" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/meterNo">
  <xs:annotation>
    <xs:documentation>Meter number (utility-defined identifier usually called a Company
Number).</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="meterReading"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/meterReading">
  <xs:annotation>
    <xs:documentation>This is one or more readings made on a single meter. The meter could be a
revenue meter or some other type of meter. The readings can be any type of value that can be
measured by the meter.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="deviceId" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/meterReading.deviceID">
          <xs:annotation>
            <xs:documentation>A unique identifier assigned by the AMR vendor. For instance this may
be an "endPointID" or a "transponderID".</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="meterID" type="meterID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/meterReading.meterID">
          <xs:annotation>
            <xs:documentation>This is the identifier for a meter. Note that the meter might be of
different service types (for example: electric, gas, propane, water, etc.). Since objectIDs are
guaranteed to be unique only over the domain of one type of noun (class or object), it is necessary to

```

specify both a serviceType and an objectID in order to ensure a unique instance of a meter.

```

</xs:documentation>
</xs:annotation>
</xs:element>
<xs:element name="readingValues" type="readingValues" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/meterReading.readingValues"/>
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="meterStatusList"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/meterStatusList">
  <xs:annotation>
    <xs:documentation>List of meter statuses.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="meterStatus" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/meterStatusList.meterStatus">
      <xs:annotation>
        <xs:documentation>Status of this meter. Suggested values for this element include: New,
InInventory, Installed, Active, Inactive, Connected, Disconnected, ConnectedOverride,
DisconnectedOverride, OnHold, InForRepair, InTesting, Retired, Other and
Unknown.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="mspAlarm" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspMeterReadings/mspAlarm">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="eventCode" type="meterEvent" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/mspAlarm.eventCode">
          <xs:annotation>
            <xs:documentation>This is an object that carries meter and/or headend event
codes. The numeric code (e.g., "7.20.7.27") would be
carried in the eventCode complex type itself. This is used as the mRID in IEC 61968-9 standard
EndDeviceEvent instances. The domain, domainPart, type, index, value, and codeString
representation could optionally be sent in
attributes on the complex type.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="eventTime" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/mspAlarm.eventTime">
          <xs:annotation>
            <xs:documentation>The date and time that the event occurred.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="sourceIdentifier" type="objectRef" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/mspAlarm.sourceIdentifier">
          <xs:annotation>
            <xs:documentation>Identifier for the source of the alarm. For instance, this might be a
meter, in which case the objectID should be the concatenated string used in the meterID (that includes
the noun and the objectID), or it could be a SCADA point, in which case the noun should be
SCADAPoint and objectID would be the SCADAPointID.</xs:documentation>

```



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</xs:annotation>
</xs:element>
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="mspDateTime"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets.
    The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

```

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).</xs:documentation>

```

</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspMeter" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspMeterReadings/mspMeter">
  <xs:annotation>
    <xs:documentation>This is an abstract class from which all meter device classes
inherit.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="meterNo" type="meterNo" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/mspMeter.meterNo">
          <xs:annotation>
            <xs:documentation>Meter number (utility-defined identifier usually called a Company
Number).</xs:documentation>
          </xs:annotation>
          </xs:element>
          <xs:element name="meterStatusList" type="meterStatusList" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/mspMeter.meterStatusList"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak V411/mspMeterReadings/mspObject">
    <xs:annotation>

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```

    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/mspObject.objectID">
      <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectID is always
required.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="objectID" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/objectID">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="objectRef" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/objectRef">
  <xs:sequence/>
</xs:complexType>
<xs:simpleType name="readingStatus"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus">
  <xs:annotation>
    <xs:documentation>This is the meter reading status as defined by ANSI
C12.19.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string">
    <xs:enumeration value="Other" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.Other"/>
    <xs:enumeration value="suspiciousRead"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.suspiciousRead"/>
    <xs:enumeration value="DSTinEffect"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.DSTinEffect"/>
    <xs:enumeration value="manuallyUpdated"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.manuallyUpdated"/>
    <xs:enumeration value="manuallyEntered"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.manuallyEntered"/>
    <xs:enumeration value="reset" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.reset"/>
    <xs:enumeration value="seasonChange"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.seasonChange"/>
    <xs:enumeration value="firstRead"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.firstRead"/>
    <xs:enumeration value="billingRead"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.billingRead"/>
    <xs:enumeration value="afterCorrection"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.afterCorrection"/>
    <xs:enumeration value="beforeCorrection"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.beforeCorrection"/>
  </xs:restriction>
</xs:simpleType>

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    <xs:enumeration value="finalRead"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatus.finalRead"/>
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="readingStatusCode"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatusCode">
  <xs:annotation>
    <xs:documentation>This is a three-part code that describes the status or quality of a meter
reading. For additional information about these reading statusCode definitions, see the definitions for
QualityCodes in IEC 61968-9:2013, Annex D. meanings. The readingStatusCode that is included in
the data field shall consist of a dotted notation where the readingStatusCode value =
originatingSystemID.codeCategory.codeIndex.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="any" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatusCode.any"/>
    <xs:element name="codeCategory" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatusCode.codeCategory">
      <xs:annotation>
        <xs:documentation>This string categorizes quality or status codes. The values of this string
should match those given in IEC61968-9, Annex D, Table D.2.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="codeIndex" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatusCode.codeIndex">
      <xs:annotation>
        <xs:documentation>These values are index numbers that further identify quality or status
codes. These values should match those included in IEC61968-9, Annex D, Table
D.3.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="originatingSystemID" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatusCode.originatingSystemID">
      <xs:annotation>
        <xs:documentation>This is a three digit code that identifies the system that is commenting on
the quality or status of the meter reading. Values should be those given in IEC61968-9, Annex D,
Table D.1. For instance, 0 – Not applicable or unavailable, 1 – End device, 2 – Metering system, and
3- Meter data management system.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="readingStatusCodeEntry"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatusCodeEntry">
  <xs:annotation>
    <xs:documentation>This is an entry that documents the status codes for meter readings and
intervalData blocks along with their meanings. For additional information about these reading
statusCode definitions, see the definitions for QualityCodes in IEC 61968-9, Annex D.
meanings.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="description" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatusCodeEntry.description"/>

```

```

    <xs:element name="readingStatusCode" type="readingStatusCode" minOccurs="1"
maxOccurs="1" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatusCodeEntry.readingStatusCode"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="readingStatusCodeList"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatusCodeList">
  <xs:annotation>
    <xs:documentation>List of the codes used to describe the status of meter readings and their
meanings. For additional details on these codes, see the definitions for QualityCode in IEC 61968-9,
Annex D.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="readingStatusCodeEntry" type="readingStatusCodeEntry" minOccurs="0"
maxOccurs="unbounded" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatusCodeList.readingStatusCodeEntry"/>
    <xs:element name="readingValue" type="readingValue" minOccurs="0"
maxOccurs="unbounded" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingStatusCodeList.readingValue"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="readingType"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingType">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="channel" type="xs:integer" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingType.channel">
          <xs:annotation>
            <xs:documentation>Logical positioning of this measurement data.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="defaultQuality" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingType.defaultQuality">
          <xs:annotation>
            <xs:documentation>This quality code is to be supplied to each block.value unless a
block.validityStatus is supplied instead. (e.g. status = "Good").</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="defaultValueDataType" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingType.defaultValueDataType">
          <xs:annotation>
            <xs:documentation>The numeric type to be expected for the associated Block.value (e.g.
dataType= unsignedInteger).</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="dynamicConfiguration" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingType.dynamicConfiguration">
          <xs:annotation>
            <xs:documentation>Demand configuration such as block, rolling, logarithmic and sizes such
as 15 minutes, 30 minutes, 5 minutes subinterval.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="forwardChronology" type="xs:boolean" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingType.forwardChronology">

```

```

<xs:annotation>
  <xs:documentation>By default (true), intervals are presented in "forward" chronological
order. This field is provided for systems that must operate in "reverse" chronological
order.</xs:documentation>
</xs:annotation>
</xs:element>
<xs:element name="measTypeCategory" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingType.measTypeCategory">
  <xs:annotation>
    <xs:documentation>This attribute can be used as a further descriptor for a TOU rate or
some other condition which characterizes the metering. Typical categories include: Energy, Power,
Demand, Voltage, Current, Voltage Angle, Current Angle, Phase Angle, Power Factor, Pressure,
Volume, Date, Time.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="measurementType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingType.measurementType">
  <xs:annotation>
    <xs:documentation>This is either a string description of the type of measurement (i.e.,
Demand, energy, indoor temperature, etc) or a pointer to an instance of a CIM
MeasurementType.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="units" type="uom" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingType.units">
  <xs:annotation>
    <xs:documentation>The units of measure that this readingType are described
in.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="channels" type="channels" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingType.channels"/>
  <xs:element name="readingTypeCode" type="readingTypeCode" minOccurs="0"
maxOccurs="1" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingType.readingTypeCode"/>
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="readingTypeCode"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode">
  <xs:annotation>
    <xs:documentation>This is the readingType identifier as recommended in Annex C of IEC 61968-
9. An example of the readingTypeCode would be "2.6.7.1.0.12.0.0.0.3.72". The readingTypeCode is
built up by concatenating the numerical values of each of the attributes (except "name") in this
element. Note that several of the attributes as defined in IEC61968-9 are compound, so that the total
number of attributes does not match the total number of concatenated values. The readingTypeCode
given above corresponds to "15-minute IntervalData Forward Energy, (kWh)".</xs:documentation>
  </xs:annotation>
</xs:sequence>
  <xs:element name="accumulationBehavior" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode.accumulationBehavior">
    <xs:annotation>
      <xs:documentation>This attribute describes how the reading was accumulated. Examples are:
1=Bulk Quality, 2=Continuous Cumulative, etc. It is suggested that these values are numerical and
match the recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
</xs:complexType>

```



```

    </xs:annotation>
  </xs:element>
  <xs:element name="any" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode.any"/>
  <xs:element name="dataQualifier" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode.dataQualifier">
    <xs:annotation>
      <xs:documentation>This attribute describes the type of data (examples are 1=Absolute,
2=Average, ..8=Maximum, etc.).It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="displayableUOM" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode.displayableUOM">
    <xs:annotation>
      <xs:documentation>It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="flowDirection" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode.flowDirection">
    <xs:annotation>
      <xs:documentation>This attribute describes the direction of flow of the quantify measured.
Examples are: 1=Forward, 2=Lagging, 3=Leading, etc. It is suggested that these values are numerical
and match the recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="measurementCategory" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode.measurementCategory">
    <xs:annotation>
      <xs:documentation>It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="name" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode.name">
    <xs:annotation>
      <xs:documentation>This is the human-readable string name for this type of
reading.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="phaseIndex" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode.phaseIndex">
    <xs:annotation>
      <xs:documentation>It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="timeAttribute" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode.timeAttribute">
    <xs:annotation>
      <xs:documentation>This attribute describes the time period of interest. Examples are: 1="10-
minute", 2="15-minute", etc. It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
    </xs:annotation>
  </xs:element>

```

```

        </xs:annotation>
      </xs:element>
      <xs:element name="unitsMultiplier" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode.unitsMultiplier">
        <xs:annotation>
          <xs:documentation>It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="UOMCategory" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeCode.UOMCategory">
        <xs:annotation>
          <xs:documentation>It is suggested that these values are numerical and match the
recommended enumerations shown in Annex C of IEC61968-9.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="readingTypeList"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeList">
    <xs:annotation>
      <xs:documentation>The definition of readingType comes from Annex C of IEC 61968-
9.</xs:documentation>
    </xs:annotation>
    <xs:sequence>
      <xs:element name="readingType" type="readingType" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingTypeList.readingType"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="readingValue"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingValue">
    <xs:annotation>
      <xs:documentation>A reading taken by a meter.</xs:documentation>
    </xs:annotation>
    <xs:sequence>
      <xs:element name="readingType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingValue.readingType">
        <xs:annotation>
          <xs:documentation>This string carries the classification of the reading. Suggested
enumeration values include:
Outage
Billing
Voltage
Phase
Water usage
Gas usage
Propane usage</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="timeStamp" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingValue.timeStamp"/>
      <xs:element name="value" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingValue.value">
        <xs:annotation>

```

```

        <xs:documentation>Value of reading. Units for this reading are contained in the "units"
        element.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="readingStatusCodeList" type="readingStatusCodeList" minOccurs="0"
    maxOccurs="unbounded" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
    V411/mspMeterReadings/readingValue.readingStatusCodeList"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="readingValues"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/readingValues">
  <xs:sequence>
    <xs:element name="readingValue" type="readingValue" minOccurs="0"
    maxOccurs="unbounded" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
    V411/mspMeterReadings/readingValues.readingValue"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="serviceLocation"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/serviceLocation">
  <xs:annotation>
    <xs:documentation>This is a location at which a customer can receive service. A serviceLocation
    can have utility services of several different serviceTypes.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="customerID" type="objectID" minOccurs="1"
        sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
        V411/mspMeterReadings/serviceLocation.customerID">
          <xs:annotation>
            <xs:documentation>Customer's identification number. This should be the objectID of the
            customer.</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:simpleType name="uom" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterReadings/uom">
  <xs:annotation>
    <xs:documentation>This is the list of units of measure from ANSI C12.19</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
</xs:schema>

```

mspMeterReadSchedule

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawsdl="http://www.w3.org/ns/sawsdl"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mspMeterReadSchedule"
xmlns:msp="http://www.multispeak.org/2012/mspMeterReadSchedule"
targetNamespace="http://www.multispeak.org/2012/mspMeterReadSchedule"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mspMeterReadSchedule">
  <xs:element name="mspMeterReadSchedule" type="mspMeterReadSchedule"/>
  <xs:complexType name="mspMeterReadSchedule">
    <xs:sequence/>

```



```

</xs:complexType>
<xs:complexType name="absoluteSchedule"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/absoluteSchedule">
  <xs:sequence/>
</xs:complexType>
<xs:complexType name="account"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/account">
  <xs:annotation>
    <xs:documentation>This is a customer account.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="serviceLocations" type="serviceLocations" minOccurs="0"
maxOccurs="1" sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/account.serviceLocations"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="associatedMeters"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/associatedMeters">
  <xs:sequence/>
</xs:complexType>
<xs:complexType name="duration"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/duration">
  <xs:annotation>
    <xs:documentation>Time duration.</xs:documentation>
  </xs:annotation>
  <xs:sequence/>
</xs:complexType>
<xs:simpleType name="floatType"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/floatType">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="meterGroupList"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/meterGroupList">
  <xs:sequence>
    <xs:element name="meterGroupID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/meterGroupList.meterGroupID"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="meterID"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/meterID">
  <xs:annotation>
    <xs:documentation>This is the identifier for a meter. Note that the meter might be of different
service types (for example: electric, gas, propane, water, etc.). Since objectIDs are guaranteed to be
unique only over the domain of one type of noun (class or object), it is necessary to specify both a
serviceType and an objectID in order to ensure a unique instance of a meter. The value of the meterID
should contain a string concatenated as utility.serviceType.objectID</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/meterID.objectID">
      <xs:annotation>
        <xs:documentation>The index or key that uniquely identifies an instance of this type of
object.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="serviceType" type="serviceType" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/meterID.serviceType">

```

```

    <xs:annotation>
      <xs:documentation>This identifies the type of service this meter records. For instance for an
electricMeter, the serviceType would be "Electric".</xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="meterList"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/meterList">
  <xs:annotation>
    <xs:documentation>The meterIDs for the meters contained in this meter
group.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="meterID" type="meterID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/meterList.meterID">
      <xs:annotation>
        <xs:documentation>This is the identifier for a meter. Note that the meter might be of different
service types (for example: electric, gas, propane, water, etc.). Since objectIDs are guaranteed to be
unique only over the domain of one type of noun (class or object), it is necessary to specify both a
serviceType and an objectID in order to ensure a unique instance of a meter.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="mspDateTime"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets.
The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

```

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).</xs:documentation>

```

  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspMeter"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/mspMeter">
  <xs:annotation>
    <xs:documentation>This is an abstract class from which all meter device classes
inherit.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence/>

```

```

</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/mspObject.objectID">
      <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectID is always
required.

```

This is the key field for this object type. The use of objectID is always required.</xs:documentation>

```

    </xs:annotation>
  </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="mspPointObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/mspPointObject">
  <xs:annotation>
    <xs:documentation>This is an abstract class for objects that exhibit point
geometry.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:complexType>
      <xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/objectID">
        <xs:restriction base="xs:string"/>
      </xs:simpleType>
      <xs:complexType name="period"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/period">
        <xs:sequence>
          <xs:element name="timeUnits" type="timeUnits" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/period.timeUnits">
            <xs:annotation>
              <xs:documentation>Units of time.</xs:documentation>
            </xs:annotation>
          </xs:element>
        </xs:sequence>
      </xs:complexType>
      <xs:complexType name="periodicSchedule"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/periodicSchedule">
        <xs:sequence>
          <xs:element name="startTime" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/periodicSchedule.startTime"/>
          <xs:element name="period" type="period" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/periodicSchedule.period"/>
        </xs:sequence>
      </xs:complexType>
      <xs:complexType name="readingSchedule"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/readingSchedule">
        <xs:annotation>
          <xs:documentation>A meter reading schedule.</xs:documentation>
        </xs:annotation>
        <xs:complexContent>
          <xs:extension base="schedule">

```

```

    <xs:sequence>
      <xs:element name="meterGroupList" type="meterGroupList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/readingSchedule.meterGroupList"/>
      <xs:element name="meterList" type="meterList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/readingSchedule.meterList"/>
    </xs:sequence>
  </xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="readingType"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/readingType">
  <xs:sequence>
    <xs:element name="measurementType" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/readingType.measurementType">
      <xs:annotation>
        <xs:documentation>This is either a string description of the type of measurement (i.e.,
Demand, energy, indoor temperature, etc) or a pointer to an instance of a CIM
MeasurementType.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="readingTypeCode" type="readingTypeCode" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/readingType.readingTypeCode">
      <xs:annotation>
        <xs:documentation>This optional element carries the information to create a CIM-compatible
readingType, based on Annex C of IEC61968-9:2013. If it is desired to create CIM-compatible
readingType identifiers (objectIDs in MultiSpeak parlance) then they can be created by concatenating
the numerical attributes carried in this element.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="readingTypeCode"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/readingTypeCode">
  <xs:annotation>
    <xs:documentation>This is the readingType identifier as recommended in Annex C of IEC 61968-
9:2013. An example of the readingTypeCode would be "2.6.7.1.0.12.0.0.0.3.72". The
readingTypeCode is built up by concatenating the numerical values of each of the attributes (except
"name") in this element. Note that several of the attributes as defined in IEC61968-9 are compound,
so that the total number of attributes does not match the total number of concatenated values. The
readingTypeCode given above corresponds to "15-minute IntervalData Forward Energy,
(kWh)".</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="name" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/readingTypeCode.name">
      <xs:annotation>
        <xs:documentation>This is the human-readable string name for this type of
reading.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="readingValues"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/readingValues">
  <xs:sequence/>
</xs:complexType>

```

```

<xs:complexType name="schedule"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/schedule">
  <xs:annotation>
    <xs:documentation>This is a general-purpose schedule object.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="isEnabled" type="xs:boolean" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/schedule.isEnabled"/>
        <xs:element name="offset" type="duration" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/schedule.offset">
          <xs:annotation>
            <xs:documentation>The offset from midnight (i.e., 0 hours, 0 minutes, 0 seconds) for the
periodic time points to begin. For example, for an interval meter that is set up for five minute intervals,
setting recurrencePeriod=300 (5 minutes) and offset=120 (2 minutes) would result in scheduled events
to read the meter executing at 2, 7, 12, 17, 22, 27, 32, 37, 42, 47, 52, and 57 minutes past each
hour.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="periodicSchedule" type="periodicSchedule" minOccurs="0"
maxOccurs="1" sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/schedule.periodicSchedule"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="serviceLocation"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/serviceLocation">
  <xs:annotation>
    <xs:documentation>This is a location at which a customer can receive service. A serviceLocation
can have utility services of several different serviceTypes.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="customerID" type="objectID" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/serviceLocation.customerID">
          <xs:annotation>
            <xs:documentation>Customer's identification number. This should be the objectID of the
customer.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="description" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/serviceLocation.description">
          <xs:annotation>
            <xs:documentation>Additional information about this serviceLocation</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="serviceLocations"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/serviceLocations">
  <xs:sequence>
    <xs:element name="serviceLocation" type="serviceLocation" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/serviceLocations.serviceLocation"/>
  </xs:sequence>

```



```

</xs:complexType>
<xs:simpleType name="serviceType"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/serviceType">
  <xs:annotation>
    <xs:documentation>The type of utility service.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="timePeriod"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/timePeriod">
  <xs:sequence>
    <xs:element name="endTime" type="mspDateTime" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/timePeriod.endTime"/>
    <xs:element name="startTime" type="mspDateTime" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/timePeriod.startTime"/>
    <xs:element name="readingType" type="readingType" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/timePeriod.readingType"/>
    <xs:element name="schedule" type="schedule" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/timePeriod.schedule"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="timePoint"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/timePoint">
  <xs:sequence>
    <xs:element name="absoluteTime" type="mspDateTime" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/timePoint.absoluteTime"/>
    <xs:element name="sequenceNumber" type="xs:integer" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/timePoint.sequenceNumber"/>
    <xs:element name="absoluteSchedule" type="absoluteSchedule" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/timePoint.absoluteSchedule"/>
    <xs:element name="readingSchedule" type="readingSchedule" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak
V411/timePoint.readingSchedule"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="timeUnits"
sawSDL:modelReference="http://www.multispeak.org/2012//MultiSpeak V411/timeUnits">
  <xs:annotation>
    <xs:documentation>Units of time</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
</xs:schema>

```

mspMeterServiceRequest

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mspMeterServiceRequest"
xmlns:msp="http://www.multispeak.org/2012/mspMeterServiceRequest"
targetNamespace="http://www.multispeak.org/2012/mspMeterServiceRequest"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mspMeterServiceRequest">
  <xs:element name="mspMeterServiceRequest" type="mspMeterServiceRequest"/>
  <xs:complexType name="mspMeterServiceRequest">
    <xs:sequence>

```

```

</xs:complexType>
<xs:simpleType name="accountNumber"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/accountNumber">
  <xs:annotation>
    <xs:documentation>Account number for this service location. Depending on the system of record,
this may be either the account.objectID or the account.objectName.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="customer" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/customer">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="servType" type="servType" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/customer.servType"/>
        <xs:element name="specialNeeds" type="specialNeeds" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/customer.specialNeeds"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="electricMeter"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/electricMeter">
  <xs:complexContent>
    <xs:extension base="mspMeter">
      <xs:sequence>
        <xs:element name="meterBaseID" type="objectID" minOccurs="0"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/electricMeter.meterBaseID">
          <xs:annotation>
            <xs:documentation>This is the objectID of the meterBase with which this electric meter is
associated, if any.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="electricNameplate" type="electricNameplate" minOccurs="0"
maxOccurs="1" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/electricMeter.electricNameplate"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="electricNameplate"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/electricNameplate">
  <xs:sequence>
    <xs:element name="form" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/electricNameplate.form">
      <xs:annotation>
        <xs:documentation>Meter form designation per ANSI C12.10. An alphanumeric designation
denoting the circuit arrangement for which the meter is applicable and its specific terminal
arrangement.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="kh" type="xs:float" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/electricNameplate.kh">

```

```

    <xs:annotation>
      <xs:documentation>Meter kh (watthour) constant. This constant is the number of watthours
that must be applied to the meter to cause one disk revolution for an electromechanical meter or the
number of watthours represented by one increment pulse for an elctronice meter.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="kr" type="xs:float" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/electricNameplate.kr">
    <xs:annotation>
      <xs:documentation>Watthour meter register constant. The multiplier used to convert the
register reading to kilowatthours (or other suitable units).</xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="existingMeter"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/existingMeter">
  <xs:sequence>
    <xs:element name="actionTaken" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/existingMeter.actionTaken">
      <xs:annotation>
        <xs:documentation>Action taken on this meter. Possible enumerations include: Installed,
Removed, Exchanged, Read, Disconnected, Checked, Other, or Unknown.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="meterID" type="meterID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/existingMeter.meterID"/>
    <xs:element name="meterType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/existingMeter.meterType"/>
    <xs:element name="serviceType" type="serviceType" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/existingMeter.serviceType">
      <xs:annotation>
        <xs:documentation>The type of utility service for this account (electric, water, gas, propane,
etc.)</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="readingValues" type="readingValues" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/existingMeter.readingValues"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="meterEvent"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterEvent">
  <xs:annotation>
    <xs:documentation>This is an object that carries
meter and/or headend event
codes. The numeric code
(e.g., "7.20.7.27") would be
carried in the meterEvent
complex type itself. This is used as the mRID in IEC 61968-9 standard EndDeviceEvent instances.
The domain, domainPart, type, index, value, and codeString representation
could optionally be sent in
attributes on the complex type.</xs:documentation>
  </xs:annotation>
</xs:sequence>

```



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    <xs:element name="any" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterEvent.any"/>
    <xs:element name="codeString" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterEvent.codeString">
        <xs:annotation>
            <xs:documentation>This would carry the string representation of the event code. For instance,
the code string for event code 7.20.7.27 is "Meter change-out" as defined in IEC 61968-9. This is
included primarily for extensibility.)</xs:documentation>
        </xs:annotation>
    </xs:element>
    <xs:element name="domain" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterEvent.domain">
        <xs:annotation>
            <xs:documentation>This would carry the domain numeric value (e.g., 7)</xs:documentation>
        </xs:annotation>
    </xs:element>
    <xs:element name="domainPart" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterEvent.domainPart">
        <xs:annotation>
            <xs:documentation>This would carry the domain part numeric value (e.g.,
20)</xs:documentation>
        </xs:annotation>
    </xs:element>
    <xs:element name="index" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterEvent.index">
        <xs:annotation>
            <xs:documentation>This would carry the index numeric value (e.g., 27)</xs:documentation>
        </xs:annotation>
    </xs:element>
    <xs:element name="type" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterEvent.type">
        <xs:annotation>
            <xs:documentation>This would carry the type part numeric value (e.g., 7)</xs:documentation>
        </xs:annotation>
    </xs:element>
    <xs:element name="value" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterEvent.value">
        <xs:annotation>
            <xs:documentation>The meterEventCode may also have an additional value here to
differentiate event conditions of a type.</xs:documentation>
        </xs:annotation>
    </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="meterID" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterID">
    <xs:annotation>
        <xs:documentation>This is the identifier for a meter. Note that the meter might be of different
service types (for example: electric, gas, propane, water, etc.). Since objectIDs are guaranteed to be
unique only over the domain of one type of noun (class or object), it is necessary to specify both a
serviceType and an objectID in order to ensure a unique instance of a meter. The value of the meterID
should contain a string concatenated as utility.serviceType.objectID</xs:documentation>
    </xs:annotation>
</xs:sequence>

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    <xs:element name="meterNo" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterID.meterNo">
    <xs:annotation>
    <xs:documentation>Meter number. This is a utility-assigned, human readable number. The
objectID of the meter is called meterID.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="objectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterID.objectID">
    <xs:annotation>
    <xs:documentation>The index or key that uniquely identifies an instance of this type of
object.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="serviceType" type="serviceType" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterID.serviceType">
    <xs:annotation>
    <xs:documentation>This identifies the type of service this meter records. For instance for an
electricMeter, the serviceType would be "Electric".</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="utility" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterID.utility">
    <xs:annotation>
    <xs:documentation>This is a string identifier for the utility that owns this
meter.</xs:documentation>
    </xs:annotation>
    </xs:element>
</xs:sequence>
</xs:complexType>
<xs:simpleType name="meterNo" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterNo">
    <xs:annotation>
    <xs:documentation>Meter number (utility-defined identifier usually called a Company
Number).</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="meterReading"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterReading">
    <xs:annotation>
    <xs:documentation>This is one or more readings made on a single meter. The meter could be a
revenue meter or some other type of meter. The readings can be any type of value that can be
measured by the meter.</xs:documentation>
    </xs:annotation>
    <xs:sequence>
    <xs:element name="deviceID" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterReading.deviceID">
    <xs:annotation>
    <xs:documentation>A unique identifier assigned by the AMR vendor. For instance this may be
an "endPointID" or a "transponderID".</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="meterID" type="meterID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterReading.meterID">

```

```

<xs:annotation>
  <xs:documentation>This is the identifier for a meter. Note that the meter might be of different
  service types (for example: electric, gas, propane, water, etc.). Since objectIDs are guaranteed to be
  unique only over the domain of one type of noun (class or object), it is necessary to specify both a
  serviceType and an objectID in order to ensure a unique instance of a meter.</xs:documentation>
</xs:annotation>
</xs:element>
<xs:element name="readingValues" type="readingValues" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/meterReading.readingValues"/>
</xs:sequence>
</xs:complexType>
<xs:complexType name="mspAlarm" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspAlarm">
  <xs:sequence>
    <xs:element name="eventCode" type="meterEvent" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspAlarm.eventCode">
      <xs:annotation>
        <xs:documentation>This is an object that carries meter and/or headend event
        codes. The numeric code (e.g., "7.20.7.27") would be
        carried in the eventCode complex type itself. This is used as the mRID in IEC 61968-9 standard
        EndDeviceEvent instances. The domain, domainPart, type, index, value, and codeString
        representation could optionally be sent in
        attributes on the complex type.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="eventTime" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspAlarm.eventTime">
      <xs:annotation>
        <xs:documentation>The date and time that the event occurred.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="sourceIdentifier" type="objectRef" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspAlarm.sourceIdentifier">
      <xs:annotation>
        <xs:documentation>Identifier for the source of the alarm. For instance, this might be a meter,
        in which case the objectID should be the concatenated string used in the meterID (that includes the
        noun and the objectID), or it could be a SCADA point, in which case the noun should be SCADAPoint
        and objectID would be the SCADAPointID.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="mspDateTime"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets.
    The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:
  
```

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All date time fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time. All dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).</xs:documentation>

```

</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspDeviceExchange" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspDeviceExchange">
  <xs:annotation>
    <xs:documentation>This is an abstract class that addresses the exchange of one piece of
equipment of a type for another piece of equipment of the same type. The objectID of this object is the
objectID of the device being added; the replaceID attribute carries the objectID of the device being
replaced.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="meterBaseID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspDeviceExchange.meterBaseID">
      <xs:annotation>
        <xs:documentation>The objectID of a meter base.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="serviceLocationID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspDeviceExchange.serviceLocationID">
      <xs:annotation>
        <xs:documentation>This is the objectID for the customer service location at which the device
exchange is taking place.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="meterID" type="meterID" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspDeviceExchange.meterID"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="mspMeter" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspMeter">
  <xs:annotation>
    <xs:documentation>This is an abstract class from which all meter device classes
inherit.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="AMRVendor" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspMeter.AMRVendor">
          <xs:annotation>
            <xs:documentation>Name of AMR vendor.</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>

```

```

    <xs:element name="meterNo" type="meterNo" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspMeter.meterNo">
    <xs:annotation>
    <xs:documentation>Meter number (utility-defined identifier usually called a Company
Number).</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="serialNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspMeter.serialNumber">
    <xs:annotation>
    <xs:documentation>Meter serial number. This is the original number assigned to the meter
by the manufacturer.</xs:documentation>
    </xs:annotation>
    </xs:element>
    </xs:sequence>
    </xs:extension>
    </xs:complexContent>
    </xs:complexType>
    <xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspObject">
    <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
    </xs:annotation>
    <xs:sequence>
    <xs:element name="objectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/mspObject.objectID">
    <xs:annotation>
    <xs:documentation>This is the key field for this object type. The use of objectID is always
required.</xs:documentation>
    </xs:annotation>
    </xs:element>
    </xs:sequence>
    </xs:complexType>
    <xs:complexType name="network" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/network">
    <xs:annotation>
    <xs:documentation>This child object of service location incorporates information typically stored
in a GIS as system of record, in contrast to the information in the parent class "serviceLocation" that is
typically stored in a customer information system as system of record.</xs:documentation>
    </xs:annotation>
    <xs:sequence/>
    </xs:complexType>
    <xs:simpleType name="objectID" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/objectID">
    <xs:restriction base="xs:string"/>
    </xs:simpleType>
    <xs:complexType name="objectRef" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/objectRef">
    <xs:sequence/>
    </xs:complexType>
    <xs:simpleType name="qualityDescription"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/qualityDescription">
    <xs:restriction base="xs:string">
    <xs:enumeration value="Calculated"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/qualityDescription.Calculated"/>

```



```

    <xs:enumeration value="Default" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/qualityDescription.Default"/>
    <xs:enumeration value="Estimated"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/qualityDescription.Estimated"/>
    <xs:enumeration value="Failed" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/qualityDescription.Failed"/>
    <xs:enumeration value="Initial" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/qualityDescription.Initial"/>
    <xs:enumeration value="Last" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/qualityDescription.Last"/>
    <xs:enumeration value="Measured"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/qualityDescription.Measured"/>
    <xs:enumeration value="Other" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/qualityDescription.Other"/>
    <xs:enumeration value="Unknown"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/qualityDescription.Unknown"/>
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="readingType"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/readingType">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="channel" type="xs:integer" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/readingType.channel">
          <xs:annotation>
            <xs:documentation>Logical positioning of this measurement data.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="defaultQuality" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/readingType.defaultQuality">
          <xs:annotation>
            <xs:documentation>This quality code is to be supplied to each block.value unless a
block.validityStatus is supplied instead. (e.g. status = "Good").</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="defaultValueDataType" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/readingType.defaultValueDataType">
          <xs:annotation>
            <xs:documentation>The numeric type to be expected for the associated Block.value (e.g.
dataType= unsignedInteger).</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="dynamicConfiguration" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/readingType.dynamicConfiguration">
          <xs:annotation>
            <xs:documentation>Demand configuration such as block, rolling, logarithmic and sizes such
as 15 minutes, 30 minutes, 5 minutes subinterval.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="forwardChronology" type="xs:boolean" minOccurs="1"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/readingType.forwardChronology">
          <xs:annotation>

```

`<xs:documentation>`By default (true), intervals are presented in "forward" chronological order. This field is provided for systems that must operate in "reverse" chronological order.`</xs:documentation>`
`</xs:annotation>`
`</xs:element>`
`<xs:element name="measTypeCategory" type="xs:string" minOccurs="1"`
`sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak`
`V411/mspMeterServiceRequest/readingType.measTypeCategory">`
`<xs:annotation>`
`<xs:documentation>`This attribute can be used as a further descriptor for a TOU rate or some other condition which characterizes the metering. Typical categories include: Energy, Power, Demand, Voltage, Current, Voltage Angle, Current Angle, Phase Angle, Power Factor, Pressure, Volume, Date, Time.`</xs:documentation>`
`</xs:annotation>`
`</xs:element>`
`<xs:element name="measurementType" type="xs:string" minOccurs="1"`
`sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak`
`V411/mspMeterServiceRequest/readingType.measurementType">`
`<xs:annotation>`
`<xs:documentation>`This is either a string description of the type of measurement (i.e., Demand, energy, indoor temperature, etc) or a pointer to an instance of a CIM MeasurementType.`</xs:documentation>`
`</xs:annotation>`
`</xs:element>`
`<xs:element name="units" type="uom" minOccurs="1"`
`sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak`
`V411/mspMeterServiceRequest/readingType.units">`
`<xs:annotation>`
`<xs:documentation>`The units of measure that this readingType are described in.`</xs:documentation>`
`</xs:annotation>`
`</xs:element>`
`<xs:element name="readingValues" type="readingValues" minOccurs="0"`
`maxOccurs="unbounded" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak`
`V411/mspMeterServiceRequest/readingType.readingValues"/>`
`</xs:sequence>`
`</xs:extension>`
`</xs:complexContent>`
`</xs:complexType>`
`<xs:complexType name="readingValues"`
`sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak`
`V411/mspMeterServiceRequest/readingValues">`
`<xs:sequence>`
`<xs:element name="readingType" type="readingType" minOccurs="0" maxOccurs="unbounded"`
`sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak`
`V411/mspMeterServiceRequest/readingValues.readingType"/>`
`</xs:sequence>`
`</xs:complexType>`
`<xs:complexType name="replacementMeter"`
`sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak`
`V411/mspMeterServiceRequest/replacementMeter">`
`<xs:sequence>`
`<xs:element name="actionTaken" type="xs:string" minOccurs="1"`
`sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak`
`V411/mspMeterServiceRequest/replacementMeter.actionTaken">`
`<xs:annotation>`
`<xs:documentation>`Action taken on this meter. Possible enumerations include: Installed, Removed, Exchanged, Read, Disconnected, Checked, Other, or Unknown.`</xs:documentation>`
`</xs:annotation>`
`</xs:element>`

```

    <xs:element name="meterID" type="meterID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/replacementMeter.meterID"/>
    <xs:element name="meterType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/replacementMeter.meterType"/>
    <xs:element name="serviceType" type="serviceType" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/replacementMeter.serviceType">
        <xs:annotation>
            <xs:documentation>The type of utility service for this account (electric, water, gas, propane,
etc.)</xs:documentation>
        </xs:annotation>
    </xs:element>
    <xs:element name="readingValues" type="readingValues" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/replacementMeter.readingValues"/>
</xs:sequence>
</xs:complexType>
<xs:complexType name="serviceOrder"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder">
    <xs:annotation>
        <xs:documentation>This is an order to work that may be location-specific, but does not include
capital construction.</xs:documentation>
    </xs:annotation>
    <xs:complexContent>
        <xs:extension base="mspObject">
            <xs:sequence>
                <xs:element name="accountNumber" type="accountNumber" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.accountNumber">
                    <xs:annotation>
                        <xs:documentation>Account number for this work order. if it exists. Depending on the
system of record, this may be either the account.objectID or the account.objectName, but is assumed
to be unique.</xs:documentation>
                    </xs:annotation>
                </xs:element>
                <xs:element name="closeDate" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.closeDate">
                    <xs:annotation>
                        <xs:documentation>Close date for this construction.</xs:documentation>
                    </xs:annotation>
                </xs:element>
                <xs:element name="crewID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.crewID">
                    <xs:annotation>
                        <xs:documentation>Identifier for crew to perform this work.</xs:documentation>
                    </xs:annotation>
                </xs:element>
                <xs:element name="customerID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.customerID">
                    <xs:annotation>
                        <xs:documentation>Customer requesting service</xs:documentation>
                    </xs:annotation>
                </xs:element>
                <xs:element name="network" type="network" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.network"/>

```



```

    <xs:element name="projectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.projectID">
    <xs:annotation>
    <xs:documentation>This is a pointer to a project, if any, that contains this service
order</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="serviceLocationID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.serviceLocationID">
    <xs:annotation>
    <xs:documentation>Identifier for the service location.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="soDescription" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.soDescription"/>
    <xs:element name="soNumber" type="soNumber" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.soNumber">
    <xs:annotation>
    <xs:documentation>This is the work order number of the associated work order, if
any.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="soType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.soType">
    <xs:annotation>
    <xs:documentation>Service order type.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="startDate" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.startDate">
    <xs:annotation>
    <xs:documentation>Start date for construction.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="woNumber" type="woNumber" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.woNumber">
    <xs:annotation>
    <xs:documentation>This is the work order number of the associated work order, if
any.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="workedBy" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.workedBy"/>
    <xs:element name="soStatusCode" type="soStatusCode" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceOrder.soStatusCode"/>
    </xs:sequence>
    </xs:extension>
    </xs:complexContent>
    </xs:complexType>
    <xs:simpleType name="serviceType" sawSDL:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/serviceType">
    <xs:annotation>
    <xs:documentation>The type of utility service.</xs:documentation>

```

```

    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="servType" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/servType">
    <xs:annotation>
      <xs:documentation>Type of customer service. Example enumerations are: "Residential",
"Commerical", "Industrial", etc.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="soNumber" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/soNumber">
    <xs:annotation>
      <xs:documentation>Service order number.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="soStatusCode"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/soStatusCode">
    <xs:annotation>
      <xs:documentation>Service order status code.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string">
      <xs:enumeration value="Active" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/soStatusCode.Active"/>
      <xs:enumeration value="Cancelled"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/soStatusCode.Cancelled"/>
      <xs:enumeration value="Closed" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/soStatusCode.Closed"/>
      <xs:enumeration value="Enroute" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/soStatusCode.Enroute"/>
      <xs:enumeration value="OnHold" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/soStatusCode.OnHold"/>
      <xs:enumeration value="Onsite" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/soStatusCode.Onsite"/>
      <xs:enumeration value="Other" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/soStatusCode.Other"/>
      <xs:enumeration value="Unknown"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/soStatusCode.Unknown"/>
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="specialNeeds"
sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/specialNeeds">
    <xs:annotation>
      <xs:documentation>A descriptor for any special needs that this customer has, such as life
support equipment.

```

A descriptor for any special needs that this customer has, such as life support equipment. THIS SIMPLE TYPE HAS BEEN DEPRECATED IN V4.1 AND WILL BE REPLACED WITH AN ARRAY OF NAME-VALUE PAIRS DESCRIBING TYPES OF SPECIAL NEEDS IN V4.2.</xs:documentation>

```

    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="uom" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/uom">
    <xs:annotation>
      <xs:documentation>This is the list of units of measure from ANSI C12.19</xs:documentation>

```

```

</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="woNumber" sawsdl:modelReference="http://www.epri.net/2012/MultiSpeak
V411/mspMeterServiceRequest/woNumber">
  <xs:annotation>
    <xs:documentation>Work order number</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
</xs:schema>

```

mspPricingStructureConfig

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawsdl="http://www.w3.org/ns/sawsdl"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mspPricingStructureConfig"
xmlns:msp="http://www.multispeak.org/2012/mspPricingStructureConfig"
targetNamespace="http://www.multispeak.org/2012/mspPricingStructureConfig"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mspPricingStructureConfig">
  <xs:element name="mspPricingStructureConfig" type="mspPricingStructureConfig"/>
  <xs:complexType name="mspPricingStructureConfig">
    <xs:sequence>
      <xs:element name="mspPricingStructure" type="mspPricingStructure" minOccurs="0"
maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="chargeItem"
sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/chargeItem">
    <xs:sequence>
      <xs:element name="charge" type="money" minOccurs="1"
sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/chargeItem.char
ge">
        <xs:annotation>
          <xs:documentation>The amount of this charge, in defined units of
currency.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="description" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/chargeItem.des
cription">
        <xs:annotation>
          <xs:documentation>This string field should carry a description of the type of charge. Examples
include:
kWh
kW
kVAR
Power cost adjustment
Tax
Service charge
Device</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="chargeItemList"
sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/chargeItemList"
>
    <xs:sequence>

```

```

    <xs:element name="chargetItem" type="chargetItem" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/chargetItemList.c
hargetItem"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="currencyCode"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/currencyCode">
  <xs:annotation>
    <xs:documentation>Three letter currency code as standardized in ISO4217-2001. Typically is the
two letter Internet domain code for the country followed by a one letter designator for the currency; for
instance the USA dollar is USD.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="description"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/description">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="money"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/money">
  <xs:sequence/>
</xs:complexType>
<xs:simpleType name="mspDateTime"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets.
The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

```

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).</xs:documentation>

```

  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectId" type="objectId" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspObject.obje
ctID">
      <xs:annotation>

```

```

        <xs:documentation>This is the key field for this object type. The use of objectID is always
required.</xs:documentation>
    </xs:annotation>
</xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="mspPricingStructure" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspPricingStruc
ture">
    <xs:complexContent>
        <xs:extension base="mspObject">
            <xs:sequence>
                <xs:element name="description" type="description" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspPricingStruc
ture.description"/>
                <xs:element name="mspTariffs" type="mspTariffs" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspPricingStruc
ture.mspTariffs"/>
                <xs:element name="serviceType" type="serviceType" minOccurs="0"
maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspPricingStruc
ture.serviceType"/>
                <xs:element name="servType" type="servType" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspPricingStruc
ture.servType"/>
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="mspTariffIntervalList"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspTariffInterva
lList">
    <xs:sequence>
        <xs:element name="startDateTime" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspTariffInterva
lList.startDateTime"/>
        <xs:element name="chargeItemList" type="chargeItemList" minOccurs="0"
maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspTariffInterva
lList.chargeItemList"/>
    </xs:sequence>
</xs:complexType>
<xs:complexType name="mspTariffs" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspTariffs">
    <xs:complexContent>
        <xs:extension base="mspObject">
            <xs:sequence>
                <xs:element name="mspTariffIntervalList" type="mspTariffIntervalList" minOccurs="0"
maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspTariffs.msp
TariffIntervalList"/>
                <xs:element name="rateInfo" type="rateInfo" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/mspTariffs.rateI
nfo"/>
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/objectID">
    <xs:restriction base="xs:string"/>
</xs:simpleType>

```



```

<xs:complexType name="rate"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rate">
  <xs:annotation>
    <xs:documentation>This is the rate for sale of a commodity. For instance, if the current rate for
    electrical energy is $0.10/kWh, then the float value 0.10 is carried in the rate base element and the
    optional attributes currency code would carry "USD" for dollars, the commodityUnits would carry
    "kWh", and the rateDescription would carry "Current".
  </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="commodityUnits" type="uom" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rate.commodity
Units">
      <xs:annotation>
        <xs:documentation>This attribute identifies the units in which the commodity being sold is
        expressed. For instance, for electrical energy sold at retail this might be "kWh".
      </xs:documentation>
      </xs:annotation>
      </xs:element>
      <xs:element name="currencyCode" type="currencyCode" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rate.currencyCo
de">
        <xs:annotation>
          <xs:documentation>This attribute identifies the currency in which this rate is expressed, as
          defined in ISO 4217. For instance, "USD" is United States Dollars.
        </xs:documentation>
        </xs:annotation>
        </xs:element>
        <xs:element name="rateCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rate.rateCode">
          <xs:annotation>
            <xs:documentation>The string identifier for this rate.
          </xs:documentation>
          </xs:annotation>
          </xs:element>
          <xs:element name="rateDescription" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rate.rateDescrip
tion"/>
        </xs:sequence>
      </xs:complexType>
    <xs:complexType name="rateInfo"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rateInfo">
      <xs:sequence>
        <xs:element name="dateTime" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rateInfo.dateTim
e"/>
        <xs:element name="rateCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rateInfo.rateCod
e">
          <xs:annotation>
            <xs:documentation>This is the string identifier for this rate.
          </xs:documentation>
          </xs:annotation>
          </xs:element>
          <xs:element name="rateCodeDescription" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rateInfo.rateCod
eDescription">
            <xs:annotation>
              <xs:documentation>This string attribute describes the typ of rate. Examples are "Current" for
              the currently effective rate. "Peak" for the critical peak pricing rate.
              "AverageThisPeriod" for the average rate during the current billing period. "AverageLastPeriod" for the

```

average during the previous billing period. "MinimumThisPeriod" "MaximumThisPeriod",
 etc.</xs:documentation>
 </xs:annotation>
 </xs:element>
 <xs:element name="rates" type="rates" minOccurs="0" maxOccurs="1"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rateInfo.rates"/>
 </xs:sequence>
 </xs:complexType>
 <xs:complexType name="rates"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rates">
 <xs:sequence>
 <xs:element name="rate" type="rate" minOccurs="0" maxOccurs="unbounded"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/rates.rate"/>
 </xs:sequence>
 </xs:complexType>
 <xs:simpleType name="serviceType"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType">
 <xs:annotation>
 <xs:documentation>The type of utility service.</xs:documentation>
 </xs:annotation>
 <xs:restriction base="xs:string">
 <xs:enumeration value="Electric"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Ele
 ctric"/>
 <xs:enumeration value="Gas"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Ga
 s"/>
 <xs:enumeration value="Water"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Wa
 ter"/>
 <xs:enumeration value="Propane"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Pro
 pane"/>
 <xs:enumeration value="Refuse"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Ref
 use"/>
 <xs:enumeration value="Sewer"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Se
 wer"/>
 <xs:enumeration value="Telecom"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Tel
 ecom"/>
 <xs:enumeration value="TV"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.TV"
 />
 <xs:enumeration value="Cable"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Ca
 ble"/>
 <xs:enumeration value="Heating"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.He
 ating"/>
 <xs:enumeration value="Steam"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Ste
 am"/>
 <xs:enumeration value="Transportation"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Tra
 nsportation"/>
 <xs:enumeration value="All"
 sawsdl:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.All"
 />


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    <xs:enumeration value="Other"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Other"/>
    <xs:enumeration value="Unknown"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/serviceType.Unknown"/>
  </xs:restriction>
</xs:simpleType>
<xs:simpleType name="servType"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/servType">
  <xs:annotation>
    <xs:documentation>Type of customer service. Example enumerations are: "Residential",
"Commerical", "Industrial", etc.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="uom"
sawSDL:modelReference="http://www.multispeak.org/2012/mspPricingStructureConfig/uom">
  <xs:annotation>
    <xs:documentation>This is the list of units of measure from ANSI C12.19</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
</xs:schema>

```

mspReceiptRecord

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mspReceiptRecord"
xmlns:msp="http://www.multispeak.org/2012/mspReceiptRecord"
targetNamespace="http://www.multispeak.org/2012/mspReceiptRecord"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mspReceiptRecord">
  <xs:element name="mspReceiptRecord" type="mspReceiptRecord"/>
  <xs:complexType name="mspReceiptRecord">
    <xs:sequence>
      <xs:element name="paymentTransaction" type="paymentTransaction" minOccurs="0"
maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="accountNumber"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/accountNumber">
    <xs:annotation>
      <xs:documentation>Account number for this service location. Depending on the system of record,
this may be either the account.objectID or the account.objectName.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="achPayment"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/achPayment">
    <xs:sequence>
      <xs:element name="checkInfo" type="checkInfo" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/achPayment.checkInfo"/>
      <xs:element name="tender" type="tender" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/achPayment.tender"/>
    </xs:sequence>
  </xs:complexType>

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</xs:complexType>
<xs:complexType name="address"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/address">
  <xs:sequence/>
</xs:complexType>
<xs:complexType name="allowableTransactionTypes"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/allowableTransactionTypes">
  <xs:sequence>
    <xs:element name="transactionType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/allowableTransactionTypes.transactionType">
      <xs:annotation>
        <xs:documentation>Transaction types are the valid payment transaction nouns allowed for
this customer (Example: creditCardPayment, eCheckPayment, achPayment, cashPayment, or
sendToOperator).</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="cashPayment"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/cashPayment">
  <xs:sequence>
    <xs:element name="tender" type="tender" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/cashPayment.tender"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="checkInfo"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/checkInfo">
  <xs:sequence>
    <xs:element name="bankRoutingNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/checkInfo.bankRoutingNumber"/>
    <xs:element name="checkingAccountNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/checkInfo.checkingAccountNumber"/>
    <xs:element name="checkNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/checkInfo.checkNumber"/>
    <xs:element name="nameOnCheck" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/checkInfo.nameOnCheck"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="creditCardInfo"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardInfo">
  <xs:sequence>
    <xs:element name="address" type="address" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardInfo.address"/>
    <xs:element name="cardVerificationValue" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardInfo.cardVerificationValue">
      <xs:annotation>
        <xs:documentation>Security value on card (cw2)</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>

```

```

    <xs:element name="creditCardExpDate" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardInfo.creditCardExpDate">
    <xs:annotation>
        <xs:documentation>month/year.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="creditCardNumber" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardInfo.creditCardNumber"/>
    <xs:element name="firstName" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardInfo.firstName">
    <xs:annotation>
        <xs:documentation>First (or given) name of the card holder.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="lastName" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardInfo.lastName">
    <xs:annotation>
        <xs:documentation>Last (or family) name – surname.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="mName" type="middleName" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardInfo.mName">
    <xs:annotation>
        <xs:documentation>Middle name of cardholder.</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="suffix" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardInfo.suffix">
    <xs:annotation>
        <xs:documentation>Suffix to name (e.g., Jr, III, etc).</xs:documentation>
    </xs:annotation>
</xs:element>
    <xs:element name="type" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardInfo.type">
    <xs:annotation>
        <xs:documentation>Visa, MasterCard, AmEx...</xs:documentation>
    </xs:annotation>
</xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="creditCardPayment"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardPayment">
    <xs:sequence>
        <xs:element name="creditCardInfo" type="creditCardInfo" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardPayment.creditCardInfo"/>
        <xs:element name="tender" type="tender" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/creditCardPayment.tender"/>
    </xs:sequence>
</xs:complexType>
<xs:complexType name="eCheckPayment"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/eCheckPayment">

```

```

<xs:sequence>
  <xs:element name="checkInfo" type="checkInfo" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/eCheckPayment.checkInfo"/>
  <xs:element name="tender" type="tender" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/eCheckPayment.tender"/>
</xs:sequence>
</xs:complexType>
<xs:simpleType name="middleName"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/middleName">
  <xs:annotation>
    <xs:documentation>Middle name of a person.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="money"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/money">
  <xs:sequence/>
</xs:complexType>
<xs:simpleType name="mspDateTime"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets.
The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

```

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).</xs:documentation>

```

</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectId" type="objectId" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/mspObject.objectID">

```

```

    <xs:annotation>
      <xs:documentation>This is the key field for this object type. The use of objectID is always
required.</xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
</xs:complexType>
<xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/objectID">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="payableItem"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItem">
  <xs:annotation>
    <xs:documentation>An item to which a customer payment could be applied.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="chargeCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItem.chargeCode"/>
        <xs:element name="currentAmount" type="money" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItem.currentAmount"/>
        <xs:element name="description" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItem.description"/>
        <xs:element name="disconnectDate" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItem.disconnectDate"/>
        <xs:element name="dueDate" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItem.dueDate"/>
        <xs:element name="lastPaymentAmount" type="money" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItem.lastPaymentAmount"/>
        <xs:element name="lastPaymentDate" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItem.lastPaymentDate"/>
        <xs:element name="pastDueAmount" type="money" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItem.pastDueAmount"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="payableItemList"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItemList">
  <xs:annotation>
    <xs:documentation>List of items to which a customer payment could be
applied.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="allowableTransactionTypes" type="allowableTransactionTypes"
minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItemList.allowableTransactionTypes"/>

```



```

    <xs:element name="payableItem" type="payableItem" minOccurs="1" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/payableItemList.payableItem"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="paymentTransaction"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/paymentTransaction">
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="accountNumber" type="accountNumber" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/paymentTransaction.accountNumber">
          <xs:annotation>
            <xs:documentation>Account number for customer's utility account.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="typeService" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/paymentTransaction.typeService"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="paymentTransactionGroup"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/paymentTransactionGroup">
  <xs:sequence>
    <xs:element name="achPayment" type="achPayment" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/paymentTransactionGroup.achPayment"/>
    <xs:element name="cashPayment" type="cashPayment" minOccurs="0"
maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/paymentTransactionGroup.cashPayment"/>
    <xs:element name="creditCardPayment" type="creditCardPayment" minOccurs="0"
maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/paymentTransactionGroup.creditCardPayment"/>
    <xs:element name="eCheckPayment" type="eCheckPayment" minOccurs="0"
maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/paymentTransactionGroup.eCheckPayment"/>
    <xs:element name="paymentTransaction" type="paymentTransaction" minOccurs="0"
maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/paymentTransactionGroup.paymentTransaction"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="responseCode"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/responseCode">
  <xs:restriction base="xs:string">
    <xs:enumeration value="Success"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/responseCode.Success"/>
    <xs:enumeration value="Failure"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/responseCode.Failure"/>
  </xs:restriction>
</xs:simpleType>

```

```

    <xs:enumeration value="Error"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/responseCode.Error"/>
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="tender"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/tender">
  <xs:annotation>
    <xs:documentation>A payment in cash or equivalents.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="amount" type="money" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/tender.amount">
          <xs:annotation>
            <xs:documentation>Payment amount, expressed in currency units.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="authorizationCode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/tender.authorizationCode">
          <xs:annotation>
            <xs:documentation>Bank or clearing house authorization code. Blank until transaction
committed</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="convenienceFee" type="money" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/tender.convenienceFee">
          <xs:annotation>
            <xs:documentation>Financial transaction convenience fee</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="financialTransactionID" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/tender.financialTransactionID">
          <xs:annotation>
            <xs:documentation>This ID cross references the transactionID returned from the credit
card vendor (if supplied) for billing.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="paymentMode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/tender.paymentMode">
          <xs:annotation>
            <xs:documentation>Payment mode (ACH, eCheck, credit card, kiosk, IVR,
etc.).</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="responseCode" type="responseCode" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/tender.responseCode"/>
        <xs:element name="total" type="money" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/tender.total">
          <xs:annotation>
            <xs:documentation>Total of payments and fees tendered.</xs:documentation>
          </xs:annotation>
        </xs:element>

```



```

        <xs:element name="utilityFee" type="money" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/tender.utilityFee">
        <xs:annotation>
        <xs:documentation>Utility added transaction fee. May be fixed or a calculated value. CB
calculated method.</xs:documentation>
        </xs:annotation>
        </xs:element>
        <xs:element name="payableItemList" type="payableItemList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspReceiptRecord/MultiSpeak
V411/tender.payableItemList"/>
        </xs:sequence>
        </xs:extension>
        </xs:complexContent>
        </xs:complexType>
    </xs:schema>

```

mspSDPLocationConfig

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mspSDPLocationConfig"
xmlns:msp="http://www.multispeak.org/2012/mspSDPLocationConfig"
targetNamespace="http://www.multispeak.org/2012/mspSDPLocationConfig"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mspSDPLocationConfig">
    <xs:element name="mspSDPLocationConfig" type="mspSDPLocationConfig"/>
    <xs:complexType name="mspSDPLocationConfig">
        <xs:sequence>
            <xs:element name="mspSDPLocation" type="mspSDPLocation" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
    </xs:complexType>
    <xs:complexType name="address">
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/address">
        <xs:sequence>
            <xs:element name="address1" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/address.address1"
>
            <xs:annotation>
            <xs:documentation>Address line 1.</xs:documentation>
            </xs:annotation>
        </xs:element>
            <xs:element name="address2" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/address.address2"
>
            <xs:annotation>
            <xs:documentation>Address line 2.</xs:documentation>
            </xs:annotation>
        </xs:element>
            <xs:element name="city" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/address.city">
            <xs:annotation>
            <xs:documentation>Name of the city or municipality.</xs:documentation>
            </xs:annotation>
        </xs:element>
            <xs:element name="country" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/address.country"/>
            <xs:element name="detailedAddressFields" type="detailedAddressFields" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/address.detailedAd
dressFields">

```

```

        <xs:annotation>
          <xs:documentation>Detailed information about the components of the composite address
lines.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="postalCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/address.postalCod
e">
        <xs:annotation>
          <xs:documentation>The postal code. For instance, in the United States this is the zip
code.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="state" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/address.state">
        <xs:annotation>
          <xs:documentation>The state or province.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="townCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/address.townCode
">
        <xs:annotation>
          <xs:documentation>Additional information about the city or town, if
required.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="addressItem"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/addressItem">
    <xs:sequence>
      <xs:element name="address" type="address" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/addressItem.addre
ss"/>
      <xs:element name="addressType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/addressItem.addre
ssType">
        <xs:annotation>
          <xs:documentation>Type of address. For instance, billing, mailing, service location,
etc.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="priorityOrder" type="priorityOrder" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/addressItem.priorit
yOrder">
        <xs:annotation>
          <xs:documentation>Order that should be used to contact this person. First = 1, second = 2,
etc.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="addressList"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/addressList">
    <xs:sequence>
      <xs:element name="addressItem" type="addressItem" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/addressList.address
Item"/>
    </xs:sequence>
  </xs:complexType>

```

```

<xs:complexType name="contactInfo"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/contactInfo">
  <xs:annotation>
    <xs:documentation>This class contains ways to contact an entity.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="addressList" type="addressList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/contactInfo.addressList"/>
    <xs:element name="eMailList" type="eMailList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/contactInfo.eMailList"/>
    <xs:element name="otherContactInformation" type="otherContactInformation" minOccurs="0"
maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/contactInfo.otherContactInformation"/>
    <xs:element name="phoneList" type="phoneList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/contactInfo.phoneList"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="description"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/description">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="detailedAddressFields"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/detailedAddressFields">
  <xs:sequence/>
</xs:complexType>
<xs:complexType name="eMailAddress"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/eMailAddress">
  <xs:sequence>
    <xs:element name="eMail" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/eMailAddress.eMail"/>
    <xs:element name="eMailType" type="eMailType" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/eMailAddress.eMailType">
      <xs:annotation>
        <xs:documentation>The type of e-mail address.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="priorityOrder" type="priorityOrder" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/eMailAddress.priorityOrder">
      <xs:annotation>
        <xs:documentation>Order in which these e-mail addresses should be used to contact this
person.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="eMailList"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/eMailList">
  <xs:sequence>
    <xs:element name="eMailAddress" type="eMailAddress" minOccurs="0"
maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/eMailList.eMailAddress"/>
  </xs:sequence>
</xs:complexType>

```

```

<xs:simpleType name="eMailType"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/eMailType">
  <xs:annotation>
    <xs:documentation>Type of e-mail address.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string">
    <xs:enumeration value="Personal"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/eMailType.Personal"/>
    <xs:enumeration value="Business"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/eMailType.Business"/>
    <xs:enumeration value="Alternate"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/eMailType.Alternate"/>
    <xs:enumeration value="Other"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/eMailType.Other"/>
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="gpsLocation"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/gpsLocation">
  <xs:annotation>
    <xs:documentation>A geographical location as collected using a global positioning system receiver.</xs:documentation>
  </xs:annotation>
  <xs:sequence/>
</xs:complexType>
<xs:complexType name="gpsPoint"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/gpsPoint">
  <xs:annotation>
    <xs:documentation>THE gpsPoint OBJECT HAS BEEN DEPRECATED. It will be replaced by the GPSTLocation in V4.2.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="altitude" type="xs:double" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/gpsPoint.altitude">
      <xs:annotation>
        <xs:documentation>The altitude above mean sea level for this geographical location, assuming the use of the WGS84 geodetic datum.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="latitude" type="xs:double" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/gpsPoint.latitude">
      <xs:annotation>
        <xs:documentation>The latitude portion of a geographical location.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="longitude" type="xs:double" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/gpsPoint.longitude">
      <xs:annotation>
        <xs:documentation>The longitude portion of a geographical location</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either directly or indirectly).</xs:documentation>
  </xs:annotation>

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```

<xs:sequence>
  <xs:element name="objectId" type="objectId" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/mspObject.objectI
D">
    <xs:annotation>
      <xs:documentation>This is the key field for this object type. The use of objectId is always
required.</xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="mspPointObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/mspPointObject">
  <xs:annotation>
    <xs:documentation>This is an abstract class for objects that exhibit point
geometry.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="facilityID" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/mspPointObject.fa
cilityID">
      <xs:annotation>
        <xs:documentation>Utility string designation for this object.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="gpsPoint" type="gpsPoint" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/mspPointObject.gp
sPoint">
      <xs:annotation>
        <xs:documentation>The gpsPoint element has been deprecated as of the V4.1.2 Release. It
will be replaced with the GPSLocation in V4.2. It is suggested that you use only the GPSLocation
element for carrying locations expressed in latitude and longitude and the WGS84
datum.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="gridLocation" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/mspPointObject.gri
dLocation">
      <xs:annotation>
        <xs:documentation>This is a utility-assigned location identifier that may or may not be tied to
any GIS coordinate system. Sometimes referred to by the utility as "map number" or "grid
number".</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="rotation" type="rotation" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/mspPointObject.rot
ation">
      <xs:annotation>
        <xs:documentation>Rotation angle for display of the feature, expressed in degrees measured
clockwise from the vertical.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="mspSDPLocation" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/mspSDPLocation"
>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>

```



```

        <xs:element name="contactInfo" type="contactInfo" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/mspSDPLocation.c
ontactInfo"/>
        <xs:element name="description" type="description" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/mspSDPLocation.d
escription"/>
        <xs:element name="serviceLocation" type="serviceLocation" minOccurs="0"
maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/mspSDPLocation.s
erviceLocation"/>
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/objectID">
    <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="otherContactInformation"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/otherContactInform
ation">
    <xs:sequence>
        <xs:element name="otherContactItem" type="otherContactItem" minOccurs="0"
maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/otherContactInform
ation.otherContactItem"/>
    </xs:sequence>
</xs:complexType>
<xs:complexType name="otherContactItem"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/otherContactItem">
    <xs:annotation>
        <xs:documentation>This is an element to carry other miscellaneous contact information, such as
web site address, truck radio number, etc.</xs:documentation>
    </xs:annotation>
    <xs:sequence>
        <xs:element name="details" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/otherContactItem.d
etails">
            <xs:annotation>
                <xs:documentation>This element carries the miscellaneous contact
information.</xs:documentation>
            </xs:annotation>
        </xs:element>
        <xs:element name="infoType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/otherContactItem.i
nfoType">
            <xs:annotation>
                <xs:documentation>This element is used to describe the type of information stored in the
"details" element. Examples might be: IP address, truck number, radio number.</xs:documentation>
            </xs:annotation>
        </xs:element>
    </xs:sequence>
</xs:complexType>
<xs:complexType name="phoneList"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/phoneList">
    <xs:sequence>
        <xs:element name="phoneNumber" type="phoneNumber" minOccurs="0"
maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/phoneList.phoneN
umber"/>
    </xs:sequence>
</xs:complexType>

```

```

<xs:complexType name="phoneNumber"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/phoneNumber">
  <xs:sequence>
    <xs:element name="phone" type="telephoneNumber" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/phoneNumber.pho
ne"/>
    <xs:element name="phoneType" type="phoneType" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/phoneNumber.pho
neType">
      <xs:annotation>
        <xs:documentation>The type of phone number.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="priorityOrder" type="priorityOrder" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/phoneNumber.priori
tyOrder"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="phoneType"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/phoneType">
  <xs:annotation>
    <xs:documentation>Type of phone number.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="priorityOrder"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/priorityOrder">
  <xs:annotation>
    <xs:documentation>Order in which this contact information type should be used to contact this
person. First = 1, second =2, etc.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="rotation"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/rotation">
  <xs:annotation>
    <xs:documentation>Rotation angle for display of the feature, expressed in degrees measured
clockwise from the vertical.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="serviceLocation"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/serviceLocation">
  <xs:annotation>
    <xs:documentation>This is a location at which a customer can receive service. A serviceLocation
can have utility services of several different serviceTypes.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="customerID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/serviceLocation.cu
stomerID">
          <xs:annotation>
            <xs:documentation>Customer's identification number. This should be the objectID of the
customer.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="description" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/serviceLocation.de
scription">
          <xs:annotation>

```



```

        <xs:documentation>Additional information about this serviceLocation</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="facilityName" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/serviceLocation.fac
ilityName">
      <xs:annotation>
        <xs:documentation>Name of facility for multi-site business customer.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="serviceAddress" type="address" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/serviceLocation.ser
viceAddress">
      <xs:annotation>
        <xs:documentation>Physical address of this service location.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="SIC" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/serviceLocation.SI
C">
      <xs:annotation>
        <xs:documentation>The standard industry code for the business activity at this service
location, if this service location is commercial,</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="siteID" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/serviceLocation.sit
eID">
      <xs:annotation>
        <xs:documentation>Identifier for site for a multi-site business
customer.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="phoneList" type="phoneList" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/serviceLocation.ph
oneList"/>
    <xs:element name="specialNeeds" type="specialNeeds" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/serviceLocation.sp
ecialNeeds"/>
  </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="specialNeeds"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/specialNeeds">
  <xs:annotation>
    <xs:documentation>A descriptor for any special needs that this customer has, such as life
support equipment.
A descriptor for any special needs that this customer has, such as life support equipment. THIS
SIMPLE TYPE HAS BEEN DEPRECATED IN V4.1 AND WILL BE REPLACED WITH AN ARRAY OF
NAME-VALUE PAIRS DESCRIBING TYPES OF SPECIAL NEEDS IN V4.2.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="telephoneNumber"
sawSDL:modelReference="http://www.multispeak.org/2012/mspSDPLocationConfig/telephoneNumber"
>
  <xs:sequence/>
</xs:complexType>
</xs:schema>

```

mSPServiceCategoryConfig

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mSPServiceCategoryConfig"
xmlns:mSP="http://www.multispeak.org/2012/mSPServiceCategoryConfig"
targetNamespace="http://www.multispeak.org/2012/mSPServiceCategoryConfig"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mSPServiceCategoryConfig">
  <xs:element name="mSPServiceCategoryConfig" type="mSPServiceCategoryConfig"/>
  <xs:complexType name="mSPServiceCategoryConfig">
    <xs:sequence>
      </xs:complexType>
      <xs:simpleType name="description"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceCategoryConfig/description">
        <xs:restriction base="xs:string"/>
      </xs:simpleType>
      <xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceCategoryConfig/objectID">
        <xs:restriction base="xs:string"/>
      </xs:simpleType>
      <xs:complexType name="objectRef"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceCategoryConfig/objectRef">
        <xs:sequence>
          <xs:element name="name" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceCategoryConfig/objectRef.name">
            <xs:annotation>
              <xs:documentation>Name of instance of the object. When used to describe an engineering
line section, the name is the name of the section and is required.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="noun" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceCategoryConfig/objectRef.noun">
            <xs:annotation>
              <xs:documentation>The noun is the name of the object type that includes this instance. For
example, if instance is described by the MultiSpeak "substation" object then this field will contain
"substation".</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="objectID" type="objectID" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceCategoryConfig/objectRef.objectID">
            <xs:annotation>
              <xs:documentation>This is the identifier for the instance of the object. objectID must be
unique over the domain of the specific type of object.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="utility" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceCategoryConfig/objectRef.utility">
            <xs:annotation>
              <xs:documentation>A string identifier for the utility that owns the item being identified by this
objectRef.</xs:documentation>
            </xs:annotation>
          </xs:element>
        </xs:sequence>
      </xs:complexType>
    </xs:sequence>
  </xs:complexType>

```

```

    <xs:element name="serviceType" type="serviceType" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/objectRef.serviceType"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="serviceType"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType">
  <xs:annotation>
    <xs:documentation>The type of utility service.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string">
    <xs:enumeration value="Electric"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Electric"/>
    <xs:enumeration value="Gas"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Gas"/>
    <xs:enumeration value="Water"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Water"/>
    <xs:enumeration value="Propane"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Propane"/>
    <xs:enumeration value="Refuse"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Refuse"/>
    <xs:enumeration value="Sewer"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Sewer"/>
    <xs:enumeration value="Telecom"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Telecom"/>
    <xs:enumeration value="TV"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.TV"/>
    <xs:enumeration value="Cable"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Cable"/>
    <xs:enumeration value="Heating"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Heating"/>
    <xs:enumeration value="Steam"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Steam"/>
    <xs:enumeration value="Transportation"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Transportation"/>
    <xs:enumeration value="All"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.All"/>
    <xs:enumeration value="Other"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Other"/>
    <xs:enumeration value="Unknown"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceCategoryConfig/serviceType.Unknown"/>
  </xs:restriction>
</xs:simpleType>
</xs:schema>

```

mSPServiceDeliveryPointConfig

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mSPServiceDeliveryPointConfig"
xmlns:mSP="http://www.multispeak.org/2012/mSPServiceDeliveryPointConfig"
targetNamespace="http://www.multispeak.org/2012/mSPServiceDeliveryPointConfig"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mSPServiceDeliveryPointConfig">
  <xs:element name="mSPServiceDeliveryPointConfig" type="mSPServiceDeliveryPointConfig"/>
  <xs:complexType name="mSPServiceDeliveryPointConfig">
    <xs:sequence>
      <xs:element name="mSPServiceDeliveryPoint" type="mSPServiceDeliveryPoint" minOccurs="0"
maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PointType"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceDeliveryPointConfig/mSPGeometry/PointType">
    <xs:annotation>
      <xs:documentation>A Point is defined by a single coordinate tuple.</xs:documentation>
    </xs:annotation>
    <xs:sequence/>
  </xs:complexType>
  <xs:complexType name="account"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceDeliveryPointConfig/MultiSpeak V411/account">
    <xs:annotation>
      <xs:documentation>This is a customer account.</xs:documentation>
    </xs:annotation>
    <xs:sequence>
      <xs:element name="customerID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceDeliveryPointConfig/MultiSpeak V411/account.customerID">
        <xs:annotation>
          <xs:documentation>Link to the objectID (foreign key) of the customer that has this
account.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="serviceLocations" type="serviceLocations" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceDeliveryPointConfig/MultiSpeak V411/account.serviceLocations"/>
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="billingCycle"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceDeliveryPointConfig/MultiSpeak V411/billingCycle">
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="budgBill"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceDeliveryPointConfig/MultiSpeak V411/budgBill">
    <xs:annotation>
      <xs:documentation>Budget billing code.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="currencyCode"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mSPServiceDeliveryPointConfig/MultiSpeak V411/currencyCode">

```

```

<xs:annotation>
  <xs:documentation>Three letter currency code as standardized in ISO4217-2001. Typically is the
  two letter Internet domain code for the country followed by a one letter designator for the currency; for
  instance the USA dollar is USD.</xs:documentation>
</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="current"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/current">
  <xs:sequence>
    <xs:element name="units" type="currentUnits" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/current.units">
      <xs:annotation>
        <xs:documentation>Units in which this current is measured.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="electricService" type="electricService" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/current.electricService"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="currentUnits"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/currentUnits">
  <xs:restriction base="xs:string">
    <xs:enumeration value="A" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/currentUnits.A"/>
    <xs:enumeration value="mA"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/currentUnits.mA"/>
    <xs:enumeration value="kA"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/currentUnits.kA"/>
    <xs:enumeration value="MA"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/currentUnits.MA"/>
    <xs:enumeration value="microA"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/currentUnits.microA"/>
    <xs:enumeration value="Per_Unit"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/currentUnits.Per Unit"/>
    <xs:enumeration value="Percent"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/currentUnits.Percent"/>
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="electricService"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricService">
  <xs:annotation>
    <xs:documentation>A point of service for receiving electrical utility service.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="billingCycle" type="billingCycle" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricService.billingCycle"/>

```



```

    <xs:element name="budgetBill" type="budgBill" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricService.budgetBill">
    <xs:annotation>
    <xs:documentation>Budget bill code. </xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="loadMgmt" type="loadMgmt" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricService.loadMgmt">
    <xs:annotation>
    <xs:documentation>Load management code.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="meterBaseID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricService.meterBaseID">
    <xs:annotation>
    <xs:documentation>This is the objectID of the meterBase object. MeterBase is the
permanent location of the meter installation.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="serviceStatus" type="servStatus" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricService.serviceStatus"/>
    <xs:element name="current" type="current" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricService.current"/>
    <xs:element name="meterBase" type="meterBase" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricService.meterBase"/>
    <xs:element name="realEnergy" type="realEnergy" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricService.realEnergy"/>
    <xs:element name="voltage" type="voltage" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricService.voltage"/>
    </xs:sequence>
    </xs:extension>
    </xs:complexContent>
    </xs:complexType>
    <xs:complexType name="electricServiceList"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricServiceList">
    <xs:sequence>
    <xs:element name="electricService" type="electricService" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/electricServiceList.electricService"/>
    </xs:sequence>
    </xs:complexType>
    <xs:complexType name="extensions"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/extensions">
    <xs:annotation>
    <xs:documentation>The extensions container is one way that MultiSpeak objects can be
extended using XSD elements.</xs:documentation>
    </xs:annotation>
    <xs:sequence/>
    </xs:complexType>
    <xs:complexType name="extensionsList"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/extensionsList">

```

```

    <xs:annotation>
      <xs:documentation>This is an optional, self-describing means to extend any MultiSpeak
object.</xs:documentation>
    </xs:annotation>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="gpsLocation"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/gpsLocation">
  <xs:annotation>
    <xs:documentation>A geographical location as collected using a global positioning system
receiver.</xs:documentation>
  </xs:annotation>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="gpsPoint"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/gpsPoint">
  <xs:annotation>
    <xs:documentation>THE gpsPoint OBJECT HAS BEEN DEPRECATED. It will be replaced by the
GPSLocation in V4.2.</xs:documentation>
  </xs:annotation>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="instrumentTransformers"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/instrumentTransformers">
  <xs:annotation>
    <xs:documentation>A set of CT and PT ratios that may be associated with an electric meter
base.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="ct" type="xs:double" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/instrumentTransformers.ct">
      <xs:annotation>
        <xs:documentation>Ratio of the current transformers installed at this meter
base.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="pt" type="xs:double" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/instrumentTransformers.pt">
      <xs:annotation>
        <xs:documentation>Ratio of the potential transformers installed at this meter
base.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="loadMgmt"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/loadMgmt">
  <xs:annotation>
    <xs:documentation>Load management code.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="meterBase"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/meterBase">
  <xs:annotation>

```


`<xs:documentation>`This is an electric meter base. Meter base encapsulates all devices and characteristics of an electric service that remain unchanged when an electric meter is replaced, such as instrument transformers, connect/disconnect devices, etc.`</xs:documentation>`

```

</xs:annotation>
<xs:complexContent>
  <xs:extension base="mspObject">
    <xs:sequence>
      <xs:element name="instrumentTransformers" type="instrumentTransformers" minOccurs="0"
maxOccurs="1" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/meterBase.instrumentTransformers"/>
    </xs:sequence>
  </xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="mspDateTime"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/mspDateTime">
  <xs:annotation>
    <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets.
The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

```

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).`</xs:documentation>`

```

</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspObject" abstract="true"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectID" type="objectID" minOccurs="1"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/mspObject.objectID">
      <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectID is always
required.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>

```

```

<xs:simpleType name="mspPhase"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/mspPhase">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspPointObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/mspPointObject">
  <xs:annotation>
    <xs:documentation>This is an abstract class for objects that exhibit point
geometry.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="facilityID" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/mspPointObject.facilityID">
          <xs:annotation>
            <xs:documentation>Utility string designation for this object.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="GMLLocation" type="PointType" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/mspPointObject.GMLLocation"/>
          <xs:element name="gpsLocation" type="gpsLocation" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/mspPointObject.gpsLocation"/>
            <xs:element name="gpsPoint" type="gpsPoint" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/mspPointObject.gpsPoint">
              <xs:annotation>
                <xs:documentation>The gpsPoint element has been deprecated as of the V4.1.2 Release.
It will be replaced with the GPSLocation in V4.2. It is suggested that you use only the GPSLocation
element for carrying locations expressed in latitude and longitude and the WGS84
datum.</xs:documentation>
              </xs:annotation>
            </xs:element>
            <xs:element name="gridLocation" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/mspPointObject.gridLocation">
              <xs:annotation>
                <xs:documentation>This is a utility-assigned location identifier that may or may not be tied
to any GIS coordinate system. Sometimes referred to by the utility as "map number" or "grid
number".</xs:documentation>
              </xs:annotation>
            </xs:element>
            <xs:element name="rotation" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/mspPointObject.rotation">
              <xs:annotation>
                <xs:documentation>Rotation angle for display of the feature, expressed in degrees
measured clockwise from the vertical.</xs:documentation>
              </xs:annotation>
            </xs:element>
          </xs:sequence>
        </xs:extension>
      </xs:complexContent>
    </xs:complexType>
  <xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/objectID">

```

```

    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:simpleType name="priority"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/priority">
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="rate"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rate">
    <xs:complexContent>
      <xs:extension base="mspObject">
        <xs:sequence>
          <xs:element name="rateCode" type="rateCode" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rate.rateCode">
            <xs:annotation>
              <xs:documentation>This is the string identifier for this rate.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="rateCodeDescription" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rate.rateCodeDescription">
            <xs:annotation>
              <xs:documentation>This string attribute describes the typ of rate. Examples are "Current"
for the currently effective rate. "Peak" for the critical peak pricing rate.
"AverageThisPeriod" for the average rate during the current billing period. "AverageLastPeriod" for the
average during the previous billing period. "MinimumThisPeriod" "MaximumThisPeriod",
etc.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="startTime" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rate.startTime">
            <xs:annotation>
              <xs:documentation>Date and time that this rate goes into effect.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="rateComponents" type="rateComponents" minOccurs="0"
maxOccurs="1" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rate.rateComponents"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="rateCode"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rateCode">
    <xs:annotation>
      <xs:documentation>This is the string identifier for this rate.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="rateComponent"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rateComponent">
    <xs:annotation>
      <xs:documentation>This is the rate for sale of a commodity. For instance, if the current rate for
electrical energy is $0.10/kWh, then the float value 0.10 is carried in the rate base element and the
optional attributes currency code would carry "USD" for dollars, the commodityUnits would carry
"kWh", and the rateDescription would carry "Current".</xs:documentation>
    </xs:annotation>

```

```

<xs:sequence>
  <xs:element name="commodityUnits" type="uom" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rateComponent.commodityUnits">
    <xs:annotation>
      <xs:documentation>This attribute identifies the units in which the commodity being sold is
expressed. For instance, for electrical energy sold at retail this might be "kWh".</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="currencyCode" type="currencyCode" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rateComponent.currencyCode">
    <xs:annotation>
      <xs:documentation>This attribute identifies the currency in which this rate is expressed, as
defined in ISO 4217. For instance, "USD" is United States Dollars.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="rateCode" type="rateCode" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rateComponent.rateCode">
    <xs:annotation>
      <xs:documentation>The string identifier for this rate.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="rateComponentDescription" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rateComponent.rateComponentDescription">
    <xs:annotation>
      <xs:documentation>This string attribute describes the type of rate component. Examples are
"Current" for the currently effective rate. "Peak" for the critical peak pricing rate.
"AverageThisPeriod" for the average rate during the current billing period. "AverageLastPeriod" for
the average during the previous billing period. "MinimumThisPeriod" "MaximumThisPeriod",
etc.</xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="rateComponents"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rateComponents">
  <xs:sequence>
    <xs:element name="rateComponent" type="rateComponent" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/rateComponents.rateComponent"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="realEnergy"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergy">
  <xs:sequence>
    <xs:element name="units" type="realEnergyUnits" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergy.units">
      <xs:annotation>
        <xs:documentation>Units in which this real energy is measured.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="electricService" type="electricService" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergy.electricService"/>
  </xs:sequence>
</xs:complexType>

```

```

<xs:simpleType name="realEnergyUnits"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits">
  <xs:restriction base="xs:string">
    <xs:enumeration value="Wh"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.Wh"/>
    <xs:enumeration value="kWh"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.kWh"/>
    <xs:enumeration value="MWh"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.MWh"/>
    <xs:enumeration value="GWh"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.GWh"/>
    <xs:enumeration value="mWh"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.mWh"/>
    <xs:enumeration value="microWh"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.microWh"/>
    <xs:enumeration value="TWh"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.TWh"/>
    <xs:enumeration value="PWh"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.PWh"/>
    <xs:enumeration value="BTU"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.BTU"/>
    <xs:enumeration value="Therm"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.Therm"/>
    <xs:enumeration value="Erg"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.Erg"/>
    <xs:enumeration value="Calorie"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.Calorie"/>
    <xs:enumeration value="eV"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.eV"/>
    <xs:enumeration value="Joule"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/realEnergyUnits.Joule"/>
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="resource"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/resource">
  <xs:annotation>
    <xs:documentation>Resource is a container object that can have people, vehicles, equipment,
and crews to perform work tasks.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence>
        <xs:element name="mspServiceDeliveryPoint" type="mspServiceDeliveryPoint"
minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/resource.mspServiceDeliveryPoint"/>

```



```

    <xs:element name="serviceLocation" type="serviceLocation" minOccurs="0"
maxOccurs="unbounded" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/resource.serviceLocation"/>
    </xs:sequence>
  </xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="serviceLocation"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceLocation">
  <xs:annotation>
    <xs:documentation>This is a location at which a customer can receive service. A serviceLocation
can have utility services of several different serviceTypes.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspPointObject">
      <xs:sequence>
        <xs:element name="customerID" type="objectID" minOccurs="1"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceLocation.customerID">
          <xs:annotation>
            <xs:documentation>Customer's identification number. This should be the objectID of the
customer.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="description" type="xs:string" minOccurs="1"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceLocation.description">
          <xs:annotation>
            <xs:documentation>Additional information about this serviceLocation</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="electricServiceList" type="electricServiceList" minOccurs="0"
maxOccurs="1" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceLocation.electricServiceList"/>
        <xs:element name="mspServiceDeliveryPoint" type="mspServiceDeliveryPoint"
minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceLocation.mspServiceDeliveryPoint"/>
        <xs:element name="priority" type="priority" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceLocation.priority"/>
        <xs:element name="resource" type="resource" minOccurs="0" maxOccurs="unbounded"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceLocation.resource"/>
        <xs:element name="serviceType" type="serviceType" minOccurs="0"
maxOccurs="unbounded" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceLocation.serviceType"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="serviceLocations"
sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceLocations">
  <xs:sequence>
    <xs:element name="serviceLocation" type="serviceLocation" minOccurs="0"
maxOccurs="unbounded" sawsdl:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceLocations.serviceLocation"/>
  </xs:sequence>
</xs:complexType>

```

```

<xs:simpleType name="serviceType"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceType">
  <xs:annotation>
    <xs:documentation>The type of utility service.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string">
    <xs:enumeration value="Electric"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceType.Electric"/>
    <xs:enumeration value="Gas"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceType.Gas"/>
    <xs:enumeration value="Water"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/serviceType.Water"/>
  </xs:restriction>
</xs:simpleType>
<xs:simpleType name="servStatus"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/servStatus">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="uom" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/uom">
  <xs:annotation>
    <xs:documentation>This is the list of units of measure from ANSI C12.19</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="voltage"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/voltage">
  <xs:sequence>
    <xs:element name="units" type="voltageUnits" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/voltage.units">
      <xs:annotation>
        <xs:documentation>Units of voltage</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="electricService" type="electricService" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/voltage.electricService"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="voltageUnits"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/MultiSpeak V411/voltageUnits">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspServiceDeliveryPoint" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspServiceDeliveryPointConfig/mspServiceDeliveryPoint">
  <xs:sequence/>
</xs:complexType>
</xs:schema>

```

mspServiceLocationConfig

```
<?xml version="1.0" encoding="utf-8"?>
```



```

<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
  xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
  xmlns="http://www.multispeak.org/2012/mspServiceLocationConfig"
  xmlns:msp="http://www.multispeak.org/2012/mspServiceLocationConfig"
  targetNamespace="http://www.multispeak.org/2012/mspServiceLocationConfig"
  elementFormDefault="qualified" attributeFormDefault="unqualified"
  xml:base="http://www.multispeak.org/2012/mspServiceLocationConfig">
  <xs:element name="mspServiceLocationConfig" type="mspServiceLocationConfig"/>
  <xs:complexType name="mspServiceLocationConfig">
    <xs:sequence>
      <xs:element name="serviceLocation" type="serviceLocation" minOccurs="0"
maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="address">
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address">
    <xs:sequence>
      <xs:element name="address1" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.address1">
        <xs:annotation>
          <xs:documentation>Address line 1.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="address2" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.address2">
        <xs:annotation>
          <xs:documentation>Address line 2.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="city" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.city">
        <xs:annotation>
          <xs:documentation>Name of the city or municipality.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="country" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.country"/>
      <xs:element name="detailedAddressFields" type="detailedAddressFields" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/address.detailedAddressFields">
        <xs:annotation>
          <xs:documentation>Detailed information about the components of the composite address
lines.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="postalCode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.postalCode">
        <xs:annotation>
          <xs:documentation>The postal code. For instance, in the United States this is the zip
code.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="state" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.state">
        <xs:annotation>
          <xs:documentation>The state or province.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="townCode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.townCode">
        <xs:annotation>

```

```

    <xs:documentation>Additional information about the city or town, if
required.</xs:documentation>
  </xs:annotation>
</xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="addressItem"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/addressItem">
  <xs:sequence>
    <xs:element name="address" type="address" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/addressItem.address"/>
    <xs:element name="addressType" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/addressItem.addressType">
      <xs:annotation>
        <xs:documentation>Type of address. For instance, billing, mailing, service location,
etc.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="priorityOrder" type="priorityOrder" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/addressItem.priorityOrder">
      <xs:annotation>
        <xs:documentation>Order that should be used to contact this person. First = 1, second = 2,
etc.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="addressList"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/addressList">
  <xs:sequence>
    <xs:element name="addressItem" type="addressItem" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/addressList.addressItem">
      <xs:annotation>
        <xs:documentation>This association is reversed from its normal representation in the model to
get this profile to generate correctly.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="objectID" type="objectID" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/addressList.objectID"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="detailedAddressFields"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/detailedAddressFields">
  <xs:sequence>
    <xs:element name="addressGeneral" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.addressGeneral"/>
    <xs:element name="buildingNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.buildingNumber"/>
    <xs:element name="streetNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.streetNumber"/>
    <xs:element name="streetPrefix" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.streetPrefix"/>
    <xs:element name="streetSuffix" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.streetSuffix"/>
  </xs:sequence>

```

```

    <xs:element name="suiteNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.suiteNumber"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="eMailAddress"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailAddress">
  <xs:sequence>
    <xs:element name="eMail" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailAddress.eMail"/>
    <xs:element name="eMailType" type="eMailType" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/eMailAddress.eMailType">
      <xs:annotation>
        <xs:documentation>The type of e-mail address.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="eMailType"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailType">
  <xs:annotation>
    <xs:documentation>Type of e-mail address.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="gpsPoint"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/gpsPoint">
  <xs:annotation>
    <xs:documentation>THE gpsPoint OBJECT HAS BEEN DEPRECATED. It will be replaced by the
GPSLocation in V4.2.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="altitude" type="xs:double" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/gpsPoint.altitude">
      <xs:annotation>
        <xs:documentation>The altitude above mean sea level for this geographical location,
assuming the use of the WGS84 geodetic datum.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="latitude" type="xs:double" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/gpsPoint.latitude">
      <xs:annotation>
        <xs:documentation>The latitude portion of a geographical location.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="longitude" type="xs:double" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/gpsPoint.longitude">
      <xs:annotation>
        <xs:documentation>The longitude portion of a geographical location</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="objectID" type="objectID" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/gpsPoint.objectID"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="gridLocation"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/gridLocation">
  <xs:annotation>
    <xs:documentation>This is a string identifier for a traditional map location that is not based on
either an x,y or lat/long coordinate system.</xs:documentation>
  </xs:annotation>

```

```

    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="inspection"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/inspection">
    <xs:sequence>
      <xs:element name="elementStatus" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/inspection.elementStatus">
        <xs:annotation>
          <xs:documentation>The status of the inspected item. For instance: Okay, Needs Attention,
Needs Immediate Attention, Not Functional, or Okay as Left.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="mspDateTime"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspDateTime">
    <xs:annotation>
      <xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets.
The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

```

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).</xs:documentation>

```

    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="mspMultiGeometryObject"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspMultiGeometryObject">
    <xs:sequence>
      <xs:element name="gridLocation" type="gridLocation" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/mspMultiGeometryObject.gridLocation">
        <xs:annotation>
          <xs:documentation>This is a utility-assigned location identifier that may or may not be tied to
any GIS coordinate system. Sometimes referred to by the utility as "map number" or "grid
number".</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/objectID">
    <xs:restriction base="xs:string"/>
  </xs:simpleType>

```

```

<xs:complexType name="otherContactItem"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/otherContactItem">
  <xs:annotation>
    <xs:documentation>This is an element to carry other miscellaneous contact information, such as
web site address, truck radio number, etc.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="details" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/otherContactItem.details">
      <xs:annotation>
        <xs:documentation>This element carries the miscellaneous contact
information.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="infoType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/otherContactItem.infoType">
      <xs:annotation>
        <xs:documentation>This element is used to describe the type of information stored in the
"details" element. Examples might be: IP address, truck number, radio number.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="priorityOrder"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/priorityOrder">
  <xs:annotation>
    <xs:documentation>Order in which this contact information type should be used to contact this
person. First = 1, second =2, etc.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="serviceLocation"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/serviceLocation">
  <xs:annotation>
    <xs:documentation>This is a location at which a customer can receive service. A serviceLocation
can have utility services of several different serviceTypes.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="customerID" type="objectID" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/serviceLocation.customerID">
      <xs:annotation>
        <xs:documentation>Customer's identification number. This should be the objectID of the
customer.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="description" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/serviceLocation.description">
      <xs:annotation>
        <xs:documentation>Additional information about this serviceLocation</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="facilityName" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/serviceLocation.facilityName">
      <xs:annotation>
        <xs:documentation>Name of facility for multi-site business customer.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>

```



```

    <xs:element name="SIC" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/serviceLocation.SIC">
    <xs:annotation>
    <xs:documentation>The standard industry code for the business activity at this service
location, if this service location is commercial.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="siteID" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/serviceLocation.siteID">
    <xs:annotation>
    <xs:documentation>Identifier for site for a multi-site business customer.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="addressList" type="addressList" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/serviceLocation.addressList"/>
    <xs:element name="eMailAddress" type="eMailAddress" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/serviceLocation.eMailAddress"/>
    <xs:element name="gpsPoint" type="gpsPoint" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/serviceLocation.gpsPoint"/>
    <xs:element name="inspection" type="inspection" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/serviceLocation.inspection"/>
    <xs:element name="mspMultiGeometryObject" type="mspMultiGeometryObject" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/serviceLocation.mspMultiGeometryObject"/>
    <xs:element name="otherContactItem" type="otherContactItem" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/serviceLocation.otherContactItem"/>
    <xs:element name="telephoneNumber" type="telephoneNumber" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/serviceLocation.telephoneNumber"/>
    </xs:sequence>
</xs:complexType>
<xs:complexType name="telephoneNumber"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/telephoneNumber">
    <xs:sequence>
    <xs:element name="areaCode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/telephoneNumber.areaCode">
    <xs:annotation>
    <xs:documentation>Area or region code.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="cityCode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/telephoneNumber.cityCode">
    <xs:annotation>
    <xs:documentation>City code for telephone number, if applicable.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="countryCode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/telephoneNumber.countryCode">
    <xs:annotation>
    <xs:documentation>Country code for telephone number, if applicable.</xs:documentation>
    </xs:annotation>
    </xs:element>

```

```

    <xs:element name="extension" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/telephoneNumber.extension">
    <xs:annotation>
    <xs:documentation>Extension for telephone number if required.</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="localNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/telephoneNumber.localNumber">
    <xs:annotation>
    <xs:documentation>The primary portion of a telephone number as dialed in a local calling
area.</xs:documentation>
    </xs:annotation>
    </xs:element>
</xs:sequence>
</xs:complexType>
</xs:schema>

```

mspSupplierConfig

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mspSupplierConfig"
xmlns:msp="http://www.multispeak.org/2012/mspSupplierConfig"
targetNamespace="http://www.multispeak.org/2012/mspSupplierConfig"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mspSupplierConfig">
  <xs:element name="mspSupplierConfig" type="mspSupplierConfig"/>
  <xs:complexType name="mspSupplierConfig">
    <xs:sequence>
      <xs:element name="resource" type="resource" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="address">
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address">
    <xs:sequence>
      <xs:element name="address1" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.address1">
        <xs:annotation>
        <xs:documentation>Address line 1.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="address2" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.address2">
        <xs:annotation>
        <xs:documentation>Address line 2.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="city" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.city">
        <xs:annotation>
        <xs:documentation>Name of the city or municipality.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="country" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.country"/>
      <xs:element name="detailedAddressFields" type="detailedAddressFields" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/address.detailedAddressFields">

```



```

        <xs:annotation>
          <xs:documentation>Detailed information about the components of the composite address
lines.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="postalCode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.postalCode">
        <xs:annotation>
          <xs:documentation>The postal code. For instance, in the United States this is the zip
code.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="state" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.state">
        <xs:annotation>
          <xs:documentation>The state or province.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="townCode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/address.townCode">
        <xs:annotation>
          <xs:documentation>Additional information about the city or town, if
required.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="addressItem"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/addressItem">
    <xs:sequence>
      <xs:element name="address" type="address" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/addressItem.address"/>
      <xs:element name="addressType" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/addressItem.addressType">
        <xs:annotation>
          <xs:documentation>Type of address. For instance, billing, mailing, service location,
etc.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="priorityOrder" type="priorityOrder" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/addressItem.priorityOrder">
        <xs:annotation>
          <xs:documentation>Order that should be used to contact this person. First = 1, second = 2,
etc.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="addressList" type="addressList" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/addressItem.addressList"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="addressList"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/addressList">
    <xs:sequence>
      <xs:element name="contactInfo" type="contactInfo" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/addressList.contactInfo"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="contactInfo"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/contactInfo">

```

```

    <xs:annotation>
      <xs:documentation>This class contains ways to contact an entity.</xs:documentation>
    </xs:annotation>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="detailedAddressFields"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/detailedAddressFields">
  <xs:sequence>
    <xs:element name="addressGeneral" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.addressGeneral"/>
    <xs:element name="buildingNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.buildingNumber"/>
    <xs:element name="postOfficeBox" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.postOfficeBox"/>
    <xs:element name="region" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.region"/>
    <xs:element name="streetNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.streetNumber"/>
    <xs:element name="streetPrefix" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.streetPrefix"/>
    <xs:element name="streetSuffix" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.streetSuffix"/>
    <xs:element name="streetType" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.streetType"/>
    <xs:element name="suiteNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/detailedAddressFields.suiteNumber"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="eMailAddress"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailAddress">
  <xs:sequence>
    <xs:element name="eMail" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailAddress.eMail"/>
    <xs:element name="eMailType" type="eMailType" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/eMailAddress.eMailType">
      <xs:annotation>
        <xs:documentation>The type of e-mail address.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="priorityOrder" type="priorityOrder" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/eMailAddress.priorityOrder">
      <xs:annotation>
        <xs:documentation>Order in which these e-mail addresses should be used to contact this
person.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="eMailList" type="eMailList" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailAddress.eMailList"/>
  </xs:sequence>
</xs:complexType>

```

```

<xs:complexType name="eMailList"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailList">
  <xs:sequence>
    <xs:element name="contactInfo" type="contactInfo" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailList.contactInfo"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="eMailType"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailType">
  <xs:annotation>
    <xs:documentation>Type of e-mail address.</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string">
    <xs:enumeration value="Personal"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailType.Personal"/>
    <xs:enumeration value="Business"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailType.Business"/>
    <xs:enumeration value="Alternate"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailType.Alternate"/>
    <xs:enumeration value="Other"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/eMailType.Other"/>
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspObject.objectID">
      <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectID is always
required.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="utility" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspObject.utility">
      <xs:annotation>
        <xs:documentation>String that is used to identify the owner of this item. It may be a text string
or a pointer to an organization.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/objectID">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="otherContactInformation"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/otherContactInformation">
  <xs:sequence>
    <xs:element name="contactInfo" type="contactInfo" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/otherContactInformation.contactInfo"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="otherContactItem"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/otherContactItem">
  <xs:annotation>

```

```

    <xs:documentation>This is an element to carry other miscellaneous contact information, such as
web site address, truck radio number, etc.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="details" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/otherContactItem.details">
      <xs:annotation>
        <xs:documentation>This element carries the miscellaneous contact
information.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="infoType" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/otherContactItem.infoType">
      <xs:annotation>
        <xs:documentation>This element is used to describe the type of information stored in the
"details" element. Examples might be: IP address, truck number, radio number.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="otherContactInformation" type="otherContactInformation" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/otherContactItem.otherContactInformation"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="phoneList"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneList">
  <xs:sequence>
    <xs:element name="contactInfo" type="contactInfo" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneList.contactInfo"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="phoneNumber"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneNumber">
  <xs:sequence>
    <xs:element name="phone" type="telephoneNumber" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneNumber.phone"/>
    <xs:element name="phoneType" type="phoneType" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/phoneNumber.phoneType">
      <xs:annotation>
        <xs:documentation>The type of phone number.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="priorityOrder" type="priorityOrder" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/phoneNumber.priorityOrder">
      <xs:annotation>
        <xs:documentation>Order in which these phone numbers should be used to contact this
person.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="phoneList" type="phoneList" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/phoneNumber.phoneList"/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="phoneType"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneType">
  <xs:annotation>
    <xs:documentation>Type of phone number.</xs:documentation>
  </xs:annotation>

```

```

    <xs:restriction base="xs:string">
      <xs:enumeration value="Home"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneType.Home"/>
      <xs:enumeration value="Business"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneType.Business"/>
      <xs:enumeration value="Mobile"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneType.Mobile"/>
      <xs:enumeration value="Business_mobile"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneType.Business
mobile"/>
      <xs:enumeration value="Pager"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneType.Pager"/>
      <xs:enumeration value="Fax"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneType.Fax"/>
      <xs:enumeration value="Other"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/phoneType.Other"/>
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="priorityOrder"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/priorityOrder">
    <xs:annotation>
      <xs:documentation>Order in which this contact information type should be used to contact this
person. First = 1, second =2, etc.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="resource"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/resource">
    <xs:annotation>
      <xs:documentation>Resource is a container object that can have people, vehicles, equipment,
and crews to perform work tasks.</xs:documentation>
    </xs:annotation>
    <xs:complexContent>
      <xs:extension base="mspObject">
        <xs:sequence>
          <xs:element name="contactInfo" type="contactInfo" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/resource.contactInfo"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="telephoneNumber"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/telephoneNumber">
    <xs:sequence>
      <xs:element name="areaCode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/telephoneNumber.areaCode">
        <xs:annotation>
          <xs:documentation>Area or region code.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="cityCode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/telephoneNumber.cityCode">
        <xs:annotation>
          <xs:documentation>City code for telephone number, if applicable.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="countryCode" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/telephoneNumber.countryCode">
        <xs:annotation>

```



```

        <xs:documentation>Country code for telephone number, if applicable.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="extension" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/telephoneNumber.extension">
      <xs:annotation>
        <xs:documentation>Extension for telephone number if required.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="localNumber" type="xs:string" minOccurs="0"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/telephoneNumber.localNumber">
      <xs:annotation>
        <xs:documentation>The primary portion of a telephone number as dialed in a local calling
area.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

mspTransactionRecord

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:sawSDL="http://www.w3.org/ns/sawSDL"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:cimp="http://www.iec.org/CimProfile"
xmlns="http://www.multispeak.org/2012/mspTransactionRecord"
xmlns:msp="http://www.multispeak.org/2012/mspTransactionRecord"
targetNamespace="http://www.multispeak.org/2012/mspTransactionRecord"
elementFormDefault="qualified" attributeFormDefault="unqualified"
xml:base="http://www.multispeak.org/2012/mspTransactionRecord">
  <xs:element name="mspTransactionRecord" type="mspTransactionRecord"/>
  <xs:complexType name="mspTransactionRecord">
    <xs:sequence>
      <xs:element name="Transaction" type="Transaction" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="account"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/account">
    <xs:annotation>
      <xs:documentation>This is a customer account.</xs:documentation>
    </xs:annotation>
    <xs:complexContent>
      <xs:extension base="mspObject">
        <xs:sequence/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="accountNumber"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/accountNumber">
    <xs:annotation>
      <xs:documentation>Account number for this service location. Depending on the system of record,
this may be either the account.objectID or the account.objectName.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
  </xs:simpleType>
  <xs:complexType name="allowableTransactionTypes"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/allowableTransactionTypes">
    <xs:sequence>

```

```

    <xs:element name="transactionType" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/allowableTransactionTypes.transactionType">
    <xs:annotation>
    <xs:documentation>Transaction types are the valid payment transaction nouns allowed for
this customer (Example: creditCardPayment, eCheckPayment, achPayment, cashPayment, or
sendToOperator).</xs:documentation>
    </xs:annotation>
    </xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="convenienceFeeItem"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/convenienceFeeItem">
    <xs:sequence>
    <xs:element name="convenienceFee" type="money" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/convenienceFeeItem.convenienceFee"/>
    <xs:element name="responseCode" type="responseCode" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/convenienceFeeItem.responseCode">
    <xs:annotation>
    <xs:documentation>Enumeration includes, "Success". "Failure" and
"Error".</xs:documentation>
    </xs:annotation>
    </xs:element>
    <xs:element name="transactionID" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/convenienceFeeItem.transactionID"/>
    <xs:element name="utilityFee" type="money" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/convenienceFeeItem.utilityFee"/>
    </xs:sequence>
</xs:complexType>
<xs:complexType name="convenienceFeeList"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/convenienceFeeList">
    <xs:sequence>
    <xs:element name="convenienceFeeItem" type="convenienceFeeItem" minOccurs="1"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/convenienceFeeList.convenienceFeeItem"/>
    </xs:sequence>
</xs:complexType>
<xs:simpleType name="currencyCode"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/currencyCode">
    <xs:annotation>
    <xs:documentation>Three letter currency code as standardized in ISO4217-2001. Typically is the
two letter Internet domain code for the country followed by a one letter designator for the currency; for
instance the USA dollar is USD.</xs:documentation>
    </xs:annotation>
    <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="description"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/description">
    <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="money"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/money">
    <xs:sequence/>
</xs:complexType>
<xs:simpleType name="mspDateTime"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspDateTime">
    <xs:annotation>

```


<xs:documentation>ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator “T”, in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

ISO 8601 date time. All dateTime fields will be formatted with UTC offsets. The use of "Z" is acceptable to denote UTC or "Zulu" time.all dates will be formatted as:

2005-06-15T11:30:22.56-07:00

where the date is in YYYY-MM-DD format and the time is in military time, beginning after the date-time separator "T", in hh:mm:ss.ss format, followed by the time zone offset. The use of fractional seconds is optional and any level of decimal precision in seconds is supported. The time zone offset is shown as hh:mm offset from GMT, with either a positive offset (east of the prime meridian) or negative offset (west of the prime meridian).

```

</xs:annotation>
<xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="mspMeter"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspMeter">
  <xs:annotation>
    <xs:documentation>This is an abstract class from which all meter device classes
inherit.</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="mspObject">
      <xs:sequence/>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="mspObject" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspObject">
  <xs:annotation>
    <xs:documentation>This is the class from which all classes that can be acted upon inherit (either
directly or indirectly).</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="objectID" type="objectID" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/mspObject.objectID">
      <xs:annotation>
        <xs:documentation>This is the key field for this object type. The use of objectID is always
required.</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="objectID"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/objectID">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="payableItem"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/payableItem">
  <xs:annotation>
    <xs:documentation>An item to which a customer payment could be applied.</xs:documentation>
  </xs:annotation>

```

```

<xs:complexContent>
  <xs:extension base="mspObject">
    <xs:sequence>
      <xs:element name="chargeCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/payableItem.chargeCode"/>
      <xs:element name="currentAmount" type="money" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/payableItem.currentAmount"/>
      <xs:element name="description" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/payableItem.description"/>
      <xs:element name="dueDate" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/payableItem.dueDate"/>
    </xs:sequence>
  </xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="payableItemList"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/payableItemList">
  <xs:annotation>
    <xs:documentation>List of items to which a customer payment could be
applied.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="allowableTransactionTypes" type="allowableTransactionTypes"
minOccurs="0" maxOccurs="1" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/payableItemList.allowableTransactionTypes"/>
    <xs:element name="payableItem" type="payableItem" minOccurs="1" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/payableItemList.payableItem"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="paymentTransaction"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/paymentTransaction">
  <xs:sequence>
    <xs:element name="accountNumber" type="accountNumber" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/paymentTransaction.accountNumber">
      <xs:annotation>
        <xs:documentation>Account number for customer's utility account.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="typeService" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/paymentTransaction.typeService"/>
    <xs:element name="account" type="account" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/paymentTransaction.account"/>
    <xs:element name="convenienceFeeList" type="convenienceFeeList" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/paymentTransaction.convenienceFeeList"/>
    <xs:element name="mspMeter" type="mspMeter" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/paymentTransaction.mspMeter"/>
    <xs:element name="payableItemList" type="payableItemList" minOccurs="0"
maxOccurs="unbounded" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/paymentTransaction.payableItemList"/>
    <xs:element name="rateInfo" type="rateInfo" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/paymentTransaction.rateInfo"/>
  </xs:sequence>

```

```

</xs:complexType>
<xs:complexType name="rate"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/rate">
  <xs:annotation>
    <xs:documentation>This is the rate for sale of a commodity. For instance, if the current rate for
    electrical energy is $0.10/kWh, then the float value 0.10 is carried in the rate base element and the
    optional attributes currency code would carry "USD" for dollars, the commodityUnits would carry
    "kWh", and the rateDescription would carry "Current".
  </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="commodityUnits" type="uom" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/rate.commodityUnits">
      <xs:annotation>
        <xs:documentation>This attribute identifies the units in which the commodity being sold is
        expressed. For instance, for electrical energy sold at retail this might be "kWh".
      </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="currencyCode" type="currencyCode" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/rate.currencyCode">
      <xs:annotation>
        <xs:documentation>This attribute identifies the currency in which this rate is expressed, as
        defined in ISO 4217. For instance, "USD" is United States Dollars.
      </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="rateCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/rate.rateCode">
      <xs:annotation>
        <xs:documentation>The string identifier for this rate.
      </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="rateDescription" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/rate.rateDescription"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="rateInfo"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/rateInfo">
  <xs:sequence>
    <xs:element name="dateTime" type="mspDateTime" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/rateInfo.dateTime"/>
    <xs:element name="rateCode" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/rateInfo.rateCode">
      <xs:annotation>
        <xs:documentation>This is the string identifier for this rate.
      </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="rateCodeDescription" type="xs:string" minOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/rateInfo.rateCodeDescription">
      <xs:annotation>
        <xs:documentation>This string attribute describes the type of rate. Examples are "Current" for
        the currently effective rate. "Peak" for the critical peak pricing rate.
        "AverageThisPeriod" for the average rate during the current billing period. "AverageLastPeriod" for the
        average during the previous billing period. "MinimumThisPeriod" "MaximumThisPeriod",
        etc.
      </xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>

```

```

    <xs:element name="rates" type="rates" minOccurs="0" maxOccurs="1"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/rateInfo.rates"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="rates"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/rates">
  <xs:sequence>
    <xs:element name="rate" type="rate" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/rates.rate"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="realEnergy"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/realEnergy">
  <xs:sequence/>
</xs:complexType>
<xs:simpleType name="responseCode"
sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak V411/responseCode">
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:simpleType name="uom" sawSDL:modelReference="http://www.multispeak.org/2012/MultiSpeak
V411/uom">
  <xs:annotation>
    <xs:documentation>This is the list of units of measure from ANSI C12.19</xs:documentation>
  </xs:annotation>
  <xs:restriction base="xs:string"/>
</xs:simpleType>
<xs:complexType name="Transaction" abstract="true"
sawSDL:modelReference="http://www.multispeak.org/2012/Transaction">
  <xs:sequence>
    <xs:element name="allowableTransactionTypes" type="allowableTransactionTypes"
minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/Transaction.allowableTransactionTypes"/>
    <xs:element name="description" type="description" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/Transaction.description"/>
    <xs:element name="mspObject" type="mspObject" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/Transaction.mspObject"/>
    <xs:element name="realEnergy" type="realEnergy" minOccurs="0" maxOccurs="unbounded"
sawSDL:modelReference="http://www.multispeak.org/2012/Transaction.realEnergy"/>
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

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INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch